AYURVEDA • POSTGRADUATE STUDIES

MD / MS AYURVEDA

Rasashastra & Bhaishajya Kalpana | Semester II

TEXTBOOK OF

Applied Basics of Rasashastra and Bhaishajya Kalpana

AYPG – AB – RB

Module-wise and Unit-wise Study Notes

Prepared strictly as per the NCISM Curriculum

8 Modules • 42 Units • 16 Credits • 480 Notional Hours

Theory • Practical Training • Experiential Learning

Complete Question Bank with ABQ, LAQ and SAQ

AUTHOR

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BAMS, MD (Ayurveda), PhD (Scholar)

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Parul Institute of Ayurved and Research, Parul University

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Parul Institute of Ayurved and Research

Parul University, Vadodara, Gujarat, Bharat

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Applied Basics of Rasashastra and Bhaishajya Kalpana

For Postgraduate Scholars of Ayurveda

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Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana
Module-wise and Unit-wise Study Notes for MD / MS (Ayurveda), Semester II
Course code AYPG-AB-RB, National Commission for Indian System of Medicine

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NOTE ON PREPARATION

This book was prepared by the author with the assistance of **Claude Opus**, a large language model developed by Anthropic. The choice of subject matter, the mapping to the NCISM curriculum, the selection and verification of references, and the review of all content were directed and carried out by the author, who accepts sole and full responsibility for its contents.

DISCLAIMER

This book is intended solely for the academic use of postgraduate scholars and teachers of Ayurveda. Rasashastra and Bhaishajya Kalpana involve substances that are toxic in their raw state, and procedures involving high temperature, corrosive media and hazardous materials. **No procedure described in this book should be attempted except in a licensed pharmacy or a recognised teaching Rasashala, under the direct supervision of a qualified teacher, and in accordance with the prescribed safety measures and statutory requirements.** Regulatory provisions, pharmacopoeial limits and official guidelines are revised from time to time; readers must verify the current position from the primary source before relying on it. Nothing in this book constitutes clinical advice or a warranty of any kind. The author and the publisher accept no liability for any loss or injury arising from its use.

TYPESET IN

Liberation Serif and Liberation Sans. Set on A4 with mirrored margins for double-sided printing and binding.

Dedicated to

all the Postgraduate Scholars of

Rasashastra and Bhaishajya Kalpana

FOREWORD



Postgraduate education in Rasashastra and Bhaishajya Kalpana has long needed a text a scholar could depend on from the first page to the last, and this book supplies exactly that. Its structure follows the National Commission's prescribed curriculum with unusual discipline, and its explanations rarely stray beyond what a postgraduate examination genuinely requires.

What distinguishes this work, in my judgement, is the care with which it separates settled knowledge from open questions. Wherever a claim about our traditional preparations can be supported by published data, the author states it directly. Wherever the supporting evidence has not yet reached that standard, he says so without hesitation. Such candour is rarer in our field than it ought to be, and a young scholar who absorbs it early will be far better equipped for genuine research than one who does not.

The book also succeeds in showing where modern instrumentation and classical technique genuinely inform one another, without collapsing the distinction between the two or claiming more agreement than the facts allow. That balance is not easily struck, and the author has struck it with maturity.

I have known Dr Milind Chatrabhuji as a scholar for a considerable time, and this book reflects the same seriousness I have long associated with him. That it is offered without cost, for any institution to copy and circulate freely, only extends its value further.

I recommend it without reservation to postgraduate scholars of Rasashastra and Bhaishajya Kalpana, and to the teachers who guide them.

Dr Rohit Gokarn

MD (Ayurveda), PhD — Rasashastra and Bhaishajya Kalpana
Professor, Department of Rasashastra and Bhaishajya Kalpana
Principal, Shree Swaminarayan Ayurvedic College
Kalol, Gujarat

PREFACE

This book began as a set of notes for my own students, and grew because the need was larger than I had assumed.

When the National Commission for Indian System of Medicine introduced the revised postgraduate curriculum, the subject *Applied Basics of Rasashastra and Bhaishajya Kalpana* arrived with a scope that surprised many of us who teach it. In eight modules it asks a scholar to move from pharmaceutics to databases, from atomic structure to metallurgy, from chromatography to animal ethics, from quality control to nanotechnology. Read as a list, it looks like eight unrelated subjects placed side by side.

It is not. Read it again and a single question runs through all of it: **what exactly are we making, and how do we know?** Module 4 gives the process, Module 3 the chemistry that explains it, Module 5 the instruments that measure it, Module 7 the standards that control it, Module 1 the law that governs it, and Module 8 asks what it all means at the scale where these materials actually work. Once a scholar sees that thread, the course stops being a burden and becomes an education.

My difficulty, and my students' difficulty, was that no single source followed this curriculum. Material existed, but it was scattered across classical texts, pharmacopoeias, instrument manuals, government guidelines and journal papers. A scholar preparing for the examination had to assemble it, and much time went into assembling rather than understanding. This book is an attempt to do that assembling once, so that it need not be done again.

Four decisions shaped the writing.

First, the language is deliberately simple. Short sentences. Plain words. Sanskrit terms in ordinary Roman letters, without diacritical marks. A scholar who reads English as a second or third language should not have to fight the prose in order to reach the subject. If a passage here seems simply written, that is not because the topic is simple.

Second, the curriculum is followed exactly. Every module, every unit, every sub-point carries the number the curriculum gives it. Nothing has been added and nothing left out. Each unit states its session learning objectives and, from Table 6F, whether it can be asked as a short answer, a long answer or an analytical question. A scholar can therefore see precisely what is expected before beginning to read.

Third, no reference has been invented. Where a source could not be verified, it has been left out rather than guessed at. Where current data was used — a market figure, the present position of a law, the name of a regulatory body — the issuing authority has been named in the text so that the reader can check it. Regulations change; a book that hides its sources cannot be corrected.

Fourth, and most important to me, the boundary between what is established and what is proposed has been marked plainly. Our subject attracts enthusiastic claims. Some are well supported; some are not yet. It is written here, for instance, that published characterisation of Bhasma reports particles in the micrometre and nanometre range, and that organic material from the Bhavana media is found associated with the particle surface. It is also written that whether this amounts to true metal-carbon bonding, and

whether the nanoscale fraction is what produces the therapeutic effect, remains under investigation. Both halves of that statement belong in a postgraduate textbook. A scholar who cannot tell a finding from a hypothesis has not yet been trained, and a tradition defended by overstatement is not defended at all.

A word on what our Acharyas achieved, since this book returns to it often. Rekhapurnata is a particle size test. Varitara is a surface property test. Apunarbhava is a test for free metal. Tantuvat is a test of concentration. The Musha is a refractory formulation with a body, a grog, a burnout material and a fibre reinforcement. Mitra Panchaka is a complete smelting charge of flux and reductant. These were not approximations of a science our predecessors lacked. They were that science, done with the instruments available. What our generation has been given is not better judgement but better instruments, and the obligation that comes with them is to verify, not merely to admire.

This book is offered free of cost. Any scholar, teacher or institution may download it, print it and pass it on, and I hope many will. It is a first edition and it will contain errors. I would be grateful to any reader who takes the trouble to point them out, so that the next edition may be better than this one.

Dr Milind Pareshkumar Chatrabhuji

Vadodara, Gujarat • 2026

ACKNOWLEDGEMENTS

My first and deepest thanks are to my wife, **Dr Smita Chatrabhuji**. This book was written in hours that belonged to her and to our family, and it was written only because she gave those hours without once being asked to. Whatever is of use in these pages exists because of her patience.

I thank **my family**, who have borne, over many years and many academic assignments, a husband, a father and a son who was often present in the house but absent from it. Their forbearance is not acknowledged often enough, and I acknowledge it here.

I record my gratitude to **Dr Rohit Gokarn**, Professor and Principal, Shree Swaminarayan Ayurvedic College, Kalol, under whose guidance I am pursuing my doctoral work. What I understand of research in this subject, and of the discipline of saying only what the evidence permits, I learned substantially from him. I am grateful to him for his Foreword.

I thank **Parul Institute of Ayurved and Research, Parul University**, and my colleagues in the PG Department of Rasashastra and Bhaishajya Kalpana, for the academic environment in which this work was possible.

I thank the **National Commission for Indian System of Medicine** for a curriculum that, whatever its demands on teacher and scholar alike, has genuinely widened the scientific training of our postgraduates.

Finally, I thank **my students**. Their questions in the classroom and in the Rasashala decided what needed explaining and what did not. This book is theirs before it is mine.

Dr Milind Pareshkumar Chatrabhuji

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ABBREVIATIONS

Regulatory, statutory and institutional

AFI	Ayurvedic Formulary of India
API	Ayurvedic Pharmacopoeia of India (also, in Module 1, Active Pharmaceutical Ingredient)
ANDA	Abbreviated New Drug Application
ASU	Ayurveda, Siddha and Unani
ASUDCC	Ayurveda, Siddha, Unani Drugs Consultative Committee
ASUDTAB	Ayurveda, Siddha, Unani Drugs Technical Advisory Board
AYUSH	Ayurveda, Yoga and Naturopathy, Unani, Siddha, Sowa-Rigpa and Homoeopathy
CCRAS	Central Council for Research in Ayurvedic Sciences
CCSEA	Committee for Control and Supervision of Experiments on Animals (formerly CPCSEA)
CDSCO	Central Drugs Standard Control Organisation
CTD	Common Technical Document
CTRI	Clinical Trials Registry — India
DBT	Department of Biotechnology
DCGI	Drugs Controller General of India
EMA	European Medicines Agency
GMP / cGMP	Good Manufacturing Practices / current Good Manufacturing Practices
IAEC	Institutional Animal Ethics Committee
ICH	International Council for Harmonisation
ICMR	Indian Council of Medical Research
IND	Investigational New Drug application
NCISM	National Commission for Indian System of Medicine
NDA	New Drug Application
NMPB	National Medicinal Plants Board
PCIM&H	Pharmacopoeia Commission for Indian Medicine and Homoeopathy
QA / QC	Quality Assurance / Quality Control
SOP	Standard Operating Procedure
TKDL	Traditional Knowledge Digital Library
USFDA	United States Food and Drug Administration
USGS	United States Geological Survey
WHO	World Health Organization

Analytical and instrumental

AAS	Atomic Absorption Spectroscopy
AFM	Atomic Force Microscope
BCS	Biopharmaceutics Classification System
DLS	Dynamic Light Scattering
DTA / DSC	Differential Thermal Analysis / Differential Scanning Calorimetry
EDAX	Energy Dispersive X-ray Analysis
FTIR	Fourier Transform Infrared Spectroscopy
GC / GC-MS	Gas Chromatography / Gas Chromatography-Mass Spectrometry
HPLC	High Performance Liquid Chromatography
HPTLC	High Performance Thin Layer Chromatography
ICP-MS	Inductively Coupled Plasma Mass Spectrometry
ICP-OES	Inductively Coupled Plasma Optical Emission Spectrometry
LC-MS	Liquid Chromatography-Mass Spectrometry
NMR	Nuclear Magnetic Resonance
PDI	Polydispersity Index
PSA	Particle Size Analyser
SEM	Scanning Electron Microscope
SLN / NLC	Solid Lipid Nanoparticle / Nanostructured Lipid Carrier
STM	Scanning Tunnelling Microscope
TEM	Transmission Electron Microscope
TGA	Thermogravimetric Analysis
TLC	Thin Layer Chromatography
UV-Vis	Ultraviolet-Visible Spectroscopy
VSM	Vibrating Sample Magnetometer
XRD	X-Ray Diffraction
XRF	X-Ray Fluorescence

Examination and curriculum

ABQ	Analytical Based Question (20 marks)
CO / PO	Course Outcome / Programme Outcome
LAQ	Long Answer Question (10 marks)
SAQ	Short Answer Question (5 marks)
M1U1	Module 1, Unit 1 (and so on throughout)

Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana

HOW TO USE THIS BOOK

These notes are written for the postgraduate student of **Rasashastra and Bhaishajya Kalpana**. They follow the NCISM curriculum of **AYPG-AB-RB — Applied Basics of Rasashastra and Bhaishajya Kalpana**, Semester II, exactly as it is printed. No topic has been added. No topic has been left out.

How the book is arranged

The book has eight chapters. One chapter for one module. Inside each module, the units come in the same order as the curriculum. Inside each unit, the sub-points carry the same numbers as the curriculum (1.1, 1.2, 1.3 and so on). So you can keep the curriculum in one hand and these notes in the other, and never lose your place.

Each unit is arranged like this.

Part of the unit	What it gives you
Syllabus points	The exact wording of the curriculum
Session learning objectives	What the curriculum expects you to be able to do
Exam weightage	Whether the unit can come as ABQ, SAQ or LAQ
Main notes	The full explanation, point by point
Key points to remember	A short revision list
Exam corner	Likely questions and how to attempt them
References used	The exact reference numbers from the NCISM list

At the end of every module you will also find:

- **Practical Training activities**, written up as procedure, requirements, observation and precautions
- **Experiential Learning activities**, with the method of doing and the method of recording
- **Viva pointers** for the practical examination
- **Module question bank**

How to read

1. First read the syllabus points. Know what you have to learn.
2. Then read the main notes once, slowly, without trying to memorise.
3. Read them a second time and mark the tables. Most examiners give marks for a well-drawn table.
4. Revise from “Key points to remember”.
5. Practise writing answers from “Exam corner”.

A word about the language

The sentences here are kept short on purpose. Long sentences hide the meaning. If a sentence looks simple, that is not because the topic is simple. It is because you should be able to read it once and understand it.

Ayurvedic terms are written in plain Roman letters. No diacritical marks are used. Shodhana is written as Shodhana, not with special symbols. This is easier to read and easier to write in the examination.

About the references

Every reference used in this book comes from the official NCISM reference list for this course. The reference numbers given at the end of each unit are the same numbers used in the curriculum. Where current data was needed — for example, market figures or the present position of a law — the source has been named in the text itself. No reference in this book is invented.

> **One request.** These notes will help you pass. They will not replace the classical texts. Keep Rasatarangini, Rasa Ratna Samuccaya, Sharangadhara Samhita and the Ayurvedic Pharmacopoeia of India beside you. Read them. This book only shows you the road.

THE COURSE AT A GLANCE

Course code and name

Course code	Name of course
AYPG-AB-RB	Applied Basics of Rasashastra and Bhaishajya Kalpana

Course Learning Outcomes (Table 1 of the curriculum)

At the end of this course, the student should be able to:

CO No	Course Learning Outcome	Mapped Programme Learning Outcomes
CO 1	Elucidate classical Ayurvedic knowledge in alignment with multidisciplinary principles applied to Ayurveda pharmaceuticals	PO1, PO3, PO7
CO 2	Evaluate the use of classical texts of Rasashastra and Bhaishajya Kalpana in research and drug development	PO1, PO5, PO7
CO 3	Classify, identify and authenticate raw drugs using both ancient Ayurvedic and contemporary methods	PO1, PO2, PO3, PO7
CO 4	Apply quality control and quality assurance processes for in-process control and finished products, integrating classical methods with modern technical advancements	PO2, PO3, PO5, PO7
CO 5	Identify global healthcare needs and research areas, and design, develop and manufacture innovative dosage forms in compliance with regulatory standards	PO1, PO2, PO3, PO7, PO8
CO 6	Justify the clinical administration of Ayurvedic formulations and adapt them for emerging health conditions	PO4, PO6, PO7, PO8
CO 7	Analyse the application of material and operational management, pharmacy practice and pharmacology in Ayurvedic pharmaceutics	PO1, PO5, PO6

CO No	Course Learning Outcome	Mapped Programme Learning Outcomes
CO 8	Evaluate current and emerging trends in Ayurvedic pharmaceutics with a focus on interdisciplinary scientific advancements	PO5, PO7, PO8

Credit framework

The course has **8 modules**, **16 credits** and **480 notional learning hours**.

One credit means 30 hours of learner engagement. These 30 hours are divided between teaching, practical training and experiential learning in the ratio **1 : 2 : 3**.

So one credit contains:

Component	Hours
Teaching (lecture)	5
Practical training	10
Experiential learning	13
Modular assessment	2
Total	**30**

Each credit of modular assessment carries **25 marks**.

Module summary

Module	Name	Credits	Hours	Modular assessment marks
M1	Pharmaceutics and Pharmaceutical Technology	2	60	50
M2	Databases and Information Systems for Metal, Mineral, and Plant Research	2	60	50
M3	Essential Chemistry — Unlocking the Rasashastra Techniques	2	60	50

Module	Name	Credits	Hours	Modular assessment marks
M4	Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures	2	60	50
M5	Principles, specification, SOPs of handling of classical and advanced instruments	3	90	75
M6	Animal house and Animal handling technique	1	30	25
M7	Classical and Advanced quality control techniques for Ayurveda Formulations	2	60	50
M8	Nano-science, nano-technology and its application in Ayurveda	2	60	50
	Total	**16**	**480**	**400**

Distribution of hours in each module

Module	Lectures	Practical training	Experiential learning including modular assessment	Total
M1	10	20	30	60
M2	10	20	30	60
M3	10	20	30	60
M4	10	20	30	60
M5	15	30	45	90
M6	5	10	15	30
M7	10	20	30	60
M8	10	20	30	60
Total	**80**	**160**	**240**	**480**

MODULE 1
PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY

Credits	Lectures	Practical Training	Experiential Learning	Total hours	Modular assessment
2	10	20	30	60	50 marks

About this module

This module gives the foundation of pharmaceutics and pharmaceutical technology. It covers the whole spectrum of pharmaceutical design, formulation, manufacture and evaluation. It also deals with the pharmaceutical industry, regulatory frameworks and the product development process. Further, the module explores the various pharmaceutical dosage forms, the principles of formulation and Good Manufacturing Practices. Finally, it introduces the fundamental principles of biopharmaceutics and pharmacokinetics.

Module Learning Objectives

At the end of this module, the student should be able to:

1. Discuss the fundamental principles and scope of pharmaceutics and pharmaceutical technology.
2. Describe the different types of pharmaceutical dosage forms and their formulation principles.
3. Discuss the importance of Good Manufacturing Practices (GMP) and quality control in pharmaceutical manufacturing.
4. Apply biopharmaceutic principles to pharmaceutical product development and optimization.
5. Illustrate the pharmaceutical product development process and regulatory frameworks governing the pharmaceutical industry.

Units in this module

Unit	Name
M1U1	Introduction to Pharmaceutics
M1U2	Pharmaceutical Dosage Forms
M1U3	Pharmaceutical Formulation
M1U4	Pharmaceutical Manufacturing and Quality Control
M1U5	Biopharmaceutics

Why this module matters to you

You are a student of Rasashastra and Bhaishajya Kalpana. Your subject is the pharmaceutics of Ayurveda. This module teaches you the modern language of the same science. Once you know both languages, you can explain a Bhasma to a chemist, and you can explain a tablet to a Vaidya. That is the real purpose of this module.

UNIT 1
INTRODUCTION TO PHARMACEUTICS

SYLLABUS POINTS

1.1 Definition and scope of pharmaceutics
1.2 Historical development of pharmaceutics
1.3 Pharmaceutical industry overview
1.4 Regulatory frameworks (FDA, etc.)
1.5 Pharmaceutical product development process

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Define pharmaceutics and describe its scope and historical development	1	Lecture	Knows how
Discuss pharmaceutical industry overview, market size (national and international) and regulatory framework	1	Lecture	Knows how
Demonstrate regulatory frameworks (FDA, etc.)	2	Practical Training 1.1	Shows how
Demonstrate pharmaceutical product development process	2	Practical Training 1.2	Shows how
Identify applications suiting various regulatory frameworks	3	Experiential Learning 1.1	Does
Enlist and evaluate challenges and solutions in the pharmaceutical product development process	3	Experiential Learning 1.2	Knows how

Exam weightage (Table 6F of the curriculum)

ABQ	SAQ	LAQ
No	Yes	Yes

1.1 Definition and scope of pharmaceutics

1.1.1 Starting point: why a drug is not a medicine

A drug is a chemical substance. It has an action on the body. But a drug cannot be given to a patient as it is.

Take an example. Paracetamol is a white powder. The dose is 500 mg. If we hand over loose powder to a patient, many problems come up:

- The patient cannot measure 500 mg correctly.
- The powder tastes bitter.
- The powder absorbs moisture and spoils.
- Some patients cannot swallow loose powder.
- The powder may irritate the stomach.

So the drug must be converted into a convenient, safe and stable form. That form is called a **dosage form**. A tablet, a capsule, a syrup, an ointment — all are dosage forms.

The science that converts a drug into a dosage form is **pharmaceutics**.

The same idea exists in Ayurveda. Guduchi is a drug. But we do not give the raw Guduchi stem to the patient. We convert it into Guduchi Churna, Guduchi Ghana, Guduchi Kwatha or Guduchi Satva. This conversion is Bhaishajya Kalpana.

1.1.2 Definition

Pharmaceutics is the branch of pharmaceutical science that deals with the design, formulation, manufacture, standardisation and evaluation of dosage forms, so that the drug can be delivered to the body in a safe, effective, stable and acceptable manner.

A simpler working definition: *Pharmaceutics is the science of dosage form design.*

It is also called the science of medicine preparation, or the art and science of dosage form design.

1.1.3 Basic terms to know

Term	Meaning
Drug	A substance used for diagnosis, treatment, mitigation or prevention of disease
Active Pharmaceutical Ingredient (API)	The component of a formulation that produces the therapeutic effect
Excipient	An inactive substance added along with the API to help in manufacture, stability, appearance or acceptability
Dosage form	The final physical form in which a drug is given to the patient
Formulation	The complete composition of a product, showing API and all excipients with quantities
Drug delivery system	An arrangement that carries the drug to the required site, at the required rate
Medicament / Medicine	The finished product ready for administration to the patient
Placebo	A dosage form with no active drug, used as a control in trials
Potency	The strength of the active drug in the product

1.1.4 Why is a dosage form necessary?

This is a favourite short-answer question. Remember these eight reasons.

1. **Accurate dosing.** A tablet of a fixed weight delivers a fixed dose. Loose powder cannot.
2. **Protection of the drug.** Coating and packing protect the drug from air, moisture and light.
3. **Masking of taste and odour.** Bitter drugs are coated or made into syrup.
4. **Protection of the body.** Enteric coating protects the stomach from irritating drugs.
5. **Convenient administration.** Injections, eye drops, ointments and inhalers reach places a plain powder cannot.
6. **Placement of drug at a required site.** Eye drops for eye, ointment for skin, suppository for rectum.
7. **Control over rate of release.** Sustained-release and controlled-release products act slowly.
8. **Patient acceptance and compliance.** Attractive, easy-to-take forms improve regular use.

1.1.5 Branches of pharmaceutics

Branch	What it deals with
Physical pharmacy	Physical and chemical principles behind dosage forms — solubility, rheology, surface tension, stability, particle size
Dispensing pharmacy	Preparation of medicines in small quantity against a prescription
Industrial pharmacy	Manufacture of medicines on a large scale, with machinery and process control
Pharmaceutical technology	Unit operations, equipment, process design and scale-up
Biopharmaceutics and pharmacokinetics	Relation between the dosage form, drug release, absorption and blood levels
Pharmaceutical microbiology	Sterility, preservation, microbial limit testing
Pharmaceutical jurisprudence	Laws, rules and ethics governing drugs
Novel drug delivery systems	Modern carriers such as liposomes, nanoparticles, transdermal patches

> Biopharmaceutics is taken up in detail in Unit 5 of this module. Novel drug delivery is discussed in Module 8.

1.1.6 Scope of pharmaceutics

The scope is wide. It covers the whole journey of a medicine, from raw material to patient.

A. Academic and research scope

- Selection and testing of raw materials
- Preformulation study of a new drug
- Design of a new dosage form
- Improvement of an existing dosage form
- Development of novel drug delivery systems

B. Industrial scope

- Production of solid, liquid, semi-solid and parenteral dosage forms
- Selection of machinery and unit operations
- Scale-up from laboratory batch to commercial batch
- Packaging and labelling
- Storage and distribution

C. Quality scope

- Quality control of raw material, in-process material and finished product
- Quality assurance and documentation
- Stability studies and fixing of shelf life
- GMP compliance and audit

D. Regulatory scope

- Preparation of drug dossiers
- Product registration and licensing
- Pharmacovigilance

E. Clinical scope

- Correct dispensing
- Patient counselling
- Drug interaction and compatibility advice

1.1.7 Relevance for a student of Rasashastra and Bhaishajya Kalpana

Rasashastra and Bhaishajya Kalpana is the Ayurvedic counterpart of pharmaceutics and pharmaceutical technology. The curriculum preface itself states that **Yoga Vijnana** and **Prayoga Vijnana** are comparable with pharmaceutics and pharmaceuticals.

The following table shows the parallel. Learn it. It is useful in both theory and viva.

Concept in pharmaceutics	Corresponding concept in Ayurveda
Drug / API	Dravya
Excipient / vehicle	Anupana, Sahapana, Prakshepaka Dravya
Dosage form	Kalpana
Formulation	Yoga
Dosage form design	Bhaishajya Kalpana
Purification or pre-treatment of raw material	Shodhana
Particle size reduction	Mardana, Bhavana, Churnikarana
Calcination	Marana, Puta
Extraction	Kwatha, Satva Patana
Fixed dose	Matra
Shelf life	Saviryata Avadhi
Quality tests	Siddhi Lakshana, Bhasma Pariksha
Good Manufacturing Practices	Schedule T

1.2 Historical development of pharmaceutics

The history can be divided into three broad phases.

1. **Ancient or traditional phase** — medicines prepared by the physician himself.
2. **Transitional or galenical phase** — pharmacy separates from medicine.
3. **Modern or industrial phase** — large-scale manufacture, science-based design, regulation.

1.2.1 Ancient phase

Mesopotamia and Egypt. Clay tablets from Mesopotamia carry some of the earliest written drug records. In Egypt, the *Ebers Papyrus* (about 1550 BC) lists a large number of formulations with methods of preparation. Egyptian medicine men prepared and dispensed their own remedies.

China. Traditional Chinese medicine developed its own systems of decoctions, pills and powders.

India. Ayurveda developed a complete and independent pharmaceutical science.

- The **Charaka Samhita** and **Sushruta Samhita** describe the basic Kalpanas — Swarasa, Kalka, Kwatha, Hima and Phanta, along with Sneha Kalpana, Sandhana Kalpana and Avaleha.
- The **Sharangadhara Samhita** (about the 13th century AD) is the landmark text for Bhaishajya Kalpana. It arranged the Kalpanas systematically and gave measurements, proportions and Paka Lakshanas. Many present-day Ayurvedic manufacturing practices still follow it.
- **Rasashastra** developed later as a separate discipline dealing with metals, minerals and their processing. Texts such as **Rasa Ratna Samuccaya**, **Rasendra Chudamani**, **Ayurveda Prakasha**, **Rasa Kamadhenu**, **Rasendra Sara Sangraha** and later **Rasatarangini** describe Shodhana, Marana, Puta, Musha, Yantra and Kupipakwa Rasayana.

So India already had a well-organised pharmaceutics long before the modern industry began.

1.2.2 Greek and Roman period

- **Hippocrates** (about 460–370 BC) separated medicine from superstition and gave it a rational base.
- **Dioscorides** (1st century AD) wrote *De Materia Medica*, describing about 600 plants and their preparations. It remained a standard reference in Europe for nearly 1500 years.
- **Galen** (about 130–200 AD) prepared medicines from plant and animal sources by extraction, and standardised the methods. Preparations made by extracting crude drugs with a solvent are still called **galenicals** in his memory. Tinctures, extracts and infusions are galenicals.

The Ayurvedic equivalents of galenicals are Kwatha, Arka, Asava, Arishta and Sneha Kalpana.

1.2.3 Arabic period and the birth of pharmacy as a profession

Arab physicians and chemists refined the techniques of distillation, crystallisation, sublimation and filtration. They introduced syrups, aromatic waters and sugar-coated pills.

- **Ibn Sina (Avicenna, 980–1037 AD)** wrote *The Canon of Medicine*, which was a standard European medical text for centuries.
- Around the 8th century, the first public apothecary shops opened in Baghdad. This was the beginning of pharmacy as a profession separate from medicine.

By the 13th century, the profession of the apothecary was legally recognised in Europe.

1.2.4 Modern phase — key milestones

Year	Event	Importance
1805	Sertuerner isolated morphine from opium	First isolation of a pure active principle from a plant; began the era of pure drugs
1843	William Brockedon patented a machine for compressing powder into a tablet	Birth of the tablet dosage form
Late 19th century	Rise of synthetic organic chemistry	Aspirin and other synthetic drugs became possible
1928	Fleming discovered penicillin	Started the antibiotic era and large-scale sterile manufacture
1937	Elixir sulfanilamide disaster in the USA — diethylene glycol used as solvent, over 100 deaths	Showed that a formulation itself can kill; led directly to modern drug law
1938	Federal Food, Drug and Cosmetic Act, USA	Proof of safety made compulsory before marketing
1961	Thalidomide tragedy — birth defects in thousands of infants	Showed the need for testing in pregnancy and for proof of efficacy
1962	Kefauver-Harris Amendment, USA	Proof of efficacy also made compulsory; GMP strengthened
1990	ICH established	Harmonisation of technical requirements between USA, Europe and Japan
Late 20th century onward	Controlled release, transdermal systems, biotechnology products, nanotechnology	Present era of targeted and smart delivery

1.2.5 Development in India

Year	Event
1901	Acharya P. C. Ray established the Bengal Chemical and Pharmaceutical Works, Kolkata — an early Indian pharmaceutical enterprise
1940	Drugs and Cosmetics Act enacted
1945	Drugs and Cosmetics Rules framed
1948	Pharmacy Act enacted; Pharmacy Council of India established
1955	First edition of the Indian Pharmacopoeia published

Year	Event
1964	Drugs and Cosmetics Act amended to bring Ayurvedic, Siddha and Unani drugs under its scope
1978	Ayurvedic Formulary of India, Part I published
1990	Ayurvedic Pharmacopoeia of India, Part I, Volume I published
2000	Schedule T, giving GMP for ASU drugs, notified
2010	Pharmacopoeia Commission for Indian Medicine, later PCIM&H, established
2014	Ministry of AYUSH created as a separate ministry
2020	National Commission for Indian System of Medicine (NCISM) Act came into force

POINT TO REMEMBER FOR THE EXAM

Two disasters shaped modern drug regulation. The elixir sulfanilamide disaster brought in the requirement of **safety**. The thalidomide tragedy brought in the requirement of **efficacy**. Nearly every question on “why regulation is needed” can be answered from these two events.

1.3 Pharmaceutical industry overview

1.3.1 What is the pharmaceutical industry?

The pharmaceutical industry is the organised sector that discovers, develops, manufactures, tests, markets and distributes medicines for human and veterinary use.

It is a **knowledge-based**, **capital-intensive** and **highly regulated** industry. These three words describe its character well.

1.3.2 Segments of the industry

Segment	Description
Bulk drugs and API	Manufacture of the active ingredient itself
Formulations	Conversion of API into finished dosage forms
Biologicals and vaccines	Products of biological origin
Biosimilars	Similar versions of approved biological products
Nutraceuticals and food supplements	Health products at the border of food and drug
Over-the-counter (OTC)	Medicines sold without prescription
Herbal and AYUSH products	Ayurvedic, Siddha, Unani and Homoeopathic medicines
Contract research and manufacturing (CRO and CDMO)	Research or manufacture done for another company
Medical devices and diagnostics	Allied segment

1.3.3 Types of companies

1. **Innovator or research-based companies.** They discover new molecules, run clinical trials and hold patents. Development cost is very high.
2. **Generic companies.** They manufacture off-patent drugs. Cost is lower. India is strong here.
3. **CDMO and CRO.** They provide research or manufacturing services to other companies.
4. **Biotechnology companies.** They work on biological and gene-based products.

1.3.4 Departments of a pharmaceutical company

A student should know these, because practical and viva questions often come from here.

Department	Main function
Research and Development (R&D)	Design of new products and improvement of existing ones
Production or Manufacturing	Actual making of the product as per the batch record
Quality Control (QC)	Testing of raw material, in-process samples and finished product
Quality Assurance (QA)	System that ensures quality is built into the process; documentation, SOPs, audits, batch release
Regulatory Affairs	Licences, registrations, dossiers, communication with authorities
Engineering and Maintenance	Machines, utilities, calibration
Stores and Warehouse	Receipt, storage and issue of materials
Packaging	Primary and secondary packing, labelling
Marketing and Sales	Distribution and promotion
Pharmacovigilance	Collection and analysis of adverse drug reactions

1.3.5 International market size

The global medicine market was valued at roughly **USD 1.6 to 1.7 trillion in 2024**. The United States alone accounted for nearly half of all worldwide prescription drug sales in that year. The IQVIA Institute for Human Data Science has projected that global medicine spending will grow at 5 to 8 per cent per year and reach about **USD 2.3 trillion by 2028**.

The largest national markets in 2024 were the United States, followed by China, Germany, Japan and France.

1.3.6 National market size — India

India is often called the **pharmacy of the world**. The reasons are clear from the figures below.

Indicator	Figure
Rank by volume of production	3rd in the world
Rank by value	Among the top fifteen; different official sources quote 11th to 14th
Domestic market size	About USD 60 billion
Pharmaceutical exports, 2024–25	USD 30.47 billion
Number of companies	More than 3,000
Number of manufacturing units	About 10,500
Generic brands produced	More than 60,000 across 60 therapeutic categories
Share of global generic medicine supply	About 20 per cent by volume
Share of global vaccine supply	About 60 per cent
Projected market size by 2030	About USD 130 billion

Indian pharmaceutical exports reached 191 countries in 2024–25. About half of these exports went to highly regulated markets such as the United States and Europe. Exports have risen nearly sixteen times from USD 1.9 billion in 2000–01.

Source: Press Information Bureau, Government of India, and the Ministry of Commerce and Industry.

Strengths of the Indian industry

- Low cost of manufacture
- Large pool of trained technical manpower
- Highest number of US FDA-approved plants outside the USA
- Strong generic and API base

Weaknesses

- Dependence on imported key starting materials
- Limited investment in new drug discovery
- Quality lapses in some small units
- Rising cost of regulatory compliance

1.3.7 The AYUSH and Ayurvedic industry in India

This part is directly relevant to your speciality.

A report published by the Forum on Indian Traditional Medicine under the Research and Information System for Developing Countries, titled *Ayush Sector in India: Prospects and Challenges* (2021), estimated the AYUSH industry size in 2020–21 at **USD 18.1 billion**. The sector had grown by about 17 per cent during 2014 to 2020.

Key institutions supporting the sector

Body	Role
Ministry of AYUSH	Nodal ministry for Ayurveda, Yoga and Naturopathy, Unani, Siddha, Sowa-Rigpa and Homoeopathy
PCIM&H	Lays down formulary specifications and pharmacopoeial standards for ASU&H drugs, which serve as official compendia for quality control under the Drugs and Cosmetics Act
CCRAS	Research council; issues drug development and clinical evaluation guidelines
NCISM	Education and curriculum regulation
AYUSHEXCIL	Export Promotion Council set up by the Ministry of AYUSH with support from the Ministry of Commerce
NMPB	National Medicinal Plants Board — raw material and cultivation

Government schemes to note. Production Linked Incentive (PLI) scheme and Strengthening of Pharmaceutical Industry (SPI) scheme for the pharmaceutical sector. For AYUSH, the **Ayush Oushadhi Gunvatta Evam Uttpadan Samvardhan Yojana (AOGUSY)**, which upgrades pharmacies and drug testing laboratories, strengthens pharmacovigilance and improves regulatory frameworks.

Well-known Ayurvedic manufacturers in India. Dabur, Baidyanath, Zandu, Himalaya, Kottakkal Arya Vaidya Sala, Vaidyaratnam, Charak, Patanjali, Kerala Ayurveda, SDP, Unjha and many regional pharmacies.

1.4 Regulatory frameworks (FDA, etc.)

1.4.1 Why is regulation needed?

A patient cannot test a medicine before taking it. He has to trust the manufacturer. Regulation replaces this blind trust with a legal guarantee.

A regulatory framework ensures four things:

1. **Quality** — the product contains what the label says.
2. **Safety** — the product does not cause unacceptable harm.
3. **Efficacy** — the product actually works.
4. **Truthful information** — the label and claims are correct.

History has shown what happens without regulation. Recall the two events from section 1.2.4.

1.4.2 Functions of a regulatory authority

- Framing rules and standards
- Licensing of manufacturing premises
- Approval of new drugs and clinical trials
- Inspection and audit of factories
- Testing of market samples
- Approval of labels and package inserts
- Recall of defective products
- Pharmacovigilance and post-marketing surveillance
- Prosecution in case of violation

1.4.3 United States Food and Drug Administration (USFDA)

The USFDA is the most influential drug regulator in the world. Indian exporters must satisfy it, so a postgraduate student should know its structure.

The main drug centre is **CDER**, the Center for Drug Evaluation and Research. Biological products go to **CBER**.

Key applications

Application	Full form	Purpose
IND	Investigational New Drug application	Permission to begin clinical trials in humans after preclinical work
NDA	New Drug Application	Approval to market a new drug
ANDA	Abbreviated New Drug Application	Approval of a generic product; requires proof of bioequivalence, not fresh clinical trials
BLA	Biologics License Application	Approval of a biological product
DMF	Drug Master File	Confidential file on API, facility or packaging

Key regulations. cGMP requirements are given in the US Code of Federal Regulations, Title 21, Parts 210 and 211. The list of approved products with therapeutic equivalence ratings is published as the **Orange Book**.

1.4.4 Other major regulatory agencies

Agency	Country or region
EMA — European Medicines Agency	European Union
MHRA — Medicines and Healthcare products Regulatory Agency	United Kingdom
TGA — Therapeutic Goods Administration	Australia
PMDA — Pharmaceuticals and Medical Devices Agency	Japan
NMPA — National Medical Products Administration	China
Health Canada	Canada
WHO	International guidance, prequalification, WHO-GMP certification

1.4.5 ICH

The **International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH)** brings regulators and industry together, so that a company need not repeat the same studies for every country.

ICH guidelines are grouped into four series.

Series	Subject	Common examples
Q	Quality	Q1 Stability, Q2 Analytical validation, Q3 Impurities, Q8 Pharmaceutical development, Q9 Quality risk management, Q10 Pharmaceutical quality system
S	Safety	Toxicology and carcinogenicity studies
E	Efficacy	E6 Good Clinical Practice
M	Multidisciplinary	M3 Non-clinical safety studies, MedDRA, CTD format

1.4.6 Regulatory framework in India — allopathic drugs

The primary law is the **Drugs and Cosmetics Act, 1940** together with the **Drugs and Cosmetics Rules, 1945**.

Other important laws

- Pharmacy Act, 1948
- Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954
- Narcotic Drugs and Psychotropic Substances Act, 1985
- Drugs (Prices Control) Order
- New Drugs and Clinical Trials Rules, 2019

Authorities

Authority	Function
CDSCO — Central Drugs Standard Control Organisation	Central drug regulator
DCGI — Drugs Controller General of India	Head of CDSCO; approves new drugs, clinical trials and imports
State Licensing Authority or State Drug Controller	Issues manufacturing and sale licences; inspects units
DTAB and DCC	Statutory advisory bodies for technical and consultative matters
NPPA — National Pharmaceutical Pricing Authority	Fixes and monitors drug prices
Indian Pharmacopoeia Commission	Publishes the Indian Pharmacopoeia

Schedules to remember

Schedule	Subject
Schedule M	GMP for allopathic drugs, and requirements of premises, plant and equipment
Schedule Y	Requirements for clinical trials and for import or manufacture of new drugs; now largely replaced by the New Drugs and Clinical Trials Rules, 2019
Schedule H, H1, X	Prescription-only and habit-forming drugs
Schedule T	GMP for Ayurvedic, Siddha and Unani drugs
Schedule E(1)	List of poisonous substances of ASU origin
First Schedule	List of authoritative books of Ayurveda, Siddha and Unani

1.4.7 Regulatory framework in India — Ayurvedic, Siddha and Unani drugs

This is the most important part of section 1.4 for your speciality.

Definitions under the Act

- **Section 3(a)** defines an Ayurvedic, Siddha or Unani drug. It includes all medicines intended for internal or external use for the diagnosis, treatment, mitigation or prevention of disease or disorder in human beings or animals, and manufactured exclusively in accordance with the formulae described in the authoritative books specified in the First Schedule.
- **Section 3(h)** defines a **patent or proprietary medicine**. In relation to the ASU systems, it covers formulations containing only such ingredients as are mentioned in the formulae of the authoritative books of the First Schedule. It excludes a medicine given by the parenteral route, and excludes a formulation already included in those books.

In simple words:

- **Classical medicine** is made exactly as per a formula in a First Schedule book. Example: Triphala Churna.
- **Patent or proprietary medicine** is a new combination, but all its ingredients come from the classical books.

Licensing and quality

- **Rule 158-B** of the Drugs and Cosmetics Rules, 1945 provides the regulatory guidelines for issue of licence to manufacture Ayurvedic, Siddha and Unani medicines. It also lays down the proof of safety and effectiveness required for the different categories of ASU drugs.
- It is mandatory for manufacturers to comply with the prescribed requirements for licensing of manufacturing units and medicines. This includes proof of safety and effectiveness, compliance with Good Manufacturing Practices as per **Schedule T**, and the quality standards laid down in the pharmacopoeia.
- Enforcement of quality control and issue of drug licences for ASU and Homoeopathic drugs is vested with the **State Drug Controllers and State Licensing Authorities** appointed by the concerned State or Union Territory Government.

Advisory bodies

The Central Government is advised by the **Ayurveda, Siddha, Unani Drugs Technical Advisory Board (ASUDTAB)** and the **Ayurveda, Siddha, Unani Drugs Consultative Committee (ASUDCC)**. Both are statutory bodies under the Drugs and Cosmetics Act.

Standards and testing

- Standardised formulations published in the respective formularies include 985 Ayurvedic, 399 Siddha and 1229 Unani formulations.
- Central and State drug laboratories for testing ASU medicines are in place. Fifty-five laboratories have been approved under the Drugs and Cosmetics Rules, 1945.
- PCIM&H also acts as an appellate drug testing laboratory and receives samples from Government agencies for ascertaining their quality.

Voluntary quality marks. The **AYUSH Premium Mark** and the **AYUSH Standard Mark**, operated by the Quality Council of India. The WHO **Certification of Pharmaceutical Products (CoPP)** is also used for export.

Clinical trials in ASU. The Drugs and Cosmetics Act and Rules do not have explicit provisions for clinical trials of Ayurvedic medicines. Rule 158-B provides the requirement of a **pilot study** to generate proof of safety and effectiveness for certain categories of Ayurvedic medicines. The Ministry of AYUSH has published Good Clinical Practice Guidelines for the conduct of clinical trials on Ayurvedic, Siddha and Unani medicines, on a voluntary basis.

Source: Press Information Bureau, Government of India, and the AYUSH Next portal of the Ministry of AYUSH.

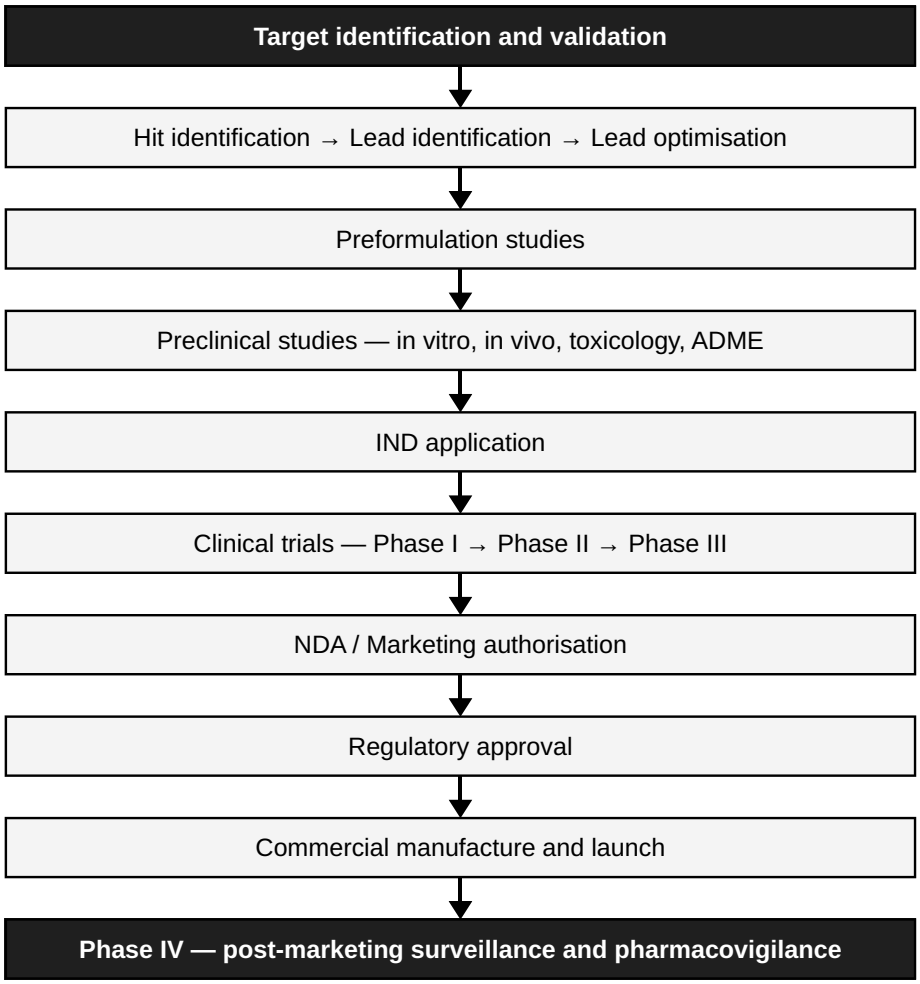
1.4.8 Comparison at a glance

Point	Allopathic drug	ASU drug
Governing law	Drugs and Cosmetics Act, 1940 and Rules, 1945	Same Act and Rules, Chapter IVA
Central regulator	CDSCO and DCGI	Ministry of AYUSH
Licensing authority	State Licensing Authority	State Licensing Authority
GMP schedule	Schedule M	Schedule T
Licensing rule	Rules 71, 76 and related rules	Rule 158-B
Standards book	Indian Pharmacopoeia	Ayurvedic Pharmacopoeia of India and Ayurvedic Formulary of India
Advisory bodies	DTAB and DCC	ASUDTAB and ASUDCC
Standards body	Indian Pharmacopoeia Commission	PCIM&H

1.5 Pharmaceutical product development process

1.5.1 Overview

Product development is the complete path from an idea to a marketed medicine. For a new chemical entity it usually takes **10 to 15 years** and costs a very large sum. Out of thousands of compounds screened, only one may finally reach the market.



1.5.2 Stage 1 — Drug discovery

Step	What happens
Target identification	A biological molecule such as an enzyme or receptor, linked to the disease, is chosen
Target validation	It is proved that acting on this target changes the disease
Hit identification	Large libraries of compounds are screened; active ones are called hits
Lead identification	The most promising hits become leads
Lead optimisation	The lead is chemically modified to improve potency, selectivity and safety

1.5.3 Stage 2 — Preformulation studies

Preformulation is the study of the physical and chemical properties of the drug substance, alone and along with excipients, before the dosage form is designed. This stage decides the fate of the product.

Parameter	Why it matters
Organoleptic properties — colour, odour, taste	Affects patient acceptance and the need for masking
Solubility and pH-solubility profile	Poor solubility means poor absorption
pKa or dissociation constant	Predicts ionisation and absorption at different pH
Partition coefficient (log P)	Predicts membrane permeability
Polymorphism	Different crystal forms have different solubility and stability
Particle size and surface area	Affects dissolution rate and flow
Bulk density, tapped density, angle of repose, Carr's index, Hausner ratio	Decide flow property for tableting
Hygroscopicity	Affects storage and packing
Melting point	Identity and purity check
Drug-excipient compatibility	Prevents interaction and degradation
Stability under heat, light, humidity, oxidation and hydrolysis	Decides shelf life and packing

1.5.4 Stage 3 — Preclinical studies

These are studies done in the laboratory and in animals, before giving the drug to human beings.

- **In vitro studies** — cell lines, enzyme assays, receptor binding
- **In vivo studies** — in suitable animal models
- **Pharmacology** — main effect and side effects
- **ADME studies** — Absorption, Distribution, Metabolism and Excretion
- **Toxicology** — acute, sub-acute and chronic toxicity; genotoxicity; reproductive toxicity; carcinogenicity

All animal work must follow the rules of the CPCSEA, now CCSEA. This is discussed in detail in Module 6.

1.5.5 Stage 4 — Clinical trials

Phase	Subjects	Number	Main purpose	Approximate duration
Phase I	Healthy volunteers; patients in the case of anticancer drugs	20–100	Safety, tolerability, pharmacokinetics, dose range	Several months
Phase II	Patients with the disease	100–300	Efficacy and short-term side effects; dose finding	Months up to about 2 years
Phase III	Patients, multicentre	300–3,000 or more	Confirm efficacy, compare with existing treatment, detect less common adverse effects	1 to 4 years
Phase IV	General population after marketing	Large	Long-term safety, rare adverse effects, new uses	Ongoing

The USFDA describes the drug development process in these stages, from discovery and preclinical research, through the three phases of clinical research, to FDA review and post-market safety monitoring.

1.5.6 Stage 5 — Formulation development and scale-up

Step	Description
Laboratory or bench batch	Small trial batch to fix the formula
Pilot batch	Usually one-tenth of the commercial batch size; the process is refined
Exhibit batch	Batch made for submission to the regulator, with full documentation
Scale-up	Moving from small to commercial batch size without loss of quality
Process validation	Documented proof that the process consistently gives the required quality; normally three consecutive batches
Technology transfer	Handing over the validated process from R&D to production
Commercial manufacture	Routine production under GMP

1.5.7 Stage 6 — Stability studies

Stability studies fix the shelf life and the storage condition. The guideline followed is **ICH Q1A(R2)**.

Type of study	Condition	Duration
Long-term	25 °C ± 2 °C and 60% RH ± 5%, or the zone-specific condition	Minimum 12 months, continued up to the proposed shelf life
Intermediate	30 °C ± 2 °C and 65% RH ± 5%	6 months
Accelerated	40 °C ± 2 °C and 75% RH ± 5%	6 months

India falls in **Climatic Zone IVA and IVB**, which is hot and humid. So stability requirements here are stricter than in Europe.

1.5.8 Stage 7 — Regulatory submission and launch

- The complete data is compiled into a **dossier**, usually in the **Common Technical Document (CTD)** format.
- The CTD has five modules: regional administrative information, summaries, quality, non-clinical study reports and clinical study reports.
- After approval, the product is launched, and pharmacovigilance begins.

1.5.9 Product development for Ayurvedic drugs

The process is different, because Ayurvedic products already have a long tradition of use.

Route 1 — Classical formulation. The product is made exactly as given in a First Schedule book. Safety and efficacy are accepted on the basis of textual evidence and long use. Licensing is comparatively simple.

Route 2 — Patent or proprietary medicine. A new combination is made from classical ingredients. Here proof of safety and effectiveness is required as per Rule 158-B, and the requirement varies with the category of the product.

A practical development path for a new Ayurvedic product

1. Identify the health need and the classical reference.
2. Select the drugs. Apply Namarupa Vijnana and authenticate the raw material.
3. Fix the formulation composition and the Kalpana.
4. Standardise the process and write the SOP.
5. Carry out pharmaceutical standardisation — organoleptic, physicochemical and phytochemical parameters as per the Ayurvedic Pharmacopoeia of India.
6. Conduct safety studies where required.
7. Conduct a pilot clinical study or clinical evaluation as applicable.
8. Develop packaging and fix the shelf life through a stability study.
9. Obtain the manufacturing licence from the State Licensing Authority.
10. Launch, and maintain pharmacovigilance records.

The **CCRAS Guideline for Drug Development** is the standard national document for this pathway. It is reference 49 of the NCISM list and is freely available on the CCRAS website.

1.5.10 Common challenges in product development

This section directly supports Experiential Learning 1.2.

Challenge	Possible solution
Poor aqueous solubility of the drug	Particle size reduction, salt formation, solid dispersion, use of surfactants, nano-formulation
Poor stability	Better packing, antioxidants, protection from light and moisture, lyophilisation
Poor flow property	Granulation, use of glidants
Bitter taste	Coating, complexation, flavouring, use of Madhura Dravya in Ayurvedic products
Drug-excipient incompatibility	Change the excipient after a compatibility study
Difficulty in scale-up	Proper pilot batches and process validation
Batch-to-batch variation of herbal raw material	Proper source, season and part; chemoprofiling with TLC or HPTLC; fixing of a marker compound
Heavy metal or pesticide contamination	Source control; testing as per API limits
High cost and long time of regulatory compliance	Early planning, good documentation, use of the CTD format

KEY POINTS TO REMEMBER

- Pharmaceutics is the science of dosage form design.
- A dosage form is needed for accurate dose, protection, taste masking, convenience, site placement, controlled release and patient compliance.
- Galen gave the term galenicals. Sharangadhara Samhita is the landmark text of Bhaishajya Kalpana.
- The 1937 elixir sulfanilamide disaster brought the safety requirement. The 1961 thalidomide tragedy brought the efficacy requirement.
- India is third in the world by volume of pharmaceutical production, with exports of USD 30.47 billion in 2024–25.
- The AYUSH industry was valued at USD 18.1 billion in 2020–21.
- For ASU drugs: Rule 158-B for licensing, Schedule T for GMP, ASUDTAB and ASUDCC for advice, PCIM&H for standards.
- USFDA applications: IND before trials, NDA for a new drug, ANDA for a generic.
- Clinical trial phases: I for safety, II for efficacy, III for confirmation, IV after marketing.
- Preformulation decides the fate of the formulation.

EXAM CORNER — UNIT 1

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). It is not marked for ABQ.

Likely short answer questions (5 marks)

1. Define pharmaceutics. Write its scope.
2. Enumerate the branches of pharmaceutics.
3. Why is a dosage form necessary? Give any five reasons.
4. Write a short note on galenicals.
5. Write a short note on the contribution of Sharangadhara Samhita to Bhaishajya Kalpana.
6. Write a short note on CDSCO and DCGI.
7. Write a short note on Rule 158-B.
8. Write a short note on Schedule T.
9. Differentiate between a classical Ayurvedic medicine and a patent or proprietary medicine.
10. Write a short note on ICH guidelines.
11. Enumerate the phases of clinical trial with the number of subjects in each.
12. Write a short note on preformulation studies.
13. Differentiate between NDA and ANDA.
14. Write a short note on PCIM&H.
15. Write a short note on stability study conditions as per ICH Q1A.

Likely long answer questions (10 marks)

1. Describe in detail the pharmaceutical product development process, from drug discovery to post-marketing surveillance.
2. Discuss the regulatory framework governing pharmaceutical products in India, with special reference to Ayurvedic, Siddha and Unani drugs.
3. Describe the historical development of pharmaceutics and correlate it with the development of Bhaishajya Kalpana in India.
4. Give an overview of the pharmaceutical industry, including its segments, departments, and national and international market size.
5. Define pharmaceutics. Describe its scope and correlate it with Rasashastra and Bhaishajya Kalpana.

How to attempt

- For a 5-mark SAQ, write a definition, then four to six points, and close with one line of relevance. About one page.
- For a 10-mark LAQ, use headings. Draw at least one table or one flow chart. The examiner gives marks for structure.
- Always add one line connecting the topic to Rasashastra and Bhaishajya Kalpana. This is a postgraduate paper of your speciality, not a general pharmacy paper.

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UNIT 2
PHARMACEUTICAL DOSAGE FORMS

SYLLABUS POINTS

2.1 Classification of dosage forms
2.2 Solid dosage forms (tablets, capsules, etc.)
2.3 Liquid dosage forms (solutions, suspensions, etc.)
2.4 Semi-solid dosage forms (ointments, creams, etc.)
2.5 Parenteral dosage forms (injections, infusions, etc.)

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Classify various pharmaceutical dosage forms	1	Lecture	Knows how
Identify and categorise classical Ayurvedic dosage forms	2	Practical Training 1.3	Knows how
Identify and categorise modified Ayurvedic dosage forms	2	Practical Training 1.4	Knows how
Explore, identify and collect various classical and modified pharmaceutical dosage forms	3	Experiential Learning 1.3	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

> This unit is asked only as a short answer question. So learn the classifications, the definitions and the examples thoroughly. You do not need to prepare a ten-mark essay from this unit. But do not skip it either — the practical examination draws heavily from here.

2.1 Classification of dosage forms

A dosage form can be classified in several ways. An examiner may ask for any one of them. Learn all four.

2.1.1 Classification according to physical state

This is the commonest classification and the one used by your syllabus.

Class	Examples in modern pharmacy	Examples in Ayurveda
Solid	Tablet, capsule, powder, granule, pill, suppository, pastille	Churna, Vati, Gutika, Guggulu, Bhasma, Pishti, Parpati, Varti
Liquid	Solution, syrup, elixir, suspension, emulsion, drops, lotion	Kwatha, Swarasa, Hima, Phanta, Arka, Asava, Arishta, Panaka
Semi-solid	Ointment, cream, paste, gel, jelly	Avaleha, Lehya, Rasakriya, Malahara, Lepa, Kalka
Gaseous	Inhalation, aerosol, spray, oxygen	Dhupana, Dhumapana

2.1.2 Classification according to route of administration

Route	Dosage forms used
Oral	Tablet, capsule, syrup, suspension, powder
Parenteral	Injection, infusion, implant
Topical	Ointment, cream, gel, lotion, paste
Rectal	Suppository, enema
Vaginal	Pessary, vaginal tablet, cream
Ophthalmic	Eye drops, eye ointment
Nasal	Nasal drops, nasal spray
Otic	Ear drops
Inhalation	Aerosol, dry powder inhaler, nebuliser solution
Transdermal	Patch

2.1.3 Classification according to release pattern

Type	Meaning
Immediate release	Drug is released as soon as the dosage form disintegrates
Modified release	Release is deliberately changed from the normal pattern
Sustained release	Drug is released slowly over a long period
Controlled release	Drug is released at a constant, predetermined rate
Delayed release	Release begins only after a set time or at a set site, such as enteric-coated tablets
Targeted release	Drug is delivered to a specific tissue or organ

2.1.4 Classification according to the system of medicine

System	Term used for the dosage form
Modern pharmacy	Dosage form
Ayurveda	Kalpana
Unani	Murakkabat
Homoeopathy	Potency and vehicle forms

2.1.5 Classical and modified Ayurvedic dosage forms

Your practical training asks you to identify and categorise both classical and modified Ayurvedic dosage forms. Learn this table well.

Category	Classical form	Modified or modern form of the same
Powder	Churna	Tablet, capsule, granule, sachet
Decoction	Kwatha	Ghana Vati, Kwatha granule, instant Kwatha powder, syrup
Fermented liquid	Asava and Arishta	Standardised bottled Arishta with fixed alcohol content
Confection	Avaleha and Lehya	Jam-type pack, chewable tablet, sugar-free Avaleha
Medicated fat	Ghrita and Taila	Soft gelatin capsule, nano-emulsion, ointment
Resin preparation	Guggulu	Coated Guggulu tablet
Mineral preparation	Bhasma and Pishti	Standardised Bhasma tablet, capsule
External application	Lepa	Cream, gel, ointment, spray
Collyrium	Anjana	Eye ointment, eye drops
Distillate	Arka	Standardised Arka, spray
Pill	Vati and Gutika	Compressed tablet, film-coated tablet

> **How to answer in the practical.** When you are shown a sample, state three things in order: the physical state, the classical name of the Kalpana, and one identifying feature. Example: “This is a solid dosage form. It is a Vati. It is uniform in size and shape, and hard to the touch.”

2.2 Solid dosage forms

Solid dosage forms are the most widely used. They are stable, easy to carry, easy to dose and cheap to make.

2.2.1 Tablets

A tablet is a solid dosage form containing a drug with or without excipients, prepared by compression or by moulding.

Advantages

- Accurate dose in each unit
- Good physical and chemical stability
- Easy to pack, transport and store
- Cheapest per unit
- Taste can be masked by coating
- Release can be modified

Disadvantages

- Difficult for children and unconscious patients to swallow
- Some drugs cannot be compressed well
- Poorly soluble drugs may be poorly absorbed
- Not suitable for emergencies where quick action is needed

Types of tablets

Type	Description
Uncoated or plain tablet	Simple compressed tablet
Film-coated tablet	Thin polymer coat for taste masking and protection
Sugar-coated tablet	Thick sugar coat, elegant but heavy
Enteric-coated tablet	Coat resists gastric acid, dissolves in the intestine
Effervescent tablet	Dissolved in water before use; contains an acid and a carbonate
Chewable tablet	Chewed before swallowing; useful for children
Dispersible tablet	Disperses quickly in water
Sublingual tablet	Kept under the tongue; drug absorbed directly into blood
Buccal tablet	Kept in the cheek pouch
Lozenge	Dissolves slowly in the mouth for local action
Sustained release tablet	Releases the drug slowly over hours
Vaginal tablet	For insertion into the vagina

Ayurvedic counterparts. Vati, Gutika, Modaka and Guggulu Vati are the classical equivalents of tablets. Today most Ayurvedic pharmacies manufacture Vati by compression, using the same tablet machines.

2.2.2 Capsules

A capsule is a solid dosage form in which the drug is enclosed in a shell made of gelatin or a plant-based polymer.

Type	Description
Hard gelatin capsule	Two-piece shell, body and cap; filled with powder, granules or pellets
Soft gelatin capsule	One-piece sealed shell; filled with liquid, oil or paste
Vegetarian capsule	Shell made of HPMC or pullulan instead of gelatin
Enteric-coated capsule	Coated to resist gastric acid
Sustained release capsule	Contains pellets that release the drug slowly

Capsule sizes run from 000, the largest, down to 5, the smallest. Size 0 and size 1 are the commonest.

Advantages. Taste and odour are completely masked. Filling is easy. The drug need not be compressed. **Disadvantages.** Costlier than tablets. Not suitable for very hygroscopic or highly soluble drugs. Gelatin shells are unacceptable to vegetarian patients unless a plant shell is used.

Ayurvedic use. Soft gelatin capsules are now widely used for Ghrita and Taila preparations, because they hide the taste and prevent rancidity.

2.2.3 Powders

A powder is a mixture of finely divided drugs or chemicals in a dry form.

Type	Description
Bulk powder	Supplied in a container, dose measured by the patient with a spoon
Divided powder	Each dose separately wrapped or packed in a sachet
Dusting powder	For external use on the skin
Insufflation	Powder blown into a body cavity
Effervescent powder	Dissolved in water before taking

Ayurvedic counterpart. Churna is the classical powder. Sieve number and particle size decide its quality. Related forms are Kwatha Churna, Nasya Churna, Manjana and Dhupana Churna.

2.2.4 Other solid dosage forms

Form	Description	Ayurvedic parallel
Granules	Small aggregates of powder particles; free flowing	Ghana granule
Pills	Small spherical masses prepared by moulding	Vati, Gutika
Suppositories	Solid form for insertion into the rectum; melts at body temperature	Varti, Phala Varti
Pessaries	Solid form for insertion into the vagina	Yoni Varti
Pastilles and lozenges	Dissolve slowly in the mouth	Khadiradi Vati
Implants	Sterile solid inserted under the skin	Not classical

2.3 Liquid dosage forms

Liquid dosage forms are useful for children, for elderly patients and for anyone who cannot swallow a tablet. Absorption is faster, because no disintegration step is needed.

2.3.1 Monophasic liquids — solutions

A solution is a clear, homogeneous liquid preparation containing one or more dissolved substances.

Form	Description
Syrup	Concentrated solution of sugar in water, with or without a drug
Elixir	Clear, sweetened, hydro-alcoholic solution
Linctus	Viscous, sweet solution used for cough
Draught	Single large dose of a liquid
Mixture	Liquid preparation for oral use, may contain dissolved or suspended matter
Gargle and mouthwash	For use in the mouth and throat, not swallowed
Douche	Solution directed into a body cavity
Enema	Solution introduced into the rectum
Liniment	Liquid or semi-liquid for external application by rubbing
Lotion	Liquid for external application without rubbing

2.3.2 Biphasic liquids — suspensions and emulsions

Suspension. A suspension is a biphasic liquid preparation in which finely divided solid particles are dispersed in a liquid, in which they are not soluble.

- Used when the drug is insoluble or unstable in solution.
- Must be shaken before use, so the label carries “Shake well before use”.
- Good suspension: settles slowly, redisperses easily, pours smoothly, no caking.
- Key excipients: suspending agent, wetting agent, flocculating agent, preservative, sweetener, flavour, colour.

Emulsion. An emulsion is a biphasic liquid preparation in which one liquid is dispersed as fine globules in another liquid, with which it does not mix. An emulsifying agent is needed to keep it stable.

Type	Description
Oil in water (o/w)	Oil dispersed in water; used for internal use; less greasy
Water in oil (w/o)	Water dispersed in oil; used externally; more greasy
Multiple emulsion	o/w/o or w/o/w; used in advanced delivery

Instability of emulsions. Creaming is reversible on shaking. Cracking is not reversible. Phase inversion means the o/w type changes to w/o or the reverse.

2.3.3 Ayurvedic liquid dosage forms

Ayurvedic form	Nature	Modern parallel
Swarasa	Fresh juice of the drug	Fresh expressed juice
Hima	Cold infusion	Cold macerate
Phanta	Hot infusion	Infusion
Kwatha or Kashaya	Decoction	Decoction
Arka	Distillate	Aromatic water, distillate
Asava and Arishta	Self-generated alcoholic preparations	Fermented tincture
Panaka and Sharkara	Sweetened liquid	Syrup
Ksheerapaka	Milk decoction	Milk-based extract
Takra and Kanji	Fermented sour liquids	Fermented preparation

2.4 Semi-solid dosage forms

Semi-solid dosage forms are mainly meant for external application to the skin and to mucous membranes. They can act locally or, in some cases, systemically.

Form	Description	Base used
Ointment	Greasy, semi-solid preparation for external use	Oleaginous, absorption, water-removable or water-soluble base
Cream	Semi-solid emulsion, either o/w or w/o	Emulsion base
Paste	Contains a high proportion, usually more than 20 per cent, of solid powder	Stiff base
Gel	Semi-solid transparent preparation of a gelling agent in a liquid	Carbopol, HPMC, tragacanth
Jelly	Water-soluble, non-greasy, transparent	Gelling agent
Poultice or cataplasm	Soft, moist mass applied hot	Kaolin, herbal paste
Suppository base products	For rectal and vaginal use	Cocoa butter, PEG

Ideal properties of a semi-solid preparation. It should spread easily. It should not be gritty. It should be non-irritant and non-sensitising. It should release the drug at the site. It should be stable and free from microbial growth.

Ayurvedic counterparts. Lepa, Malahara, Upanaha, Kalka, Rasakriya, Avaleha and Anjana. Note that Avaleha is taken internally, while Lepa and Malahara are used externally. Both are semi-solid.

2.5 Parenteral dosage forms

The word parenteral means “beside the intestine”. A parenteral dosage form is one that is given by injection, bypassing the gastrointestinal tract.

2.5.1 Why parenterals are used

- When a quick action is needed, as in an emergency
- When the patient is unconscious or vomiting
- When the drug is destroyed in the stomach, as with insulin
- When the drug is poorly absorbed orally
- When an exact blood level is required
- For local action, as in a joint injection

2.5.2 Routes of parenteral administration

Route	Site	Usual volume	Onset
Intravenous (IV)	Directly into a vein	Few mL to litres	Immediate
Intramuscular (IM)	Into a muscle	Up to about 3 mL	Rapid
Subcutaneous (SC)	Under the skin	Up to about 2 mL	Moderate
Intradermal (ID)	Into the skin layer	About 0.1 mL	Slow, used for tests
Intrathecal	Into the spinal fluid	Small	Rapid
Intra-articular	Into a joint	Small	Local
Intracardiac	Into the heart	Small	Emergency only

2.5.3 Types of parenteral products

Type	Description
Small volume parenteral	Ampoule or vial, usually up to 100 mL
Large volume parenteral	Infusion of 100 mL or more, such as dextrose or normal saline
Powder for reconstitution	Dry powder in a vial, dissolved just before use; used for unstable drugs
Suspension for injection	For a depot effect and slow release
Emulsion for injection	Such as fat emulsions for nutrition
Implant	Sterile solid placed under the skin

2.5.4 Requirements of a parenteral product

This is a very common short answer question. Remember the six requirements.

1. **Sterility.** No living micro-organism should be present.
2. **Freedom from pyrogens.** Pyrogens are fever-producing substances, mainly bacterial endotoxins.
3. **Freedom from particulate matter.** No visible particles.
4. **Isotonicity.** The product should have the same osmotic pressure as blood, wherever possible.
5. **Correct pH.** Close to the pH of blood, to avoid pain and tissue damage.
6. **Stability and correct labelling.** The container must also be suitable and sealed.

2.5.5 Containers

Container	Use
Ampoule	Single dose, sealed by fusion of glass
Vial	Single or multiple dose, closed with a rubber bung and aluminium seal
Infusion bottle or bag	Large volume parenterals
Prefilled syringe	Ready-to-use single dose
Cartridge	Used in pen injectors

2.5.6 Position in Ayurveda

Classical Ayurveda has no parenteral dosage form. This is important to remember. The Drugs and Cosmetics Act reflects this: **the definition of a patent or proprietary ASU medicine under Section 3(h) specifically excludes a medicine administered by the parenteral route**. So an Ayurvedic injection cannot be licensed as an ASU drug.

The nearest classical parallels are procedures, not dosage forms — Nasya through the nose, Basti through the rectum, and Raktamokshana. These deliver the drug outside the oral route, but they are not injections.

KEY POINTS TO REMEMBER

- Dosage forms are classified by physical state, route, release pattern and system of medicine.
- Physical state classification: solid, liquid, semi-solid and gaseous.
- Tablet is the commonest and cheapest dosage form. Capsule masks taste completely.
- Suspension and emulsion are biphasic. Solution, syrup and elixir are monophasic.
- Creaming is reversible; cracking is not.
- The six requirements of a parenteral: sterility, freedom from pyrogens, freedom from particulate matter, isotonicity, correct pH, and stability with correct labelling.
- Ayurveda has no classical parenteral dosage form, and Section 3(h) excludes parenteral products from ASU patent or proprietary medicines.

EXAM CORNER — UNIT 2

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Classify pharmaceutical dosage forms according to physical state, with examples.
2. Classify dosage forms according to the route of administration.
3. Write a short note on tablets and their types.
4. Write a short note on capsules. Give their advantages and disadvantages.
5. Differentiate between a suspension and an emulsion.

6. Write a short note on the types of emulsion and their instability.
7. Write a short note on semi-solid dosage forms.
8. Differentiate between an ointment, a cream and a paste.
9. Enumerate the requirements of a parenteral preparation.
10. Enumerate the routes of parenteral administration with the usual volume for each.
11. Correlate classical Ayurvedic dosage forms with modern dosage forms.
12. Write a short note on modified Ayurvedic dosage forms.

How to attempt

- Every question in this unit can be answered with a table. Draw one. It saves time and gains marks.
- Always give at least one Ayurvedic example along with the modern example.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **50, 51, 52, 53, 54**

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UNIT 3
PHARMACEUTICAL FORMULATION

SYLLABUS POINTS

3.1 Principles of pharmaceutical formulation
3.2 Excipients and their functions
3.3 Solid dosage forms
3.4 Liquid and semi-solid dosage forms
3.5 Parenteral dosage forms

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe the principles of pharmaceutical formulations, excipients and their functions	1	Lecture	Knows how
Discuss formulation of parenteral dosage forms	1	Lecture	Knows how
Formulate solid dosage forms selecting appropriate excipients	4	Practical Training 1.5	Shows how
Formulate liquid dosage forms selecting appropriate excipients	4	Practical Training 1.6	Shows how
Identify an appropriate excipient according to solid and liquid dosage form	6	Experiential Learning 1.4	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	Yes

> This is the only unit in Module 1 that can be asked in all three formats, including the **Analytical Based Question of 20 marks**. Prepare this unit most carefully of all.

3.1 Principles of pharmaceutical formulation

3.1.1 What is formulation?

Formulation is the process of combining the active drug with suitable excipients to produce a final dosage form that is safe, effective, stable and acceptable to the patient.

A formula is not just a list. It is a balance. Every ingredient is there for a reason, and every quantity has been decided by trial.

3.1.2 The four aims of every formulation

1. **Deliver the correct dose** to the correct site, every time.
2. **Protect the drug** from air, moisture, light and microbial attack.
3. **Make the product acceptable** in taste, appearance, smell and ease of use.
4. **Keep the cost reasonable** so that the patient can afford it.

3.1.3 Factors to consider before formulating

Factor	What must be examined
Drug factors	Solubility, stability, particle size, polymorphism, taste, dose, compatibility
Route of administration	Oral, topical, parenteral, ocular and so on
Patient factors	Age, condition, ability to swallow, compliance
Disease factors	Acute or chronic, local or systemic action needed
Excipient factors	Availability, cost, compatibility, regulatory acceptance
Manufacturing factors	Equipment available, batch size, scale-up feasibility
Regulatory factors	Pharmacopoeial limits, permitted excipients, labelling requirements
Stability and packaging	Shelf life required, container type, storage condition

3.1.4 General steps in formulation development

1. Fix the target product profile — what the product should do.
2. Complete the preformulation study (see Unit 1, section 1.5.3).
3. Select the dosage form.
4. Select excipients and check compatibility.
5. Prepare trial batches in the laboratory.
6. Evaluate each trial batch.
7. Optimise the formula.
8. Take a pilot batch.
9. Conduct a stability study.
10. Scale up and validate the process.

3.1.5 The Ayurvedic principle of formulation

Ayurveda formulated on the same logic, but expressed it in its own terms.

Modern formulation principle	Ayurvedic expression
Choice of active ingredient	Dravya selection by Rasa, Guna, Virya, Vipaka, Prabhava
Choice of excipient or vehicle	Anupana and Sahapana
Additive to improve taste or action	Prakshepaka Dravya such as Sita, Madhu, Twak, Ela
Processing aid	Bhavana Dravya
Ratio and proportion	Given in Sharangadhara Samhita for each Kalpana
Dose	Matra, fixed for each Kalpana and each age
Stability	Saviryata Avadhi
Synergy and correction of toxicity	Samskara and Yogavahi property

> **Worth noting.** Madhu and Ghrita in an Avaleha are not merely sweeteners. They act as preservative, as taste corrector and as Yogavahi. One ingredient serving three functions is exactly what modern formulation calls a multifunctional excipient.

3.2 Excipients and their functions

3.2.1 Definition

An excipient is any substance, other than the active drug, that is included in a dosage form to help in its manufacture, to protect or support it, to improve its stability, bioavailability or patient acceptability, or to assist in identification.

An excipient is often called “inactive”. That word is misleading. An excipient does not treat the disease, but it decides whether the drug will work at all.

3.2.2 Ideal properties of an excipient

- Chemically inert and compatible with the drug
- Non-toxic and non-irritant
- Physiologically inert
- Stable, and should not affect drug stability
- Free from micro-organisms
- Should not interfere with assay or analysis
- Acceptable to the regulator and to the pharmacopoeia
- Cheap and easily available
- Acceptable in colour, odour and taste

3.2.3 Excipients used in solid dosage forms

This table is the heart of the unit. Learn it completely.

Excipient class	Function	Common examples
Diluent or filler	Adds bulk when the drug dose is small	Lactose, starch, microcrystalline cellulose, dicalcium phosphate, mannitol
Binder or adhesive	Holds the powder particles together	Starch paste, gelatin, acacia, PVP, HPMC
Disintegrant	Breaks the tablet in the body so the drug is released	Starch, sodium starch glycolate, croscarmellose sodium, crospovidone
Lubricant	Prevents the tablet sticking to punches and reduces friction	Magnesium stearate, talc, stearic acid
Glidant	Improves flow of the powder	Colloidal silicon dioxide, talc
Anti-adherent	Prevents sticking to punch faces	Talc, magnesium stearate
Colouring agent	Identification and elegance	Permitted food colours, iron oxides
Flavouring and sweetening agent	Improves taste	Sugar, mannitol, saccharin, permitted flavours
Coating material	Protection, taste masking, modified release	HPMC, ethyl cellulose, shellac, sugar
Capsule shell material	Encloses the fill	Gelatin, HPMC, pullulan

> **A common examination trap.** Magnesium stearate is a lubricant. Talc can act as a lubricant, a glidant and an anti-adherent. Starch can act as a diluent, a binder and a disintegrant. When a question asks for “one example”, give the one where that function is the main one.

3.2.4 Excipients used in liquid dosage forms

Excipient class	Function	Common examples
Vehicle or solvent	Dissolves or carries the drug	Purified water, alcohol, glycerin, propylene glycol, fixed oils
Co-solvent	Increases solubility	Alcohol, glycerin, propylene glycol
Suspending agent	Keeps solid particles dispersed	Acacia, tragacanth, sodium CMC, bentonite, MCC
Emulsifying agent	Keeps oil and water mixed	Acacia, tween, span, lecithin
Wetting agent	Helps water to wet hydrophobic powder	Surfactants, glycerin, alcohol
Preservative	Prevents microbial growth	Methyl paraben, propyl paraben, benzoic acid, sodium benzoate

Excipient class	Function	Common examples
Antioxidant	Prevents oxidation	Sodium metabisulphite, ascorbic acid, BHA, BHT
Buffer	Maintains pH	Citrate, phosphate, acetate buffers
Sweetening agent	Improves taste	Sucrose, sorbitol, saccharin sodium
Flavour and colour	Improves acceptability	Permitted flavours and colours
Viscosity builder	Improves pourability and stability	Sodium CMC, HPMC, tragacanth
Chelating agent	Binds trace metals that cause degradation	Disodium EDTA

3.2.5 Excipients used in semi-solid dosage forms

Class	Function	Examples
Base	The main carrier	White soft paraffin, wool fat, PEG, emulsifying wax
Emulsifier	For creams	Cetostearyl alcohol, emulsifying wax
Gelling agent	For gels	Carbopol, HPMC, tragacanth, sodium alginate
Humectant	Prevents drying	Glycerin, propylene glycol
Preservative	Prevents microbial growth	Parabens, benzyl alcohol
Penetration enhancer	Improves absorption through skin	Propylene glycol, DMSO, oleic acid
Antioxidant	Prevents rancidity	BHA, BHT, tocopherol

Types of ointment base

Base	Character	Example
Oleaginous or hydrocarbon	Greasy, occlusive, does not wash off	White soft paraffin
Absorption base	Can take up water; anhydrous	Wool fat, hydrophilic petrolatum
Water-removable or emulsion base	o/w type, washes off with water	Hydrophilic ointment
Water-soluble base	Greaseless, completely water soluble	Polyethylene glycol ointment

3.2.6 Excipients used in parenteral dosage forms

Here the list is short, and every entry has strict limits.

Class	Function	Examples
Vehicle	Aqueous or non-aqueous carrier	Water for Injection, fixed oils such as sesame oil
Tonicity adjuster	Makes the solution isotonic with blood	Sodium chloride, dextrose
Buffer	Maintains pH	Citrate, phosphate, acetate
Antioxidant	Prevents oxidation	Sodium metabisulphite, ascorbic acid
Preservative	Only for multiple dose vials; never for IV in large volume or for intrathecal use	Benzyl alcohol, phenol, chlorocresol
Chelating agent	Binds metal ions	Disodium EDTA
Solubiliser and co-solvent	Improves solubility	Propylene glycol, polysorbate 80
Cryoprotectant	Protects during freeze drying	Mannitol, lactose, sucrose

3.2.7 Excipients in Ayurvedic formulations

Ayurveda has used excipients for centuries, under different names.

Ayurvedic material	Function in modern terms
Madhu (honey)	Sweetener, preservative, Yogavahi
Sita and Guda (sugar and jaggery)	Sweetener, bulking agent, preservative in Avaleha
Ghrita and Taila	Vehicle, solvent for lipid-soluble principles
Godugdha (milk)	Vehicle, extraction medium in Ksheerapaka
Jala (water)	Vehicle and solvent
Gomutra	Processing medium in Shodhana
Kanji and Takra	Acidic media used in Shodhana
Guggulu	Binder in Guggulu Kalpana
Babbula and Khadira Kwatha	Binder and granulating fluid for Vati
Talc and Sphatika	Glidant and astringent
Bhavana Dravya	Wetting agent, granulating fluid, particle size reducer, potency enhancer
Prakshepaka Dravya	Flavour, taste corrector, adjuvant

3.3 Formulation of solid dosage forms

3.3.1 Methods of tablet manufacture

Method	When used	Steps
Wet granulation	Most common; for poorly flowing and poorly compressible powders	Weighing, sieving, mixing, preparation of binder solution, wet massing, wet screening, drying, dry screening, lubrication, compression
Dry granulation or slugging	For moisture-sensitive and heat-sensitive drugs	Weighing, mixing, slugging or roller compaction, milling, sieving, lubrication, compression
Direct compression	When the powder flows and compresses well	Weighing, sieving, mixing with directly compressible excipients, lubrication, compression

Advantages of direct compression. Fewer steps, less equipment, no heat, no moisture, lower cost, faster. **Limitation.** Only possible if the drug has good flow and good compressibility, and the dose is not too large.

3.3.2 Evaluation of tablets

Test	Purpose
Description and appearance	Shape, colour, surface, marking
Uniformity of weight	Each tablet contains the intended weight
Uniformity of content	Each tablet contains the intended amount of drug
Hardness or crushing strength	Should withstand handling
Friability	Loss on tumbling should generally be under 1 per cent
Disintegration time	Time to break into particles
Dissolution	Rate at which the drug goes into solution
Assay	Actual drug content

3.3.3 Common tablet defects

Defect	Meaning	Usual cause
Capping	The top of the tablet separates	Air entrapment, too much fines, over-drying
Lamination	The tablet splits into layers	Air entrapment, excess lubricant
Sticking	Material sticks to the die wall	Excess moisture, insufficient lubricant
Picking	Material sticks to the punch face and lifts out	Excess moisture, deep engraving
Mottling	Uneven colour	Uneven distribution of colour or drug
Chipping and cracking	Edges break	Worn punches, brittle granules
Double impression	Extra impression on the surface	Free rotation of the punch

> This defects table is a very likely SAQ. It is also a favourite viva question at the tablet machine.

3.3.4 Formulation of capsules

Hard gelatin capsule. The powder or granule blend is prepared with a diluent, a glidant and a lubricant. The blend must flow well, because filling depends on volume, not weight. The shell size is selected from the fill volume.

Soft gelatin capsule. The fill is a liquid, an oil, a suspension or a paste. The shell contains gelatin, a plasticiser such as glycerin or sorbitol, water, colour and opacifier. Manufacture is usually by the rotary die process.

3.4 Formulation of liquid and semi-solid dosage forms

3.4.1 Formulating a solution or syrup

1. Select the vehicle. Purified water is the first choice.
2. Add a co-solvent if the drug will not dissolve.
3. Add a buffer to hold the pH where the drug is most stable.
4. Add an antioxidant and a chelating agent if the drug oxidises.
5. Add a preservative, because water supports microbial growth.
6. Add sweetener, flavour and colour last.
7. Make up to volume, filter, fill and seal.

3.4.2 Formulating a suspension

The main problem is settling. The solution is to control the particle size and to control the viscosity.

- Reduce particle size, but not so much that caking follows.
- Wet the powder with a wetting agent.
- Add a suspending agent to raise viscosity.

• Decide between a flocculated and a deflocculated system.

Property	Flocculated	Deflocculated
Particles	Loose aggregates	Separate particles
Rate of settling	Fast	Slow
Sediment volume	Large and loose	Small and compact
Redispersion	Easy	Difficult; cakes form
Appearance	Slightly cloudy supernatant	Clear supernatant

A good pharmaceutical suspension is usually a **controlled flocculated** system — it settles, but redisperses on shaking.

3.4.3 Formulating an emulsion

Methods of preparation:

Method	Description
Dry gum or continental method	Gum is mixed with oil first; ratio of oil, water and gum is 4 : 2 : 1
Wet gum or English method	Gum is mixed with water first, then oil added
Bottle or Forbes bottle method	For volatile oils; shaken in a bottle
Insitu soap method	Soap is formed during preparation, as in lime water and oil
Mechanical method	Homogeniser, colloid mill or ultrasonicator used

The **HLB system** (Hydrophile-Lipophile Balance) is used to select an emulsifier. A low HLB value, roughly 3 to 6, suits w/o emulsions. A high HLB value, roughly 8 to 18, suits o/w emulsions.

3.4.4 Formulating a semi-solid preparation

Method	Description	Suitable for
Fusion	Bases are melted together, the drug is added, then cooled with stirring	Ointments with waxy bases
Trituration or levigation	The drug is triturated with a small quantity of base, then the rest is added geometrically	Ointments with a soft base
Emulsification	Oil phase and water phase are heated separately, then mixed with stirring	Creams
Dispersion	Gelling agent is dispersed in the vehicle and allowed to swell	Gels

Evaluation of semi-solids. Appearance, pH, viscosity and rheology, spreadability, extrudability, drug content, in vitro drug release, microbial limit, and stability including phase separation.

3.5 Formulation of parenteral dosage forms

3.5.1 The formulation sequence

1. Select the vehicle. **Water for Injection** is the standard aqueous vehicle. It must be pyrogen free.
2. Adjust tonicity, generally to be isotonic with blood.
3. Adjust and buffer the pH.
4. Add an antioxidant or chelating agent if needed.
5. Add a preservative **only** for multiple dose containers.
6. Filter through a 0.22 micron membrane.
7. Fill into ampoules, vials or bottles.
8. Seal, then sterilise.
9. Carry out leak testing, visual inspection, sterility testing and pyrogen or endotoxin testing.

3.5.2 Methods of sterilisation

Method	Condition	Suitable for
Moist heat (autoclaving)	121 °C for 15 minutes at 15 psi	Aqueous solutions, glassware, rubber closures
Dry heat	160 °C for 2 hours, or 180 °C for 30 minutes	Glassware, oils, powders; also depyrogenates
Filtration	0.22 micron membrane	Heat-sensitive solutions
Gaseous	Ethylene oxide	Plastic devices, heat-sensitive materials
Radiation	Gamma rays	Disposable devices, some powders
Aseptic processing	Sterile filtration and filling in a sterile area	Products that cannot be terminally sterilised

3.5.3 Environment and testing

Parenteral manufacture takes place in **clean rooms**, classified as Grade A, B, C and D. Grade A, the laminar air flow zone, is used for filling.

Two important tests:

- **Sterility test** — confirms the absence of viable micro-organisms.
- **Pyrogen or bacterial endotoxin test** — the rabbit test, or the LAL (Limulus Amoebocyte Lysate) test.

3.5.4 Position of parenterals in Ayurveda

As stated in Unit 2, Ayurveda has no classical parenteral dosage form, and Section 3(h) of the Drugs and Cosmetics Act excludes parenteral products from the definition of a patent or proprietary ASU medicine. A postgraduate student should know parenteral formulation for two reasons: to understand sterile manufacture as a scientific discipline, and to be able to work in a research setting where Ayurvedic extracts are tested by injectable routes in animals.

KEY POINTS TO REMEMBER

- Formulation balances four aims: correct dose, protection, acceptability and cost.
- An excipient is not truly inactive. It decides whether the drug will work.
- Tablet excipients: diluent, binder, disintegrant, lubricant, glidant, anti-adherent, colour, flavour, coating.
- Three methods of tablet manufacture: wet granulation, dry granulation, direct compression.
- Tablet defects: capping, lamination, sticking, picking, mottling, chipping, double impression.
- Controlled flocculation gives the best suspension — it settles but redisperses.
- Dry gum method ratio for emulsion: oil 4, water 2, gum 1.
- Low HLB suits w/o; high HLB suits o/w.
- Parenterals: Water for Injection, isotonic, buffered, sterile, pyrogen free, particle free.
- Preservatives are never used in large volume intravenous or intrathecal products.
- Ayurvedic excipients: Madhu, Sita, Ghrita, Taila, Bhavana Dravya, Prakshepaka Dravya.

EXAM CORNER — UNIT 3

This unit can be asked as ABQ (20 marks), SAQ (5 marks) and LAQ (10 marks). Prepare it fully.

Likely short answer questions (5 marks)

1. Define an excipient. Write the ideal properties of an excipient.
2. Enumerate the excipients used in tablet formulation with one example of each.
3. Write a short note on disintegrants.
4. Write a short note on tablet defects.
5. Differentiate between a flocculated and a deflocculated suspension.
6. Write a short note on the HLB system.
7. Enumerate the methods of preparation of emulsion.
8. Write a short note on ointment bases.
9. Enumerate the methods of sterilisation with conditions.
10. Write a short note on Water for Injection.
11. Write a short note on Bhavana as a pharmaceutical process.
12. Correlate Ayurvedic Prakshepaka Dravya with modern excipients.

Likely long answer questions (10 marks)

1. Describe the excipients used in solid dosage forms with their functions and examples.
2. Describe the methods of tablet manufacture. Add a note on the evaluation of tablets.
3. Describe the formulation and evaluation of a pharmaceutical suspension.
4. Describe the formulation of parenteral dosage forms, including sterilisation and testing.
5. Discuss the principles of pharmaceutical formulation and correlate them with the principles of Bhaishajya Kalpana.

Likely analytical based question (20 marks)

1. A pharmacy wishes to convert a classical Kwatha into a tablet, so that patients can carry it easily. Analyse the pharmaceutical problems that will arise. Select suitable excipients, describe the method of manufacture you will adopt, and list the evaluation parameters you will apply.
2. An Ayurvedic Churna is bitter, hygroscopic and flows poorly. It has to be marketed in a convenient dosage form. Analyse the situation and justify your choice of dosage form, excipients and packaging.
3. A batch of tablets shows capping and lamination during compression. Analyse the possible causes, and give the corrective steps you would take at the formulation stage and at the machine stage.

How to attempt an ABQ

- An ABQ carries 20 marks. It tests analysis, not memory. Do not simply write everything you know.
- Structure the answer in four parts: identify the problem, analyse the causes, propose the solution, justify the choice.
- Give at least one table and one flow chart.
- End with a short paragraph on how this applies to Ayurvedic pharmaceutics.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **55, 56, 57, 58, 59**

1. **55.** Pharmaceutical Formulation: The Science and Technology of Dosage Forms. United Kingdom: Royal Society of Chemistry; 2018.
2. **56.** Excipient Applications in Formulation Design and Drug Delivery. Germany: Springer International Publishing; 2015.
3. **57.** Handbook of Pharmaceutical Excipients. United Kingdom: Pharmaceutical Press; 2009.
4. **58.** Innovative Dosage Forms: Design and Development at Early Stage. Germany: Wiley; 2019.
5. **59.** Pharmaceutical Dosage Forms: Parenteral Medications. United Kingdom: CRC Press; 2018.

UNIT 4
PHARMACEUTICAL MANUFACTURING AND QUALITY CONTROL

SYLLABUS POINTS

4.1 Good Manufacturing Practices (GMP and cGMP)
4.2 Quality control and quality assurance
4.3 Packaging and labelling
4.4 Portability and storage

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Discuss Good Manufacturing Practices (GMP and cGMP)	1	Lecture	Knows how
Discuss pharmaceutical stability and shelf life	1	Lecture	Knows how
Evaluate GMP audit and compliance	2	Practical Training 1.7	Does
Demonstrate, design and evaluate pharmaceutical packing and labelling	2	Practical Training 1.8	Shows how
Design and develop pharmaceutical packaging systems and labelling	4	Experiential Learning 1.5	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

4.1 Good Manufacturing Practices (GMP and cGMP)

4.1.1 Definition

Good Manufacturing Practice (GMP) is that part of quality assurance which ensures that products are consistently produced and controlled to the quality standards appropriate to their intended use.

The key word is **consistently**. Making one good batch is not GMP. Making every batch good is GMP.

4.1.2 What does “cGMP” mean?

The small letter “c” stands for **current**. It means the manufacturer must follow the practices that are current today, not the practices that were acceptable twenty years ago. Technology changes. Standards rise. A manufacturer cannot say that his method was good enough in the past.

4.1.3 Objectives of GMP

1. To ensure that the product is of the quality required for its intended use.
2. To prevent mix-up of products and materials.
3. To prevent contamination and cross-contamination.
4. To ensure that errors are detected and corrected.
5. To ensure that every step is documented and traceable.
6. To protect the patient.

4.1.4 The main elements of GMP

Remember these ten headings. Almost every GMP question can be answered from them.

Element	What it covers
Personnel	Adequate number, qualified, trained, healthy, in proper uniform
Premises	Suitable location, proper layout, unidirectional flow, easy to clean
Equipment	Suitable design, non-reactive surface, easy to clean, calibrated, qualified
Materials	Approved suppliers, quarantine, testing, correct storage, labelled status
Documentation	Master formula, batch manufacturing record, SOPs, logbooks, records retained
Production	Defined process, in-process controls, line clearance, yield reconciliation
Quality control	Independent department, sampling, testing, release or rejection
Sanitation and hygiene	Cleaning schedules, pest control, personal hygiene, waste disposal
Validation and qualification	Proof that the process, equipment and methods work as intended
Complaints, recall and self-inspection	Handling of market complaints, product recall, internal audits

4.1.5 The four qualifications

Stage	Full form	What it proves
DQ	Design Qualification	The design is suitable for the purpose
IQ	Installation Qualification	The equipment has been installed correctly
OQ	Operational Qualification	The equipment works over the operating range
PQ	Performance Qualification	The equipment performs consistently in actual use

4.1.6 GMP for Ayurvedic drugs — Schedule T

For Ayurvedic, Siddha and Unani drugs, GMP is laid down in **Schedule T** of the Drugs and Cosmetics Rules, 1945, read with Rule 157. Schedule T has two parts.

- **Part I** — Good Manufacturing Practices: requirements of the factory premises, the sections needed, hygiene, machinery, water supply, quality control, records, and the qualification of technical staff.
- **Part II** — the list of recommended machinery, equipment and minimum manufacturing premises required for each category of ASU dosage form.

The stated aims of Schedule T are:

1. Raw materials used must be authentic, of prescribed quality and free from contamination.
2. The manufacturing process must be as prescribed, to maintain the standards.
3. Adequate quality control measures must be adopted.
4. The drug released for sale must be of acceptable quality.
5. Each licensee must document the methodology and procedures in a manual, kept for reference and inspection.

Minimum area under Schedule T. For an Ayurveda or Siddha manufacturing unit, the minimum covered area is 1200 square feet, with separate cabins or partitions for each activity. If Unani medicines are made in the same premises, an additional area is required.

Technical staff. The person in charge must hold a recognised qualification in Ayurveda, or be an Ayurvedic pharmacist, as prescribed in Schedule T.

> **Compare and remember.** Schedule M is the GMP for allopathic drugs. Schedule T is the GMP for ASU drugs. Rule 157 relates to Schedule T. Rule 158-B relates to licensing of ASU medicines.

4.2 Quality control and quality assurance

4.2.1 Definitions

Quality control (QC) is that part of GMP which is concerned with sampling, specifications and testing, and with the organisation, documentation and release procedures which ensure that the necessary tests are carried out, and that materials are not released for use, nor products released for sale, until their quality has been judged satisfactory.

Quality assurance (QA) is the sum total of the organised arrangements made with the object of ensuring that products are of the quality required for their intended use.

Put simply: QC **tests** the product. QA **builds** the system that makes the product right.

4.2.2 Difference between QC and QA

Point	Quality Control	Quality Assurance
Nature	Product oriented	Process and system oriented
Aim	To detect defects	To prevent defects
Activity	Testing and inspection	Planning, documentation, audit, training
Question asked	Is this batch correct?	Is our system capable of always being correct?
Timing	Mainly after manufacture	Throughout, before, during and after
Responsibility	QC department	Everyone in the organisation
Relation	QC is a part of QA	QA is the wider concept

4.2.3 Functions of the quality control department

- Sampling of raw materials, packing materials, in-process material and finished goods
- Testing against the specification
- Approval or rejection, with reasons recorded
- Stability testing
- Maintenance of reference samples and reference standards
- Calibration of laboratory instruments
- Investigation of out-of-specification results
- Retention of records

4.2.4 Levels of quality checking

Level	What is checked
Raw material control	Identity, purity, assay of every incoming material
In-process control	Checks during manufacture, such as weight variation, hardness, pH, viscosity
Finished product control	Full testing against the finished product specification
Packaging material control	Dimensions, printing, integrity
Stability control	Behaviour of the product over its shelf life

4.2.5 Quality control in Ayurveda

For Ayurvedic products, quality control operates at three levels.

Level	Classical method	Modern method
Raw drug	Namarupa Vijnana, Grahya-Agrahya Lakshana, Desha and Kala of collection	Botanical authentication, macroscopy, microscopy, TLC, HPTLC, heavy metal and pesticide limits
In-process	Paka Lakshana, Siddhi Lakshana, Varna, Gandha, Rasa, Sparsha	pH, moisture, temperature, particle size, loss on drying
Finished product	Bhasma Pariksha such as Rekhapurnata, Varitara, Nishchandratva, Apunarbhava, Niruttha	Ash value, acid insoluble ash, XRD, SEM-EDAX, particle size analysis, assay

Detailed quality control parameters for each Ayurvedic dosage form are dealt with in **Module 7** of this course.

4.3 Packaging and labelling

4.3.1 Functions of packaging

1. **Protection** — from moisture, oxygen, light, heat, microbes and mechanical damage.
2. **Containment** — holds the product without leaking.
3. **Identification** — tells the user what the product is.
4. **Information** — carries the label, dose, warnings and expiry.
5. **Convenience** — easy to open, use, measure and reclose.
6. **Presentation and marketing** — attracts and reassures the buyer.
7. **Tamper evidence** — shows if the pack has been opened.

4.3.2 Types of packaging

Level	Meaning	Examples
Primary	In direct contact with the product	Bottle, blister, strip, ampoule, vial, tube, sachet
Secondary	Encloses the primary pack	Carton, box, with the package insert
Tertiary	For bulk transport	Shipper carton, corrugated box, pallet

4.3.3 Packaging materials

Material	Advantages	Disadvantages
Glass	Inert, transparent, impermeable, can be sterilised	Heavy, breakable, costly to transport
Plastic	Light, unbreakable, cheap, mouldable	May allow permeation, may leach, may sorb the drug, environmental problem
Metal	Strong, impermeable, light in the case of aluminium	May corrode, needs lacquer coating
Paper and board	Cheap, printable, recyclable	No barrier property; used for secondary packing only
Rubber	Elastic, reseals after needle puncture	May leach and may sorb preservatives

Types of glass. Type I is neutral borosilicate, the best, used for parenterals. Type II is treated soda-lime. Type III is ordinary soda-lime. Type IV, or NP, is for oral and topical use only.

4.3.4 Labelling

The label is a legal document. What it must carry is prescribed under the Drugs and Cosmetics Rules, 1945.

General particulars on the label of a medicine

- Name of the drug or formulation
- Complete list of ingredients with quantities
- Net content — weight, volume or number of units
- Batch number
- Manufacturing date
- Expiry date, or date before which the drug is to be used
- Manufacturing licence number
- Name and full address of the manufacturer
- Maximum retail price
- Storage conditions
- Directions for use and dose

Additional particulars for Ayurvedic medicines

- The words “Ayurvedic Medicine”
- The reference to the classical text, in the case of a classical medicine
- For a patent or proprietary medicine, the true list of all ingredients
- The caution “To be taken under medical supervision” where applicable
- For medicines containing Schedule E(1) substances, the warning as prescribed

4.3.5 Package insert

The package insert is the leaflet placed inside the carton. It gives the composition, indications, dose, mode of administration, contraindications, side effects, precautions, drug interactions, storage and the manufacturer’s details.

4.4 Portability and storage

4.4.1 Portability

Portability means how easily a medicine can be carried, handled and transported without damage or loss of quality.

Factors that decide portability

- Physical state — solids travel better than liquids
- Weight and volume of the pack
- Strength of the container — glass breaks, plastic does not
- Sensitivity to temperature and humidity during transit
- Need for a cold chain
- Unit-dose packing, which improves portability greatly

Ayurvedic relevance. Portability is one main reason why classical Kalpanas are converted into modified forms. A Kwatha must be prepared fresh and cannot be carried. Convert it into Ghana Vati, and the patient can carry a month’s treatment in a small bottle. Similarly, Ghrita in a soft gelatin capsule travels better than Ghrita in a jar.

4.4.2 Storage

Correct storage protects the product for its full shelf life.

Standard storage conditions

Term used on the label	Meaning
Store in a cool place	Between 8 °C and 25 °C
Store at room temperature	Usually between 20 °C and 30 °C
Store in a refrigerator	Between 2 °C and 8 °C
Store in a freezer	Below 0 °C
Protect from light	Store in a light-resistant container
Store in a dry place	Protect from moisture
Store in a well-closed container	Protects from extraneous solids
Store in a tightly closed container	Protects from loss and from moisture
Store in a hermetically sealed container	Impervious to air

4.4.3 Stability and shelf life

Stability is the capacity of a product to remain within its specifications for identity, strength, quality and purity, throughout its period of storage and use.

Shelf life is the time during which the product remains within its specification, when stored under the stated conditions. The expiry date is fixed from the shelf life.

Types of stability

Type	What it concerns
Chemical	Assay remains within the limit; no harmful degradation product
Physical	Appearance, hardness, dissolution and other physical properties unchanged
Microbiological	Sterility or microbial limits maintained; preservative remains effective
Therapeutic	The therapeutic effect is unchanged
Toxicological	No significant increase in toxicity

Factors affecting stability. Temperature, moisture and humidity, light, oxygen, pH, container material, microbial contamination, and the excipients themselves.

Stability study conditions are given in Unit 1, section 1.5.7.

4.4.4 Saviryata Avadhi — the Ayurvedic concept of shelf life

Ayurveda has its own concept of shelf life, called **Saviryata Avadhi**. Sharangadhara Samhita gives the period for which each Kalpana retains its potency.

Kalpana	Traditional Saviryata Avadhi as per Sharangadhara
Swarasa, Kalka, Kwatha	To be used fresh, on the same day
Churna	About 2 months
Gutika and Vati	About 1 year
Ghrita and Taila	About 16 months
Avaleha and Guda paka	About 1 year
Asava and Arishta	Improve with age; no fixed limit
Bhasma and Rasaushadhi	Considered to remain potent for a long period

Important for the exam. The Government of India has notified shelf-life periods for Ayurvedic, Siddha and Unani drugs under the Drugs and Cosmetics Rules. For a licensed product, the legal date of expiry printed on the label is what applies. The classical Saviryata Avadhi is the traditional concept. Always mention both in an answer, and state clearly which one is legally binding.

KEY POINTS TO REMEMBER

- GMP ensures products are **consistently** produced and controlled. The “c” in cGMP stands for current.
- Ten elements of GMP: personnel, premises, equipment, materials, documentation, production, quality control, sanitation, validation, complaints and self-inspection.
- Qualification stages: DQ, IQ, OQ, PQ.
- Schedule T is GMP for ASU drugs, under Rule 157. Part I gives GMP; Part II gives machinery and premises.
- QC detects defects; QA prevents them. QC is a part of QA.
- Packaging levels: primary, secondary, tertiary.
- Glass Type I is neutral borosilicate, used for parenterals.
- Saviryata Avadhi is the Ayurvedic concept of shelf life. The legal expiry date is what binds a licensed product.

EXAM CORNER — UNIT 4

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Define GMP. Enumerate its main elements.

2. What is cGMP? Why is the word “current” important?
3. Write a short note on Schedule T.
4. Differentiate between quality control and quality assurance.
5. Write a short note on DQ, IQ, OQ and PQ.
6. Enumerate the functions of packaging.
7. Write a short note on the types of packaging.
8. Write a short note on the types of glass used in pharmaceutical packaging.
9. Enumerate the particulars that must appear on the label of an Ayurvedic medicine.
10. Write a short note on the types of stability.
11. Write a short note on Saviryata Avadhi.
12. Write a short note on storage conditions and their meaning.

How to attempt

- For any GMP question, begin with the definition, then the ten elements as a list. That alone secures most of the marks.
- For the QC and QA question, always draw the comparison table.
- Whenever the question touches Ayurveda, mention Schedule T and Saviryata Avadhi.

REFERENCES USED IN THIS UNIT

Sources for this unit: Drugs and Cosmetics Act, 1940 and Drugs and Cosmetics Rules, 1945, Schedule T and Rule 157 (CDSCO); Press Information Bureau, Government of India, on quality control of AYUSH drugs; Ministry of AYUSH, AYUSH Next portal; ICH Q1A(R2), Q9 and Q10 guidelines. Reference **163** of the NCISM list — Lohar D, Protocol for Testing of Ayurvedic, Siddha and Unani Medicines, Department of AYUSH, Government of India (2008) — also supports this unit.

UNIT 5
BIOPHARMACEUTICS

SYLLABUS POINTS

5.1 Biopharmaceutics and pharmacokinetics
5.2 Bioavailability and bioequivalence
5.3 Pharmaceutical dissolution and absorption
5.4 Pharmaceutical product development and optimization

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe biopharmaceutics and pharmacokinetics	1	Lecture	Knows how
Describe bioavailability and bioequivalence	1	Lecture	Knows how
Discuss pharmaceutical product development and optimization	1	Lecture	Knows how
Design and optimize pharmaceutical formulation for enhanced bioavailability	7	Experiential Learning 1.6	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	Yes

5.1 Biopharmaceutics and pharmacokinetics

5.1.1 Definitions

Biopharmaceutics is the study of the relationship between the physicochemical properties of a drug, the dosage form in which it is given, the route of administration, and the rate and extent of drug absorption into the body.

In plain words, biopharmaceutics answers one question: *does the dosage form actually deliver the drug into the blood?*

Pharmacokinetics is the study of what the body does to the drug. It covers Absorption, Distribution, Metabolism and Excretion, together called **ADME**.

Pharmacodynamics is the study of what the drug does to the body.

> An easy way to remember: **Pharmacokinetics** — body acts on drug. **Pharmacodynamics** — drug acts on body. **Biopharmaceutics** — the dosage form connects the two.

5.1.2 The LADMER sequence

The full journey of a drug given orally is often written as LADMER.

Letter	Step
L	Liberation — release of the drug from the dosage form
A	Absorption — passage into the systemic circulation
D	Distribution — spread into tissues
M	Metabolism — chemical change, mainly in the liver
E	Excretion — removal from the body
R	Response — the therapeutic effect

5.1.3 Basic pharmacokinetic parameters

Parameter	Meaning
C_{max}	The highest concentration reached in the blood
T_{max}	The time at which C_{max} is reached
AUC	Area under the plasma concentration-time curve; represents the total amount of drug that reached the blood
$t_{1/2}$	Half life; the time in which the concentration falls by half
V_d	Volume of distribution
Cl	Clearance; the volume of plasma cleared of the drug per unit time
MEC	Minimum effective concentration
MTC	Minimum toxic concentration
Therapeutic window	The range between MEC and MTC

> **C_{max} and T_{max} tell you about the *rate* of absorption. AUC tells you about the *extent* of absorption.** This one sentence answers half the questions on this unit.

5.1.4 Routes and orders of kinetics

- **Zero order kinetics.** A constant amount of drug is eliminated per unit time. The graph is a straight line.
- **First order kinetics.** A constant fraction of the drug is eliminated per unit time. Most drugs follow this. The graph is exponential.

5.2 Bioavailability and bioequivalence

5.2.1 Bioavailability

Bioavailability is the rate and extent to which the active drug is absorbed from a dosage form and becomes available at the site of action.

Bioavailability of an intravenous injection is taken as 100 per cent, because the whole dose enters the blood directly.

Types of bioavailability

Type	Comparison made against
Absolute bioavailability	The intravenous dose of the same drug
Relative or comparative bioavailability	Another dosage form or another product of the same drug

5.2.2 Factors affecting bioavailability

This is a very likely long answer question. Organise it under three headings.

A. Pharmaceutical factors — related to the drug and the dosage form

- Particle size and surface area — smaller particles dissolve faster
- Solubility and dissolution rate
- Salt form and polymorphic form
- Degree of crystallinity; amorphous forms dissolve faster
- Nature and quantity of excipients
- Type of dosage form; solutions are absorbed faster than suspensions, which are faster than capsules, which are faster than tablets
- Manufacturing variables such as compression force and granulation method
- Disintegration and dissolution time

B. Physiological factors — related to the patient

- pH of the gastrointestinal tract
- Gastric emptying rate and intestinal transit time
- Presence of food
- Blood flow to the gut
- Surface area available for absorption
- Age, sex, disease state and pregnancy
- Presence of other drugs

C. Pharmacokinetic factors

- First-pass metabolism in the liver and gut wall
- Protein binding
- Efflux transporters such as P-glycoprotein
- Enterohepatic circulation

5.2.3 Methods of assessing bioavailability

Method	What is measured
Plasma level study	Drug concentration in blood over time; gives C_{max}, T_{max} and AUC
Urinary excretion study	Cumulative amount of drug excreted in urine
Pharmacological response	Measurable effect, such as change in blood pressure
Clinical response	Therapeutic outcome; least accurate but most relevant

5.2.4 Bioequivalence

Two products are **bioequivalent** when they contain the same active ingredient in the same dose and dosage form, and when their rate and extent of absorption are not significantly different after administration in the same molar dose under similar conditions.

Related terms

Term	Meaning
Pharmaceutical equivalence	Same active ingredient, same dose, same dosage form, same route; excipients may differ
Pharmaceutical alternative	Same active moiety, but different salt, ester or dosage form
Bioequivalence	Equivalent rate and extent of absorption
Therapeutic equivalence	Pharmaceutically equivalent and bioequivalent, and therefore expected to give the same clinical effect

Why bioequivalence matters. A generic company does not repeat full clinical trials. It proves that its product behaves in the body like the innovator product. That proof is the bioequivalence study. This is the basis of the ANDA route described in Unit 1.

Design of a bioequivalence study. Usually an open, randomised, two-treatment, two-period, two-sequence, single-dose, crossover study in healthy adult volunteers, with a suitable washout period between the two periods.

5.2.5 Biopharmaceutics Classification System (BCS)

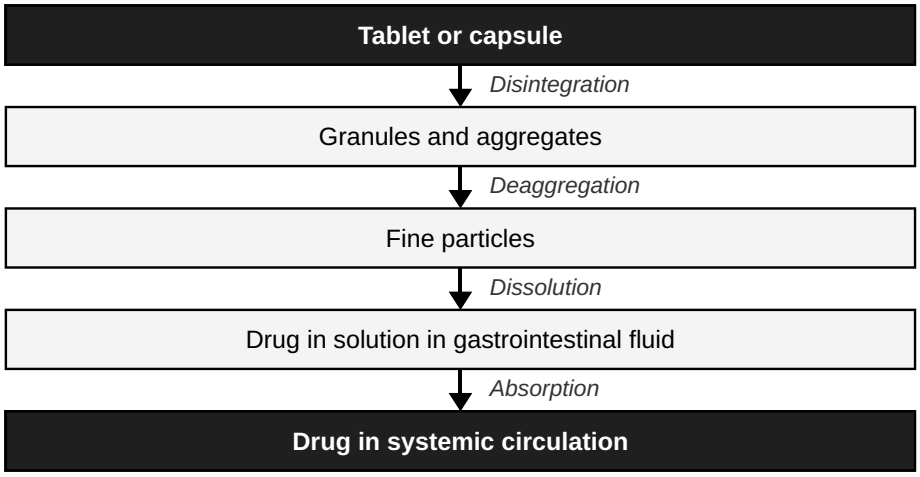
The BCS divides drugs into four classes on the basis of aqueous solubility and intestinal permeability.

Class	Solubility	Permeability	Rate-limiting step	Example approach
I	High	High	Gastric emptying	Usually no problem
II	Low	High	Dissolution	Improve dissolution
III	High	Low	Permeability	Use permeation enhancers
IV	Low	Low	Both	Most difficult; needs advanced delivery

Most plant-derived constituents of interest in Ayurveda — for example curcumin — behave like **Class IV** compounds. They dissolve poorly and permeate poorly. This is exactly why nanotechnology has become important in Ayurvedic research, and it is the link between this unit and Module 8.

5.3 Pharmaceutical dissolution and absorption

5.3.1 The sequence for a solid oral dosage form



For most poorly soluble drugs, **dissolution is the slowest step**, and therefore it controls the rate of absorption. This is why dissolution testing matters so much.

5.3.2 The Noyes-Whitney principle

The rate of dissolution increases when:

- the surface area of the particles is increased, that is, particle size is reduced
- the solubility of the drug in the medium is increased
- the concentration of dissolved drug around the particle is kept low, by good sink conditions
- the diffusion layer around the particle is made thinner, by stirring
- the temperature is raised

> **The Ayurvedic connection.** Bhavana, Mardana and Puta all reduce particle size and increase surface area. Classical Rasashastra was, in effect, applying the same principle centuries before it was written as an equation. This point is worth writing in any long answer on this unit.

5.3.3 Dissolution testing

Apparatus	Name	Typical use
USP Apparatus I	Rotating basket	Capsules, floating dosage forms
USP Apparatus II	Paddle	Tablets; the commonest
USP Apparatus III	Reciprocating cylinder	Modified release
USP Apparatus IV	Flow-through cell	Poorly soluble drugs, implants

Common dissolution media are 0.1 N hydrochloric acid, which simulates gastric fluid, and phosphate buffer at pH 6.8, which simulates intestinal fluid. The usual volume is 900 mL at 37 °C.

Dissolution profile comparison uses two factors: the difference factor **f1** and the similarity factor **f2**. Profiles are generally accepted as similar when f2 lies between 50 and 100.

5.3.4 Mechanisms of drug absorption

Mechanism	Description	Energy needed	Carrier needed
Passive diffusion	Movement from high to low concentration	No	No
Pore transport	Passage of small water-soluble molecules through aqueous pores	No	No
Facilitated diffusion	Carrier-mediated, down the gradient	No	Yes
Active transport	Carrier-mediated, against the gradient	Yes	Yes
Ion pair transport	Charged drug pairs with an opposite ion and crosses	No	No
Endocytosis	Engulfing of large molecules or particles	Yes	No

Most drugs are absorbed by **passive diffusion**. Only the unionised, lipid-soluble form crosses the membrane easily. This is the pH-partition hypothesis: weak acids are better absorbed from the stomach, and weak bases from the intestine. In practice, however, the intestine absorbs most drugs, because its surface area is enormous.

5.4 Pharmaceutical product development and optimization

5.4.1 What is optimisation?

Optimisation is the process of finding the best possible combination of formulation and process variables, so that the product meets all the required quality attributes at the lowest cost and with the least variability.

Trial and error is not optimisation. Optimisation is systematic.

5.4.2 Quality by Design (QbD)

Under the ICH Q8 guideline, product development follows a designed approach rather than a trial approach.

Term	Meaning
QTPP — Quality Target Product Profile	What the finished product must achieve
CQA — Critical Quality Attribute	A property that must be within a limit, such as dissolution or assay
CMA — Critical Material Attribute	A raw material property that affects a CQA
CPP — Critical Process Parameter	A process setting that affects a CQA, such as compression force
Design space	The proven range of variables within which quality is assured
Control strategy	The set of controls that keeps the process inside the design space

5.4.3 Tools used in optimisation

- **Factorial design** — studies several factors and their interactions at the same time
- **Response surface methodology** — maps the response over the whole range
- **Central composite design and Box-Behnken design** — efficient experimental designs
- **Risk assessment tools** — Ishikawa or fishbone diagram, FMEA, as per ICH Q9
- **Process Analytical Technology (PAT)** — measurement of quality during the process, not after it

5.4.4 Strategies to improve bioavailability

This is directly required by Experiential Learning 1.6, and is a very likely long answer question.

Strategy	Method	Example or note
Particle size reduction	Micronisation, nanosizing	Increases surface area
Solid dispersion	Drug dispersed in a hydrophilic carrier	Converts crystalline drug to amorphous
Salt formation	Convert to a more soluble salt	Only for ionisable drugs
Complexation	Cyclodextrin inclusion complexes	Improves solubility and stability
Use of surfactants	Lowers interfacial tension	Improves wetting
Co-solvency	Alcohol, glycerin, propylene glycol	Increases solubility
Lipid-based systems	SMEDDS, SNEDDS, emulsions	Useful for lipophilic drugs
Nanotechnology	Nanoparticles, liposomes, niosomes, phytosomes	Detailed in Module 8
Permeation enhancers	Improve membrane crossing	For BCS Class III drugs
Metabolism inhibitors	Reduce first-pass loss	Piperine is a classical example
Change of route	Sublingual, transdermal, parenteral	Avoids first-pass metabolism

5.4.5 The Ayurvedic view of bioavailability

Ayurveda did not use the word bioavailability, but it worked on the same problem, and it had its own tools.

Ayurvedic concept	Modern interpretation
Yogavahi	A substance that carries other drugs and enhances their action; Madhu and Guggulu are examples
Anupana	The vehicle taken with the medicine; it modifies absorption and directs the drug
Bhavana	Repeated levigation; reduces particle size, increases surface area, adds active principles
Samskara	Processing that changes the properties of a substance
Deepana and Pachana	Improve digestion and absorption before the main drug is given
Trikatu and Pippali	Classical bioavailability enhancers; piperine inhibits first-pass metabolism
Sneha Kalpana	Lipid vehicle for lipid-soluble principles; comparable to a lipid-based delivery system
Bhasma	Extreme particle size reduction to the micro and nano range

Worth remembering. Trikatu is the best documented Ayurvedic example of a bioavailability enhancer. Piperine from Pippali and Maricha inhibits certain drug-metabolising enzymes and slows intestinal transit. So the classical instruction to give Trikatu along with a heavy formulation has a clear modern explanation.

KEY POINTS TO REMEMBER

- Biopharmaceutics links the dosage form to absorption. Pharmacokinetics is what the body does to the drug. Pharmacodynamics is what the drug does to the body.
- LADMER: Liberation, Absorption, Distribution, Metabolism, Excretion, Response.
- C_{max} and T_{max} show the rate of absorption. AUC shows the extent.
- Bioavailability of an intravenous dose is 100 per cent.
- Bioequivalence is the basis of generic approval through the ANDA route.
- BCS has four classes based on solubility and permeability. Most Ayurvedic phytoconstituents behave like Class IV.
- For poorly soluble drugs, dissolution is the rate-limiting step of absorption.
- USP Apparatus II, the paddle, is the commonest dissolution apparatus. Similarity factor f2 should be 50 to 100.
- Most drugs are absorbed by passive diffusion, in the unionised lipid-soluble form.
- Bhavana, Yogavahi, Anupana and Trikatu are the Ayurvedic tools for improving bioavailability.

EXAM CORNER — UNIT 5

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). It is not marked for ABQ.

Likely short answer questions (5 marks)

1. Define biopharmaceutics and pharmacokinetics. Differentiate between them.
2. Write a short note on the LADMER sequence.
3. Define C_{max}, T_{max} and AUC. State what each indicates.
4. Define bioavailability. Write its types.
5. Define bioequivalence. Differentiate it from pharmaceutical equivalence.
6. Write a short note on the Biopharmaceutics Classification System.
7. Write a short note on dissolution testing apparatus.
8. Enumerate the mechanisms of drug absorption.
9. Write a short note on the pH-partition hypothesis.
10. Write a short note on Quality by Design.
11. Write a short note on Yogavahi.
12. Write a short note on Trikatu as a bioavailability enhancer.

Likely long answer questions (10 marks)

1. Define bioavailability. Discuss in detail the factors affecting bioavailability.
2. Describe the methods of assessing bioavailability and the design of a bioequivalence study.
3. Describe the process of dissolution and absorption of a solid oral dosage form. Add a note on factors affecting dissolution.
4. Discuss the strategies used to enhance the bioavailability of a poorly soluble drug, and correlate them with Ayurvedic concepts.
5. Discuss pharmaceutical product development and optimisation with reference to Quality by Design.

How to attempt

- For the “factors affecting bioavailability” question, always use the three headings: pharmaceutical, physiological and pharmacokinetic. Marks are given for the structure.
- Draw the disintegration-to-absorption flow chart. It is quick and it earns marks.
- Close every answer in this unit with the Ayurvedic correlation. Bhavana and Trikatu are the two easiest examples to quote.

REFERENCES USED IN THIS UNIT

Sources for this unit: ICH Q8(R2) Pharmaceutical Development, ICH Q9 Quality Risk Management, and ICH Q1A(R2) Stability Testing; United States Pharmacopeia, general chapters on dissolution; CDSCO Guidelines for Bioavailability and Bioequivalence Studies; Ayurvedic Pharmacopoeia of India, Part I, published by PCIM&H, Ministry of AYUSH (NCISM reference **156**); and reference **49**, CCRAS Guideline of Drug Development.

MODULE 1
PRACTICAL TRAINING ACTIVITIES

Module 1 carries **20 hours of practical training**, spread over eight activities. Each is written below in the same order as Table 4 of the curriculum. Record every one of these in your practical record book.

No	Practical name	Hours
1.1	Demonstration of regulatory frameworks (FDA, etc.)	2
1.2	Demonstration of pharmaceutical product development process	2
1.3	Identify and categorise classical Ayurvedic dosage forms	2
1.4	Identification and categorisation of modified Ayurvedic dosage forms	2
1.5	Formulation of solid dosage forms selecting appropriate excipients	4
1.6	Formulation of liquid dosage forms selecting appropriate excipients	4
1.7	GMP audit and compliance evaluation	2
1.8	Demonstration, designing and evaluation of pharmaceutical packaging and labelling	2
	Total	**20**

Practical Training 1.1 — Demonstration of regulatory frameworks

Aim. To understand and demonstrate the regulatory frameworks that a pharmaceutical company must follow while developing a new drug.

As per the curriculum. The teacher demonstrates a scenario in which a pharmaceutical company is developing a new drug for a specific disease condition. He identifies and explains the key regulations and guidelines that the company must follow for submission of an IND (Investigational New Drug) application. Students then explore the specific regulatory guidelines applicable to different regulatory bodies, construct the process of application, and present it.

Requirements. Internet-enabled computer, projector, copies of the Drugs and Cosmetics Act 1940 and Rules 1945, official websites of USFDA, EMA, CDSCO and the Ministry of AYUSH, and the ICH guideline list.

Procedure

1. Note the disease condition and the drug given in the scenario.
2. Decide whether the product is an allopathic new drug or an ASU drug. The whole regulatory path changes at this point.

3. For an allopathic new drug, list the preclinical data needed before an IND can be filed.
4. Construct the sequence: preclinical data, IND submission, permission, Phase I, Phase II, Phase III, NDA, approval, Phase IV.
5. Compare the requirement of the same product under USFDA, EMA and CDSCO. Prepare a comparison table.
6. For an ASU product, work out the parallel path: Rule 158-B category, proof of safety and effectiveness, Schedule T compliance, and the State Licensing Authority.
7. Present your findings to the class.

Observation to record. Prepare a table with four columns — Regulatory body, Country or region, Application to be filed, Key guideline followed.

Result. Record the complete regulatory pathway constructed for the given scenario.

Precautions. Use only official websites for the guidelines. Guidelines are revised often, so record the date of access. Do not mix the allopathic pathway with the ASU pathway.

Practical Training 1.2 — Demonstration of pharmaceutical product development process

Aim. To demonstrate and construct the complete pharmaceutical product development process for a given drug.

Requirements. Case scenario of a drug under development, preformulation data sheets, product development flow charts, projector.

Procedure

1. Study the drug profile given — solubility, stability, dose, taste and intended route.
2. List the preformulation parameters that must be studied. Refer to Unit 1, section 1.5.3.
3. Decide the dosage form on the basis of that data, and justify the choice.
4. Construct the stage-wise development plan: discovery, preformulation, preclinical, IND, clinical phases, NDA, launch, Phase IV.
5. Add the pharmaceutical development stages: laboratory batch, pilot batch, exhibit batch, scale-up, process validation, technology transfer.
6. Draw the full flow chart on a chart paper or slide.
7. Present, and defend your choice of dosage form.

Observation to record. The complete flow chart, with the approximate time required at each stage.

Result. Record the development plan constructed for the given drug.

Precautions. Do not skip the preformulation stage. Justify every decision with data, not with preference.

Practical Training 1.3 — Identify and categorise classical Ayurvedic dosage forms

Aim. To identify classical Ayurvedic dosage forms and categorise them according to physical state and method of preparation.

Requirements. Samples of classical Kalpanas from the departmental museum or pharmacy — Churna, Vati, Gutika, Guggulu, Bhasma, Pishti, Parpati, Kwatha, Arka, Asava, Arishta, Avaleha, Ghrita, Taila, Lepa, Anjana, Varti. Also Sharangadhara Samhita, the Ayurvedic Formulary of India, a hand lens, a watch glass and a spatula.

Procedure

1. Take one sample at a time in a watch glass.
2. Examine the physical state — solid, liquid, semi-solid or gaseous.
3. Note the organoleptic characters — Varna (colour), Gandha (odour), Rasa (taste), Sparsha (touch) and consistency.
4. Identify the Kalpana by these characters.
5. Confirm the identification from the classical text.
6. Categorise it in the appropriate group.
7. Record the classical reference of that Kalpana.

Observation table to draw

Sample No	Physical state	Organoleptic characters	Name of Kalpana	Category	Classical reference

Result. Record the number of samples identified and the categories to which they belong.

Precautions. Do not taste Bhasma, Rasaushadhi or any Schedule E(1) substance. Handle Bhasma with a spatula, never with wet fingers. Keep hygroscopic Churna closed. Return every sample to its correct container.

Practical Training 1.4 — Identification and categorisation of modified Ayurvedic dosage forms

Aim. To identify modified and modern Ayurvedic dosage forms and to correlate each with its classical parent form.

Requirements. Market samples of modified Ayurvedic products — Ghana Vati, coated Guggulu tablet, Ayurvedic syrup, soft gelatin capsule of Ghrita, Ayurvedic cream and gel, granule sachet, Arka spray, Ayurvedic eye drops, effervescent Ayurvedic powder.

Procedure

1. Examine each product with its pack and label.
2. Note the dosage form, the ingredients and the claimed indication.
3. Identify the classical Kalpana from which it has been modified.
4. Note the reason for the modification — taste, portability, stability, dose accuracy or patient acceptance.
5. Read the label and note whether it declares a classical reference or is a patent or proprietary medicine.
6. Categorise the sample.

Observation table to draw

Sample	Modified dosage form	Classical parent Kalpana	Reason for modification	Classical or proprietary

Result. Record the correlation established between the modified and the classical forms.

Precautions. Read the label carefully. Do not assume a product is classical merely because it carries a Sanskrit name. Check the licence number and the declaration on the pack.

Practical Training 1.5 — Formulation of solid dosage forms selecting appropriate excipients

Aim. To formulate a solid dosage form by selecting suitable excipients, and to evaluate it.

Requirements. Model drug or Ayurvedic extract, diluent (lactose or microcrystalline cellulose), binder (starch paste or PVP), disintegrant (starch or sodium starch glycolate), lubricant (magnesium stearate), glidant (talc or colloidal silicon dioxide), sieves, mortar and pestle, granulator, tray drier, tablet compression machine, weighing balance, Monsanto hardness tester, Roche friabilator, disintegration test apparatus.

Procedure

1. Fix the dose of the active ingredient and the intended tablet weight.
2. Select each excipient and write the reason for the selection.
3. Prepare the master formula for a batch, showing the quantity and percentage of each ingredient.
4. Weigh all ingredients accurately.
5. Sieve and mix the drug, diluent and disintegrant.
6. Prepare the binder solution and carry out wet massing.
7. Pass the wet mass through a suitable sieve to form granules.
8. Dry the granules in the tray drier. Check loss on drying.
9. Pass the dried granules through a finer sieve.
10. Add the lubricant and glidant and blend.
11. Evaluate the granules — bulk density, tapped density, angle of repose, Carr’s index and Hausner ratio.
12. Compress the tablets.
13. Evaluate the tablets — appearance, weight variation, hardness, friability and disintegration time.

Observation tables to draw. One table for the master formula with the function of each excipient. One for granule evaluation. One for tablet evaluation.

Result. Record whether the tablets comply with the pharmacopoeial limits.

Precautions. Weigh accurately. Do not over-wet the mass. Do not over-mix after adding the lubricant, as this delays disintegration. Clean the punches and dies before and after use. Observe any capping or lamination and record it.

Practical Training 1.6 — Formulation of liquid dosage forms selecting appropriate excipients

Aim. To formulate a liquid dosage form by selecting suitable excipients, and to evaluate it.

Requirements. Model drug or Ayurvedic extract, purified water, co-solvent, suspending agent, wetting agent, preservative, buffer salts, sweetener, flavour, colour; beakers, measuring cylinders, glass rod, mechanical stirrer, homogeniser, pH meter, viscometer, amber bottles.

Procedure

1. Decide the type of liquid to be prepared — solution, syrup, suspension or emulsion.
2. Select each excipient and record the reason.
3. Prepare the master formula.
4. Dissolve the buffer salts and the preservative in a part of the vehicle.
5. Disperse or dissolve the drug, using a wetting agent if it is a suspension.
6. Prepare the suspending agent or emulsifier separately and allow it to hydrate fully.
7. Combine the phases with continuous stirring.
8. Add the sweetener, flavour and colour.
9. Make up to the final volume.
10. Homogenise if required.
11. Evaluate — appearance, pH, viscosity, sedimentation volume, redispersibility, globule size in the case of an emulsion, and drug content.
12. Fill into a clean, dry amber bottle and label it.

Observation tables to draw. Master formula with the function of each excipient, and an evaluation table.

Result. Record whether the preparation is stable and acceptable.

Precautions. Always allow gums and polymers to hydrate fully before adding the drug. Never add electrolytes rapidly to a hydrocolloid. Use a preservative in every aqueous liquid. Label the bottle “Shake well before use” if it is a suspension.

Practical Training 1.7 — GMP audit and compliance evaluation

Aim. To carry out a GMP audit of a pharmaceutical or Ayurvedic manufacturing unit and evaluate its compliance.

Requirements. Copy of Schedule T and Schedule M, a prepared audit checklist, the departmental pharmacy or a visit to a licensed Ayurvedic manufacturing unit, and a notebook.

Procedure

1. Prepare an audit checklist under the ten GMP headings given in Unit 4, section 4.1.4.
2. Obtain permission and observe the entry procedure of the unit.
3. Observe the layout, and check whether the flow of material and personnel is unidirectional.
4. Check the separate sections required under Schedule T Part II for the dosage forms made there.
5. Check the water supply, drainage, ventilation and pest control.
6. Examine the documentation — master formula record, batch manufacturing record, SOPs and logbooks.
7. Check the quality control laboratory and its instruments, and their calibration status.
8. Check the storage of raw materials, and the labelling of quarantine, approved and rejected material.
9. Check personnel hygiene, uniform and training records.
10. Record every observation as compliance or non-compliance.
11. Classify non-compliances as critical, major or minor.
12. Suggest corrective and preventive action for each.

Observation table to draw

No	GMP element observed	Requirement as per Schedule T or M	Actual observation	Compliant or not	Suggested CAPA

Result. Record the overall compliance status and the number of observations in each category.

Precautions. Never photograph or record anything without written permission. Do not touch running machinery. Maintain confidentiality about what you see in a commercial unit. Report facts only, without personal comment.

Practical Training 1.8 — Demonstration, designing and evaluation of pharmaceutical packaging and labelling

Aim. To study, design and evaluate pharmaceutical packaging and labelling.

Requirements. Samples of packaging — glass bottles, plastic containers, blister and strip packs, ampoules, vials, collapsible tubes, cartons, package inserts; labels of allopathic and Ayurvedic medicines; the labelling provisions of the Drugs and Cosmetics Rules, 1945; drawing sheets or design software.

Procedure

1. Examine each packaging sample. Note the material and classify it as primary, secondary or tertiary.
2. Record the advantages and disadvantages of each material.
3. Collect at least five labels, including at least two of Ayurvedic medicines.
4. Check every label against the list of required particulars given in Unit 4, section 4.3.4.
5. Record any deficiency found.
6. Now design a complete pack for a given Ayurvedic product. Select the primary container with justification. Design the label with all mandatory particulars. Draft the package insert.

7. Evaluate your own design against the legal requirements.

Observation tables to draw. One table for packaging material evaluation, and one label compliance checklist.

Result. Record the designed packaging system and the compliance status of the labels examined.

Precautions. Handle glass ampoules carefully. Include every mandatory particular in the design, especially batch number, manufacturing date, expiry date and licence number. Do not make any therapeutic claim that is prohibited under the Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954.

MODULE 1
EXPERIENTIAL LEARNING ACTIVITIES

Module 1 carries **30 hours** under experiential learning, including the modular assessment. Six activities are prescribed in Table 5 of the curriculum.

No	Experiential learning activity	Hours
1.1	Recognising applications suiting various regulatory frameworks	3
1.2	Listing out challenges and solutions in the pharmaceutical product development process	3
1.3	Exploration, identification and collection of various classical and modified pharmaceutical dosage forms	3
1.4	Selection of an appropriate excipient according to solid and liquid dosage form	6
1.5	Designing and development of pharmaceutical packaging systems and labelling	4
1.6	Optimization of pharmaceutical formulation for enhanced bioavailability	7
	Total	**26**

Experiential Learning 1.1 — Recognising applications suiting various regulatory frameworks

What you have to do. Take a set of different products — a new chemical entity, a generic tablet, a biosimilar, a classical Ayurvedic Churna, an Ayurvedic patent or proprietary medicine, and a nutraceutical. For each one, decide which regulatory framework applies and which application must be filed.

Method. Work in a small group. Use only official sources. Prepare a matrix. Present it, and defend your reasoning when questioned.

Matrix to prepare

Product	System	Regulatory authority	Application or licence needed	Governing rule or schedule	Data required

How to record it in the log book. Write the aim, the method, the completed matrix, the difficulties you faced, and what you learnt. Attach the printed guideline pages you used.

Experiential Learning 1.2 — Listing out challenges and solutions in the pharmaceutical product development process

What you have to do. Take one real product development problem. List every challenge that arises at each stage, and propose a workable solution for each.

Method. Choose an Ayurvedic product for greater relevance — for example, converting Dashamoola Kwatha into a tablet. Move stage by stage from raw material to marketing. Refer to Unit 1, section 1.5.10 for the table of common challenges. Then add the challenges specific to herbal material.

Points you must cover

- Raw material — authentication, seasonal variation, adulteration, heavy metals, pesticide residue
- Extraction — solvent choice, yield, loss of volatile principles
- Formulation — hygroscopicity, poor flow, bitter taste, large dose
- Standardisation — choice of marker compound, absence of a reference standard
- Stability — microbial growth, colour change, loss of potency
- Regulatory — category under Rule 158-B, proof of safety and effectiveness
- Cost and scale-up

How to record it. Prepare a two-column table of challenge and solution. Add one flow chart. State your conclusion in five lines.

Experiential Learning 1.3 — Exploration, identification and collection of classical and modified dosage forms

What you have to do. Visit an Ayurvedic pharmacy, a retail medical store and the departmental museum. Collect or photograph as many dosage forms as possible, both classical and modified. Identify each and build a catalogue.

Method. Prepare a collection register. For each entry, record the name, the dosage form, the classical parent form, the manufacturer, the licence number and the claimed indication. Arrange the collection by physical state.

Deliverable. A labelled specimen board or an album, along with the register. This is a strong addition to your record book, and examiners look for it.

Experiential Learning 1.4 — Selection of an appropriate excipient according to solid and liquid dosage form

What you have to do. Given a set of drugs with different properties, select the correct excipients for each, and justify every choice.

Method. Take at least four different situations. For example: a bitter, poorly flowing herbal extract to be made into a tablet; a hygroscopic Churna; an insoluble Bhasma to be suspended in a liquid; and an oily Taila to be made into an emulsion. For each, prepare the excipient selection table.

Selection table to prepare

Property of the drug	Problem it creates	Excipient class needed	Excipient selected	Justification

Then prepare one trial formula for each, and test it in the laboratory if facilities permit.

How to record it. Write the full reasoning, not only the conclusion. The purpose of this activity is to show that you can think like a formulator.

Experiential Learning 1.5 — Designing and development of pharmaceutical packaging systems and labelling

What you have to do. Design a complete packaging system for one Ayurvedic product of your choice, from primary container to shipper carton, along with the label and package insert.

Method. Choose the product. State its properties — is it hygroscopic, light sensitive, oily or volatile? Select the primary packaging material and justify it against those properties. Design the secondary carton. Draft the label with every mandatory particular. Write the package insert. Finally, evaluate your design for protection, convenience, cost and legal compliance.

Deliverable. A dummy pack or a full-size printed design, the label, and the package insert.

Experiential Learning 1.6 — Optimization of pharmaceutical formulation for enhanced bioavailability

What you have to do. Take one poorly bioavailable drug or phytoconstituent. Design a formulation strategy to improve its bioavailability, and justify your design.

Method. Curcumin, Guggulsterone or a Bhasma are good choices, and all are relevant to your speciality.

1. State the problem. Classify the compound under the BCS. Most phytoconstituents fall in Class II or Class IV.
2. Identify why the bioavailability is poor — poor solubility, poor permeability, first-pass metabolism, efflux, or instability at gastric pH.
3. List the possible strategies from Unit 5, section 5.4.4.
4. Select two strategies and justify them.
5. Design the formulation, giving the composition.
6. State how you would evaluate the improvement — dissolution study, permeability study, or a plasma level study in animals.
7. Add the classical Ayurvedic approach to the same problem — Bhavana, Yogavahi, Anupana, Trikatu, Sneha Kalpana — and compare it with your modern strategy.

How to record it. Present it as a short project report: title, objective, background, strategy, proposed formulation, evaluation plan, expected outcome, and the Ayurvedic correlation.

MODULE 1
VIVA POINTERS AND QUESTION BANK

Viva pointers for the practical examination

The practical examination of this course carries 200 marks, and viva voce is a substantial part of it. The questions below are the ones most often asked from Module 1 at the table.

On dosage forms and samples shown to you

- Identify this sample. What is its physical state and its Kalpana?
- Which classical text describes this preparation?
- What is the Saviryata Avadhi of this Kalpana?
- What is the modified form of this classical preparation, and why was it modified?
- Why can this Churna not be given as such to the patient?

On formulation

- Name the excipients used in a tablet, with the function of each.
- Why is magnesium stearate added, and why should it not be over-mixed?
- What is the difference between a binder and a disintegrant?
- Which method of granulation would you use for a moisture-sensitive drug, and why?
- What is the dry gum method ratio for an emulsion?
- What is the role of Bhavana in Rasashastra, and what is its modern equivalent?

On evaluation and equipment

- Show me the friabilator. What is the acceptable friability limit?
- What is the difference between disintegration and dissolution?
- Which dissolution apparatus is used for tablets?
- What is being tested by weight variation, and what by content uniformity?
- Name the tablet defects you can see in this sample, and give the cause of each.

On GMP and regulation

- What does the letter “c” in cGMP stand for?
- Under which schedule is GMP for Ayurvedic drugs given?
- Which rule governs the licensing of an Ayurvedic medicine?
- What is the difference between quality control and quality assurance?
- Which authority issues the manufacturing licence for an Ayurvedic medicine?
- What is Schedule E(1)?

On packaging and labelling

- Show me the mandatory particulars on this label.
- Which type of glass is used for parenterals, and why?
- What is the difference between primary and secondary packaging?
- Why is an amber bottle used for this preparation?

On biopharmaceutics

- Define bioavailability. What is the bioavailability of an intravenous injection?
- What does AUC represent?
- What is bioequivalence, and why does a generic company need it?
- Give one classical Ayurvedic bioavailability enhancer, and explain how it works.

Module 1 question bank

Analytical Based Questions (20 marks) — from Unit 3 only

1. A pharmacy wishes to convert a classical Kwatha into a tablet, so that patients can carry it easily. Analyse the pharmaceutical problems that will arise. Select suitable excipients, describe the method of manufacture you will adopt, and list the evaluation parameters you will apply.
2. An Ayurvedic Churna is bitter, hygroscopic and flows poorly. It has to be marketed in a convenient dosage form. Analyse the situation and justify your choice of dosage form, excipients and packaging.
3. A batch of tablets shows capping and lamination during compression. Analyse the possible causes, and give the corrective steps you would take at the formulation stage and at the machine stage.

Long Answer Questions (10 marks) — from Units 1, 3 and 5 only

1. Describe in detail the pharmaceutical product development process, from drug discovery to post-marketing surveillance.
2. Discuss the regulatory framework governing pharmaceutical products in India, with special reference to Ayurvedic, Siddha and Unani drugs.
3. Describe the historical development of pharmaceutics and correlate it with the development of Bhaishajya Kalpana in India.
4. Give an overview of the pharmaceutical industry, including its segments, departments, and national and international market size.
5. Define pharmaceutics. Describe its scope and correlate it with Rasashastra and Bhaishajya Kalpana.
6. Describe the excipients used in solid dosage forms with their functions and examples.
7. Describe the methods of tablet manufacture. Add a note on the evaluation of tablets.
8. Describe the formulation and evaluation of a pharmaceutical suspension.
9. Describe the formulation of parenteral dosage forms, including sterilisation and testing.
10. Discuss the principles of pharmaceutical formulation and correlate them with the principles of Bhaishajya Kalpana.
11. Define bioavailability. Discuss in detail the factors affecting bioavailability.
12. Describe the methods of assessing bioavailability and the design of a bioequivalence study.

13. Describe the process of dissolution and absorption of a solid oral dosage form. Add a note on the factors affecting dissolution.
14. Discuss the strategies used to enhance the bioavailability of a poorly soluble drug, and correlate them with Ayurvedic concepts.
15. Discuss pharmaceutical product development and optimisation with reference to Quality by Design.

Short Answer Questions (5 marks) — from all five units

From Unit 1

1. Define pharmaceutics and write its scope.
2. Enumerate the branches of pharmaceutics.
3. Why is a dosage form necessary?
4. Write a short note on galenicals.
5. Write a short note on Rule 158-B.
6. Write a short note on Schedule T.
7. Differentiate between a classical Ayurvedic medicine and a patent or proprietary medicine.
8. Write a short note on ICH guidelines.
9. Enumerate the phases of clinical trial.
10. Write a short note on preformulation studies.
11. Differentiate between NDA and ANDA.
12. Write a short note on PCIM&H.

From Unit 2

1. Classify pharmaceutical dosage forms according to physical state, with examples.
2. Classify dosage forms according to the route of administration.
3. Write a short note on tablets and their types.
4. Write a short note on capsules.
5. Differentiate between a suspension and an emulsion.
6. Write a short note on semi-solid dosage forms.
7. Enumerate the requirements of a parenteral preparation.
8. Correlate classical Ayurvedic dosage forms with modern dosage forms.

From Unit 3

1. Define an excipient and write its ideal properties.
2. Enumerate the excipients used in tablet formulation with one example of each.
3. Write a short note on disintegrants.
4. Write a short note on tablet defects.
5. Differentiate between a flocculated and a deflocculated suspension.
6. Write a short note on the HLB system.
7. Write a short note on ointment bases.
8. Enumerate the methods of sterilisation with conditions.

9. Write a short note on Bhavana as a pharmaceutical process.
10. Correlate Ayurvedic Prakshepaka Dravya with modern excipients.

From Unit 4

1. Define GMP and enumerate its main elements.
2. What is cGMP?
3. Differentiate between quality control and quality assurance.
4. Write a short note on DQ, IQ, OQ and PQ.
5. Enumerate the functions of packaging.
6. Write a short note on the types of glass used in pharmaceutical packaging.
7. Enumerate the particulars required on the label of an Ayurvedic medicine.
8. Write a short note on Saviryata Avadhi.

From Unit 5

1. Differentiate between biopharmaceutics and pharmacokinetics.
2. Write a short note on the LADMER sequence.
3. Define C_{max}, T_{max} and AUC.
4. Define bioavailability and write its types.
5. Define bioequivalence and differentiate it from pharmaceutical equivalence.
6. Write a short note on the Biopharmaceutics Classification System.
7. Enumerate the mechanisms of drug absorption.
8. Write a short note on Quality by Design.
9. Write a short note on Yogavahi.
10. Write a short note on Trikatu as a bioavailability enhancer.

> **Note on the marks range.** Table 6F of the curriculum places Module 1 in the marks range of **5 to 20** in the theory paper. Prepare Unit 3 most thoroughly, since it is the only unit in this module that can appear as an Analytical Based Question.

MODULE 2
DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL AND PLANT RESEARCH

Credits	Lectures	Practical Training	Experiential Learning	Total hours	Modular assessment
2	10	20	30	60	50 marks

About this module

This module gives a complete introduction to databases and information systems, with a focus on metal, mineral and plant research. It covers the fundamental principles of databases, data retrieval languages and the various databases relevant to the field. It also covers advanced database searching, database integration, and applications in research and industry.

Module Learning Objectives

At the end of this module, the student should be able to:

1. Describe the fundamental concepts of databases and information systems, including types of databases and database management systems.
2. Access metal, mineral and plant databases for research and development, including advanced searching techniques and data analysis.
3. Integrate multiple databases and apply database applications in research, industry and decision-making.
4. Evaluate the challenges and limitations of databases in metal, mineral and plant research.
5. Design and develop a database application for metal, mineral or plant research, incorporating emerging trends and technologies.

Units in this module

Unit	Name
M2U1	Introduction to Databases and Information Systems
M2U2	Metal and Mineral Databases
M2U3	Plant Databases
M2U4	Database Application
M2U5	Challenges and Limitation of Databases

Why a Rasashastra student must study this

At first sight this module looks out of place in an Ayurveda syllabus. It is not.

Your subject deals with metals, minerals and plants. Every research paper you write will need three things — the correct botanical name of a plant, the correct mineralogical identity of a Rasa Dravya, and evidence from published work. All three now live in databases.

If you write “Guduchi” in a thesis, no reviewer abroad will know the plant. If you write *Tinospora cordifolia* (Willd.) Hook.f. & Thomson, and cite the source from which you verified the name, your work becomes citable anywhere in the world. That verification comes from a plant database.

The same is true of minerals. Abhraka is not just Abhraka. It is mica, a group of sheet silicates, and its exact species matters. A mineral database tells you that.

> Module 2 is the smallest module in marks and the easiest to score in. Table 6F places it in the range of **5 to 15** marks. Four of its five units are asked only as short answers. Learn the names, the full forms and the purposes accurately, and this module will give you easy marks.

UNIT 1
INTRODUCTION TO DATABASES AND INFORMATION SYSTEMS

SYLLABUS POINTS

1.1 Overview of databases and information systems
1.2 Types of databases (relational, object-oriented, etc.)
1.3 Database management systems (DBMS)
1.4 Data retrieval and querying languages (SQL, etc.)
1.5 Introduction to metal, mineral and plant databases

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe overview of databases and information systems, types of databases and database management systems	1	Lecture	Knows how
Describe data retrieval and querying languages and an overview of metal, mineral and plant databases	1	Lecture	Knows how
Demonstrate various searching techniques of databases	2	Practical Training 2.1	Does
Demonstrate modelling, querying and analytics of data	2	Practical Training 2.2	Does
Design and implement a simple database for a plant nursery	6	Experiential Learning 2.1	Shows how

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

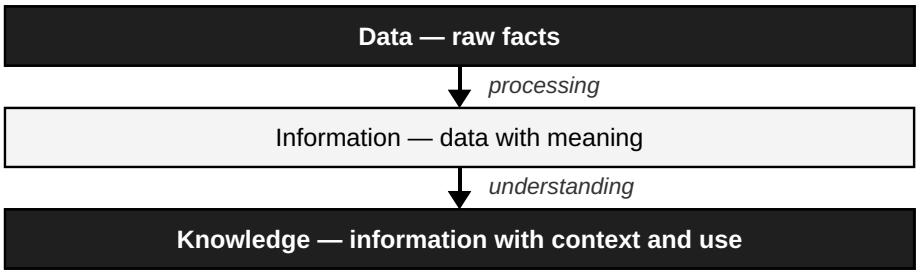
1.1 Overview of databases and information systems

1.1.1 What is data?

Data means raw, unprocessed facts. The number 7.5 is data. By itself it means nothing.

Information is data that has been processed and given meaning. “The Mohs hardness of quartz is 7” is information.

Knowledge is information understood along with its context and use. “Quartz is harder than steel, so a steel needle cannot scratch it, and this can be used as an identification test” is knowledge.



1.1.2 Definition of a database

A **database** is an organised collection of related data, stored in a systematic way, so that it can be easily accessed, managed and updated.

Two words in this definition matter most. **Organised** — the data has a structure. **Related** — the items belong together.

A telephone directory is a database. A hospital record room is a database. The Ayurvedic Pharmacopoeia of India is, in effect, a printed database of drug standards.

1.1.3 Basic terms

Term	Meaning
Field or attribute	A single item of information, such as “Botanical name”
Record or tuple	One complete set of fields about one item, such as all details of Guduchi
Table or relation	A collection of records of the same type
Primary key	A field that uniquely identifies each record, such as an accession number
Foreign key	A field in one table that refers to the primary key of another table
Query	A request made to the database to retrieve or change data
Schema	The overall design or structure of the database
Metadata	Data about the data, such as when a record was created and by whom

1.1.4 Why a database is better than a file

Before databases, data was kept in separate files. That system had clear problems.

Problem with file systems	How a database solves it
Data redundancy — the same data stored many times	Data stored once, referenced everywhere
Data inconsistency — one copy updated, another not	A single update reaches every user
Difficulty in access	A query gives the answer at once
Poor security	Access rights can be set user by user
No standard for integrity	Rules and constraints enforce correctness
Concurrent use causes conflict	Transaction control manages many users

1.1.5 Information systems

An **information system** is the complete arrangement of people, hardware, software, data and procedures that collects, stores, processes and delivers information for a defined purpose.

A database is only one part of an information system. The database holds the data. The information system decides who puts data in, who takes it out, and what is done with it.

Components of an information system

1. **Hardware** — computers, servers, storage, network
2. **Software** — operating system, DBMS, application programs
3. **Data** — the content itself
4. **People** — administrators, data entry staff, end users
5. **Procedures** — the rules and SOPs for using the system

1.2 Types of databases

1.2.1 Classification by data model

This is the classification most often asked.

Type	How data is arranged	Example or note
Hierarchical database	Data in a tree, parent to child, one parent to many children	IBM IMS; early file systems
Network database	Like hierarchical, but a child may have many parents	IDMS
Relational database	Data in tables of rows and columns, linked by keys	MySQL, PostgreSQL, Oracle, MS SQL Server, MS Access
Object-oriented database	Data stored as objects, with attributes and methods together	ObjectDB, db4o
Object-relational database	Relational base extended to hold objects	PostgreSQL, Oracle
NoSQL database	Non-tabular; stores documents, key-value pairs, graphs or wide columns	MongoDB, Cassandra, Neo4j

The relational database is the most important type for you. Almost every scientific database you will use is relational underneath. Its strength is simplicity — everything is a table, and tables are joined by keys.

1.2.2 Classification by other criteria

Basis	Types
By location	Centralised (one place) and Distributed (spread over several places)
By number of users	Single-user and Multi-user
By content	Bibliographic, Numeric, Full-text, Image, Chemical structure
By access	Open access and Subscription based
By purpose	Operational (day-to-day) and Analytical (data warehouse)
By hosting	On-premises and Cloud based

> **Useful for your research.** PubMed is a **bibliographic** database. It gives references, not full articles. The Ayurvedic Pharmacopoeia of India, when digitised, is a **numeric and text** database. Mindat is an **image plus numeric** database. Knowing the type tells you what to expect before you search.

1.3 Database Management Systems (DBMS)

1.3.1 Definition

A **Database Management System (DBMS)** is the software that allows a user to create, store, modify, retrieve and control access to a database.

The database is the data. The DBMS is the software that manages it. A student often confuses the two. Remember the difference.

A **Relational Database Management System (RDBMS)** is a DBMS based on the relational model. MySQL, Oracle, PostgreSQL, Microsoft SQL Server and Microsoft Access are all RDBMS.

1.3.2 Functions of a DBMS

1. **Data definition** — creating and changing the structure
2. **Data storage** — physically storing the data efficiently
3. **Data manipulation** — inserting, updating, deleting and retrieving
4. **Data security** — controlling who may see and change what
5. **Data integrity** — enforcing rules so that wrong data cannot enter
6. **Concurrency control** — allowing many users at once without conflict
7. **Backup and recovery** — restoring data after failure
8. **Transaction management** — ensuring a set of operations succeeds or fails as one

1.3.3 Advantages and disadvantages

Advantages	Disadvantages
Controls redundancy	Costly software and hardware
Maintains consistency	Needs trained personnel
Enforces standards and integrity	Complex to design and maintain
Provides security	Failure of the system affects all users
Allows sharing among many users	Performance may suffer for very simple tasks
Provides backup and recovery	Conversion from an old system is difficult

1.3.4 The ACID properties of a transaction

A transaction is a single logical unit of work. Every reliable DBMS guarantees four properties, remembered as **ACID**.

Property	Meaning
Atomicity	All operations succeed, or none do
Consistency	The database moves from one valid state to another valid state
Isolation	Concurrent transactions do not interfere with each other
Durability	Once committed, the change survives even a system failure

1.3.5 Data models used in design

Model	Purpose
Entity-Relationship (ER) model	A diagram showing entities, their attributes and the relationships between them; used at the design stage
Relational model	Converts the ER design into tables
Normalisation	The process of organising tables to remove redundancy; the common forms are 1NF, 2NF and 3NF

1.4 Data retrieval and querying languages

1.4.1 SQL — Structured Query Language

SQL is the standard language used to communicate with a relational database. It is not a programming language in the full sense. It is a query language — you state what you want, and the DBMS works out how to get it.

SQL commands fall into groups.

Group	Full form	Common commands	Purpose
DDL	Data Definition Language	CREATE, ALTER, DROP, TRUNCATE	Defines and changes the structure
DML	Data Manipulation Language	SELECT, INSERT, UPDATE, DELETE	Works with the data itself
DCL	Data Control Language	GRANT, REVOKE	Controls permissions
TCL	Transaction Control Language	COMMIT, ROLLBACK, SAVEPOINT	Controls transactions

1.4.2 The shape of a simple query

A SELECT statement has a fixed order of clauses.

Clause	What it does
SELECT	Which columns you want
FROM	Which table they are in
WHERE	Which rows to include
GROUP BY	How to group the rows
HAVING	Which groups to include
ORDER BY	How to sort the result

A worked example. Suppose a table named `raw_drugs` holds the columns `sanskrit_name`, `botanical_name`, `family` and `part_used`. To list every drug of the family Menispermaceae, arranged alphabetically:

```
SELECT sanskrit_name, botanical_name FROM raw_drugs WHERE family = 'Menispermaceae' ORDER BY sanskrit_name;
```

1.4.3 Search techniques used in scientific databases

These techniques apply to almost every database and search engine you will use. Practical Training 2.1 is built on them.

Technique	How it works	Example
Boolean operators	AND narrows, OR widens, NOT excludes	Guduchi AND immunomodulator
Phrase search	Quotation marks find the exact phrase	“Tinospora cordifolia”
Truncation or wildcard	A symbol stands for any ending	immun* finds immunity, immune, immunomodulator
Field-specific search	Restricts the search to one field	author, title, abstract
Proximity search	Finds words within a set distance of each other	varies by database
Filters and limits	Narrow by year, language, article type, species	last 5 years, review articles
Controlled vocabulary	Uses the database’s own index terms	MeSH terms in PubMed
Nesting	Brackets set the order of operations	(Guduchi OR Amrita) AND fever

> **The single most useful habit.** Always search by the accepted botanical name *and* its synonyms, joined by OR. A drug may be indexed under an old name in older papers. Searching only the current name will hide half the literature.

1.5 Introduction to metal, mineral and plant databases

This section is only an overview. Units 2 and 3 deal with these databases in detail.

Category	Purpose	Examples covered in this module
Mineral databases	Identity, chemistry, crystal structure, physical properties and localities of minerals	Webmineral, Mindat, Minerals Education Coalition database
Metal and geological survey databases	Production, reserves, deposits, trade and economic data on metals and minerals	USGS Mineral Resources Program, USGS Minerals Yearbook, USMIN
Metal-protein databases	Metal-binding sites in biological molecules	MetalPDB
Plant name databases	Correct scientific names, authorities and synonyms	World Flora Online, IPNI, POWO
Plant genetic databases	Genome sequences, genes and molecular data	PlantGDB and successor resources
Medicinal plant databases	Ayurvedic and Indian medicinal plant information	IMPPAT, BSI Medicinal Plant Database, ICMR resources

KEY POINTS TO REMEMBER

- Data is raw facts. Information is processed data. Knowledge is information with context.
- A database is an organised collection of related data. A DBMS is the software that manages it.
- Data models: hierarchical, network, relational, object-oriented, object-relational and NoSQL.
- The relational model stores data in tables linked by primary and foreign keys.
- SQL groups: DDL, DML, DCL and TCL.
- ACID: Atomicity, Consistency, Isolation, Durability.
- Boolean operators: AND narrows, OR widens, NOT excludes.
- Always search a plant by its accepted name and all its synonyms.

EXAM CORNER — UNIT 1

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Define data, information and knowledge. Differentiate between them.
2. Define a database. Write its advantages over a file system.
3. Enumerate the types of databases with one example of each.
4. Write a short note on the relational database model.
5. Define DBMS. Enumerate its functions.

6. Write the advantages and disadvantages of a DBMS.
7. Write a short note on the ACID properties.
8. Write a short note on SQL and its command groups.
9. Write a short note on Boolean operators in database searching.
10. Enumerate the search techniques used in scientific databases.
11. Write a short note on primary key and foreign key.
12. Write a short note on the components of an information system.

How to attempt

- Give the definition first, in one clean sentence. Then the points. Then one example connected to Ayurveda.
- For “types of databases”, the table format is the fastest way to full marks.

REFERENCES USED IN THIS UNIT

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3. **71.** Kline K, Hunt B, Kline D. SQL in a Nutshell: A Desktop Quick Reference. O’Reilly Media; 2004.
4. **72.** Minerals Education Coalition — Minerals Database.
5. **73.** eGyanKosh: Database Management Systems.
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7. **75.** NCERT textbook resource on database concepts.

UNIT 2
METAL AND MINERAL DATABASES

SYLLABUS POINTS

2.1 Overview of metal and mineral databases
2.2 USGS Mineral Database
2.3 Mineral Database
2.4 Webmineral Mineral Database
2.5 Introduction to Metal Databases

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe overview of metal and mineral databases, and USGS	1	Lecture	Knows how
Discuss mineral and metal databases, including their applications and significance	1	Lecture	Knows how
Analyse USGS Mineral Database	2	Practical Training 2.3	Does
Analysis and integration of Webmineral Database	3	Practical Training 2.4	Does
Search, explore and analyse Metal Database and applications	3	Practical Training 2.5	Does
Analyse Webmineral Database and Metal Database	6	Experiential Learning 2.2	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	No	Yes

> This is the **only unit in Module 2 that carries a ten-mark long answer question**, and it carries no short answer. So prepare it as a full essay. Learn the databases in a fixed order, with their custodian, content, and use in Rasashastra.

2.1 Overview of metal and mineral databases

2.1.1 Why a Rasashastra student needs them

Rasashastra uses Dhatu, Upadhatu, Ratna, Uparatna, Sadharana Rasa, Maharasa and Uparasa. Almost all of these are minerals or metals in the modern sense.

Classical texts identify them by Varna, Sparsha, Gaurava, Rekha and other physical characters. Modern mineralogy identifies them by chemical formula, crystal system, hardness, specific gravity and diffraction pattern. A mineral database is the bridge between the two languages.

Practical uses for you:

1. **Correct identification.** To state exactly which mineral species your Maharasa sample is.
2. **Chemical composition.** To know what elements will be present, and therefore what the Bhasma will contain.
3. **Physical constants.** Hardness and specific gravity help to detect adulteration.
4. **Crystal data.** Needed for interpreting XRD results, discussed in Module 5.
5. **Locality data.** To know from where a mineral is genuinely obtained.
6. **Economic and production data.** For availability, cost and policy questions.

2.1.2 The classical to modern correlation

Learn a few of these. They are asked in viva and they make any answer stronger.

Rasashastra Dravya	Mineral or chemical identity	Chemical formula
Hingula	Cinnabar	HgS
Gandhaka	Native sulphur	S
Manahshila	Realgar	As_4S_4
Haratala	Orpiment	As_2S_3
Tuttha	Chalcanthite, copper sulphate	$CuSO_4 \cdot 5H_2O$
Kasisa	Melanterite, ferrous sulphate	$FeSO_4 \cdot 7H_2O$
Abhraka	Mica group, chiefly biotite	Sheet silicate
Makshika	Chalcopyrite	$CuFeS_2$
Shilajatu	Mineral exudate, not a single species	Mixed
Sphatika	Alum	$KAl(SO_4)_2 \cdot 12H_2O$
Gairika	Haematite, red ochre	Fe_2O_3
Kankshi	Biotite or alum group, as per text	Varies

2.1.3 What a mineral database record contains

Field	Meaning
Mineral name and IMA status	Whether the species is approved by the International Mineralogical Association
Chemical formula	The ideal composition
Classification	Nickel-Strunz and Dana classification codes
Crystal system and space group	Symmetry of the crystal
Unit cell parameters	a, b, c and the angles
Physical properties	Colour, streak, lustre, hardness, cleavage, fracture, specific gravity
Optical properties	Refractive index, birefringence, pleochroism
X-ray diffraction data	The strongest d-spacings, used to confirm identity
Type locality and occurrence	Where it is found
References	Original literature

2.2 USGS Mineral Database

2.2.1 What USGS is

The **United States Geological Survey (USGS)** is the scientific agency of the United States Government dealing with geology, water, biology and geography. Its **National Minerals Information Center** is the world’s most widely used source of statistics on the production and consumption of minerals.

USGS data is free, official and citable. That is why the curriculum names it.

2.2.2 The main USGS resources

Resource	Content
Mineral Commodity Summaries	An annual volume giving, for about 90 mineral commodities, the production, reserves, imports, exports, prices and world data
Minerals Yearbook	A detailed yearly report in three volumes — Metals and Minerals, Area Reports (Domestic), and Area Reports (International)
USMIN Mineral Deposit Database	Locations and characteristics of mineral deposits and mines in the United States
Mineral Resources Data System (MRDS)	Records of mineral occurrences worldwide, with commodity, location and deposit type
Mineral Resources Online Spatial Data	Maps and geospatial layers

2.2.3 What you can obtain from USGS

- World and country-wise production figures for a metal, year by year
- Estimated reserves
- Price trends over time
- Import and export dependence of a country
- Location and geology of deposits
- Reports on by-products and recycling

2.2.4 How this is used in Rasashastra research

Suppose your research topic is Naga Bhasma. Naga is lead. From USGS you can obtain the world production of lead, the countries that produce it, India's position, the price trend and the principal ores. That single page of data strengthens the introduction of your dissertation, and it is a citable Government source.

The same applies to Vanga (tin), Tamra (copper), Yashada (zinc), Loha (iron), Swarna (gold) and Rajata (silver).

2.3 Mineral Database

The curriculum lists "Mineral Database" as a separate point. In practice this refers to the general-purpose mineral information databases, of which two are important.

2.3.1 Mindat

Mindat.org is the largest and most used mineral database in the world. It is an outreach project of the Hudson Institute of Mineralogy, a not-for-profit organisation, and was founded in 2000 by Jolyon Ralph.

Feature	Description
Content	Mineral species, varieties, chemical and physical data, photographs
Localities	A very large worldwide database of where each mineral is found
Classification	Uses the Nickel-Strunz classification, based on the 9th edition and extended by Mindat, and the Dana classification
Photographs	A very large collection contributed by members
Search	By name, chemistry, properties, locality, and by visual explorer
Access	Free

2.3.2 Minerals Education Coalition Minerals Database

This is a teaching-oriented database. It gives simple, reliable entries for common minerals, arranged by mineral category, with the chemical formula, properties, uses and an image. It is well suited for a postgraduate student who wants a quick, dependable summary rather than full research data. It is reference 72 and 81 of your NCISM list.

2.4 Webmineral Mineral Database

Webmineral.com is a mineralogy database that presents each mineral species in a standard data sheet. It is compact, technical and free, and it is especially convenient for looking up crystallographic and X-ray data.

2.4.1 What a Webmineral record gives

Section	Content
General information	Chemical formula, molecular weight, composition by element
Classification	Dana classification and Nickel-Strunz classification codes
Crystallography	Crystal system, class, space group, cell dimensions, axial ratios
X-ray diffraction	The strongest lines, with d-spacings and intensities
Physical properties	Colour, streak, lustre, hardness, density, cleavage, fracture, tenacity
Optical properties	Refractive indices, birefringence, dispersion, pleochroism
Calculated properties	Electron density, photoelectric value, radioactivity
Occurrence	Type locality and associated minerals

2.4.2 Why this matters for Bhasma research

The X-ray diffraction section is the most useful part for you. When you run an XRD on a Bhasma sample in Module 5, you obtain a set of peaks. To identify what compound has formed, you match your d-spacings against reference data. Webmineral gives that reference data free.

For example, after Marana of Tamra, the XRD may show peaks matching cuprite or tenorite. Confirming that from a mineral database turns an observation into evidence.

2.5 Introduction to Metal Databases

2.5.1 What a metal database contains

A metal database is different from a mineral database. A mineral database describes naturally occurring solid species. A metal database describes the element or the metal itself — its properties, its production, its compounds, or its behaviour in biological systems.

Type of metal database	Content	Example
Elemental property databases	Atomic number, mass, melting point, density, oxidation states	Periodic table databases
Production and economic databases	Mining, refining, trade, price	USGS, national mineral bureaux
Crystallographic databases	Structures of metals and alloys	Crystallographic Open Database
Metalloprotein databases	Metal-binding sites in biological macromolecules	MetalPDB
Toxicology databases	Toxic limits, exposure data	Public health agency databases

2.5.2 MetalPDB

MetalPDB is a database of metal-binding sites found in biological macromolecular structures. It is reference 82 of your NCISM list.

It is relevant to your subject in one specific and important way. Rasashastra converts metals into Bhasma, and a central research question is how those metals interact with biological molecules. MetalPDB shows how metals such as iron, copper, zinc and calcium bind naturally within proteins. It gives a scientific basis for discussing the biological handling of metals.

2.5.3 Indian sources

Source	Content
Indian Bureau of Mines	Indian Minerals Yearbook, production and reserve statistics for India
Geological Survey of India	Geology, mineral resources and maps of India
Ministry of Mines	Policy, statistics and annual reports

KEY POINTS TO REMEMBER

- USGS is the United States Geological Survey. Its National Minerals Information Center publishes the Mineral Commodity Summaries and the Minerals Yearbook.
- Mindat is the largest mineral database, run by the Hudson Institute of Mineralogy, founded in 2000.
- Webmineral gives a standard data sheet for each species, and is the quickest free source of XRD reference data.
- Minerals are classified by the Nickel-Strunz and the Dana systems.

- MetalPDB is a database of metal-binding sites in biological macromolecules.
- For India, use the Indian Bureau of Mines and the Geological Survey of India.
- Hingula is cinnabar, Manahshila is realgar, Haratala is orpiment, Makshika is chalcopyrite, Gairika is haematite.

EXAM CORNER — UNIT 2

This unit is asked as LAQ only (10 marks). There is no SAQ and no ABQ from this unit.

Likely long answer questions (10 marks)

1. Give an overview of metal and mineral databases. Describe the USGS mineral database and its applications in Rasashastra research.
2. Describe the Webmineral mineral database in detail. Explain how the data it provides is used in the standardisation of Bhasma.
3. Describe the important mineral databases available today. Discuss their significance in the identification and authentication of Rasa Dravyas.
4. Discuss metal databases, their types and their applications, with special reference to research in Rasashastra.
5. Explain how mineral and metal databases can be integrated to build the introduction and discussion of a research study on a Bhasma.

How to attempt

- Open with why the databases are needed in Rasashastra. Two or three sentences only.
- Then take each database in turn: custodian, content, how to search, what it gives.
- Draw the record-content table. It fills half a page and earns marks.
- Close with a worked example, such as Naga Bhasma or Tamra Bhasma. This is what separates a good answer from an average one.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **76, 77, 78, 79, 80, 81, 82**

1. **76.** Haldar SK. Mineral Exploration: Principles and Applications. Netherlands: Elsevier Science; 2018.
2. **77.** Minerant — mineralogical databases listing.
3. **78.** Zeng X, Xiao Y, Ji X, Wang G. Mineral Identification Based on Deep Learning That Combines Image and Mohs Hardness. 2021.
4. **79.** USGS National Minerals Information Center — Minerals Yearbook: Metals and Minerals.
5. **80.** USGS — USMIN Mineral Deposit Database.
6. **81.** Minerals Education Coalition — Minerals Database.
7. **82.** MetalPDB — database of metal sites in biological macromolecular structures.

UNIT 3
PLANT DATABASES

SYLLABUS POINTS

3.1 Overview of plant databases
3.2 The Plant List (TPL)
3.3 International Plant Names Index (IPNI)
3.4 Royal Botanic Gardens, Kew (RBG Kew) databases
3.5 Introduction to plant genetic databases (e.g., PlantGDB)

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Illustrate overview of plant databases, The Plant List and the International Plant Names Index	1	Lecture	Knows how
Describe Royal Botanic Gardens Kew databases and introduction to plant genetic databases	1	Lecture	Knows how
Demonstrate techniques to explore plant databases	4	Practical Training 2.6	Does
Exploration and analysis of plant database	6	Experiential Learning 2.3	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

3.1 Overview of plant databases

3.1.1 Why correct plant names matter

An Ayurvedic drug has many names. Guduchi is also Amrita, Chhinnaruha, Vatsadani and Madhuparni. In Hindi it is Giloy. In Gujarati it is Galo.

None of these names can be used in an international scientific paper. Only the **botanical name** can. And the botanical name itself changes as taxonomists revise their views. A name correct in 1950 may be a synonym today.

A plant database solves three problems:

1. It gives the **accepted name** today.
2. It lists every **synonym** under which the plant has been published.
3. It gives the **author citation** and the place of first publication, so the name is fully traceable.

> **An example you will meet constantly.** Guduchi was long written as *Tinospora cordifolia* (Willd.) Miers. The currently accepted authority citation is *Tinospora cordifolia* (Willd.) Hook.f. & Thomson. Both forms appear in the literature. Only a database will tell you which is accepted at the time you write.

3.1.2 Types of plant databases

Type	What it gives	Example
Nomenclatural database	Published names, authors, place of publication	IPNI
Taxonomic database	Accepted names and synonyms	World Flora Online, POWO
Floristic database	Which plants grow where	eFlora of India, Flora of China
Specimen and herbarium database	Digitised herbarium sheets	Kew herbarium catalogue, JSTOR Global Plants
Medicinal plant database	Traditional uses, phytochemistry	IMPPAT, BSI Medicinal Plant Database
Genetic and genomic database	Sequences, genes, genomes	PlantGDB, NCBI, Phytozome
Chemical database	Phytoconstituents and structures	Dr Duke’s database, IMPPAT

3.2 The Plant List (TPL)

3.2.1 What it was

The Plant List was a working list of all known plant species, created jointly by the **Royal Botanic Gardens, Kew** and the **Missouri Botanical Garden**. *It was launched in December 2010, and was produced in response to Target 1 of the Global Strategy for Plant Conservation, which sought an online flora of all known plants.*

For each name it gave the status — Accepted, Synonym or Unresolved — together with the accepted name where the entry was a synonym.

3.2.2 Its present status — this is important

The Plant List has not been updated since 2013, when version 1.1 was released, and it has been superseded by World Flora Online. The Plant List is now deprecated, and its successor, the WFO Plant List, was launched in May 2021 as a next-generation replacement, recognising the continuing need for a user-friendly, citable static list of all plant species.

> **Write this in the examination and in your thesis.** The Plant List is of historical importance and is still cited in older papers. For any new work, use **World Flora Online**. Quoting a name as verified from The Plant List in a paper written today can be challenged by a reviewer.

3.2.3 World Flora Online — the successor

World Flora Online (WFO) is a consortium of taxonomic institutions building a global resource of plant descriptions linked to a consensus classification.

The WFO Plant List is the most comprehensive and authoritative list of vascular plants and bryophytes. It is maintained by the global taxonomic community as a free and open-access resource, and a citable static version is generated on 21 June and 21 December each year from the dynamic WFO Taxonomic Backbone.

The two halves of WFO are worth distinguishing:

Part	Nature	Use
WFO Taxonomic Backbone	Dynamic; changes continuously as names are revised	The living classification
WFO Plant List	A static, dated snapshot, issued twice a year	The version you cite in a paper

This is why WFO is citable and The Plant List is not sufficient — you can state exactly which dated snapshot you consulted.

3.3 International Plant Names Index (IPNI)

3.3.1 What it is

The **International Plant Names Index (IPNI)** is a database of the published names of seed plants, ferns and lycophytes. It is a collaboration between the Royal Botanic Gardens Kew, the Harvard University Herbaria and the Australian National Herbarium.

3.3.2 What IPNI does and does not do

This distinction is the key point of the whole section, and it is a favourite question.

IPNI does	IPNI does not
Record every validly published name	Say which name is accepted
Give the author or authors of the name	Give synonymy
Give the place and date of publication	Give a taxonomic opinion
Give the type specimen details where known	Give descriptions or uses

In one line: IPNI is a **nomenclatural** database, not a **taxonomic** one. It tells you that a name exists and who published it. It does not tell you whether that name is the correct one to use.

For the accepted name you go to World Flora Online or POWO. For the authority and the publication details you go to IPNI. A complete citation usually needs both.

3.3.3 Understanding an author citation

In *Tinospora cordifolia* (Willd.) Hook.f. & Thomson:

Part	Meaning
Tinospora	Genus
cordifolia	Species epithet
(Willd.)	The basionym author — Willdenow first described this plant, under another genus
Hook.f. & Thomson	The authors who transferred it to the present genus

IPNI provides these standardised author abbreviations. Using them correctly is expected in a postgraduate thesis.

3.4 Royal Botanic Gardens, Kew (RBG Kew) databases

The Royal Botanic Gardens, Kew, in the United Kingdom, holds one of the world’s largest plant collections and runs several databases that you will use.

Resource	Content	Use to you
Plants of the World Online (POWO)	Accepted names, synonyms, distribution maps, descriptions and images	The quickest single check on a name and where the plant grows
World Checklist of Selected Plant Families (WCSP)	Peer-reviewed accepted names for selected families	Highest authority where the family is covered
International Plant Names Index (IPNI)	Published names and authorities	Nomenclature
Kew Herbarium Catalogue	Digitised herbarium specimens	Verification against an actual specimen
Seed Information Database	Seed weight, storage behaviour, germination	Cultivation and conservation work
Medicinal Plant Names Services (MPNS)	Links pharmaceutical and trade names to accepted botanical names	Very useful when a classical or trade name must be matched to a botanical name
Kew Science data and digital resources	Datasets and downloadable data	Bulk data for analysis

> **MPNS deserves your attention.** It exists precisely for the problem you face — connecting a drug name used in trade or in traditional medicine to a verified botanical name. When you cannot resolve a market name, this is the resource to try.

3.5 Plant genetic databases

3.5.1 What they are

A plant genetic or genomic database stores molecular data — DNA and RNA sequences, genes, proteins, genome assemblies and expression data.

3.5.2 PlantGDB

PlantGDB is the resource named in your syllabus. It was developed as a database of plant genome sequence data, providing assembled unique transcripts, genome browsers and tools for comparing plant genomes across species.

Its purpose was to bring together sequence data from many plant species in one place, so that a gene found in one species could be compared with related genes in another.

> **Point of caution for your thesis.** Genomic resources are replaced far more quickly than nomenclatural ones. Before citing any genomic database, open it and confirm it is still being maintained. If it is not, use a current resource such as NCBI, Phytozome or Ensembl Plants, and say which you used and on what date.

3.5.3 Other genetic and molecular resources

Resource	Content
NCBI GenBank	The primary international repository of nucleotide sequences
Phytozome	Plant genome sequences and comparative genomics tools
Ensembl Plants	Plant genomes with annotation and variation data
BOLD Systems	DNA barcode sequences; used for species identification

DNA barcoding deserves a mention. It uses a short standard DNA region to identify a species. For plants, the regions commonly used are *rbcL* and *matK*. This is directly relevant to your work — it can confirm the identity of a raw drug or detect substitution in a market sample, where morphology alone fails because the material is powdered.

3.5.4 Indian medicinal plant databases

These are the most directly useful databases for an Ayurveda postgraduate.

Database	Custodian	Content
IMPPAT	Institute of Mathematical Sciences, Chennai	Indian Medicinal Plants, Phytochemistry And Therapeutics; links plants, their phytochemicals and therapeutic uses
Medicinal Plant Database	Botanical Survey of India	Indian medicinal plants with botanical and distribution data
ICMR medicinal plant resources	Indian Council of Medical Research	Reviews and monographs on Indian medicinal plants
eFlora of India	Community project	Distribution and images of Indian plants
WHO monographs on selected medicinal plants	World Health Organization	Quality, safety and efficacy monographs

KEY POINTS TO REMEMBER

- Plant databases give the accepted name, the synonyms and the author citation.
- The Plant List was launched in 2010 by Kew and Missouri Botanical Garden, has not been updated since 2013, and is superseded by World Flora Online.
- The WFO Plant List was launched in May 2021, and a citable static version is issued on 21 June and 21 December each year.
- IPNI is nomenclatural, not taxonomic. It says a name was published; it does not say the name is accepted.
- POWO is the quickest Kew resource for an accepted name with distribution.
- MPNS links trade and pharmaceutical names to accepted botanical names.

- PlantGDB is the plant genome resource named in the syllabus; verify any genomic database is current before citing it.
- DNA barcoding of plants commonly uses *rbcL* and *matK*.
- IMPPAT is the most useful Indian database linking medicinal plants, phytochemicals and therapeutics.

EXAM CORNER — UNIT 3

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Write a short note on plant databases and their importance in Ayurvedic research.
2. Write a short note on The Plant List. What is its present status?
3. Write a short note on World Flora Online.
4. Write a short note on the International Plant Names Index.
5. Differentiate between a nomenclatural database and a taxonomic database.
6. Enumerate the databases maintained by the Royal Botanic Gardens, Kew.
7. Write a short note on Plants of the World Online.
8. Write a short note on Medicinal Plant Names Services.
9. Write a short note on plant genetic databases.
10. Write a short note on DNA barcoding in the authentication of raw drugs.
11. Write a short note on IMPPAT.
12. Explain the parts of a botanical author citation with an example.

How to attempt

- For any named database, use a fixed four-line pattern: what it is, who maintains it, what it contains, what you use it for.
- For The Plant List, you must mention that it is superseded by World Flora Online. Leaving that out will cost marks.
- Always finish with one Ayurvedic example, such as verifying the name of Guduchi.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95**

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2. **84.** Medicinal Plants: Biodiversity, Biotechnology and Conservation. Springer Nature Singapore; 2023.
3. **85.** Brummitt RK, Pando F. World Geographical Scheme for Recording Plant Distributions. 2001.
4. **86.** International Code of Nomenclature for algae, fungi, and plants (Melbourne Code).
5. **87.** Botanical Survey of India — Medicinal Plant Database.
6. **88.** IMPPAT — Indian Medicinal Plants, Phytochemistry And Therapeutics.
7. **89.** Indian Council of Medical Research — medicinal plants resources.

8. **90.** World Health Organization — monographs on selected medicinal plants.
9. **91.** World Flora Online.
10. **92.** eFlora of India — note on the second release of The Plant List.
11. **93.** International Plant Names Index.
12. **94.** Royal Botanic Gardens, Kew — collections, data and digital resources.
13. **95.** Database Commons entry for PlantGDB.

UNIT 4
DATABASE APPLICATION

SYLLABUS POINTS

4.1 Database applications in research and industry
4.2 Using databases for decision-making and policy development
4.3 Future directions: emerging trends and technologies in database development
4.4 Group project: developing a database application for metal, mineral or plant research

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Discuss database applications in research and industry, and use databases for decision-making and policy development	1	Lecture	Knows how
Analyse future directions in database development and applications, including emerging trends and technologies, and group project guidelines	1	Lecture	Knows how
Analyse case studies of database applications in metal, mineral or plant research	4	Practical Training 2.7	Shows how
Apply and justify case studies of database applications in metal, mineral or plant research	6	Experiential Learning 2.4	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

4.1 Database applications in research and industry

4.1.1 Applications in research

Stage of research	How a database is used
Selecting the topic	Literature databases show what has been done and where the gap lies
Reviewing literature	PubMed, DHARA, AYUSH Research Portal, Google Scholar
Authenticating raw drugs	Plant and mineral databases confirm identity
Designing the study	Clinical trial registries show similar completed and ongoing trials
Recording data	A study database holds subject data with validation rules
Analysing data	Statistical software works directly from the database
Publishing	Reference managers draw citations from bibliographic databases
Archiving	Repositories preserve the data for future workers

> **Two Indian databases you must know by name. DHARA** is the Digital Helpline for Ayurveda Research Articles, an index of published Ayurvedic research. **CTRI** is the Clinical Trials Registry — India, maintained by ICMR, where every clinical trial conducted in India must be registered before the first subject is enrolled. You will need CTRI registration for your own clinical work.

4.1.2 Applications in industry

Function	Use of the database
Procurement	Supplier records, rates, approved vendor list
Raw material control	Identity, source, test results, batch history
Production	Batch manufacturing records, process parameters, yields
Quality control	Specifications, test results, out-of-specification records
Inventory	Stock levels, expiry, reorder level
Regulatory affairs	Licences, product registrations, renewal dates
Sales and distribution	Orders, dispatch, distributor network
Pharmacovigilance	Adverse event reports
Traceability and recall	Tracing a batch from raw material to the patient

Traceability is the point to remember. If one batch of a Bhasma is found defective, a database can identify which raw material lot it came from, which other batches used that lot, and where every unit was despatched. Without a database, that recall is impossible.

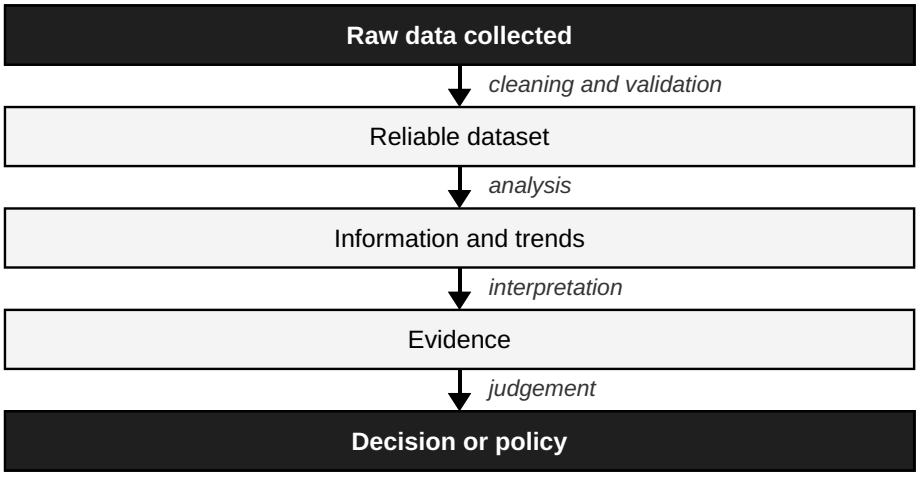
4.1.3 Applications specific to Ayurveda

Application	Description
Traditional Knowledge Digital Library (TKDL)	A digital record of Indian traditional medicine formulations, created to prevent wrong grant of patents on existing traditional knowledge
e-Aushadhi	Online portal for licensing of AYUSH manufacturers and products
AYUSH Research Portal	Consolidated research evidence on AYUSH systems
NAMASTE portal	Terminology and morbidity codes for Ayurveda, Siddha and Unani
IMPPAT	Plant, phytochemical and therapeutic links
National Medicinal Plants Board databases	Cultivation, trade and conservation data

> **TKDL is worth a full sentence in any answer.** It was created after cases in which patents were granted abroad on uses already recorded in Indian classical texts. TKDL converts those textual records into a searchable form in several languages, and makes them available to patent offices, so that prior art can be found. It is the clearest example of a database defending traditional knowledge.

4.2 Using databases for decision-making and policy development

4.2.1 The chain from data to decision



4.2.2 Levels of decision-making

Level	Who decides	Example of a database-supported decision
Operational	Production or store in-charge	When to reorder Guduchi stem
Tactical	Department or plant head	Which supplier to approve for the coming year
Strategic	Management	Which new product line to develop
Policy	Government	Whether to place a medicinal plant under trade restriction

4.2.3 Examples of policy shaped by databases

- **Conservation policy.** Red List and distribution data decide which medicinal plants are threatened, and lead to restrictions on wild collection and to cultivation programmes.
- **Mineral policy.** USGS and Indian Bureau of Mines production and reserve figures inform decisions on import dependence and on strategic mineral stockpiles.
- **Drug regulation.** Pharmacovigilance databases record adverse events. If a pattern appears, the regulator can change the label, restrict the product or withdraw it.
- **Public health.** Disease surveillance databases decide where a programme is directed.
- **Standard setting.** Analytical data collected across many batches decides the limits printed in a pharmacopoeial monograph.

4.3 Future directions and emerging technologies

Trend	What it means	Relevance to your field
Cloud databases	The database is hosted on remote servers and accessed over the internet	Small Ayurvedic units can use good systems without owning servers
Big data	Datasets too large or too varied for ordinary tools; described by volume, velocity, variety, veracity and value	Combining phytochemical, clinical and genomic data
Artificial intelligence and machine learning	Software that learns patterns from data	Predicting activity of a phytoconstituent; image-based mineral identification
Data mining	Finding previously unknown patterns in large data	Discovering repeated ingredient combinations across classical formulations
NoSQL and graph databases	Flexible, non-tabular storage	Mapping relationships between drug, constituent, target and disease
Blockchain	A distributed, tamper-evident record	Traceability of herbal raw material from farm to finished product
Internet of Things	Instruments that record data automatically	Continuous temperature logging during Puta or during storage
FAIR data principles	Data should be Findable, Accessible, Interoperable and Reusable	The standard expected of research data today
Open access and data sharing	Data published for others to reuse	Journals increasingly require it
Semantic web and ontologies	Machine-readable meaning, not just text	NAMASTE codes make Ayurvedic terms machine-readable

> **Remember the five V's of big data:** Volume, Velocity, Variety, Veracity and Value. This is a very common one-line question.

4.4 Group project — developing a database application

The curriculum prescribes a group project. Below is a practical method for it, which also serves as your answer if a question asks how a database is designed.

4.4.1 Steps in developing a database application

1. **Define the purpose.** State in one sentence what the database is for.
2. **Identify the users.** Decide who will enter data and who will read it.
3. **Collect requirements.** List every question the database must be able to answer.
4. **Identify entities and attributes.** An entity is a thing you store data about. An attribute is a fact about it.
5. **Draw the ER diagram.** Show entities, attributes and relationships.
6. **Convert to tables.** Each entity becomes a table. Assign a primary key.
7. **Normalise.** Remove repeated data; apply 1NF, 2NF and 3NF.
8. **Define relationships.** Link tables with foreign keys.
9. **Choose the software.** MS Access or MySQL is sufficient at this level.
10. **Create the tables and enter sample data.**
11. **Build queries, forms and reports.**
12. **Test with real questions.** Check that every requirement is answered.
13. **Document.** Write the schema, the data dictionary and the user instructions.

4.4.2 A worked example — a Rasa Dravya database

Purpose. To hold the identity, source, processing and analysis data of every Rasa Dravya used in the department.

Entities and their tables

Table	Key fields
DRAVYA	dravya_id (primary key), sanskrit_name, mineral_name, chemical_formula, group (Maharasa, Uparasa, Dhatu, etc.)
SOURCE	source_id (primary key), supplier, place_of_collection, date, quantity, dravya_id (foreign key)
SHODHANA	shodhana_id (primary key), dravya_id (foreign key), method, media_used, number_of_cycles, date, loss_percentage
MARANA	marana_id (primary key), dravya_id (foreign key), puta_type, number_of_puta, temperature, duration
ANALYSIS	analysis_id (primary key), sample_id, test_name, result, date, instrument
REFERENCE	ref_id (primary key), dravya_id (foreign key), text_name, chapter, verse

Questions this database can answer

- Which Dravyas were subjected to more than ten Putas?
- What was the average loss during Shodhana of Hingula?
- Which supplier gave material that failed a purity test?
- Which classical text describes the method used for a given sample?

KEY POINTS TO REMEMBER

- Databases support every stage of research, from topic selection to archiving.
- In industry, the most important database function is traceability, which makes recall possible.
- TKDL prevents wrong grant of patents on Indian traditional knowledge.
- CTRI is the Clinical Trials Registry — India; trials must be registered before enrolment begins.
- DHARA indexes published Ayurvedic research.
- The five V's of big data: Volume, Velocity, Variety, Veracity, Value.
- FAIR data: Findable, Accessible, Interoperable, Reusable.
- Database design sequence: purpose, requirements, entities, ER diagram, tables, normalisation, queries, testing, documentation.

EXAM CORNER — UNIT 4

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Enumerate the applications of databases in research.
2. Enumerate the applications of databases in the pharmaceutical industry.
3. Write a short note on the Traditional Knowledge Digital Library.
4. Write a short note on CTRI.
5. Write a short note on DHARA and the AYUSH Research Portal.
6. Explain how databases support decision-making and policy development.
7. Write a short note on big data and its five V's.
8. Write a short note on the FAIR principles of data.
9. Write a short note on the application of artificial intelligence in database research.
10. Describe the steps in developing a database application.
11. Write a short note on traceability in pharmaceutical manufacture.
12. Write a short note on emerging trends in database technology.

How to attempt

- Answer with a numbered list wherever the question says "enumerate". Do not write paragraphs.
- For any application question, give one Ayurvedic example. TKDL and CTRI are the two easiest.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **96, 97, 98, 99, 100, 101, 102**

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3. **98.** University of London Online Library — advanced search techniques.
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5. **100.** Coursera — Analyse and visualise data.
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UNIT 5
CHALLENGES AND LIMITATION OF DATABASES

SYLLABUS POINTS

5.1 Challenges in Pharmaceutical Formulation Development
5.2 Limitations of In Vitro and In Vivo Testing
5.3 Regulatory Challenges in Pharmaceutical Product Approval
5.4 Scalability and Manufacturing Challenges

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Identify database-related challenges in pharmaceutical development	1	Lecture	Knows
Discuss data quality, integration, security, scalability and regulatory compliance	1	Lecture	Knows how
Perform a role play related to the challenges and limitations of databases	2	Experiential Learning 2.5	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

> Read the session learning objectives above carefully. They set the angle of this unit. The four syllabus points name four areas of pharmaceutical work. In each, this unit asks what goes wrong **with the data** — its quality, integration, security, scalability and regulatory compliance.

5.0 The five recurring data problems

Before the four areas, learn these five. Every question in this unit can be answered from them.

Problem	What it means	Typical consequence
Data quality	Data is incomplete, wrong, outdated or inconsistently recorded	Wrong conclusions drawn
Data integration	Data sits in separate systems that cannot talk to each other	The same drug appears under different names in different systems
Data security	Unauthorised access, loss or tampering	Breach of confidentiality; loss of trial data
Scalability	The system cannot cope as data grows	Slow performance, failure at scale
Regulatory compliance	Records do not meet the standard the law requires	Audit failure, rejection of the application

5.1 Database challenges in pharmaceutical formulation development

Challenge	Explanation
Scattered formulation data	Trial batch records are kept in notebooks and spreadsheets rather than a database, so past work cannot be searched
Incomplete excipient data	Compatibility and supplier data missing, so past failures are repeated
No standard vocabulary	The same excipient recorded under trade name, chemical name and abbreviation
Loss of negative results	Failed batches are not recorded, so the same failure occurs again
Poor version control	It is unclear which version of the formula is current
Weak link between formulation and analysis	Batch data and test results held separately, so cause and effect cannot be traced

Specific to Ayurvedic formulation. Herbal raw material varies with season, region, plant part and age. Unless every one of these is recorded as a separate field, the batch-to-batch variation of a finished product can never be explained. Most classical pharmacies record only the drug name and quantity. That is the single largest data gap in Ayurvedic manufacture.

5.2 Database limitations in in vitro and in vivo testing

Limitation	Explanation
Non-comparable results	Different laboratories use different cell lines, strains, doses and endpoints, so results cannot be pooled
Missing experimental detail	Solvent, extract concentration and vehicle not recorded, making the study impossible to repeat
Publication bias	Only positive results are published, so any database built on published work is skewed
Poor species translation	Animal data recorded without the parameters needed to extrapolate to humans
Extract characterisation absent	A study reports “Guduchi extract” without stating solvent, marker content or plant part
No unique identifiers	Plant material not linked to a herbarium voucher, so the species tested cannot be verified afterwards
Data not machine-readable	Results published only as text in a PDF, so they cannot be collected automatically

> **The voucher specimen point is worth marks.** A herbarium voucher number in a paper allows any later worker to check exactly which plant was tested. Without it, the study cannot be relied on. Always deposit a voucher and quote its number in your own work.

5.3 Regulatory challenges in pharmaceutical product approval

Challenge	Explanation
Data integrity requirements	Regulators expect records to satisfy the ALCOA principles
Electronic record rules	Electronic records and signatures must meet defined standards, with audit trails
Audit trail	Every change must record who changed what, when and why
Differing requirements	Each country’s dossier requirements differ, so the same data must be reorganised
Long retention periods	Records must be preserved for many years, in a readable form
Validation of systems	The software itself must be validated before it can hold regulated data
Traceability	The regulator must be able to trace a finished batch back to raw material

The ALCOA principles of data integrity. This is a very likely short-answer question.

Letter	Requirement
A	Attributable — it is known who recorded it
L	Legible — it can be read
C	Contemporaneous — recorded at the time of the activity
O	Original — the first record, or a certified true copy
A	Accurate — it is correct

The extended form, **ALCOA+**, adds Complete, Consistent, Enduring and Available.

For ASU products. Rule 158-B requires proof of safety and effectiveness for several categories of Ayurvedic medicine. That proof is data. If the underlying records are incomplete or cannot be traced, the licence application fails, however good the product is.

5.4 Scalability and manufacturing challenges

Challenge	Explanation
Growth of data volume	Instruments such as HPLC, XRD and SEM produce very large files; storage and retrieval become difficult
System performance	A database designed for a small unit slows down as batches accumulate
Integration across a plant	Production, quality control, stores and dispatch running separate systems
Real-time data capture	Continuous process data from sensors needs different handling from batch records
Cost of upgrading	Small and medium Ayurvedic units cannot easily afford validated systems
Trained manpower	Few staff are trained in both pharmacy and data handling
Legacy data	Old paper records cannot be searched and are expensive to digitise
Backup and disaster recovery	Failure of a single server can stop production

How to close any answer in this unit. Do not end on the problem. End on the remedy: define standard fields, use controlled vocabulary, record negative results, keep an audit trail, deposit voucher specimens, follow ALCOA, validate the system, and back up the data. Two lines of remedy lift the whole answer.

KEY POINTS TO REMEMBER

- The five recurring data problems: quality, integration, security, scalability, regulatory compliance.
- ALCOA: Attributable, Legible, Contemporaneous, Original, Accurate. ALCOA+ adds Complete, Consistent, Enduring, Available.
- An audit trail records who changed what, when and why.
- Publication bias makes any database built only on published results incomplete.
- A herbarium voucher number makes a study verifiable afterwards.
- For Ayurvedic material, season, region, plant part and age must be recorded as separate fields, or batch variation can never be explained.
- Rule 158-B requires proof of safety and effectiveness — and that proof is only as good as the underlying records.

EXAM CORNER — UNIT 5

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Enumerate the challenges and limitations of databases in pharmaceutical research.
2. Write a short note on data quality and data integration problems.
3. Write a short note on data security in pharmaceutical databases.
4. Write a short note on scalability challenges in pharmaceutical databases.
5. Write a short note on the ALCOA principles of data integrity.
6. Write a short note on the audit trail.
7. Discuss the database-related challenges in pharmaceutical formulation development.
8. Discuss the limitations of databases built on in vitro and in vivo test results.
9. Write a short note on publication bias.
10. Write a short note on the importance of a voucher specimen.
11. Discuss the regulatory challenges in maintaining pharmaceutical data.
12. What are the data-related difficulties peculiar to Ayurvedic manufacturing units?

How to attempt

- Use the five recurring problems as your skeleton. They fit almost every question in this unit.
- Always give one Ayurveda-specific difficulty, such as unrecorded seasonal variation of raw material.
- End with two lines of remedy.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **103, 104**

1. **103.** Recent Challenges in Intelligent Information and Database Systems: 15th Asian Conference, ACIIDS 2023, Phuket, Thailand.
2. **104.** Digital Transformation in Healthcare 5.0, Volume 2: Metaverse, Nanorobots and Machine Learning. De Gruyter; 2024.

MODULE 2
PRACTICAL TRAINING ACTIVITIES

Module 2 carries **20 hours of practical training**, over seven activities, as listed in Table 4 of the curriculum.

No	Practical name	Hours
2.1	Demonstration of various searching techniques for databases	2
2.2	Demonstration of modelling, querying and analytics of data	2
2.3	Demonstration of techniques to search and extract data from the USGS Mineral Database	2
2.4	Analyse and integrate Webmineral Database	3
2.5	Search and explore Metal Database and applications	3
2.6	Demonstration of techniques to explore plant databases	4
2.7	Analysis of case studies of database applications in metal, mineral or plant research	4
	Total	**20**

Practical Training 2.1 — Demonstration of various searching techniques for databases

Aim. To learn and demonstrate the search techniques used in scientific databases.

Requirements. Computer with internet, access to PubMed, Google Scholar, DHARA and the AYUSH Research Portal, notebook.

Procedure

1. Choose one research question. For example, the immunomodulatory activity of Guduchi.
2. Break the question into concepts. Concept one: the drug. Concept two: the activity.
3. List all synonyms for each concept, including the botanical name and its synonyms.
4. Build a search string using OR within a concept and AND between concepts.
5. Run the search. Record the number of results.
6. Apply truncation and note the change in number.
7. Apply a phrase search with quotation marks and note the change.
8. Apply filters — years, article type, language — and note the change.
9. Repeat the same search in a second database and compare.

10. Select the ten most relevant results and export the citations.

Observation table to draw

Search string used	Technique applied	Database	Number of results	Relevant results

Result. Record the final search string that gave the most relevant results, and the number retrieved.

Precautions. Record the exact date of the search, since databases change daily. Use the botanical name along with the Sanskrit name. Do not rely on a single database.

Practical Training 2.2 — Demonstration of modelling, querying and analytics of data

Aim. To model a small dataset, query it and analyse the results.

Requirements. Computer with MS Access, MySQL or a spreadsheet application, sample dataset.

Procedure

1. Take a small real dataset, such as twenty raw drugs with their botanical names, families, parts used and rates.
2. Identify the entities and attributes.
3. Draw the ER diagram on paper.
4. Create the table or tables in the software. Set the primary key.
5. Enter the data.
6. Write and run queries — select all drugs of one family; sort by rate; count the drugs in each family; find drugs above a given rate.
7. Create a form for data entry and a report for output.
8. Produce a simple chart from the query result.

Observation to record. The ER diagram, the table structure with field names and data types, each query written out, and the result of each query.

Result. Record that the database was created and that the queries returned correct results.

Precautions. Fix the data type of each field before entering data. Never leave the primary key blank. Save your work at every stage.

Practical Training 2.3 — Searching and extracting data from the USGS Mineral Database

Aim. To search the USGS mineral resources and extract data useful for Rasashastra research.

Requirements. Computer with internet, access to the USGS National Minerals Information Center.

Procedure

1. Select one metal used in Rasashastra — lead, tin, copper, zinc, iron, gold or silver.
2. Open the Mineral Commodity Summaries for that metal.
3. Record world production, the leading producing countries, India’s production, reserves and price.
4. Open the Minerals Yearbook chapter for the same metal and note the principal ores.
5. Search the mineral deposit database for occurrences.
6. Download the data table.
7. Plot production over the last ten years.
8. Write a short paragraph interpreting the trend.

Observation table to draw

Metal	Rasashastra name	Principal ore	World production	India’s production	Reserves	Price trend

Result. Record the data extracted and the interpretation drawn.

Precautions. Note the year of the report, since figures are revised. Note the unit of measurement. Record the exact URL and date of access for citation.

Practical Training 2.4 — Analysis and integration of the Webmineral Database

Aim. To analyse mineral records in Webmineral and integrate the data with classical Rasashastra descriptions.

Requirements. Computer with internet, access to Webmineral, classical texts — Rasa Ratna Samuccaya and Rasatarangini.

Procedure

1. Select five Rasa Dravyas, such as Hingula, Manahshila, Haratala, Makshika and Gairika.
2. Identify the mineral name for each.
3. Open the Webmineral record for each mineral.
4. Record the chemical formula, composition by element, crystal system, hardness, specific gravity, colour and streak.
5. Record the strongest X-ray diffraction lines.
6. Now read the classical description of the same Dravya — Varna, Gaurava, Grahya and Agrahya Lakshana.
7. Compare the two descriptions and prepare an integration table.
8. State which classical characters correspond to which modern parameter.

Observation table to draw

Rasa Dravya	Mineral	Formula	Crystal system	Hardness	Specific gravity	Classical Lakshana	Modern parameter matched

Result. Record the correlation established between the classical and modern descriptions.

Precautions. Some classical Dravyas are not a single mineral species. Say so where it is the case, rather than forcing a match.

Practical Training 2.5 — Search and explore Metal Database and applications

Aim. To search metal databases and study their applications.

Requirements. Computer with internet, access to MetalPDB, elemental property resources, and Indian Bureau of Mines publications.

Procedure

1. Select one metal used in Bhasma preparation.
2. From an elemental property source, record atomic number, atomic mass, melting point, boiling point, density and common oxidation states.
3. Open MetalPDB and search for that metal.
4. Record how many structures contain it, and in what kind of site it binds.
5. Note the ligands that commonly bind it.
6. Open the Indian Bureau of Mines resource and record India’s production and reserves.
7. Prepare a single profile sheet for that metal.
8. Write a short note on how these properties relate to the behaviour of the metal during Marana.

Observation to record. The completed metal profile sheet.

Result. Record the profile prepared and the inference drawn.

Precautions. Distinguish clearly between the free metal, its ore and its compounds. A Bhasma is not the free metal.

Practical Training 2.6 — Techniques to explore plant databases

Aim. To explore plant databases and verify the identity of medicinal plants.

Requirements. Computer with internet, access to World Flora Online, IPNI, POWO, MPNS, IMPPAT and the BSI Medicinal Plant Database.

Procedure

1. Select ten Ayurvedic drugs.
2. For each, note the Sanskrit name and the botanical name given in your textbook.
3. Search that botanical name in World Flora Online. Record whether it is accepted or a synonym. If a synonym, record the accepted name.
4. Search the same name in IPNI. Record the author citation and the place and year of publication.
5. Open POWO and record the native distribution.
6. Search MPNS for the trade or pharmaceutical name.
7. Search IMPPAT and record the principal phytoconstituents.
8. Prepare a verification sheet for each drug.

Observation table to draw

Sanskrit name	Name in textbook	Accepted name (WFO)	Status	Author citation (IPNI)	Family	Distribution (POWO)

Result. Record how many textbook names were accepted names, how many were synonyms, and how many could not be resolved.

Precautions. Record the date of access. Do not correct a classical text — record what the text says and what the database says, side by side.

Practical Training 2.7 — Analysis of case studies of database applications

Aim. To analyse published case studies in which databases were used in metal, mineral or plant research.

Requirements. Selected published papers, computer with internet, notebook.

Procedure

1. Obtain three case studies — one on a metal or mineral, one on a plant, and one on a formulation.
2. For each, identify which databases were used.
3. Identify at which stage of the study each database was used.
4. Note what the database contributed that could not have been obtained otherwise.
5. Note any limitation the authors mention.
6. Prepare a comparison table of the three studies.
7. Present your analysis.

Observation table to draw

Case study	Field	Databases used	Stage of use	Contribution	Limitation noted

Result. Record the analysis and the general conclusions drawn about database use in research.

Precautions. Use peer-reviewed sources. Do not confuse the authors' conclusions with your own analysis; keep the two separate in your record.

MODULE 2
EXPERIENTIAL LEARNING ACTIVITIES

Module 2 carries **30 hours** under experiential learning, including modular assessment. Five activities are prescribed in Table 5 of the curriculum.

No	Experiential learning activity	Hours
2.1	Design and implement a simple database for a plant nursery	6
2.2	Analyse Webmineral Database and Metal Database	6
2.3	Exploration and analysis of plant database	6
2.4	Apply and justify case studies of database applications in metal, mineral or plant research	6
2.5	Perform a role play related to the challenges and limitations of databases	2
	Total	**26**

Experiential Learning 2.1 — Design and implement a simple database for a plant nursery

What you have to do. Design and build a working database for a medicinal plant nursery, and demonstrate it.

Suggested structure

Table	Fields
PLANT	plant_id, sanskrit_name, botanical_name, family, part_used, habit
STOCK	stock_id, plant_id, bed_number, number_of_saplings, date_of_sowing, source_of_seed
CARE	care_id, stock_id, activity, date, done_by, remarks
SALE	sale_id, plant_id, quantity, rate, date, buyer

Method. Follow the thirteen steps in Unit 4, section 4.4.1. Draw the ER diagram first, on paper, before touching the computer.

Queries your database must answer

- Which plants are ready for sale this month?
- How many saplings of each family are in stock?
- What was the survival percentage for each batch?
- Which supplier gave the seed with the best germination?

How to record it. Submit the ER diagram, the table structures with data types, at least six queries with their outputs, one form, one report, and a short user manual.

Experiential Learning 2.2 — Analyse Webmineral Database and Metal Database

What you have to do. Build a complete data profile for a group of Rasa Dravyas, drawing on both mineral and metal databases, and interpret it.

Method. Choose one group — for example the Maharasa, or the Shuddha Dhatu. For each member, extract mineral data from Webmineral and metal data from the sources in Practical 2.5. Then go further than the practical: interpret the data.

Questions to answer in your report

- Do the members of this classical group share any modern chemical feature?
- Does hardness or specific gravity explain any classical instruction about grinding or Puta?
- Does the chemistry explain the classical Shodhana media prescribed?
- Where does the classical grouping cut across modern mineralogical classification?

How to record it. A written report with the data tables, your interpretation, and a clear statement of where the classical and modern systems agree and where they do not.

Experiential Learning 2.3 — Exploration and analysis of plant database

What you have to do. Take one classical Ayurvedic formulation and verify every plant ingredient in it.

Method. Choose a formulation with eight to fifteen plant ingredients — Dashamoola, Triphala with substitutes, or Chyavanaprasha. For every ingredient, resolve the accepted botanical name, the synonyms, the family, the distribution and the part used. Then record any controversy.

The point of this activity. You will find that some ingredients have more than one botanical source in use, and that different texts and different states use different plants. Documenting that disagreement, with database evidence, is a genuine piece of research work.

How to record it. A verification table for the whole formulation, a short note on every disputed ingredient, and a conclusion stating which sources you would recommend and why.

Experiential Learning 2.4 — Apply and justify case studies of database applications

What you have to do. Move from analysing published case studies to designing your own application, and defend it.

Method. Identify a real problem in your department or in a nearby pharmacy that a database would solve. Examples: tracking Puta records, tracking raw drug purchases, or maintaining herbarium and museum specimens. Design the database, build a prototype, and present it to the group. Then justify every design decision when questioned.

How to record it. The problem statement, the design, the prototype, the presentation, and a record of the questions asked and your answers.

Experiential Learning 2.5 — Role play on challenges and limitations of databases

What you have to do. Act out, in a group, a situation in which database failure causes a real problem.

Suggested situations

Situation	Roles
A regulatory audit finds that batch records cannot be traced	Inspector, production head, QA head, company owner
A published study cannot be repeated because the plant material was never identified	Reviewer, author, taxonomist, journal editor
Two departments hold the same drug under different names, and stock is double-ordered	Store keeper, purchase officer, production head
A server fails and the day’s manufacturing data is lost	IT staff, QA head, regulator

Method. Take one situation. Assign roles. Play it out for ten to fifteen minutes. Then step out of role and discuss as a group: what went wrong, what data problem caused it, and what system would have prevented it.

How to record it. The situation chosen, the roles, a summary of what happened, the data problem identified, and the preventive measure agreed by the group.

MODULE 2
VIVA POINTERS AND QUESTION BANK

Viva pointers

On database concepts

- What is the difference between a database and a DBMS?
- What is a primary key? What is a foreign key?
- Name the types of databases you know.
- What does ACID stand for?
- What are the four groups of SQL commands?

On searching

- Show me how you would search for papers on Guduchi and immunity.
- What does AND do to your result? What does OR do?
- Why must you search a plant by its synonyms as well?
- Which database would you use to check whether a botanical name is accepted?

On mineral and metal databases

- What is USGS? What does it publish?
- Which database would you use to find the XRD reference lines for cinnabar?
- What is the mineral identity of Manahshila? Of Makshika?
- What is MetalPDB?
- Which Indian body publishes the Indian Minerals Yearbook?

On plant databases

- What is the present status of The Plant List?
- What is the difference between IPNI and World Flora Online?
- What is POWO?
- What is a voucher specimen and why is it needed?
- Which regions are used for DNA barcoding of plants?

On applications and challenges

- What is TKDL and why was it created?
- Where must an Indian clinical trial be registered?
- What does ALCOA stand for?
- What is an audit trail?
- Give one data problem peculiar to Ayurvedic manufacturing.

Module 2 question bank

Long Answer Questions (10 marks) — from Unit 2 only

1. Give an overview of metal and mineral databases. Describe the USGS mineral database and its applications in Rasashastra research.
2. Describe the Webmineral mineral database in detail. Explain how its data is used in the standardisation of Bhasma.
3. Describe the important mineral databases available today, and discuss their significance in the identification and authentication of Rasa Dravyas.
4. Discuss metal databases, their types and their applications, with special reference to research in Rasashastra.
5. Explain how mineral and metal databases can be integrated to build the introduction and discussion of a research study on a Bhasma.

Short Answer Questions (5 marks) — from Units 1, 3, 4 and 5

From Unit 1

1. Define data, information and knowledge, and differentiate between them.
2. Define a database and write its advantages over a file system.
3. Enumerate the types of databases with one example of each.
4. Define DBMS and enumerate its functions.
5. Write a short note on the ACID properties.
6. Write a short note on SQL and its command groups.
7. Write a short note on Boolean operators in database searching.
8. Write a short note on primary key and foreign key.

From Unit 3

1. Write a short note on The Plant List and its present status.
2. Write a short note on World Flora Online.
3. Write a short note on the International Plant Names Index.
4. Differentiate between a nomenclatural database and a taxonomic database.
5. Enumerate the databases maintained by the Royal Botanic Gardens, Kew.
6. Write a short note on Medicinal Plant Names Services.
7. Write a short note on DNA barcoding in the authentication of raw drugs.
8. Write a short note on IMPPAT.

From Unit 4

1. Enumerate the applications of databases in research.
2. Enumerate the applications of databases in the pharmaceutical industry.
3. Write a short note on the Traditional Knowledge Digital Library.
4. Write a short note on CTRI.
5. Write a short note on big data and its five V’s.

6. Write a short note on the FAIR principles of data.
7. Describe the steps in developing a database application.
8. Explain how databases support decision-making and policy development.

From Unit 5

1. Enumerate the challenges and limitations of databases in pharmaceutical research.
2. Write a short note on the ALCOA principles of data integrity.
3. Write a short note on the audit trail.
4. Write a short note on publication bias.
5. Write a short note on the importance of a voucher specimen.
6. What are the data-related difficulties peculiar to Ayurvedic manufacturing units?

> **Note on the marks range.** Table 6F places Module 2 in the marks range of **5 to 15** in the theory paper. Unit 2 is the only unit carrying a long answer question. Units 1, 3, 4 and 5 carry short answers only. No unit of this module is set as an Analytical Based Question.

MODULE 3
ESSENTIAL CHEMISTRY — UNLOCKING THE RASASHASTRA TECHNIQUES

Credits	Lectures	Practical Training	Experiential Learning	Total hours	Modular assessment
2	10	20	30	60	50 marks

About this module

This module integrates fundamental chemistry principles with traditional Ayurvedic practice, to improve the understanding and application of Rasashastra techniques. By exploring the connections between chemistry and Ayurveda, the learner gains a deeper understanding of the processing methods and transformations involved in Rasashastra, and becomes able to develop and optimise Ayurvedic formulations.

Module Learning Objectives

At the end of this module, the student should be able to:

1. Describe the role of basic chemistry principles in Rasashastra.
2. Apply chemical concepts effectively in Rasashastra.
3. Integrate chemical concepts with traditional Ayurvedic practices.
4. Correlate essential chemistry concepts with Ayurvedic principles, to enhance comprehension of Ayurvedic principles and practices.
5. Demonstrate the ability to apply this knowledge in the field of Ayurveda.

Units in this module

Unit	Name
M3U1	Atomic, Molecular Structure and Periodic Table Elements
M3U2	Specialised Chemistry Fields — An Outline
M3U3	Organic Chemistry
M3U4	Phytochemistry
M3U5	Metallurgy

How this module is examined

Unit	ABQ	SAQ	LAQ
U1 Atomic, Molecular Structure and Periodic Table	Yes	Yes	No
U2 Specialised Chemistry Fields	Yes	Yes	No
U3 Organic Chemistry	Yes	Yes	No
U4 Phytochemistry	No	Yes	Yes
U5 Metallurgy	No	Yes	Yes

Module 3 is placed in the marks range of **5 to 20**. Note the pattern carefully. **Three units can come as a twenty-mark Analytical Based Question**, and two units can come as a ten-mark long answer. No module in this course has more ABQ-eligible units than this one. Prepare Units 1, 2 and 3 for analysis, not for recall.

A word before you begin

Many students open this module and feel it belongs to a chemistry course, not to Ayurveda. That feeling is worth addressing at the start.

Our Acharyas described what they observed. They said that Parada must be subjected to Samskara. They said that Abhraka must be given a hundred Putas. They said that Shodhana of Haratala must be done in Kushmanda Swarasa. They recorded these instructions with great precision, and they were right.

What they did not have was a vocabulary for *why*. Chemistry gives us that vocabulary. It does not replace the classical instruction. It explains it.

When you learn in this module that a metal becomes an oxide on repeated heating, you are not being told that Marana is oxidation and nothing more. You are being given one honest explanation of one part of what happens. Hold both — the classical method in full, and the chemical explanation as far as it goes. That is the correct scientific attitude, and it is what a postgraduate examiner is looking for.

UNIT 1
ATOMIC, MOLECULAR STRUCTURE AND PERIODIC TABLE ELEMENTS

SYLLABUS POINTS

1.1 Importance of knowledge of chemistry for understanding various processing methods of Rasashastra and Oushadha Nirmana
1.2 Atomic structure, Periodic table, Chemical bonding, Isomerism — application of these principles in Shodhana, Marana, Amritikarana and Chaturvidha Rasayana
1.3 Periodic table, main group elements, transition metals: understanding of chemical properties of non-metals, metals, variable oxidation states, thermal stability, and the ability to form complex compounds

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe the role of basic chemistry principles and their application in Rasashastra	1	Lecture	Know
Analyse chemical bonding types and principles and isomerism, and their significance in better understanding of Rasashastra processes such as Shodhana and Marana	5	Practical Training 3.1	Shows how
Apply the knowledge of the periodic table for chemical properties of main group and transition metals	6	Experiential Learning 3.1	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

1.1 Importance of chemistry for Rasashastra and Oushadha Nirmana

1.1.1 Six reasons a Rasashastra student must know chemistry

1. **To understand what a Dravya is.** Hingula is not merely a red mineral. It is mercuric sulphide. Once you know that, you know it contains mercury, you know it will decompose on heating, and you know why the classical texts insist on closed vessels.

2. **To understand why a process is done.** Shodhana is not a ritual. Each medium is chosen for a chemical reason. Kanji is acidic. Nimbu Swarasa contains citric acid. Godugdha carries protein and fat. These are different chemical environments, and they act differently.
3. **To predict what will form.** If a metal is heated repeatedly in air, an oxide forms. If it is heated with sulphur, a sulphide forms. Knowing this lets you anticipate the product before you analyse it.
4. **To design and read an analysis.** XRD, SEM-EDAX, FTIR and TGA all report chemical information. Without chemistry you cannot interpret your own results.
5. **To ensure safety.** Arsenic and mercury compounds are the working materials of this subject. Understanding their chemistry is a safety requirement, not an academic luxury.
6. **To communicate and to publish.** A paper that says only “Bhasma was prepared as per classical method” will not be accepted. A paper that identifies the phases formed and explains the transformation will be.

1.1.2 The correlation at a glance

Learn this table thoroughly. It is the spine of the whole module, and it can be used in almost any answer.

Rasashastra process	Chemical process involved
Shodhana	Purification, removal of soluble impurities, surface reaction, chemical modification, particle disaggregation
Bhavana	Wet grinding; particle size reduction, adsorption of organic matter, surface modification
Mardana	Trituration; mechanical size reduction, amalgamation in the case of Parada
Marana	Repeated heating; oxidation, sulphide formation, thermal decomposition, sintering
Puta	Controlled heating cycle; a defined temperature and time programme
Amritikarana	Post-Marana treatment; removal of residual reactive species, surface stabilisation
Satvapatana	Reduction smelting; extraction of the free metal from its ore
Druti	Liquefaction; formation of a low melting phase
Jarana	Assimilation of a metal or mineral into Parada
Kupipakwa	Controlled sublimation and condensation in a glass vessel
Parpati	Melting and rapid solidification into a thin flake

1.2 Atomic structure, chemical bonding and isomerism

1.2.1 Atomic structure — the essentials

Term	Meaning
Proton	Positively charged particle in the nucleus; decides the element
Neutron	Neutral particle in the nucleus
Electron	Negatively charged particle outside the nucleus; decides chemical behaviour
Atomic number (Z)	Number of protons
Mass number (A)	Protons plus neutrons
Isotopes	Same atomic number, different mass number
Valence electrons	Electrons in the outermost shell; these take part in bonding

Orbitals and shells. Electrons occupy orbitals — s, p, d and f. Their filling follows three rules: the Aufbau principle (lowest energy first), the Pauli exclusion principle (no two electrons alike in one orbital), and Hund’s rule (orbitals of equal energy fill singly first).

The **d orbitals** matter most to you. The metals of Rasashastra — Loha, Tamra, Yashada, Naga, Vanga, Swarna, Rajata and Parada — are all d-block elements. Their chemistry is d-orbital chemistry.

1.2.2 Chemical bonding

Bond type	How it forms	Character of the compound	Example from Rasashastra
Ionic or electrovalent	Complete transfer of electrons from metal to non-metal	High melting point, soluble in water, conducts when molten	Common salt in Lavana; Sphatika
Covalent	Sharing of electrons	Lower melting point, often insoluble in water	Organic constituents of herbs; sulphur
Coordinate or dative	Both electrons supplied by one atom	Basis of all complexes	Metal-ligand bonds in Bhasma and in the body
Metallic	Positive cores in a sea of delocalised electrons	Lustre, malleability, conduction	All Shuddha Dhatu
Hydrogen bond	Attraction between H bonded to N, O or F and a lone pair	Raises boiling point, affects solubility	Water, alcohols, sugars in herbal media
Van der Waals forces	Weak attraction between molecules	Very weak, but decisive at the nanoscale	Aggregation of Bhasma particles

> **The coordinate bond is the one to master.** Every metal-organic interaction in this subject — a herbal molecule binding to a metal during Bhavana, a metal binding to a protein in the body, chelation therapy for metal poisoning — is coordinate bonding. If you understand it well, Units 2 and 3 become easy.

1.2.3 Isomerism

Isomers are compounds with the same molecular formula but different arrangements of atoms. They can have completely different properties, and in medicine they may have completely different actions.

Type	Sub-type	Difference
Structural isomerism	Chain	Different carbon skeleton
	Position	Same group at a different position
	Functional	Different functional group altogether
	Metamerism	Different alkyl groups on either side of a functional group
	Tautomerism	Rapid interconversion, as in keto and enol forms
Stereoisomerism	Geometrical (cis-trans)	Different arrangement about a double bond or ring
	Optical	Non-superimposable mirror images; rotate plane-polarised light

Why this matters to Ayurveda. Many plant constituents are optically active. One enantiomer may be active and the other inactive or even harmful. Menthol, camphor and many terpenoids exist in more than one form, and the natural form is usually a single one. When a synthetic version is substituted for a natural one, this is exactly what changes.

1.2.4 Applying these principles to the Rasashastra processes

Shodhana. Three things can happen, and often all three do:

- *Physical.* Repeated heating and quenching, called Nirvapa, causes thermal shock. The metal cracks, becomes brittle and breaks into smaller pieces. Surface impurities separate.
- *Chemical.* The medium reacts with the impurity or with the surface. An acidic medium such as Kanji or Amlavetasa dissolves surface oxides and carbonates. An alkaline medium acts differently.

- *Surface modification.* Organic molecules from the medium attach to the fresh surface exposed by cracking.

Marana. The core change is a chemical conversion of the metal into a compound, usually assisted by repeated heating.

If heated	Product formed	Example
In air, with free access of oxygen	Metal oxide	Tamra to copper oxides; Naga to lead oxide
With sulphur	Metal sulphide	Parada with Gandhaka to mercuric sulphide
In a closed vessel	Sublimation and controlled reaction	Kupipakwa Rasayana

Repeated Puta does three things at once — it completes the reaction, it reduces particle size progressively, and it burns away the organic Bhavana material, leaving a fine residue.

Amritikarana and Chaturvidha Rasayana are dealt with in Unit 2.

1.3 Periodic table, main group elements and transition metals

1.3.1 Structure of the periodic table

Feature	Description
Period	A horizontal row; the number of the period is the outermost shell number
Group	A vertical column; elements in a group have similar valence configuration
s-block	Groups 1 and 2; alkali and alkaline earth metals
p-block	Groups 13 to 18; contains metals, metalloids and non-metals
d-block	Groups 3 to 12; the transition metals
f-block	Lanthanides and actinides

1.3.2 Periodic trends

Property	Across a period, left to right	Down a group
Atomic radius	Decreases	Increases
Ionisation energy	Increases	Decreases
Electronegativity	Increases	Decreases
Metallic character	Decreases	Increases
Non-metallic character	Increases	Decreases

1.3.3 Metals, non-metals and metalloids

Property	Metals	Non-metals
Lustre	Present	Usually absent
Malleability and ductility	Yes	No, they are brittle
Conduction of heat and electricity	Good	Poor, except graphite
Electron behaviour	Lose electrons, form cations	Gain electrons, form anions
Nature of oxide	Basic	Acidic
Examples in Rasashastra	Swarna, Rajata, Tamra, Loha, Naga, Vanga, Yashada	Gandhaka, carbon

Metalloids show properties of both. Arsenic is the important one for you — Haratala and Manahshila are arsenic sulphides.

1.3.4 Transition metals — the four properties you must know

The curriculum names these specifically. Learn all four, with the reason.

1. Variable oxidation states. In transition metals the (n-1)d and ns orbitals are close in energy. So a variable number of electrons can be lost. This is why iron exists as Fe(II) and Fe(III), and copper as Cu(I) and Cu(II).

Why it matters to you. A Bhasma of the same metal can contain the metal in more than one oxidation state, depending on how it was prepared. Two batches of Lauha Bhasma with different Fe(II) to Fe(III) ratios are not the same product. This is a genuine and researchable question in your subject.

2. Formation of coloured compounds. Partly filled d orbitals allow electrons to move between d levels of slightly different energy, absorbing visible light. The transmitted light gives the colour.

Why it matters to you. The classical texts assess a Bhasma by its Varna. Tamra Bhasma should be a particular red-brown; Lauha Bhasma a particular reddish black. That colour is a direct consequence of the oxidation state and the compound formed. **The classical Varna Pariksha is, in effect, a visual test of chemical composition.** This sentence is worth writing in an examination.

3. Ability to form complex compounds. Transition metal ions are small, highly charged and have vacant d orbitals that can accept lone pairs. So they readily accept ligands and form coordination complexes.

Why it matters to you. During Bhavana with herbal juices, organic molecules with oxygen and nitrogen donor atoms are present in quantity. Complexation and surface binding are entirely plausible. This is the chemical basis of the classical requirement that Marana must be done with Bhavana, not by simple heating.

4. Catalytic activity and thermal stability. Variable oxidation states allow transition metals to act as catalysts. Their compounds also differ widely in thermal stability — some oxides decompose easily on heating, others do not.

Why it matters to you. Thermal stability decides what survives a Puta. Carbonates decompose at moderate temperature; many oxides do not. The number and intensity of Putas prescribed for a given Dravya is, in chemical terms, a statement about the thermal stability of the compounds involved.

1.3.5 The Rasashastra metals in the periodic table

Rasashastra name	Metal	Symbol	Block	Common oxidation states
Swarna	Gold	Au	d	+1, +3
Rajata	Silver	Ag	d	+1
Tamra	Copper	Cu	d	+1, +2
Loha	Iron	Fe	d	+2, +3
Naga	Lead	Pb	p	+2, +4
Vanga	Tin	Sn	p	+2, +4
Yashada	Zinc	Zn	d	+2
Parada	Mercury	Hg	d	+1, +2

A point often missed. Naga and Vanga — lead and tin — are **p-block** elements, not transition metals, although they are classified as Dhatu in Rasashastra. Zinc, though in the d-block, has a full d shell and so does not show variable oxidation states or coloured compounds like a true transition metal. Stating this correctly shows the examiner that you understand the periodic table and are not simply reciting a list.

KEY POINTS TO REMEMBER

- Chemistry explains the classical method; it does not replace it.
- Shodhana works by thermal shock, chemical reaction with the medium, and surface modification.
- Marana is chiefly conversion of the metal into an oxide or a sulphide, with progressive particle size reduction.
- Bond types: ionic, covalent, coordinate, metallic, hydrogen bond, van der Waals. The coordinate bond underlies every metal-organic interaction in this subject.
- Isomerism: structural (chain, position, functional, metamerism, tautomerism) and stereo (geometrical, optical).
- Four properties of transition metals: variable oxidation states, coloured compounds, complex formation, catalytic activity.
- Variable oxidation states arise because (n-1)d and ns orbitals are close in energy.
- The colour of a Bhasma reflects its oxidation state and phase, so Varna Pariksha is an indirect chemical test.
- Naga and Vanga are p-block elements, not transition metals.

EXAM CORNER — UNIT 1

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Explain the importance of chemistry in understanding Rasashastra processes.
2. Enumerate the types of chemical bonding with one example of each.
3. Write a short note on the coordinate bond and its relevance to Rasashastra.
4. Define isomerism. Enumerate its types.
5. Write a short note on optical isomerism and its importance in herbal drugs.
6. Write the periodic trends across a period and down a group.
7. Differentiate between metals and non-metals.
8. Write a short note on the variable oxidation states of transition metals.
9. Why do transition metals form coloured compounds? Correlate with Bhasma Pariksha.
10. Write a short note on complex formation by transition metals.
11. Explain the chemical changes occurring during Shodhana.
12. Explain the chemical changes occurring during Marana.

Likely analytical based questions (20 marks)

1. Two batches of Lauha Bhasma prepared by the same classical method differ in colour and in their results on analysis. Analyse the possible chemical reasons, relate them to the oxidation states of iron, and suggest what process controls would reduce this variation.
2. Analyse the classical process of Shodhana of a metal in the light of atomic structure and chemical bonding. Explain, step by step, what happens physically and chemically at each stage, and justify the choice of the media prescribed.
3. The classical texts prescribe a different number of Putas for different Dravyas. Analyse this instruction in terms of thermal stability, oxidation state and particle size, and explain how you would determine the correct number of Putas experimentally.

How to attempt an ABQ in this module

- Begin with the classical position, stated accurately, in three or four lines.
- Then give the chemical explanation, with the principle named.
- Then analyse — what varies, why, and what would happen if it were changed.
- Then propose — the control, the experiment, or the correction.
- Close by stating clearly what is established and what is still under investigation. An examiner values that honesty highly at postgraduate level.

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NCISM reference numbers for this unit: **105, 106**

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UNIT 2
SPECIALISED CHEMISTRY FIELDS — AN OUTLINE

SYLLABUS POINTS

2.1 Crystallography — detoxification, transformation of compound
2.2 Organometallic chemistry — Chaturvidha Rasayana
2.3 Bioinorganic chemistry — Amritikarana (metal-carbon bond, stable compound formation)
2.4 Application in Shodhana, Marana, Amritikarana, Chaturvidha Rasayana

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe the principles of crystallography and an overview of organometallic and bioinorganic chemistry	1	Lecture	Know
Demonstrate the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures	5	Practical Training 3.2	Knows how

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

2.1 Crystallography

2.1.1 What crystallography is

Crystallography is the science that studies the arrangement of atoms in crystalline solids.

Term	Meaning
Crystalline solid	Atoms arranged in a regular, repeating three-dimensional pattern
Amorphous solid	No long-range order; glass is the standard example
Space lattice	The regular array of points representing the repeating pattern
Unit cell	The smallest repeating unit that builds the whole lattice
Lattice parameters	The edge lengths a, b, c and the angles alpha, beta, gamma
Polymorphism	The same substance existing in more than one crystal form
Isomorphism	Different substances having the same crystal form

2.1.2 The seven crystal systems

System	Axes	Angles
Cubic	a = b = c	all 90 degrees
Tetragonal	a = b, not equal to c	all 90 degrees
Orthorhombic	a, b, c all different	all 90 degrees
Hexagonal	a = b, not equal to c	two at 90, one at 120 degrees
Trigonal or rhombohedral	a = b = c	all equal, none 90 degrees
Monoclinic	a, b, c all different	two at 90, one not
Triclinic	a, b, c all different	none 90 degrees

From these seven systems arise the **fourteen Bravais lattices**.

2.1.3 X-ray diffraction and Bragg's law

X-ray diffraction is the principal method of crystallography, and the principal method of characterising a Bhasma.

When X-rays strike a crystal, the regularly spaced planes of atoms scatter them. At certain angles the scattered waves reinforce one another and a peak is recorded. The condition is **Bragg's law**:

$n\lambda = 2d \sin \theta$

where n is the order, λ the wavelength of the X-rays, d the spacing between the planes, and θ the angle of incidence.

Each crystalline compound gives a unique set of d-spacings. That set is a fingerprint. By matching the pattern of your sample against reference data, you identify the compound present. This is why Unit 2 of Module 2 taught you to obtain XRD reference data from a mineral database.

> **What XRD can and cannot tell you about a Bhasma.** It identifies the crystalline phases present, and it gives an estimate of crystallite size from peak broadening. It does not detect amorphous material, and it does not directly identify organic matter present in small amount. So XRD alone never gives the full picture of a Bhasma. It must be read alongside SEM-EDAX, FTIR and TGA. Module 5 deals with all of these.

2.1.4 Crystallography in detoxification and transformation

This is the syllabus wording, and it must be answered directly.

Transformation of compound. During Shodhana and Marana the starting material is converted into a different compound with a different crystal structure. Crystallography is how that transformation is demonstrated.

Stage	What crystallography shows
Raw Dravya	The starting mineral phase, matching the natural mineral
After Shodhana	Often the same principal phase, with soluble impurity phases removed
After Marana	New phases — typically oxides or sulphides — replacing the original
After repeated Puta	Progressive narrowing to a stable phase, with reducing crystallite size

Detoxification. In chemical terms, the toxicity of a metal compound depends on the form in which it is present, its solubility and its bioavailability — not on the element alone.

Consider Parada. Free mercury and soluble mercury salts are highly toxic. Mercuric sulphide, the compound of Hingula and of Rasa Sindura, is very poorly soluble in water and has a stable crystal structure. Crystallography allows you to demonstrate which form is actually present in the finished product.

> **State this carefully in the examination.** The correct statement is that Shodhana and Marana **convert the material into a different chemical form, with different solubility and different biological behaviour**. It is scientifically sound to say the form changes and to show it by XRD. It is not sound to claim that all toxicity is abolished. Ayurvedic products containing metals are still regulated, still require proof of safety under Rule 158-B, and still carry the requirement of correct preparation and correct dose. Say what the evidence shows, and no more.

2.2 Organometallic chemistry — Chaturvidha Rasayana

2.2.1 What organometallic chemistry is

Organometallic chemistry is the study of compounds containing at least one direct **metal-carbon bond**.

The definition is strict. A metal bound to oxygen or nitrogen of an organic molecule is a **coordination compound**, not an organometallic compound. The bond must be metal to carbon.

Term	Definition	Example
Organometallic compound	Contains a direct metal-carbon bond	Grignard reagent, ferrocene
Coordination compound	Metal bound to a donor atom such as N, O or S	Haemoglobin, chlorophyll
Ligand	The species donating electrons to the metal	Water, ammonia, chloride, organic acids
Chelate	A ligand binding through two or more atoms, forming a ring	EDTA, citrate, tannins

Well-known organometallic and related compounds include **ferrocene**, in which an iron atom sits between two five-membered carbon rings, and **vitamin B12**, which contains a cobalt-carbon bond and is one of very few naturally occurring organometallic compounds.

2.2.2 Chaturvidha Rasayana

Chaturvidha Rasayana means the four types of Rasa preparation. These are the four principal Kalpanas of Rasashastra.

Rasayana	Method of preparation	Character of the product
Kharaliya Rasayana	Prepared in the Khalva Yantra by trituration alone, without fire	Fine powder; example, Rasa Parpati group preparations of the simple kind, Kajjali-based Yogas
Parpati Rasayana	Kajjali is melted and poured on a surface, then pressed to form a thin flake	Thin, flake-like, brittle; example, Rasa Parpati, Tamra Parpati
Kupipakwa Rasayana	Prepared by heating in a Kachakupi, a glass bottle, in a Valuka Yantra, with a graded fire	Product collected from the neck or the bottom of the bottle; example, Rasa Sindura, Malla Sindura
Pottali Rasayana	Ingredients tied in a bundle and subjected to heat, often in a bath of sulphur or Abhraka	Dense, compact; example, Hemagarbha Pottali, Ratnagarbha Pottali

The chemistry of each, in outline

- **Kharaliya** — mechanical. Prolonged trituration reduces particle size, brings reactants into intimate contact, and forms Kajjali, in which mercury and sulphur combine at room temperature into black mercuric sulphide.
- **Parpati** — melting and rapid solidification. Rapid cooling gives a fine-grained, partly disordered solid rather than large crystals.
- **Kupipakwa** — controlled sublimation and condensation in a closed glass vessel. The graded fire allows the product to vaporise and redeposit in a purer, crystalline form on the neck of the bottle.
- **Pottali** — heating under restricted access to air, within a protective bundle, so that reaction proceeds without free oxidation.

2.2.3 The organometallic interpretation — stated honestly

The curriculum links organometallic chemistry to Chaturvidha Rasayana. This connection needs care, and a postgraduate student should understand the position exactly.

The proposal. Some researchers have suggested that during repeated Bhavana with herbal juices and repeated Puta, organic molecules from the herbs become bound to the metal, producing organometallic or metal-organic species. This is offered as an explanation for why herb-processed Bhasma behaves differently from a simple metal oxide.

What the analytical evidence generally shows. Published characterisation of Bhasma by XRD, SEM-EDAX, FTIR and TGA commonly demonstrates:

- Metal oxides and sulphides as the principal crystalline phases
- Particles in the micrometre to nanometre range
- The presence of organic material, shown by FTIR bands and by weight loss on heating
- Elements such as carbon, and sometimes other elements derived from the herbal media

Where the honest boundary lies. The presence of organic matter associated with the particles is well documented. Whether that organic matter is bound through true **metal-carbon covalent bonds**, in the strict organometallic sense, or is adsorbed and coordinated at the particle surface, is **not settled**. Most of the evidence is consistent with surface association and coordination rather than with classical organometallic bonding.

> **How to write this in an examination.** Present the classical process fully. Present the organometallic proposal as a proposal, naming it as such. Present what the analytical evidence shows. Then state that the exact nature of the metal-organic linkage is an active area of research. That answer will score higher than one which asserts more than the evidence supports, because it shows you can distinguish a hypothesis from a finding. That distinction is precisely what postgraduate study is for.

2.3 Bioinorganic chemistry — Amritikarana

2.3.1 What bioinorganic chemistry is

Bioinorganic chemistry is the study of the role of metals and other inorganic elements in living systems.

It is the natural scientific home of Rasashastra. Rasashastra administers metals and minerals to human beings. Bioinorganic chemistry studies how living systems handle metals. The two subjects ask the same question from opposite directions.

2.3.2 Metals in the body

Category	Elements	Note
Bulk elements	Na, K, Ca, Mg, Cl, P, S	Required in gram quantities
Essential trace elements	Fe, Zn, Cu, Mn, Co, Mo, Se, Cr, I	Required in milligram or microgram quantities
Non-essential and toxic	Pb, Hg, Cd, As	No known physiological requirement

Metalloproteins and metalloenzymes. Many proteins need a metal to function.

Metalloprotein	Metal	Function
Haemoglobin and myoglobin	Iron	Oxygen transport and storage
Cytochromes	Iron	Electron transport
Carbonic anhydrase	Zinc	Carbon dioxide handling
Superoxide dismutase	Copper and zinc, or manganese	Antioxidant defence
Ceruloplasmin	Copper	Copper transport, ferroxidase activity
Vitamin B12	Cobalt	Contains a cobalt-carbon bond
Chlorophyll	Magnesium	Photosynthesis in plants

> This table gives you a strong opening line for any answer on Bhasma and the body. Metals are not foreign to the body. Iron, zinc and copper are essential, are handled by specific transport proteins, and are regulated within narrow limits. The question in Rasashastra is not whether the body can handle a metal, but **in what form, in what quantity, and how it is absorbed, distributed and excreted**.

2.3.3 Chelation

Chelation is the binding of a metal ion by a ligand at two or more points, forming a ring.

Chelates are far more stable than complexes formed by single-point binding. This is called the chelate effect.

Chelating agent	Source	Note
EDTA	Synthetic	Used in analysis and in metal poisoning
Citric acid	Nimbu, Amlavetasa	Present in many Shodhana media
Tannins and gallic acid	Triphala, Khadira	Strong metal binders
Amino acids and proteins	Godugdha, Kulattha	Bind metals through N and O
Phytic acid	Grains	Binds iron and zinc

Why this matters directly. Every classical Shodhana medium is rich in chelating molecules. Kanji, Nimbu Swarasa, Triphala Kwatha, Godugdha and Gomutra all contain organic acids, phenolics or proteins that bind metal ions. This is one of the strongest and best-supported chemical explanations for the classical choice of media, and it belongs in your answers.

2.3.4 Amritikarana

Amritikarana, literally “making it like Amrita”, is a Samskara performed after Marana. Its stated purpose in the classical texts is to remove any residual harmful quality of the Bhasma and to make it wholly wholesome and fit for administration.

It is generally carried out by treating the prepared Bhasma with prescribed media — commonly Ghrita, Godugdha, herbal juices or decoctions — with mild heat, as directed for the particular Bhasma. The exact procedure varies with the text and with the Dravya, so always cite the text you are following.

The chemical reading of Amritikarana

Classical purpose	Possible chemical basis
Removal of residual harmful quality	Removal or conversion of unreacted free metal and reactive intermediates left after Marana
Making the Bhasma “wholesome”	Binding of reactive surface sites by organic molecules, reducing surface reactivity
Improving the Bhasma’s acceptability and action	Surface association with lipids or proteins, possibly affecting dispersion and absorption
Final stabilisation	A more stable surface, less prone to further change on storage

The curriculum’s phrase “metal-carbon bond, stable compound formation” refers to this proposed surface stabilisation. Apply the same honesty here as in section 2.2.3 — describe the classical procedure fully, give the chemical interpretation as an interpretation, and mark clearly where the evidence ends.

2.4 Application in Shodhana, Marana, Amritikarana and Chaturvidha Rasayana

This section brings the three fields together. Learn this integration table. It answers section 2.4 directly.

Process	Crystallography contributes	Organometallic and coordination chemistry contributes	Bioinorganic chemistry contributes
Shodhana	Shows the phase before and after; confirms removal of impurity phases	Explains chelation of impurities by the organic acids in the media	Explains why the media chosen are biologically compatible
Marana	Identifies the new phases formed; measures crystallite size	Explains binding of organic matter from Bhavana at the particle surface	Relates the product to the metal’s handling in the body
Amritikarana	Confirms that the phase is unchanged and stable	Explains surface stabilisation by organic molecules	Explains reduction of reactive free metal
Chaturvidha Rasayana	Distinguishes the products of the four methods by phase and crystallinity	Explains the role of organic media in each method	Provides the framework for studying absorption and action

KEY POINTS TO REMEMBER

- Seven crystal systems; fourteen Bravais lattices.
- Bragg’s law: $n\lambda = 2d \sin\theta$. Each crystalline compound has a unique set of d-spacings.
- XRD identifies crystalline phases and estimates crystallite size. It does not detect amorphous material or small quantities of organic matter.
- Organometallic means a direct metal-carbon bond. Metal bound to O or N is a coordination compound, not organometallic.

- Chaturvidha Rasayana: Kharaliya, Parpati, Kupipakwa and Pottali.
- Kajjali is black mercuric sulphide formed by trituration of Parada with Gandhaka, without fire.
- Vitamin B12 contains a cobalt-carbon bond and is a rare natural organometallic compound.
- Chelation is multi-point binding; chelates are more stable than single-point complexes.
- Classical Shodhana media are rich in chelating molecules — citric acid, tannins, amino acids.
- Amritikarana is a post-Marana Samskara to remove residual harmful quality and stabilise the Bhasma.
- The organic matter associated with Bhasma particles is documented. Whether it is bound by true metal-carbon bonds is not settled.

EXAM CORNER — UNIT 2

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Define crystallography. Enumerate the seven crystal systems.
2. State Bragg's law and explain its application in the analysis of Bhasma.
3. Differentiate between crystalline and amorphous solids.
4. Write a short note on polymorphism.
5. Define organometallic chemistry. Differentiate an organometallic compound from a coordination compound.
6. Enumerate the Chaturvidha Rasayana with one example of each.
7. Write a short note on Kupipakwa Rasayana and its chemistry.
8. Write a short note on Kajjali.
9. Define bioinorganic chemistry. Enumerate the essential trace elements.
10. Write a short note on metalloenzymes with examples.
11. Define chelation. Give the chelating agents present in classical Shodhana media.
12. Write a short note on Amritikarana and its chemical basis.

Likely analytical based questions (20 marks)

1. You are asked to prove, by analysis, that Shodhana and Marana have transformed a Rasa Dravya into a different chemical form. Analyse which instrumental methods you would use, what each would show, and what conclusion each would and would not permit you to draw.
2. Analyse the four Rasayana Kalpanas in terms of their physical and chemical processes. Compare them on the basis of the temperature used, the vessel, the nature of the product, and the type of transformation involved.
3. The classical texts prescribe specific media for Shodhana and for Bhavana. Analyse this instruction in the light of coordination and bioinorganic chemistry, and design an experiment to test whether the choice of medium changes the finished Bhasma.

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NCISM reference numbers for this unit: **107, 108, 109, 110, 111**

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UNIT 3
ORGANIC CHEMISTRY

SYLLABUS POINTS

3.1 Structure of organic molecules: understanding of herbal components
3.2 Interaction of herbs and animal products with metals for detoxification, compound formation and compound stability
3.3 Functional groups: reactivity and properties of organic molecules influencing compound action

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe the structure of organic molecules	1	Lecture	Know
Discuss the interaction of herbs and animal products with metals	1	Lecture	Knows how
Explore and identify different functional groups and discuss their influence on molecular interactions and therapeutic efficacy	6	Experiential Learning 3.2	Knows how

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

3.1 Structure of organic molecules

3.1.1 Why carbon is special

Property of carbon	Consequence
Tetravalency — four bonds	Every carbon can link to four other atoms
Catenation — carbon bonds to carbon	Chains and rings of any length can form
Forms single, double and triple bonds	Enormous structural variety
Small size, strong bonds	The compounds formed are stable
Bonds readily to H, O, N, S, P and halogens	Allows the functional groups of biology

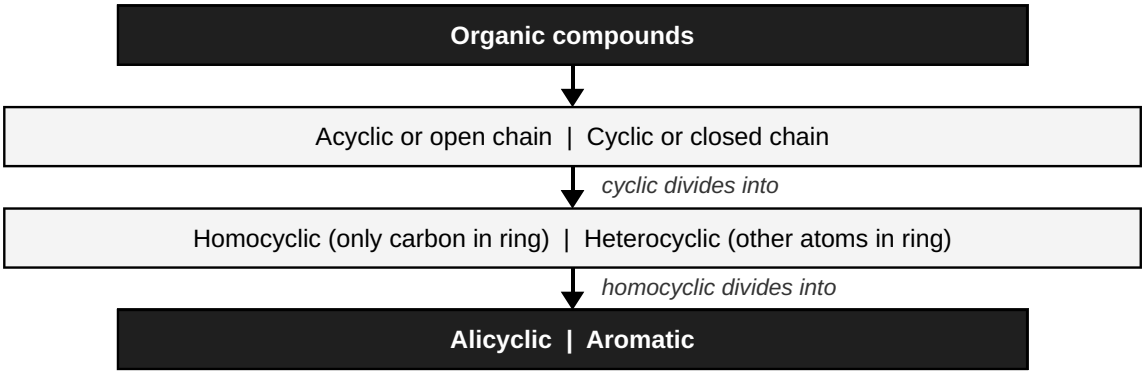
This is why the whole of plant chemistry, and therefore the whole of Bhaishajya Kalpana, is organic chemistry.

3.1.2 Hybridisation and shape

Hybridisation	Bonds	Shape	Angle	Example
sp^3	Four single bonds	Tetrahedral	109.5 degrees	Methane, alkanes
sp^2	One double, two single	Trigonal planar	120 degrees	Alkenes, benzene
sp	One triple or two doubles	Linear	180 degrees	Alkynes

Shape is not a detail. A molecule acts on a receptor by fitting it. Shape decides fit, and fit decides action.

3.1.3 Classification of organic compounds



Heterocyclic compounds deserve attention. A ring containing nitrogen, oxygen or sulphur is heterocyclic. Almost every alkaloid is a nitrogen heterocycle. Pyridine, pyrrole, indole, quinoline and imidazole rings appear again and again in medicinal plants.

3.1.4 The main classes of herbal components

This is the bridge to Unit 4, where these are treated in full.

Class	Chemical nature	Example
Carbohydrates	Polyhydroxy aldehydes and ketones	Starch, gums, mucilage
Proteins and amino acids	Polymers of amino acids	Enzymes, seed proteins
Lipids	Esters of fatty acids with glycerol	Fixed oils, Taila
Alkaloids	Nitrogenous, mostly heterocyclic, basic	Berberine, reserpine
Glycosides	Sugar joined to a non-sugar aglycone	Glycyrrhizin, sennosides
Phenolics and flavonoids	Aromatic rings with hydroxyl groups	Quercetin, gallic acid
Tannins	High molecular weight polyphenols	Triphala tannins
Terpenoids	Built from isoprene units	Menthol, camphor
Steroids	Four fused rings	Diosgenin, phytosterols
Volatile oils	Mixtures of low-boiling terpenoids	Ela, Lavanga oil
Resins	Complex amorphous mixtures	Guggulu

3.2 Interaction of herbs and animal products with metals

This is the heart of Unit 3 for a Rasashastra student, and the most likely source of an ABQ.

3.2.1 The question

Classical texts never say to purify a metal by heating it in water. They always prescribe a specific plant juice, decoction, or animal product. Why?

Chemistry gives four answers, and they operate together.

3.2.2 The four mechanisms

1. Chelation and complexation. Organic acids, phenolics, tannins and amino acids in the media bind metal ions at two or more points. This removes unwanted soluble metal ions, holds them in a stable form, and modifies the surface.

2. Reduction and oxidation. Many plant constituents are reducing agents. Phenolics, flavonoids, ascorbic acid and reducing sugars can donate electrons and change the oxidation state of a metal. This is the same chemistry used today to make metal nanoparticles by so-called green synthesis, using plant extracts as the reducing agent.

3. Surface capping and size control. When particles form in the presence of large organic molecules, those molecules attach to the growing particle surface. They limit further growth and prevent the small particles from clumping together. This is called capping. It is a well-established mechanism in nanoparticle chemistry, and it explains, in modern terms, why Bhavana with herbal juice before each Puta yields a finer product than dry heating.

4. Acid and alkali action. Acidic media such as Kanji, Amlavetasa, Nimbu Swarasa and Takra dissolve surface oxides and carbonates. Alkaline media such as Gomutra and Kshara act on other impurities. The medium creates the chemical environment in which the impurity, and not the wanted material, is attacked.

3.2.3 The media and what they contain

Medium	Principal chemical constituents	Probable role
Kanji	Lactic acid, acetic acid	Acidic; dissolves surface oxides; mild chelation
Nimbu Swarasa	Citric acid, ascorbic acid	Chelation and reduction
Amlavetasa, Chincha	Tartaric acid, citric acid	Chelation
Takra	Lactic acid, protein, fat	Acidic and protein binding
Triphala Kwatha	Tannins, gallic acid, chebulic acid	Strong chelation; reduction
Kumari Swarasa	Polysaccharides, anthraquinones	Capping; viscous medium for grinding
Nirgundi, Chitraka, Bhringaraja Swarasa	Flavonoids, phenolics, alkaloids	Reduction and capping
Kulattha Kwatha	Protein, amino acids	Chelation through N and O donors
Gomutra	Urea, ammonia, mineral salts, uric acid	Mildly alkaline; nitrogen donors
Godugdha	Casein, lactose, calcium, fat	Protein binding; buffering; capping
Ghrita	Triglycerides, fatty acids	Lipid coating; heat transfer medium
Madhu	Sugars, organic acids, enzymes	Reducing sugars; viscous medium
Ajamutra, other Mutra	Urea and related nitrogen compounds	Nitrogen donors, alkaline

> **The single strongest correlation in this whole module.** Modern green synthesis of metal nanoparticles uses a plant extract to reduce a metal salt and simultaneously to cap the particles formed. Classical Bhavana uses a plant juice, ground repeatedly with a metal or its compound, before firing. The reagents and the mechanism are the same. This is a well-supported point and it belongs in every ABQ answer on this unit.

3.2.4 Detoxification, compound formation and compound stability

The syllabus names these three outcomes. Answer them separately.

Outcome named in the syllabus	Chemical account
Detoxification	Conversion of soluble, reactive and readily absorbed forms into insoluble, stable forms; chelation and removal of free metal ions; removal of surface impurities
Compound formation	Formation of new phases — oxides, sulphides — and formation of metal-organic associations at the surface during Bhavana
Compound stability	Organic capping reduces surface reactivity and prevents re-aggregation; the crystalline phase formed is thermodynamically stable; the product resists change on storage

3.3 Functional groups

3.3.1 What a functional group is

A **functional group** is an atom or group of atoms in a molecule that determines its characteristic chemical reactions.

The carbon skeleton carries the molecule. The functional group does the work. Two molecules with the same skeleton but different functional groups behave completely differently.

3.3.2 The functional groups you must know

Functional group	Structure	Class of compound	Occurrence in herbal drugs
Hydroxyl	—OH	Alcohol, phenol	Sugars, phenolics, flavonoids, menthol
Aldehyde	—CHO	Aldehyde	Cinnamaldehyde in Twak; reducing sugars
Ketone	>C=O	Ketone	Camphor, curcumin
Carboxyl	—COOH	Carboxylic acid	Citric, tartaric, gallic acid; fatty acids

Functional group	Structure	Class of compound	Occurrence in herbal drugs
Ester	—COO—	Ester	Fixed oils, waxes, many volatile oil components
Ether	—O—	Ether	Eugenol methyl ether, many terpenoids
Amino	—NH_2	Amine	Amino acids, alkaloids
Amide	—$CONH_2$	Amide	Proteins, piperine
Thiol	—SH	Thiol	Cysteine, allicin precursors in Lashuna
Nitro	—NO_2	Nitro compound	Rare in plants
Glycosidic linkage	—O— joining sugar to aglycone	Glycoside	Sennosides, glycyrrhizin
Methoxy	—OCH_3	Methoxy compound	Curcumin, eugenol

3.3.3 Functional groups that bind metals

For your subject, this is the important sub-table. A group binds a metal if it carries a lone pair on oxygen, nitrogen or sulphur.

Group	Donor atom	Strength of metal binding	Example
Carboxyl	O	Strong, especially when two are present	Citric acid, tartaric acid
Phenolic hydroxyl	O	Strong, especially adjacent pairs	Tannins, gallic acid, catechol
Hydroxyl	O	Moderate	Sugars, polyols
Amino	N	Strong	Amino acids, proteins
Thiol	S	Very strong with soft metals such as mercury and lead	Cysteine, glutathione
Carbonyl	O	Moderate	Ketones, aldehydes

> **The thiol group deserves special attention.** Sulphur binds mercury and lead very strongly. This is why Gandhaka is the classical partner of Parada, why Kajjali forms on simple trituration, and why sulphur-containing compounds are used in modern treatment of heavy metal poisoning. If a question asks you to explain one classical practice in chemical terms, the Parada-Gandhaka combination is the clearest example available to you.

3.3.4 How functional groups influence action

1. **Solubility.** Polar groups such as —OH and —COOH increase water solubility. Long carbon chains decrease it. This decides which solvent will extract a constituent, and how it will be absorbed.
2. **Acidity and basicity.** —COOH makes a molecule acidic; —NH_2 makes it basic. This decides its ionisation at a given pH, and therefore its absorption. Recall the pH-partition hypothesis from Module 1, Unit 5.
3. **Reactivity and stability.** Some groups oxidise readily, and the compound degrades on storage. Phenolics are the common example.
4. **Receptor binding.** Hydrogen bonding through —OH, —NH and —C=O is how most drug molecules attach to their targets.
5. **Metal binding.** As set out above.
6. **Taste and odour.** Esters give pleasant odours; many alkaloids are bitter; tannins are astringent. The classical Rasa of a drug often reflects its chemical class.

KEY POINTS TO REMEMBER

- Carbon's tetravalency and catenation make organic chemistry possible.
- sp^3 is tetrahedral at 109.5 degrees, sp^2 is trigonal planar at 120 degrees, sp is linear at 180 degrees.
- Most alkaloids are nitrogen heterocycles.
- Four mechanisms by which herbs act on metals: chelation and complexation, reduction, surface capping, and acid or alkali action.
- Classical Bhavana parallels modern green synthesis of nanoparticles — the plant extract reduces and caps.
- Groups that bind metals carry a lone pair on O, N or S: carboxyl, phenolic hydroxyl, amino and thiol.
- Thiol binds mercury and lead very strongly, which explains the Parada-Gandhaka combination.
- Functional groups decide solubility, ionisation, stability, receptor binding, metal binding, taste and odour.

EXAM CORNER — UNIT 3

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Why is carbon able to form so many compounds? Explain catenation and tetravalency.
2. Write a short note on hybridisation, with the shape and bond angle of each type.
3. Classify organic compounds with a chart.
4. Write a short note on heterocyclic compounds and their occurrence in medicinal plants.
5. Define a functional group. Enumerate the important functional groups with one example of each.
6. Enumerate the functional groups capable of binding metals, with the donor atom in each.
7. Write a short note on chelation by herbal constituents.

8. Explain the role of Bhavana Dravya in Marana on chemical grounds.
9. Write a short note on the chemical constituents of Kanji, Gomutra and Godugdha, and their role in Shodhana.
10. Explain, chemically, why Gandhaka is used along with Parada.
11. How do functional groups influence the solubility and absorption of a drug?
12. Write a short note on green synthesis of nanoparticles and its parallel with Bhavana.

Likely analytical based questions (20 marks)

1. Analyse the classical instruction that Marana must be carried out with repeated Bhavana using specified herbal juices. Explain the chemistry involved, compare it with modern green synthesis of nanoparticles, and design a study to test whether the herbal medium affects the finished Bhasma.
2. A student proposes to replace the classical Shodhana medium of a metal with distilled water, to save cost. Analyse this proposal chemically. Explain what would be lost, predict what the product would be like, and justify your conclusion.
3. Analyse the interaction of herbal and animal products with metals under the three headings named in the syllabus — detoxification, compound formation, and compound stability. Support each with the chemistry involved and with examples from classical practice.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **112, 113, 114**

1. **112.** Ouellette RJ, Rawn JD. Principles of Organic Chemistry. Elsevier; 2015.
2. **113.** Walker M. Organic Chemistry 1: An Open Textbook. State University of New York; 2018.
3. **114.** Ouellette RJ, Rawn JD. Organic Chemistry: Structure, Mechanism, Synthesis. Academic Press; 2018.

UNIT 4
PHYTOCHEMISTRY

SYLLABUS POINTS

4.1 Definition and significance of phytochemistry
4.2 Overview of plant-based bioactive compounds
4.3 Classes of phytochemicals
4.4 Extraction methods
4.5 Phytochemical screening

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe the significance of phytochemistry	1	Lecture	Knows how
Describe the overview of plant-based bioactive components	1	Lecture	Knows how
Discuss different types of extraction methods including solvent selection	1	Lecture	Knows how
Demonstrate different types of extraction methods including solvent selection	4	Practical Training 3.3	Shows how
Demonstrate qualitative identification of phytoconstituents in single herbs and herbal formulations	3	Practical Training 3.4	Shows how
Extraction techniques in herbal formulations, a hands-on approach	3	Experiential Learning 3.3	Does
Identify phytoconstituents in herbs and herbal formulations	3	Experiential Learning 3.4	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	Yes

4.1 Definition and significance of phytochemistry

4.1.1 Definition

Phytochemistry is the branch of chemistry that deals with the chemical constituents of plants — their isolation, structure, biosynthesis, distribution, and biological function.

The word comes from *phyton*, meaning plant.

4.1.2 Significance for Ayurveda

Purpose	How phytochemistry serves it
Authentication of a raw drug	Chemical profile confirms identity where morphology fails, as with powders
Detection of adulteration and substitution	A missing or extra constituent reveals the substitution
Standardisation	Fixing a marker compound and its permitted limits
Quality control	Assay of the active constituent in every batch
Understanding the mechanism of action	Identifying which molecule produces which effect
Explaining classical concepts	Relating Rasa, Guna, Virya and Vipaka to chemical classes
Drug development	Isolating a lead compound for further work
Deciding the process	Knowing whether a constituent is heat-labile or water-soluble decides the Kalpana
Fixing shelf life	Following the decline of a marker over time

> **The single most practical use.** A **marker compound** is a constituent chosen to represent the drug in analysis. It may or may not be the active principle. The Ayurvedic Pharmacopoeia of India fixes markers and their limits for many drugs. Without a marker you cannot standardise, and without standardisation you cannot make the same product twice.

4.1.3 A caution worth stating

Ayurveda gives a whole drug, and usually a combination of drugs. Phytochemistry, by its nature, isolates single molecules.

Both approaches are valid, but they answer different questions. Isolating curcumin from Haridra tells you what curcumin does. It does not tell you what Haridra does, because Haridra contains volatile oil, other curcuminoids, polysaccharides and much else, acting together.

A postgraduate answer should acknowledge this. Use phytochemistry as a tool for standardisation and for understanding, without reducing a classical drug to one molecule.

4.2 Overview of plant-based bioactive compounds

4.2.1 Primary and secondary metabolites

Feature	Primary metabolites	Secondary metabolites
Role in the plant	Essential for growth, development and reproduction	Not essential for basic life; used for defence, attraction, protection
Distribution	Present in all plants	Restricted to certain families, genera or species
Quantity	Large	Usually small
Examples	Carbohydrates, proteins, lipids, nucleic acids, chlorophyll	Alkaloids, glycosides, flavonoids, tannins, terpenoids, volatile oils, resins
Medicinal importance	Nutritional, some therapeutic	Most therapeutic activity lies here

The point to remember. Most of the therapeutic action of a medicinal plant lies in its **secondary metabolites**. This is also why the same species grown in different soil, season and altitude can differ in potency — secondary metabolite production responds to stress and environment. That is the chemical basis of the classical rules regarding Desha and Kala of collection.

4.2.2 Factors affecting the phytochemical content

Factor	Effect
Species and variety	Different chemotypes produce different constituents
Part of the plant	Root, stem, leaf, flower, fruit and seed differ greatly
Age and maturity	Content rises and falls with the stage of growth
Season of collection	The classical Kala of Sangrahana
Geography, soil and altitude	The classical Desha
Method of drying	Heat and light destroy volatile and heat-labile constituents
Method of storage	Moisture, light and time cause degradation
Method of processing	Shodhana, Bhavana and Paka can alter or remove constituents

4.3 Classes of phytochemicals

This section is the most likely source of a ten-mark question in this unit. Learn each class with its chemical nature, its test and its example.

4.3.1 Alkaloids

- **Nature.** Nitrogenous compounds, mostly heterocyclic, basic in reaction, bitter in taste, usually optically active.
- **Occurrence.** Chiefly in dicotyledons — Apocynaceae, Solanaceae, Papaveraceae, Rutaceae, Berberidaceae.
- **Solubility.** The free base dissolves in organic solvents; the salt dissolves in water. This is the basis of the classical acid-base extraction.
- **Examples.** Berberine in Daruharidra, reserpine in Sarpagandha, piperine in Maricha and Pippali, vasicine in Vasa, atropine in Dhattura.
- **Tests.** Dragendorff’s, Mayer’s, Wagner’s and Hager’s reagents.

4.3.2 Glycosides

- **Nature.** A sugar portion, the glycone, joined to a non-sugar portion, the aglycone or genin, through a glycosidic linkage. On hydrolysis the two separate.
- **Classification by aglycone.**

Type	Example	Source
Anthraquinone glycoside	Sennosides	Sanaya
Cardiac glycoside	Digitoxin	Digitalis
Saponin glycoside	Glycyrrhizin	Yashtimadhu
Cyanogenetic glycoside	Amygdalin	Bitter almond
Isothiocyanate glycoside	Sinigrin	Sarshapa
Flavonoid glycoside	Rutin	Many plants
Bitter glycoside	Picroside	Katuki

- **Tests.** Borntrager’s test for anthraquinone glycosides; Keller-Killiani and Legal’s tests for cardiac glycosides.

4.3.3 Flavonoids and phenolics

- **Nature.** Aromatic compounds carrying hydroxyl groups. Flavonoids have a fifteen-carbon skeleton of two aromatic rings joined by a three-carbon bridge.
- **Action.** Antioxidant, anti-inflammatory, capillary protective; strong metal binders.
- **Examples.** Quercetin, rutin, hesperidin, catechin.
- **Tests.** Shinoda test with magnesium and hydrochloric acid; alkaline reagent test; lead acetate test.

4.3.4 Tannins

- **Nature.** High molecular weight polyphenols. Astringent in taste. Precipitate proteins.
- **Types.** Hydrolysable tannins, such as gallotannins; and condensed tannins, the proanthocyanidins.
- **Examples.** Triphala, Khadira, Arjuna, Ashoka.
- **Relevance to Rasashastra.** Tannins are among the strongest metal-binding constituents in the classical media. This is why Triphala Kwatha appears so often in Shodhana and Bhavana.
- **Tests.** Ferric chloride gives blue-black or green-black; gelatin test gives a precipitate; lead acetate gives a precipitate.

4.3.5 Terpenoids and volatile oils

- **Nature.** Built from five-carbon isoprene units. Classified by the number of units — monoterpene has two, sesquiterpene three, diterpene four, triterpene six.
- **Volatile oils** are mixtures of low-boiling terpenoids, responsible for the odour of aromatic drugs.
- **Examples.** Menthol in Pudina, camphor in Karpura, eugenol in Lavanga, curcumin-associated oils in Haridra, andrographolide in Kalmegha.
- **Relevance.** Volatile oils are lost on prolonged heating. This is why an aromatic drug is added as Prakshepaka Dravya at the end of a preparation, and not boiled with it. The classical instruction and the chemistry agree exactly.
- **Extraction.** Steam distillation or hydrodistillation, usually in a Clevenger apparatus.

4.3.6 Steroids and saponins

- **Steroids.** Four fused rings. Examples are diosgenin in Shatavari and other Dioscorea species, and phytosterols such as beta-sitosterol.
- **Saponins.** Glycosides that froth in water and haemolyse red blood cells. They lower surface tension and act as natural surfactants.
- **Examples.** Glycyrrhizin in Yashtimadhu, saponins in Shatavari and Shikakai.
- **Tests.** Foam test; haemolysis test; Salkowski and Liebermann-Burchard tests for steroids and triterpenoids.

4.3.7 Resins, gums and mucilage

Class	Nature	Example
Resins	Complex amorphous mixtures of resin acids, alcohols and esters; insoluble in water, soluble in alcohol	Guggulu, Shallaki
Oleo-gum-resins	Resin with gum and volatile oil	Guggulu, Hingu
Gums	Polysaccharides that swell in water	Babbula gum, tragacanth
Mucilage	Polysaccharides forming a slimy solution	Isabgol, Kumari

4.3.8 Summary table

Class	Chemical nature	Confirmatory test	Ayurvedic example
Alkaloids	Basic nitrogenous heterocycles	Dragendorff’s, Mayer’s	Sarpagandha, Daruharidra
Glycosides	Sugar plus aglycone	Borntrager’s, Keller-Killiani	Sanaya, Katuki
Flavonoids	Polyphenolic, C6-C3-C6	Shinoda test	Amalaki, Nagakeshara
Tannins	High molecular weight polyphenols	Ferric chloride, gelatin	Triphala, Khadira
Terpenoids	Isoprene units	Salkowski	Karpura, Lavanga
Steroids	Four fused rings	Liebermann-Burchard	Shatavari
Saponins	Glycosides, surface active	Foam test	Yashtimadhu, Shikakai
Carbohydrates	Polyhydroxy aldehyde or ketone	Molisch, Fehling’s	Gums, mucilage
Proteins	Amino acid polymers	Biuret, Ninhydrin	Seeds, Kulattha
Fixed oils	Glycerides of fatty acids	Spot test, saponification	Tila Taila, Eranda Taila

4.4 Extraction methods

4.4.1 What extraction is

Extraction is the process of separating the medicinally active portion of a plant from its inactive or inert material, using a suitable solvent.

The three requirements are a properly prepared drug, a correctly chosen solvent, and a suitable method.

4.4.2 Choice of solvent

Solvents are chosen on the principle that **like dissolves like**. Learn the polarity series in order.

Solvent	Polarity	Extracts chiefly
Water	Highest	Sugars, gums, mucilage, tannins, glycosides, some alkaloid salts
Methanol	Very high	Almost everything; used for total extract
Ethanol	High	Alkaloids, glycosides, flavonoids, tannins, resins
Acetone	Moderate to high	Flavonoids, some glycosides
Ethyl acetate	Moderate	Flavonoid aglycones, moderately polar constituents
Chloroform	Low to moderate	Alkaloid bases, terpenoids, steroids
Diethyl ether	Low	Fats, volatile oils, alkaloid bases
Petroleum ether and hexane	Lowest	Fixed oils, waxes, sterols, chlorophyll

Successive solvent extraction uses these in order, from least polar to most polar, so that each fraction contains a defined group of constituents. This is the standard approach in a research study.

Ideal properties of a solvent. It should dissolve the desired constituent selectively; be non-toxic; be non-reactive; be easily removed; be cheap and readily available; and be safe to handle.

> **The classical solvents were chosen on the same principle.** Jala extracts water-soluble constituents in Kwatha. Ghrita and Taila extract lipid-soluble constituents in Sneha Kalpana. Madya, containing alcohol, extracts a wider range in Asava and Arishta. The classical texts arrived at a polarity-based solvent system without the vocabulary of polarity.

4.4.3 Methods of extraction

A. Cold methods

Method	Description	Suitable for
Maceration	Drug soaked in solvent in a closed vessel with occasional shaking, then strained	Heat-labile constituents; classical Hima
Percolation	Solvent passed slowly and continuously through a column of drug	More efficient than maceration; used for tinctures
Expression	Mechanical pressing	Fresh juice, Swarasa; fixed oils
Infusion, cold	Drug steeped briefly in cold water	Classical Hima

B. Hot methods

Method	Description	Suitable for
Infusion, hot	Drug steeped in hot water for a short time	Classical Phanta
Decoction	Drug boiled with water and reduced	Classical Kwatha; heat-stable, water-soluble constituents
Digestion	Maceration with gentle heat	When mild heat improves yield
Reflux extraction	Solvent boiled with the drug and condensed back	Efficient; solvent is conserved
Soxhlet extraction	Continuous hot percolation with fresh condensed solvent	Complete extraction with a small volume of solvent

C. Distillation methods

Method	Description
Hydrodistillation	Drug boiled with water; vapour condensed and oil separated
Steam distillation	Steam passed through the drug; used for heat-sensitive volatile oils
Water and steam distillation	Combination of the two

Volatile oil content is measured in a **Clevenger apparatus**.

D. Modern methods

Method	Principle	Advantage
Supercritical fluid extraction	Carbon dioxide above its critical point acts as solvent	No solvent residue; low temperature; excellent for volatile oils
Ultrasound-assisted extraction	Cavitation disrupts cell walls	Faster, less solvent, lower temperature
Microwave-assisted extraction	Microwaves heat the cell contents directly	Very fast; high yield
Accelerated solvent extraction	Solvent at raised temperature and pressure	Fast, automated, less solvent
Enzyme-assisted extraction	Enzymes break the cell wall	Improves yield of bound constituents
Counter-current extraction	Drug and solvent move in opposite directions	Continuous, efficient, industrial scale

4.4.4 Types of extract

Extract	Description
Infusion and decoction	Dilute, aqueous, prepared fresh
Tincture	Alcoholic or hydro-alcoholic; typically dilute
Fluid extract	Concentrated, usually one millilitre representing one gram of drug
Soft extract	Semi-solid, partially evaporated
Dry extract	Fully evaporated to a solid; Ghana in classical terms
Standardised extract	Adjusted to a fixed content of a marker

4.5 Phytochemical screening

4.5.1 What it is

Preliminary phytochemical screening is a set of simple qualitative chemical tests carried out on an extract to detect which classes of constituent are present.

It tells you the class, not the individual compound. It is a first step, followed by chromatography and spectroscopy.

4.5.2 The standard tests

Constituent	Test	Reagent	Positive result
Alkaloids	Dragendorff’s	Potassium bismuth iodide	Orange-red precipitate
	Mayer’s	Potassium mercuric iodide	Cream or white precipitate
	Wagner’s	Iodine in potassium iodide	Reddish brown precipitate
	Hager’s	Saturated picric acid	Yellow precipitate
Anthraquinone glycosides	Borntrager’s	Ammonia after hydrolysis	Pink to red colour in the ammoniacal layer
Cardiac glycosides	Keller-Killiani	Glacial acetic acid, ferric chloride, sulphuric acid	Reddish brown ring at the junction
	Legal’s	Sodium nitroprusside in pyridine and alkali	Pink to red colour
Saponins	Foam test	Shake with water	Persistent froth for over fifteen minutes
	Haemolysis	Blood agar	Zone of haemolysis
Flavonoids	Shinoda	Magnesium turnings and concentrated hydrochloric acid	Pink, crimson or magenta

Constituent	Test	Reagent	Positive result
	Alkaline reagent	Sodium hydroxide, then acid	Yellow, disappearing on adding acid
Tannins and phenolics	Ferric chloride	Ferric chloride solution	Blue-black or green-black
	Gelatin	One per cent gelatin with sodium chloride	White precipitate
Steroids	Salkowski	Chloroform and concentrated sulphuric acid	Red colour in the lower layer
	Liebermann-Burchard	Acetic anhydride and sulphuric acid	Green to blue colour
Triterpenoids	Salkowski	As above	Golden yellow lower layer
Carbohydrates	Molisch	Alpha-naphthol and sulphuric acid	Violet ring at the junction
Reducing sugars	Fehling’s	Fehling A and B, heated	Brick red precipitate
	Benedict’s	Benedict’s reagent, heated	Red, yellow or green precipitate
Proteins	Biuret	Copper sulphate in alkali	Violet or pink colour
Amino acids	Ninhydrin	Ninhydrin solution, heated	Purple or blue colour
Fixed oils and fats	Spot test	Press between filter paper	Permanent translucent stain
Resins	Turbidity test	Acetone extract poured into water	Turbidity

4.5.3 Screening a herbal formulation

A single herb is straightforward. A formulation is harder, and the curriculum asks for both.

Difficulty	How to handle it
Many drugs contribute the same class	Screening tells you the class is present, not which drug gave it
Excipients interfere	Sugar, jaggery and honey give a false positive for carbohydrate
Metals in the formulation interfere	Metal ions can react with the reagents; note this in the record
Processing may have destroyed a constituent	Heat-labile constituents may be absent even though the drug was used

The correct approach is to screen the formulation, screen each ingredient separately, and then compare. Follow this with TLC or HPTLC, which separates before it detects, and which is dealt with in Module 5.

KEY POINTS TO REMEMBER

- Phytochemistry studies the chemical constituents of plants.
- Most therapeutic activity lies in secondary metabolites, which vary with Desha and Kala.
- A marker compound represents the drug in analysis; it need not be the active principle.
- Alkaloids are basic and nitrogenous; glycosides give a sugar and an aglycone on hydrolysis.
- Tannins are the strongest metal binders among the common classes, which is why Triphala is so often used in Shodhana.
- Volatile oils are lost on prolonged heating, which is why aromatics are added as Prakshepaka at the end.
- Solvent polarity order: water, methanol, ethanol, acetone, ethyl acetate, chloroform, ether, petroleum ether.
- Soxhlet is continuous hot percolation. Clevenger measures volatile oil.
- Key tests: Dragendorff's and Mayer's for alkaloids, Borntrager's for anthraquinones, Shinoda for flavonoids, ferric chloride for tannins, foam test for saponins, Salkowski and Liebermann-Burchard for steroids, Molisch for carbohydrates.

EXAM CORNER — UNIT 4

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). There is no ABQ from this unit.

Likely short answer questions (5 marks)

1. Define phytochemistry and write its significance in Ayurveda.
2. Differentiate between primary and secondary metabolites.
3. Write a short note on alkaloids with their tests and examples.
4. Write a short note on glycosides and their classification.
5. Write a short note on tannins and their relevance to Rasashastra.
6. Write a short note on volatile oils and their extraction.
7. Write a short note on saponins.
8. Enumerate the solvents used in extraction in order of polarity.
9. Write a short note on Soxhlet extraction.
10. Write a short note on supercritical fluid extraction.
11. Write a short note on preliminary phytochemical screening.
12. Write a short note on the marker compound.

Likely long answer questions (10 marks)

1. Define phytochemistry. Describe the classes of phytochemicals with their chemical nature, tests and examples.
2. Describe in detail the methods of extraction of plant material, with the principle, advantages and limitations of each.

3. Discuss the selection of a solvent for extraction. Correlate the modern solvent system with the classical Kalpanas.
4. Describe preliminary phytochemical screening. Give the tests for each class of constituent, and discuss the difficulties in screening a herbal formulation.
5. Discuss the significance of phytochemistry in the standardisation and quality control of Ayurvedic formulations.

How to attempt

- For any class of phytochemical, use a fixed four-part pattern: chemical nature, occurrence, test, example.
- For extraction, always divide into cold, hot, distillation and modern. The examiner is looking for that structure.
- Draw the solvent polarity series. It takes one line and it is always relevant.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **115, 116, 117, 118**

1. **115.** Phytochemistry of Medicinal Plants. Springer US; 2013.
2. **116.** Shah B. Textbook of Pharmacognosy and Phytochemistry. India: Elsevier India.
3. **117.** Harborne JB. Phytochemical Methods: A Guide to Modern Techniques of Plant Analysis. Springer; 2012.
4. **118.** Plant-derived Bioactives: Production, Properties and Therapeutic Applications. Springer Nature Singapore; 2020.

UNIT 5
METALLURGY

SYLLABUS POINTS

5.1 Introduction to metallurgy and general physicochemical methods of extraction of metals from ores
5.2 Common ores used in Rasashastra and their Satvapatana from the metallurgical point of view

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Define metallurgy and describe its scope	1	Lecture	Know
Describe general physicochemical methods of extraction of metals from ores	1	Lecture	Know
Discuss common ores used in Rasashastra and their Satvapatana from the metallurgical point of view	1	Lecture	Knows how
Demonstrate different types of metals and their properties	3	Practical Training 3.5	Shows how
Differentiate various types of metals based on their properties	4	Experiential Learning 3.5	Does
Differentiate metals using basic metal identification techniques	4	Experiential Learning 3.6	Does

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	Yes

5.1 Introduction to metallurgy

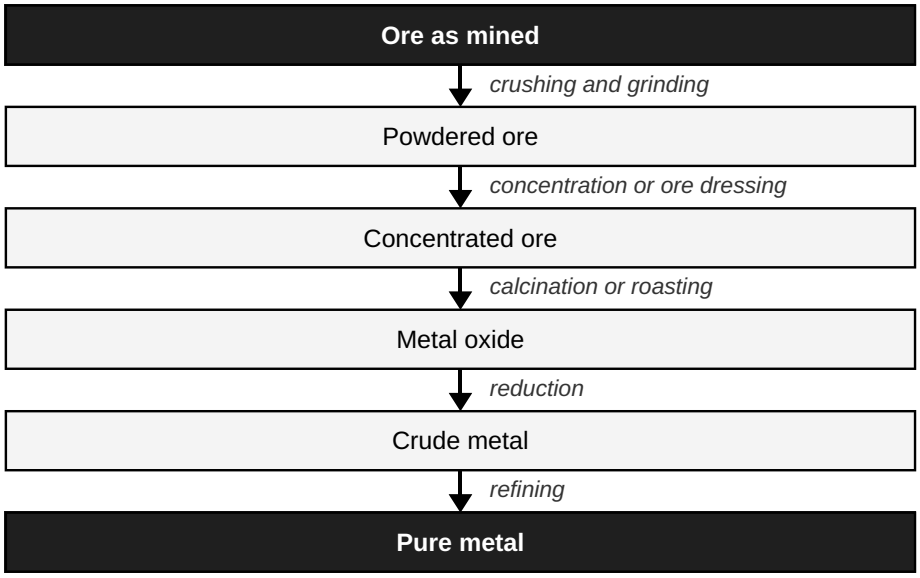
5.1.1 Definitions

Term	Definition
Metallurgy	The science and technology of extracting metals from their ores, refining them, and preparing them for use
Mineral	A naturally occurring substance with a definite chemical composition and crystal structure
Ore	A mineral from which a metal can be extracted profitably. **Every ore is a mineral, but every mineral is not an ore.**
Gangue or matrix	The unwanted earthy material associated with the ore
Flux	A substance added to combine with the gangue and make it fusible
Slag	The fusible product formed by flux and gangue, which is removed
Metallurgy of a metal	The whole sequence of operations for that metal

5.1.2 Types of ores

Type	Anion present	Example
Native	The free metal	Gold, silver, sometimes copper
Oxide	O	Haematite, cuprite, zincite
Sulphide	S	Chalcopyrite, galena, cinnabar, sphalerite
Carbonate	CO_3	Calamine, malachite, siderite
Sulphate	SO_4	Anglesite, gypsum
Halide	Cl, F	Rock salt, fluorspar
Silicate	SiO_4	Mica, feldspar

5.1.3 The general steps of metallurgy



5.1.4 Step 1 — Crushing and grinding

The ore is broken in jaw crushers and then ground in ball mills, to increase the surface area and to free the ore particles from the gangue. This is called **pulverisation**.

The classical parallel is Churnikarana, and the classical instrument is the Khalva Yantra.

5.1.5 Step 2 — Concentration or ore dressing

Removal of the gangue. The method is chosen according to the physical difference between ore and gangue.

Method	Principle	Suitable for
Hand picking	Visual difference	Coarse, obvious differences
Gravity separation or levigation	Difference in density; lighter gangue washed away	Heavy oxide ores
Froth flotation	Ore particles wetted by oil, gangue by water; ore rises in the froth	**Sulphide ores**
Magnetic separation	Difference in magnetic behaviour	Magnetite, chromite, wolframite
Leaching	Ore dissolved selectively by a chemical reagent	Bauxite, gold, silver

> **Classical parallel.** Gravity separation is exactly what Dhavana and Prakshalana achieve in the classical treatment of a Dravya. Magnetic separation of iron ore is paralleled by the classical use of Kanta Loha, magnetite, which the texts describe as having attractive power.

5.1.6 Step 3 — Conversion to oxide

The metal is easier to reduce from an oxide than from a sulphide or carbonate. So both are first converted to the oxide.

Process	Condition	Applied to	What happens
Calcination	Heating below the melting point, in **absence or limited supply of air**	Carbonate and hydrated ores	Carbon dioxide and water driven off; oxide left
Roasting	Heating **in a free supply of air**	Sulphide ores	Sulphur burnt off as sulphur dioxide; oxide left

This distinction is a certain examination question. Calcination is without air, for carbonates. Roasting is with air, for sulphides.

> **The Rasashastra parallel is direct and important.** Puta is a controlled heating operation. Depending on whether the crucible is sealed with Sandhi Bandhana, and on the fuel and the arrangement, the material may be heated with restricted air, resembling calcination, or with free air, resembling roasting. The classical distinction between Puta done in a closed Sharava Samputa and heating in an open vessel is, in metallurgical terms, exactly the calcination-roasting distinction.

5.1.7 Step 4 — Reduction to the metal

Method	Description	Example
Smelting with carbon	Oxide heated with coke or charcoal; carbon takes the oxygen	Iron, lead, zinc, tin
Self-reduction or auto-reduction	Part of the sulphide is roasted to oxide, which then reacts with the remaining sulphide	Copper, lead, mercury
Aluminothermy or thermite process	Aluminium reduces the oxide, with great heat	Chromium, manganese
Hydrogen reduction	Hydrogen used as reducing agent	Tungsten, molybdenum
Electrolytic reduction	Molten salt electrolysed	Aluminium, sodium, magnesium
Displacement	A more reactive metal displaces a less reactive one from solution	Silver, gold from cyanide solution by zinc

The Ellingham diagram plots the free energy of formation of oxides against temperature. It shows which reducing agent can reduce which oxide, and at what temperature. Its practical message is simple — carbon becomes a better reducing agent as temperature rises, which is why smelting works.

5.1.8 Step 5 — Refining

Method	Principle	Used for
Liquation	Low melting metal flows away from the higher melting impurity	Tin, lead
Distillation	Volatile metal distilled	**Zinc, mercury**
Poling	Green wood stirred into the molten metal; hydrocarbons reduce the oxide	Copper
Cupellation	Impurity oxidised and absorbed into a bone-ash cup	Silver, gold
Electrolytic refining	Impure metal as anode, pure metal deposits at the cathode	Copper, silver, gold, zinc
Zone refining	A molten zone travels along a bar, carrying impurities	Very high purity semiconductors
Van Arkel process	Metal converted to a volatile iodide and decomposed on a hot filament	Titanium, zirconium

5.2 Common ores in Rasashastra and their Satvapatana

5.2.1 What Satvapatana is

Satvapatana is the classical Rasashastra process of extracting the metallic essence, the **Satva**, from a mineral or an ore.

In metallurgical terms it is **reduction smelting** — the ore is mixed with reducing agents and fluxes, packed into a crucible, and heated strongly until the metal separates and collects.

The classical process and the modern process are the same operation, described in two vocabularies.

5.2.2 The apparatus and materials

Classical item	Function	Modern equivalent
Koshthi Yantra	The furnace in which Satvapatana is carried out; a pit or chamber giving intense, sustained heat with bellows	Blast furnace or crucible furnace
Musha	The crucible holding the charge; made of refractory clay with additives	Refractory crucible
Vayu Yantra or Bhastrika	Bellows to force air and raise the temperature	Forced draught
Dravaka Gana	Substances that promote melting and flow	Flux
Mitra Panchaka	The five substances added to assist Satvapatana	Flux plus reducing agent
Kshara and Amla Dravya	Alkaline and acidic materials used in pre-treatment	Chemical pre-treatment

Mitra Panchaka is commonly given as **Tankana, Guda, Guggulu, Gunja and Madhu**. The list varies a little between texts, so always cite the text you follow.

The chemistry of Mitra Panchaka

Substance	Chemical nature	Function in metallurgical terms
Tankana (borax)	Sodium borate	Flux; dissolves metal oxides; lowers the melting point; protects the surface from re-oxidation
Guda (jaggery)	Sugars	Carbon source on charring; reducing agent
Madhu (honey)	Sugars	Reducing agent; binder
Guggulu	Oleo-gum-resin	Carbon source; binder for the pellet
Gunja	Seed containing protein and oil	Organic carbon and nitrogen source; reducing agent

> **This table is the single best answer in the whole unit.** Every member of Mitra Panchaka is either a flux or a source of carbon. Borax is used as a flux in metallurgy to this day. Jaggery, honey, resin and seed all char to carbon, which is the reducing agent of every smelting operation. The Acharyas assembled a complete smelting charge — ore, flux, and reductant — and the composition is metallurgically correct.

5.2.3 The general procedure of Satvapatana

1. **Shodhana** of the raw mineral, by the method prescribed for it.
2. **Churnikarana** — reduction to a fine powder.
3. **Bhavana** with the prescribed Kshara, Amla or herbal juices, sometimes repeatedly. This introduces further reducing and fluxing material and makes an intimate mixture.
4. **Chakrika or Vataka** — small cakes or pellets are formed and dried.
5. **Charging** — the pellets are placed in the Musha, with Mitra Panchaka.
6. **Sealing** the Musha and drying it.
7. **Heating** in the Koshthi Yantra with bellows, to a strong and sustained fire.
8. **Cooling** — self-cooling, Swangasheeta.
9. **Collection** — the Musha is broken and the Satva collected from the bottom.
10. **Assessment** of yield and quality.

5.2.4 The important ores and their Satva

Rasashastra Dravya	Mineral identity	Chemical formula	Satva obtained
Makshika	Chalcopyrite	$CuFeS_2$	Copper, with iron
Rasaka or Kharpara	Calamine group — smithsonite and hemimorphite	$ZnCO_3$, zinc silicate	**Zinc, Yashada**
Abhraka	Mica group, chiefly biotite	Sheet silicate	Abhraka Satva
Vaikranta	Described variously; identity disputed between texts	Varies	Vaikranta Satva
Manahshila	Realgar	As_4S_4	Arsenic
Haratala	Orpiment	As_2S_3	Arsenic
Hingula	Cinnabar	HgS	**Mercury, Parada**
Gairika	Haematite	Fe_2O_3	Iron
Anjana	Antimony and lead sulphide group	Sb_2S_3, PbS	Antimony or lead
Tuttha	Chalcanthite	$CuSO_4 \cdot 5H_2O$	Copper
Kasisa	Melanterite	$FeSO_4 \cdot 7H_2O$	Iron

> **Note on Vaikranta.** Its botanical and mineralogical identity is genuinely disputed. Different authors have proposed tourmaline, and various other minerals. Do not assert a single identity as settled. State that the identity is controversial and name the proposals. That is the correct scholarly position and an examiner will credit it.

5.2.5 Two worked examples

A. Hingula to Parada — extraction of mercury from cinnabar

The classical method uses a Damaru Yantra or Vidyadhara Yantra — a closed arrangement in which the material is heated below and the vapour condenses above.

Step	Classical	Metallurgical account
Charge	Hingula, ground and mixed as prescribed	Cinnabar, HgS
Heating	Strong heat in the lower vessel	Roasting; mercuric sulphide reacts with oxygen, releasing mercury vapour and sulphur dioxide
Condensation	Mercury collects on the cooler upper surface	Distillation and condensation of mercury vapour
Collection	Parada scraped from the upper vessel	Crude mercury, requiring further purification

This is a **roasting followed by distillation**, and it is essentially the method by which mercury is still obtained industrially.

B. Rasaka to Yashada — extraction of zinc

This is the most historically important example in the whole subject, and it deserves emphasis.

Zinc is difficult to extract. Its boiling point is lower than the temperature needed to reduce its oxide. So when zinc oxide is reduced with carbon, the zinc comes off as a **vapour**, and if air reaches it, it immediately re-oxidises. Zinc therefore cannot be smelted in an open furnace like iron or copper. It must be reduced and condensed in a **closed, downward-directed system**, out of contact with air.

The classical Satvapatana of Rasaka does exactly this. The charge of Rasaka with organic reducing material is heated in a sealed vessel, and the zinc vapour is led downward to condense away from the fire.

Zawar in Rajasthan is documented archaeologically as a site of large-scale zinc production by downward distillation in closed retorts, some centuries before the process was patented in Europe. This is a well-established finding, and it is the strongest single piece of evidence for the metallurgical sophistication of the Indian tradition. It belongs in any answer on Satvapatana.

Step	Classical	Metallurgical account
Pre-treatment	Shodhana and Bhavana of Rasaka	Preparation and mixing with reductant
Charge	Rasaka with organic material, in a sealed Musha	Zinc ore with carbon in a closed retort
Heating	Strong sustained fire in the Koshthi	Carbothermic reduction above 900 degrees Celsius
Product	Zinc vapour formed	Zinc boils at 907 degrees Celsius, so it leaves as vapour
Condensation	Vapour led away from the fire and condensed	Downward distillation, out of contact with air
Collection	Yashada Satva collected	Metallic zinc

5.2.6 Comparison of Satvapatana with modern smelting

Point	Satvapatana	Modern smelting
Purpose	To obtain the Satva, the metallic essence	To obtain the metal
Furnace	Koshthi Yantra with bellows	Blast or electric furnace
Vessel	Musha of refractory clay	Refractory crucible or lining
Reducing agent	Guda, Madhu, Guggulu, Gunja and other organic matter, charring to carbon	Coke or charcoal
Flux	Tankana and other Dravaka	Borax, limestone, silica
Pre-treatment	Shodhana, Bhavana with Kshara and Amla	Roasting, calcination, chemical pre-treatment
Temperature control	By fuel quantity, arrangement and bellows; judged by experience	Instrumented and measured
Scale	Small, batch	Large, continuous
Yield	Low, variable	High, consistent
Basic chemistry	**Identical**	**Identical**

KEY POINTS TO REMEMBER

- Every ore is a mineral, but every mineral is not an ore.
- Gangue is the unwanted material; flux combines with it to form slag.
- Froth flotation is for sulphide ores. Magnetic separation is for magnetic ores.
- **Calcination is heating in absence of air, for carbonates. Roasting is heating in free air, for sulphides.**

- Reduction methods: carbon smelting, self-reduction, aluminothermy, hydrogen reduction, electrolysis, displacement.
- Refining methods: liquation, distillation, poling, cupellation, electrolytic refining, zone refining, van Arkel.
- Satvapatana is reduction smelting. Koshthi Yantra is the furnace; Musha is the crucible.
- Mitra Panchaka is commonly Tankana, Guda, Guggulu, Gunja and Madhu. Tankana is the flux; the rest are carbon sources and reducing agents.
- Hingula to Parada is roasting followed by distillation.
- Rasaka to Yashada requires downward distillation in a closed vessel, because zinc boils at 907 degrees Celsius and re-oxidises in air. Zawar in Rajasthan is documented as an early site of this process.
- Vaikranta's mineral identity is disputed. Do not state one as settled.

EXAM CORNER — UNIT 5

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). There is no ABQ from this unit.

Likely short answer questions (5 marks)

1. Define metallurgy, ore, gangue, flux and slag.
2. Differentiate between a mineral and an ore.
3. Enumerate the types of ores with examples.
4. Write a short note on the methods of concentration of ore.
5. Differentiate between calcination and roasting, and correlate them with Puta.
6. Enumerate the methods of reduction of an ore.
7. Enumerate the methods of refining of a metal.
8. Define Satvapatana. Describe the apparatus used.
9. Write a short note on Mitra Panchaka and its chemical role.
10. Write a short note on Koshthi Yantra.
11. Describe the Satvapatana of Rasaka and explain why a closed vessel is required.
12. Write a short note on the extraction of Parada from Hingula.

Likely long answer questions (10 marks)

1. Define metallurgy. Describe the general physicochemical methods of extraction of metals from their ores.
2. Describe Satvapatana in detail, and compare it with modern smelting from the metallurgical point of view.
3. Enumerate the common ores used in Rasashastra. Describe the Satvapatana of any two, with their chemistry.
4. Describe the Satvapatana of Rasaka. Explain the metallurgical difficulty of zinc extraction and how the classical method overcomes it.

5. Discuss the classical concepts of Musha, Koshthi and Mitra Panchaka in the light of modern metallurgy.

How to attempt

- For the general metallurgy question, follow the five steps in order, and draw the flow chart. It is worth several marks by itself.
- For any Satvapatana question, always give both vocabularies side by side in a table. That is what the question is really asking.
- Mention Zawar when you write on zinc. It is documented and it strengthens the answer.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **119, 120, 121**

1. **119.** Cottrell A. An Introduction to Metallurgy. United Kingdom: St. Martin's Press; 1967.
2. **120.** Ormandy PG. An Introduction to Metallurgical Laboratory Techniques. Pergamon; 2016.
3. **121.** Alexander P. A Handbook of Minerals, Crystals, Rocks and Ores. India: New India Publishing Agency; 2009.

Classical sources for Satvapatana: Rasa Ratna Samuccaya (NCISM references 2 and 3) and Rasatarangini (NCISM reference 1).

MODULE 3
PRACTICAL TRAINING ACTIVITIES

Module 3 carries **20 hours of practical training**, over five activities, as listed in Table 4 of the curriculum.

No	Practical name	Hours
3.1	Analysis of chemical bonding types, principles and isomerism	5
3.2	Demonstration of the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures	5
3.3	Demonstration of different types of extraction methods including solvent selection	4
3.4	Demonstration of qualitative identification of phytoconstituents in single herbs and herbal formulations	3
3.5	Demonstration of different types of metals and their properties	3
	Total	**20**

Practical Training 3.1 — Analysis of chemical bonding types, principles and isomerism

Aim. To analyse the types of chemical bonding and isomerism, and to relate them to Rasashastra processes.

Requirements. Molecular model kit, periodic table chart, samples of ionic and covalent compounds, structural formulae of common phytoconstituents, classical texts.

Procedure

1. Build models of representative molecules — water, methane, benzene, ammonia — and record the shape and bond angle of each.
2. Take a set of given compounds and classify the bonding in each as ionic, covalent, coordinate or metallic. Justify each classification.
3. Test the physical properties that follow from the bonding — solubility in water and in an organic solvent, melting behaviour, and conductivity of the solution.
4. Build models of a pair of structural isomers and a pair of geometrical isomers. Record the difference.
5. Take the structure of a chelating molecule such as citric acid or gallic acid. Identify the donor atoms. Build the chelate with a metal ion using the model kit.

6. Now take one classical Shodhana procedure. Identify the medium, identify its principal chemical constituents, and state which bonding interaction is involved.

Observation tables to draw. One for molecular shape and bond angle. One for bonding type against observed physical property. One relating the classical procedure to the bonding interaction.

Result. Record the classification made and the correlation established.

Precautions. Handle solvents in a ventilated area. Do not taste any sample. State the reason for every classification; a classification without a reason earns no marks.

Practical Training 3.2 — Crystallography and organometallic principles in Rasashastra

Aim. To demonstrate the fundamental principles of crystallography and of organometallic chemistry, with reference to Rasashastra procedures.

Requirements. Crystal models of the seven systems, mineral specimens, hand lens, sample XRD patterns of a raw Dravya and of its Bhasma, reference data from a mineral database, molecular models.

Procedure

1. Examine the crystal models. Record the axes and angles of each of the seven systems.
2. Examine mineral specimens with a hand lens. Assign each to a crystal system where the habit permits.
3. Take a sample XRD pattern of a raw mineral. Read off the positions of the strongest peaks. Convert the angles to d-spacings using Bragg’s law.
4. Match the d-spacings against reference data from Webmineral or an equivalent source, and identify the phase.
5. Now take the XRD pattern of the corresponding Bhasma. Identify the phases present.
6. Compare the two patterns. Record which peaks have disappeared, which are new, and which have broadened.
7. Estimate the change in crystallite size from the peak broadening, qualitatively.
8. Build models of a coordination complex and of an organometallic compound. State the difference in the bond involved.
9. Prepare a written statement of what the XRD evidence does and does not prove about the transformation.

Observation tables to draw. One for the seven crystal systems. One for peak position, d-spacing and identified phase, before and after Marana.

Result. Record the phases identified before and after, and the conclusion drawn.

Precautions. Do not overstate the conclusion. XRD shows crystalline phases only. State clearly what would be needed in addition — FTIR, TGA, SEM-EDAX — to describe the product fully.

Practical Training 3.3 — Extraction methods and solvent selection

Aim. To demonstrate the methods of extraction and the principles of solvent selection.

Requirements. Coarsely powdered drug, solvents in order of polarity, Soxhlet apparatus, reflux assembly, percolator, conical flasks, water bath, rotary evaporator or evaporating dish, Clevenger apparatus, weighing balance.

Procedure

1. Weigh the powdered drug accurately and record its moisture content.
2. **Maceration.** Soak a weighed quantity in ethanol in a closed flask for the prescribed period, with occasional shaking. Filter and evaporate. Record the yield.
3. **Decoction.** Boil a weighed quantity with water and reduce as prescribed. Filter, evaporate, record the yield.
4. **Soxhlet extraction.** Carry out successive extraction with petroleum ether, then chloroform, then ethanol, then water. Record the yield of each fraction separately.
5. **Distillation.** If the drug is aromatic, determine the volatile oil content in the Clevenger apparatus.
6. Calculate the percentage yield for every method.
7. Compare the yields and the appearance of each extract.
8. State which method and which solvent you would recommend for this drug, and why.

Observation table to draw

Method	Solvent	Weight of drug	Weight of extract	Percentage yield	Colour and consistency

Result. Record the yields obtained and the recommended method with justification.

Precautions. Petroleum ether and diethyl ether are highly inflammable — never use a naked flame. Use a water bath. Check that all joints are tight. Do not let the Soxhlet run dry. Record the moisture content, or the yields cannot be compared.

Practical Training 3.4 — Qualitative identification of phytoconstituents

Aim. To identify the classes of phytoconstituents present in a single herb and in a herbal formulation.

Requirements. Extracts prepared in Practical 3.3, test tubes, reagents for the standard tests, water bath, filter paper, blood agar plate if the haemolysis test is done.

Procedure

1. Prepare a small quantity of each extract for testing.
2. Carry out the tests for each class in turn, following the table in Unit 4, section 4.5.2 — alkaloids, glycosides, flavonoids, tannins, saponins, steroids, triterpenoids, carbohydrates, proteins, fixed oils and resins.
3. Run a blank alongside every test.
4. Record the result of each as present or absent, with the observation seen.
5. Repeat the whole screening on a herbal formulation containing that drug.

6. Screen each remaining ingredient of the formulation separately.
7. Compare, and state which ingredient is likely to have contributed each class detected in the formulation.

Observation table to draw

Constituent class	Test used	Reagent	Observation	Inference

Result. Record the classes detected in the single drug and in the formulation.

Precautions. Some reagents contain mercury, bismuth or picric acid — handle with care and dispose of correctly. Always run a blank. Remember that sugar, jaggery and honey in a formulation give a positive carbohydrate test that does not come from the drug. Record that as a limitation.

Practical Training 3.5 — Demonstration of different types of metals and their properties

Aim. To demonstrate the physical and chemical properties of the metals used in Rasashastra.

Requirements. Samples of Swarna, Rajata, Tamra, Loha, Naga, Vanga and Yashada, Parada in a sealed container, hammer, file, magnet, burner, tongs, dilute acids, test tubes, balance, measuring cylinder.

Procedure

1. Record the colour, lustre and appearance of each metal.
2. Test malleability by hammering a small piece, and ductility where a wire is available.
3. Test hardness by scratching one metal with another, and arrange them in order.
4. Test the response to a magnet.
5. Determine the specific gravity of each by the displacement method.
6. Heat a small quantity of each carefully and record the change in colour or surface. Record the relative ease of melting.
7. Test the action of dilute acid on a small piece of each, and record the observation.
8. Arrange the metals in order of reactivity from your observations.
9. Compare your order with the standard reactivity series.
10. Record the classical Grahya and Agrahya Lakshana of each metal from the text, and compare with your observations.

Observation table to draw

Metal	Rasashastra name	Colour	Malleability	Magnetic	Specific gravity	Action of heat	Action of dilute acid

Result. Record the properties determined and the reactivity order obtained.

Precautions. Parada must be handled only in a closed container and never heated in an open vessel. Mercury vapour is toxic. Use tongs for hot metal. Add acid to the metal, never the reverse. Work in a ventilated area. Dispose of all metal residues correctly and never down the sink.

MODULE 3
EXPERIENTIAL LEARNING ACTIVITIES

Module 3 carries **30 hours** under experiential learning, including modular assessment. Six activities are prescribed in Table 5 of the curriculum.

No	Experiential learning activity	Hours
3.1	Exploration of the periodic table: chemical properties of main group and transition metals	6
3.2	Exploration and identification of different functional groups and their influence on molecular interactions and therapeutic efficacy	6
3.3	Extraction techniques in herbal formulations: a hands-on approach	3
3.4	Identification of phytoconstituents in herbs and herbal formulations: a practical approach	3
3.5	Test different types of metals for their properties	4
3.6	Hands-on experience with basic metal identification techniques	4
	Total	**26**

Experiential Learning 3.1 — Exploration of the periodic table

What you have to do. Build a complete periodic-table profile of the metals of Rasashastra, and use it to explain classical practice.

Method. For each of the eight metals — Swarna, Rajata, Tamra, Loha, Naga, Vanga, Yashada and Parada — record the atomic number, block, group, period, electronic configuration, common oxidation states, melting and boiling points, density, and standard electrode potential.

Then answer these questions in your report

- Which of the eight are true transition metals, and which are not? Justify with the electronic configuration.
- Which show variable oxidation states, and what does that imply for the Bhasma of that metal?
- Arrange them by melting point. Does the classical difficulty of Marana of each metal follow that order?
- Arrange them by reactivity. Does the classical statement of which metals are easily "killed" follow that order?
- Parada is a liquid at room temperature. Explain this in terms of its electronic configuration, and relate it to why Parada alone among the metals has a separate Samskara sequence.

How to record it. A single large data table, followed by a written answer to each question, with the chemical reasoning stated.

Experiential Learning 3.2 — Exploration and identification of functional groups

What you have to do. Take a set of phytoconstituents, identify their functional groups, and discuss how those groups govern their molecular interactions and their therapeutic effect.

Method. Choose ten molecules relevant to your subject — curcumin, berberine, piperine, gallic acid, quercetin, glycyrrhizin, andrographolide, eugenol, menthol and citric acid are a good set. For each, draw the structure, mark every functional group, and complete the analysis.

Analysis table to prepare

Molecule	Source drug	Functional groups present	Polarity and solubility	Acidic or basic	Metal-binding capacity	Likely therapeutic relevance

Then answer these questions

- Which of these molecules would you expect to bind a metal most strongly, and why?
- Which would be extracted by water, and which by petroleum ether?
- Which would be lost on prolonged boiling, and which would survive?
- For one molecule, explain how its functional groups would allow it to attach to a receptor.

How to record it. Structures drawn by hand, the completed analysis table, and the written answers.

Experiential Learning 3.3 — Extraction techniques in herbal formulations

What you have to do. Move from extracting a single drug to extracting a complete classical formulation, and account for what changes.

Method. Take one classical formulation. Extract it by two different methods. Extract each of its ingredients separately by the same methods. Compare the yields.

Questions to answer

- Is the yield from the formulation equal to the sum of the yields from its ingredients? If not, why not?
- Does any ingredient interfere with the extraction of another?
- Does the processing already applied to the formulation, such as Paka, change what can be extracted?
- Which method would you recommend for routine standardisation of this formulation?

How to record it. Yield tables, comparison, and a recommended standard operating procedure for extracting this formulation.

Experiential Learning 3.4 — Identification of phytoconstituents in herbs and formulations

What you have to do. Build a complete phytochemical profile of one classical formulation and interpret it.

Method. Screen the formulation and every ingredient, as in Practical 3.4. Then go further — attempt to assign each class detected to its probable source ingredient, and identify which classes are absent that you would have expected.

The value of this exercise. A class expected but not detected is the most interesting result. It may mean the constituent was destroyed during processing, or that the ingredient was substituted, or that the test was interfered with. Investigating which of these it is, is genuine research work.

How to record it. The full screening matrix of formulation against ingredients, the assignment of each class to its source, and a discussion of every unexpected result.

Experiential Learning 3.5 — Test different types of metals for their properties

What you have to do. Extend Practical 3.5 into a complete comparative study, and relate it to the classical descriptions.

Method. Determine, for each metal, the physical properties measured in the practical, and add the classical characters given in the texts — Grahya Lakshana, Agrahya Lakshana, Varna, Gaurava and Snigdhata. Then correlate.

Questions to answer

- Which classical Lakshana corresponds to which measurable modern property?
- Which classical characters cannot be reduced to a modern measurement, and what does that tell you?
- On the basis of your measurements, could you distinguish a genuine sample from a substituted one?

How to record it. The measurement table, the classical description table, and the correlation between them, with a clear statement of where the correlation fails.

Experiential Learning 3.6 — Basic metal identification techniques

What you have to do. Learn and apply the basic techniques by which an unknown metal sample is identified.

Techniques to apply

Technique	What it shows
Physical examination	Colour, lustre, malleability, hardness
Specific gravity determination	A characteristic constant for each metal
Magnetic test	Distinguishes iron and its alloys
Flame test	Characteristic colours for certain metals
Action of dilute and concentrated acids	Distinguishes by reactivity
Confirmatory wet tests for the cation	Standard qualitative inorganic analysis
Streak and scratch tests	For mineral samples

Method. Obtain a set of coded unknown samples from your teacher. Apply the techniques in a logical order — non-destructive first, destructive last. Record each observation, and narrow down the possibilities at each step. Identify each sample and state your confidence.

How to record it. A dichotomous identification chart, the observation record for every sample, the identification made, and the reasoning that led to it.

> **The habit this activity builds.** Identify by a sequence, not by a guess. Begin with what costs nothing and destroys nothing, and only then move to tests that consume the sample. This is exactly the discipline required when an unknown Rasa Dravya is brought to your department, and it is what a practical examiner is testing.

MODULE 3
VIVA POINTERS AND QUESTION BANK

Viva pointers

On atomic structure and the periodic table

- Which block do the Rasashastra metals belong to?
- Why do transition metals show variable oxidation states?
- Why are transition metal compounds coloured? How does this relate to Bhasma Pariksha?
- Is Naga a transition metal? Is Yashada?
- Why is Parada a liquid at room temperature?

On bonding and crystallography

- Name the types of chemical bonding.
- What is a coordinate bond? Where does it occur in Rasashastra?
- Name the seven crystal systems.
- State Bragg’s law. What does XRD tell you about a Bhasma?
- What can XRD not tell you?

On organic and phytochemistry

- What is a functional group? Which groups bind metals?
- Why is Gandhaka used with Parada? Explain chemically.
- Name four mechanisms by which a herbal medium acts on a metal.
- Name the test for alkaloids. For tannins. For flavonoids.
- Which solvent extracts fixed oils? Which extracts tannins?
- Why is a volatile oil drug added at the end of a preparation?

On metallurgy

- Differentiate a mineral from an ore.
- Differentiate calcination from roasting.
- What is Satvapatana in metallurgical terms?
- What is Mitra Panchaka? What is the function of each member?
- Why must zinc be extracted in a closed vessel?
- What is the mineral identity of Makshika? Of Rasaka?

Module 3 question bank

Analytical Based Questions (20 marks) — from Units 1, 2 and 3

1. Two batches of Lauha Bhasma prepared by the same classical method differ in colour and on analysis. Analyse the chemical reasons, relate them to the oxidation states of iron, and suggest process controls.
2. Analyse the classical process of Shodhana of a metal in the light of atomic structure and chemical bonding, and justify the media prescribed.
3. The classical texts prescribe different numbers of Putas for different Dravyas. Analyse this in terms of thermal stability, oxidation state and particle size, and design an experiment to fix the correct number.
4. You are asked to prove by analysis that Shodhana and Marana have transformed a Rasa Dravya chemically. Analyse which instrumental methods you would use, what each would show, and what each would not permit you to conclude.
5. Analyse the four Rasayana Kalpanas in terms of their physical and chemical processes, comparing temperature, vessel, product and transformation.
6. Analyse the classical instruction that Marana be carried out with repeated Bhavana using herbal juices. Compare it with modern green synthesis of nanoparticles and design a study to test whether the medium affects the product.

7. A student proposes to replace a classical Shodhana medium with distilled water to save cost. Analyse this proposal chemically and justify your conclusion.
8. Analyse the interaction of herbal and animal products with metals under the headings of detoxification, compound formation and compound stability.

Long Answer Questions (10 marks) — from Units 4 and 5

1. Define phytochemistry. Describe the classes of phytochemicals with their chemical nature, tests and examples.
2. Describe in detail the methods of extraction of plant material, with the principle, advantages and limitations of each.
3. Discuss the selection of a solvent for extraction, and correlate the modern solvent system with the classical Kalpanas.
4. Describe preliminary phytochemical screening, with the tests for each class, and the difficulties in screening a formulation.
5. Discuss the significance of phytochemistry in the standardisation and quality control of Ayurvedic formulations.
6. Define metallurgy. Describe the general physicochemical methods of extraction of metals from their ores.
7. Describe Satvapatana in detail and compare it with modern smelting.
8. Enumerate the common ores used in Rasashastra. Describe the Satvapatana of any two, with their chemistry.
9. Describe the Satvapatana of Rasaka, and explain the metallurgical difficulty of zinc extraction.
10. Discuss Musha, Koshthi and Mitra Panchaka in the light of modern metallurgy.

Short Answer Questions (5 marks) — from all five units

From Unit 1

1. Explain the importance of chemistry in understanding Rasashastra processes.
2. Enumerate the types of chemical bonding with examples.
3. Define isomerism and enumerate its types.
4. Write the periodic trends across a period and down a group.
5. Write a short note on the variable oxidation states of transition metals.
6. Why do transition metals form coloured compounds? Correlate with Bhasma Pariksha.
7. Explain the chemical changes occurring during Shodhana.
8. Explain the chemical changes occurring during Marana.

From Unit 2

1. Define crystallography and enumerate the seven crystal systems.
2. State Bragg’s law and explain its application to Bhasma analysis.
3. Differentiate an organometallic compound from a coordination compound.
4. Enumerate the Chaturvidha Rasayana with examples.
5. Write a short note on Kajjali.

6. Define bioinorganic chemistry and enumerate the essential trace elements.
7. Define chelation and give the chelating agents in classical Shodhana media.
8. Write a short note on Amritikarana and its chemical basis.

From Unit 3

1. Explain catenation and tetravalency of carbon.
2. Write a short note on hybridisation with shapes and bond angles.
3. Define a functional group and enumerate the important ones.
4. Enumerate the functional groups that bind metals, with the donor atom.
5. Explain chemically the role of Bhavana Dravya in Marana.
6. Explain why Gandhaka is used along with Parada.
7. Write a short note on green synthesis of nanoparticles and its parallel with Bhavana.

From Unit 4

1. Differentiate between primary and secondary metabolites.
2. Write a short note on alkaloids with tests and examples.
3. Write a short note on tannins and their relevance to Rasashastra.
4. Enumerate the solvents used in extraction in order of polarity.
5. Write a short note on Soxhlet extraction.
6. Write a short note on the marker compound.

From Unit 5

1. Define metallurgy, ore, gangue, flux and slag.
2. Differentiate between calcination and roasting, and correlate with Puta.
3. Enumerate the methods of concentration of ore.
4. Define Satvapatana and describe the apparatus used.
5. Write a short note on Mitra Panchaka and its chemical role.
6. Write a short note on the extraction of Parada from Hingula.

> **Note on the marks range.** Table 6F places Module 3 in the marks range of **5 to 20**. Units 1, 2 and 3 are ABQ-eligible; Units 4 and 5 carry the long answers. This is the most analytically demanding module in the course. Prepare Units 1, 2 and 3 by understanding, not by memorising, and prepare Units 4 and 5 by structure.

MODULE 4
CONCEPT, UNDERSTANDING, APPLICATION, SOPs AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES

Credits	Lectures	Practical Training	Experiential Learning	Total hours	Modular assessment
2	10	20	30	60	50 marks

About this module

This module provides a comprehensive understanding of Shodhana and Marana procedures, including their principles, applications and standard operating procedures. It covers the chemical changes involved in these processes, and it examines case studies, troubleshooting, optimisation and emerging trends.

Module Learning Objectives

At the end of this module, the student should be able to:

1. Demonstrate a comprehensive understanding of Shodhana — purification, detoxification, modification and potentiation — and of Marana, incineration.
2. Describe the fundamental principles, applications and standard operating procedures of Shodhana and Marana processes.
3. Describe the chemical changes involved in the Shodhana and Marana processes.
4. Analyse real-world case studies of Shodhana and Marana procedures.
5. Apply troubleshooting techniques for common issues in Shodhana and Marana processes.
6. Plan future directions in Shodhana and Marana procedures.

Units in this module

Unit	Name
M4U1	Shodhana and Marana Procedures
M4U2	SOP of Shodhana Processes
M4U3	SOP of Marana Process
M4U4	Chemical Changes involved in Shodhana and Marana Processes
M4U5	Shodhana and Marana — Case Studies, Troubleshooting and Emerging Trends

How this module is examined

Unit	ABQ	SAQ	LAQ
U1 Shodhana and Marana Procedures	Yes	Yes	No
U2 SOP of Shodhana Processes	No	Yes	Yes
U3 SOP of Marana Process	No	Yes	Yes
U4 Chemical Changes	Yes	Yes	No
U5 Case Studies, Troubleshooting, Emerging Trends	No	Yes	No

Module 4 is placed in the marks range of **5 to 20**.

A note on how this module should be studied

This is the module of your own speciality. You already know these procedures from your undergraduate study and from the Rasashala. What this module asks of you is different.

It asks you to write them as **standard operating procedures** — precise, repeatable, documented — and to explain the **chemical changes** behind them. That shift, from knowing how to do a thing to being able to specify and justify it, is the whole purpose of the module.

Whenever you write about a procedure here, always name the classical text you are following. Different texts prescribe different media, different numbers of repetitions and different Putas for the same Dravya. A procedure quoted without its source is incomplete at postgraduate level.

> **Safety comes first in this module, in the examination hall and in the Rasashala.** The materials of this module include mercury, arsenic sulphides, lead and other toxic substances. In every practical answer, in every SOP you write, and in every session you attend, the safety precautions must be stated and followed. An SOP without a safety section is an incomplete SOP, and an examiner will mark it as such.

UNIT 1
SHODHANA AND MARANA PROCEDURES

SYLLABUS POINTS

1.1 Overview of Shodhana and Marana procedures
1.2 Importance of purification and incineration in Ayurvedic pharmaceutical manufacturing
1.3 Principles of Shodhana and Marana procedures
1.4 Applications of Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Discuss Shodhana and Marana procedures	1	Lecture	Knows how
Discuss the importance of purification and incineration in Ayurvedic pharmaceutical manufacturing	1	Lecture	Knows how
Analyse principles of Shodhana and Marana procedures with their application in Ayurvedic pharmaceutical manufacturing	1	Lecture	Knows how

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

1.1 Overview of Shodhana and Marana

1.1.1 Shodhana — meaning and definition

The word **Shodhana** comes from the root meaning to cleanse or to make pure.

Definition. Shodhana is the process by which the unwanted, harmful or unsuitable qualities of a Dravya are removed, and the Dravya is made fit for further pharmaceutical processing and for therapeutic use.

The English word “purification” is only a partial translation. The curriculum itself gives four senses, and you should use all four:

Sense	What it means
Purification	Removal of physical impurities — earth, sand, gangue, foreign matter
Detoxification	Removal or conversion of the harmful constituent
Modification	Change of the physical and chemical nature of the Dravya, so that it becomes brittle, fine and reactive
Potentiation	Addition of desirable qualities from the media used, so that the Dravya becomes therapeutically more suitable

> **Write all four in the examination.** A student who translates Shodhana simply as purification has given a quarter of the answer. The modification and potentiation senses are what distinguish Shodhana from ordinary cleaning, and they are what an examiner is looking for.

1.1.2 Types of Shodhana

Type	Meaning	Example
Samanya Shodhana	General purification, common to a whole group of Dravyas	The general Shodhana of metals by Nirvapa in the five media
Vishesha Shodhana	Specific purification, prescribed for one particular Dravya	The specific Shodhana of Tamra, of Abhraka, of Haratala

Samanya Shodhana is generally performed first, and Vishesha Shodhana after it, as prescribed for that Dravya.

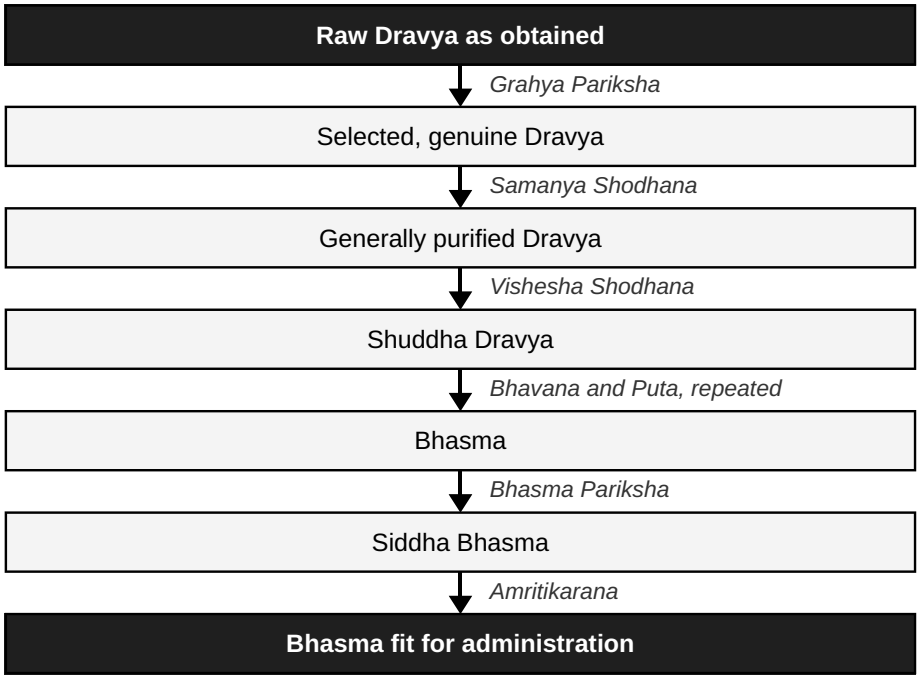
1.1.3 Marana — meaning and definition

The word **Marana** means to kill. In Rasashastra it means to destroy the metallic character of a metal.

Definition. Marana is the process by which a metal or mineral is converted, by repeated trituration with prescribed media and repeated heating, into a fine, lustreless, therapeutically suitable powder called **Bhasma**.

The word “killing” is precise in its own way. What is killed is the metallic state — the lustre, the malleability, the ability to be melted back and drawn into a wire. The classical tests of a completed Bhasma are, in essence, tests that these metallic properties have gone.

1.1.4 The place of these processes in the sequence



1.2 Importance of Shodhana and Marana

1.2.1 Why Shodhana is essential

1. Removal of physical impurities. A mined mineral carries sand, clay and gangue. A metal carries slag and oxide scale. These have no therapeutic value and may be harmful.

2. Removal or conversion of toxic constituents. Many Rasa Dravyas are toxic in their raw state. Shodhana converts, binds or removes the harmful component.

3. Making the Dravya brittle and easy to powder. A metal cannot be triturated. Repeated heating and quenching cracks it, and it becomes brittle. Without this step Marana is not possible at all.

4. Removal of the classical Doshas. The texts describe specific Doshas in each Dravya, which Shodhana is intended to remove. Parada, for example, is described as carrying **Naisargika Dosha** — Visha, Vahni and Mala; **Yaugika Dosha** — Naga and Vanga; and **Aupadhika or Kanchuka Dosha**. The number and names of the Kanchukas vary between texts, so cite the text you follow rather than giving one list as universal.

5. Potentiation. The media used are not inert. Their constituents act on the Dravya and contribute to its final character.

6. Standardisation. A Shodhana procedure carried out to a fixed SOP gives a starting material of consistent quality for every batch.

1.2.2 Why Marana is essential

1. To make the metal absorbable. A metal as such cannot be absorbed. Conversion into a very fine compound, with an enormous surface area, changes this entirely.

2. To destroy the metallic properties. Lustre, malleability and the ability to be reconverted to the metal must all go. This is exactly what the classical tests examine.

3. To reduce particle size to the extreme. Repeated Bhavana and Puta reduce the particle size progressively. Studies characterising Bhasma have repeatedly reported particles in the micrometre to nanometre range.

4. To convert the metal into a stable compound. The product is typically an oxide or a sulphide, which is stable and does not readily revert.

5. To make the Dravya suitable for a small dose. Bhasma is given in Ratti or milligram doses. That is possible only because the material is so finely divided and so uniform.

6. To impart the qualities of the media used. Repeated Bhavana with prescribed juices is part of the therapeutic design, not merely a grinding aid.

1.2.3 The importance in manufacturing terms

Manufacturing requirement	How Shodhana and Marana serve it
Consistent raw material	Shodhana removes variable impurities
Batch-to-batch uniformity	A fixed SOP gives a reproducible product
Safety of the finished product	Toxic constituents removed or converted
Defined particle size	Progressive reduction through repeated Puta
Stability and shelf life	The final compound is stable and does not revert
Regulatory compliance	Schedule T requires documented, standardised processes
Analytical verification	Each stage can be checked by XRD, SEM-EDAX and other methods

1.3 Principles of Shodhana and Marana

1.3.1 The principle of Samskara

Both processes rest on the classical principle of **Samskara** — that the qualities of a substance can be changed by processing.

The Samhita states the principle plainly: Samskara means the imparting of new qualities to a substance. The agents through which it acts include water, fire, cleansing, churning, place, time, impregnation, and the vessel used.

Every element of Shodhana and Marana is one of these agents. Swedana is the action of water and fire. Mardana is churning. Bhavana is impregnation. The Musha and the Sharava are the vessels. The classical framework and the actual procedure fit exactly.

1.3.2 Principles underlying Shodhana

Principle	How it operates
Thermal shock	Repeated heating to red heat and quenching cracks the metal and makes it brittle
Chemical action of the medium	Acidic, alkaline and reducing media react with the surface and with the impurities
Dissolution and washing away	Soluble impurities pass into the medium and are discarded with it
Mechanical action	Trituration and rubbing separate adherent impurities
Difference in physical property	Sedimentation and flotation separate materials of different density
Difference in volatility	Sublimation and distillation separate volatile from non-volatile
Impregnation	Constituents of the medium associate with the Dravya

1.3.3 Principles underlying Marana

Principle	How it operates
Repeated heating in a defined cycle	Each Puta advances the chemical conversion further
Chemical conversion	The metal becomes an oxide or a sulphide
Intimate mixing before heating	Bhavana brings the reactants into contact at the finest scale
Progressive size reduction	Each cycle of grinding and heating gives finer particles
Surface modification	Organic matter from the Bhavana media associates with the particle surface
Controlled temperature	The type of Puta fixes the temperature reached, and therefore the product formed
Repetition to completion	The process continues until the classical tests are satisfied

1.3.4 The role of the Marana Dravya

Marana is rarely done by heating alone. A Marana Dravya is added, and its nature decides the result.

Marana Dravya used	Type of Marana	Note
Parada, alone or with Gandhaka	Rasa Marana	Widely regarded as producing a superior Bhasma
Gandhaka	Gandhaka Marana	Forms sulphides
Herbal juices and decoctions alone	Herbal Marana	No Parada or Gandhaka used
Combination of the above	Mixed	Common in practice

1.4 Applications in Ayurvedic pharmaceutical manufacturing

1.4.1 Where these processes are applied

Group of Dravyas	Examples	Shodhana required	Marana required
Dhatu — metals	Swarna, Rajata, Tamra, Loha, Naga, Vanga, Yashada	Yes	Yes
Upadhatu	Kansya, Pittala, Sindura, Mandura	Yes	Yes
Maharasa	Abhraka, Vaikranta, Makshika, Vimala, Adrija, Sasyaka, Chapala, Rasaka	Yes	Yes for most
Uparasa	Gandhaka, Gairika, Kasisa, Kankshi, Haratala, Manahshila, Anjana, Kankushtha	Yes	For some
Sadharana Rasa	Kampillaka, Gauripashana, Navasadara, Kaparda, Agnijara, Girisindura, Hingula, Mruddarashringa	Yes	For some
Ratna and Uparatna	Manikya, Mukta, Pravala, Vaidurya and others	Yes	Yes
Visha and Upavisha	Vatsanabha, Bhallataka, Dhattura, Gunja, Jayapala, Karaveera, Ahiphena	Yes	No
Sudha Varga	Kaparda, Shukti, Shankha, Mrigashringa, Godanti	Yes	Yes

> **An important distinction.** Visha and Upavisha Dravyas undergo Shodhana but not Marana. They are plant and animal products, and the purpose of their Shodhana is detoxification, not conversion into a Bhasma. If a question asks which Dravyas require Marana, remember to exclude this group.

1.4.2 Application in a manufacturing unit

Stage in the unit	Activity
Raw material store	Receipt, sampling, Grahya Pariksha, quarantine
Shodhana section	Samanya and Vishesha Shodhana as per SOP, with records
Bhavana section	Grinding with prescribed media, in a Khalva Yantra or mechanical end-runner mill
Puta section	Preparation of Sharava Samputa, Sandhi Bandhana, firing, cooling
In-process quality control	Colour, weight, particle size, Bhasma Pariksha at each stage
Finishing	Amritikarana, sieving, final testing
Packing	Weighing, filling, labelling
Documentation	Batch manufacturing record for every stage

The Schedule T requirements discussed in Module 1 apply to every one of these stages. A Rasashala operating under a licence must have separate, defined areas, written SOPs, and records that allow any finished batch to be traced back to its raw material.

KEY POINTS TO REMEMBER

- Shodhana has four senses: purification, detoxification, modification and potentiation. Give all four.
- Samanya Shodhana is general and common to a group; Vishesha Shodhana is specific to one Dravya.
- Marana destroys the metallic character and converts the metal into a fine, stable, absorbable Bhasma.
- The classical framework is Samskara — the imparting of new qualities by processing.
- Parada is described as having Naisargika Dosha (Visha, Vahni, Mala), Yaugika Dosha (Naga, Vanga) and Aupadhika or Kanchuka Dosha; the Kanchuka enumeration varies between texts.
- Marana Dravya may be Parada, Gandhaka, herbal media, or a combination.
- Visha and Upavisha Dravyas undergo Shodhana but not Marana.
- Without Shodhana making the metal brittle, Marana is not possible at all.

EXAM CORNER — UNIT 1

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Define Shodhana. Explain its four senses.
2. Differentiate between Samanya and Vishesha Shodhana.

3. Define Marana. Why is it called “killing” of the metal?
4. Enumerate the importance of Shodhana.
5. Enumerate the importance of Marana.
6. Write a short note on the principle of Samskara.
7. Enumerate the principles underlying Shodhana.
8. Enumerate the principles underlying Marana.
9. Write a short note on the Doshas of Parada.
10. Write a short note on Marana Dravya and its types.
11. Which groups of Dravyas require Shodhana but not Marana? Give reasons.
12. Describe the sequence from raw Dravya to Bhasma fit for administration.

Likely analytical based questions (20 marks)

1. A pharmacy proposes to shorten the Shodhana of a metal by omitting the Nirvapa steps and grinding the metal mechanically instead. Analyse this proposal on classical and chemical grounds. Predict the effect on the finished Bhasma and justify your conclusion.
2. Analyse the principles underlying Shodhana and Marana. Explain how each principle operates, correlate each with the classical concept of Samskara, and discuss how these principles guide the design of a standard operating procedure.
3. Two Rasashalas prepare the same Bhasma from the same raw material and obtain products that differ in colour, particle size and in the results of Bhasma Pariksha. Analyse every stage at which a difference could arise, and design a control system to prevent it.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **1, 2, 7, 9**

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UNIT 2
SOP OF SHODHANA PROCESSES

SYLLABUS POINTS

2.1 Types of Shodhana procedures (physical, chemical, biological)
2.2 Methods of Shodhana (filtration, sedimentation, distillation, etc.)
2.3 SOPs for Shodhana procedures

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe types of Shodhana procedures (physical, chemical, biological)	1	Lecture	Knows how
Discuss methods of Shodhana and its SOPs	1	Lecture	Knows how
Demonstrate Shodhana procedure of Rasa Dravyas using various methods	10	Practical Training 4.1	Does
Perform Shodhana of Rasa Dravyas with various methods	10	Experiential Learning 4.1	Shows how

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	Yes

2.1 Types of Shodhana procedures

The curriculum classifies Shodhana as physical, chemical and biological. Learn this classification, and be able to place every classical method within it.

2.1.1 Physical Shodhana

Separation depends on a difference in physical property. No chemical change is intended.

Method	Physical property used	Classical example
Hand picking and sorting	Appearance	Removal of visible stones and foreign matter
Washing, Prakshalana	Solubility and density	Washing away adherent earth
Sedimentation and decantation	Density	Settling of heavy matter, pouring off the liquid
Filtration	Particle size	Straining a liquid through cloth
Levigation, Dhavana	Density in a stream of water	Separation of light gangue
Sieving, Chalana	Particle size	Grading of powder
Magnetic separation	Magnetic behaviour	Removal of iron particles
Heating and quenching, Nirvapa	Thermal expansion and contraction	Making a metal brittle
Sublimation and distillation, Patana	Volatility	Purification of volatile Dravyas
Melting and straining	Melting point	Gandhaka melted and strained through cloth

2.1.2 Chemical Shodhana

A chemical reaction is intended. The medium reacts with the Dravya or with its impurity.

Method	Chemical action	Classical example
Treatment with acidic media	Dissolution of oxides and carbonates; chelation	Kanji, Nimbu Swarasa, Amlavetasa
Treatment with alkaline media	Reaction with acidic impurities	Gomutra, Kshara Jala
Treatment with reducing media	Change of oxidation state	Herbal juices rich in phenolics
Formation of a stable compound	Conversion of the toxic form	Parada with Gandhaka
Repeated Bhavana	Impregnation and surface reaction	Bhavana with prescribed Swarasa
Swedana in Dolayantra	Sustained contact with a hot liquid medium	Swedana of Parada, of Gandhaka

2.1.3 Biological Shodhana

Living organisms, or products of biological origin, are used. Enzymes, microbial action and biological macromolecules act on the Dravya.

Method	Biological agent	Classical example
Use of fermented media	Microbial acids from fermentation	Kanji, Takra, Sura
Use of animal products	Proteins, urea, lipids, enzymes	Godugdha, Gomutra, Ghrita, Takra
Use of fresh plant juices	Plant enzymes and living constituents	Swarasa of Kumari, Nirgundi, Bhringaraja
Use of Kulattha and other pulses	Plant proteins and amino acids	Kulattha Kwatha in metal Shodhana

> **How to answer the classification question.** Most classical procedures use more than one type at once. Nirvapa of Tamra in Kanji is physical, through thermal shock, and chemical, through the acid attacking the oxide layer, and biological, because Kanji is a fermented product. State this. A procedure is rarely purely one type, and saying so shows understanding rather than rote learning.

2.2 Methods of Shodhana

2.2.1 The classical methods, defined

Method	Definition
Swedana	Heating the Dravya, tied in a cloth bundle, suspended in a boiling liquid medium in a Dolayantra, for a prescribed period
Mardana	Trituration of the Dravya with a liquid or semi-solid medium in a Khalva Yantra
Nirvapa or Dhalana	Heating the Dravya, usually a metal, to red heat and quenching it in a prescribed liquid; repeated a prescribed number of times
Bharjana	Frying the Dravya, usually in a pan, with or without a medium
Prakshalana	Washing the Dravya with a liquid
Dhavana	Repeated washing, generally with a stream of water
Patana	Sublimation or distillation, in Urdhwa, Adhah or Tiryak direction, using the appropriate Yantra
Avapa	Adding a Dravya into a molten mass
Nimajjana	Immersing the Dravya in a liquid for a prescribed period
Galana	Filtration or straining

Method	Definition
Bhavana	Wet trituration with a liquid medium, repeated and dried between each cycle

2.2.2 The general Shodhana of metals

The standard **Samanya Shodhana of Loha**, the metals, is by **Nirvapa**. The metal is made into thin sheets or small pieces, heated to red heat, and quenched. This is done seven times in each of five media, in the following order.

Order	Medium	Chemical nature
1	Tila Taila	Fixed oil; lipid
2	Takra	Fermented, acidic, protein-containing
3	Gomutra	Alkaline, nitrogen-containing
4	Kanji	Fermented, acidic
5	Kulattha Kwatha	Protein and amino acid rich

Why this sequence works, in chemical terms. The metal is repeatedly cracked by thermal shock, exposing fresh surface each time. That fresh surface is then attacked by media of different chemical character in turn — a lipid, then an acid, then an alkali, then another acid, then a protein solution. Impurities of different kinds are removed at different stages, and organic material from every medium is left associated with the surface.

2.2.3 Specific Shodhana of important Dravyas

The following are given in outline. **Always follow the method of the text prescribed by your department, and cite it.**

Dravya	Outline of the classical method	Chief purpose
Parada	A graded series of Samskaras — Svedana, Mardana, Murchana, Utthapana and others as prescribed, using Kanji, Kshara, Amla, Lashuna, Saindhava and other media	Removal of the described Doshas, including the metallic impurities Naga and Vanga
Gandhaka	Melted in Ghrita and poured through cloth into Godugdha; repeated as prescribed	Removal of earthy and volatile impurities; lipid impregnation
Hingula	Levigated with Nimbu Swarasa or Ardraka Swarasa for the prescribed period	Removal of adherent impurities
Abhraka	Heated to red heat and quenched repeatedly in the prescribed media; then Dhanyabhraka is prepared by rubbing with paddy in a cloth bundle	Making it brittle; separating the flakes from the matrix

Dravya	Outline of the classical method	Chief purpose
Tamra	Samanya Shodhana by Nirvapa, followed by the Vishesha method prescribed, commonly involving Nirvapa in prescribed media	Removal of surface impurities; making it brittle
Loha	Samanya Shodhana by Nirvapa in the five media, followed by Vishesha Shodhana as prescribed, commonly Triphala Kwatha	Removal of impurities and scale
Haratala and Manahshila	Swedana in Dolayantra with the prescribed media, such as Kushmanda Swarasa, Churnodaka or Ardraka Swarasa	Detoxification; removal of free arsenic species
Kasisa and Tuttha	Bhavana or Swedana with the prescribed media	Removal of soluble impurities
Naga and Vanga	Melted and poured into the prescribed medium, repeatedly	Removal of dross; making brittle
Shankha, Shukti, Kaparda	Swedana in Dolayantra with Kanji or Nimbu Swarasa	Removal of organic and adherent matter
Vatsanabha	Swedana in Godugdha in Dolayantra for the prescribed period	Detoxification
Bhallataka	Rubbing with brick powder, then Swedana in Godugdha	Removal of the irritant oil
Gunja	Swedana in Godugdha or Kanji	Detoxification
Jayapala	Removal of the seed coat and embryo, then Swedana in Godugdha and pressing between filter paper	Removal of the drastic principle

2.3 Standard Operating Procedure for Shodhana

2.3.1 What an SOP must contain

An SOP is a written instruction detailed enough that a trained person, following it exactly, produces the same result every time. This is the format you should use in the examination and in the record book.

Section	Content
1. Title and SOP number	Name of the procedure, unique number, version, date
2. Objective	What the procedure achieves
3. Scope	Which Dravya and which stage it applies to
4. Responsibility	Who performs, who supervises, who checks

Section	Content
5. Reference	The classical text with chapter and verse; the pharmacopoeial reference
6. Materials	Dravya with quantity and specification; media with quantity and specification
7. Equipment	Every apparatus with its specification
8. Safety precautions	Hazards, protective equipment, ventilation, disposal
9. Procedure	Numbered steps, with quantity, temperature, duration and repetitions stated
10. In-process checks	What is observed and recorded at each stage
11. Completion criteria	The classical and analytical tests that show the process is complete
12. Yield and reconciliation	Expected yield, permitted range, calculation
13. Storage	Container, condition, labelling
14. Records	The formats to be filled
15. Deviation handling	What to do and record if a step cannot be followed

2.3.2 A worked SOP — Samanya Shodhana of a metal

Title. Samanya Shodhana of Loha by Nirvapa. **SOP number** and **version** as allotted by the department.

Objective. To carry out the general purification of a metal by repeated heating and quenching, so that it becomes brittle and free from surface impurities, and is fit for Vishesha Shodhana.

Scope. Applies to the metals for which Samanya Shodhana by Nirvapa is prescribed.

Responsibility. Performed by the postgraduate scholar; supervised by the teacher in charge of the Rasashala; records checked by the head of department.

Reference. Cite the classical text with chapter and verse, and the Ayurvedic Formulary of India where applicable.

Materials. Metal in thin sheets or small pieces, of stated weight and specification. Tila Taila, Takra, Gomutra, Kanji and Kulattha Kwatha, each of stated quantity and freshly prepared or verified.

Equipment. Musha or crucible; Koshthi or furnace; tongs, Sandamsha; five separate vessels of stated capacity for the media; weighing balance; drying tray; storage containers.

Safety precautions. Heat-resistant gloves and apron; face shield when quenching; adequate ventilation, because quenching red-hot metal in oil and other media produces dense fumes; a fire extinguisher within reach; no water near the hot oil; the vessel must be deep enough to prevent splashing; the metal must be added to the medium, and the medium never poured onto the metal.

Procedure

1. Weigh the metal accurately and record the initial weight.
2. Verify that the metal has been examined for Grahya Lakshana and record the observation.

3. Prepare all five media and keep them ready in separate labelled vessels.
4. Heat the metal in the Musha until it becomes uniformly red hot.
5. Using the Sandamsha, transfer it immediately and completely into Tila Taila.
6. Allow it to cool. Remove, wash and dry.
7. Repeat steps 4 to 6 until seven Nirvapas in Tila Taila are complete.
8. Repeat the whole cycle of seven Nirvapas in Takra, then in Gomutra, then in Kanji, then in Kulattha Kwatha, in that order.
9. Wash the metal thoroughly with hot water after the final Nirvapa.
10. Dry completely.
11. Weigh and record the final weight.
12. Store in a clean, dry, labelled container.

In-process checks. For each Nirvapa, record the appearance of the metal, the appearance of the medium after quenching, and any observation of fumes or reaction. Record the weight after each set of seven.

Completion criteria. The metal has become brittle and can be broken or powdered. The surface is free from scale. The medium at the last Nirvapa shows little change compared with the first.

Yield and reconciliation. Record the initial weight, the final weight, the loss, and the percentage loss. Investigate and record the reason if the loss falls outside the range set by the department.

Records. Batch record with date, operator, quantities, observations and signatures at every stage.

Deviation handling. Any step not followed as written must be recorded, with the reason and the signature of the supervisor, before the batch proceeds.

> **How to reproduce this in the examination.** You will not have time to write all fifteen sections in full. Write the section headings, and fill in the ones that carry the marks — objective, reference, materials, equipment, safety, procedure, in-process checks and completion criteria. The headings themselves show that you know what an SOP is, and that is half of what is being tested.

KEY POINTS TO REMEMBER

- Shodhana is classified as physical, chemical and biological. Most classical procedures are more than one at once.
- Classical methods: Swedana, Mardana, Nirvapa or Dhalana, Bharjana, Prakshalana, Dhavana, Patana, Avapa, Nimajjana, Galana, Bhavana.
- Samanya Shodhana of metals is Nirvapa, seven times each, in Tila Taila, Takra, Gomutra, Kanji and Kulattha Kwatha, in that order.
- Gandhaka Shodhana is by melting in Ghrita and pouring through cloth into Godugdha.
- Dhanyabhraka is prepared by rubbing Shodhita Abhraka with paddy in a cloth bundle.
- Haratala and Manahshila are subjected to Swedana in Dolayantra with the prescribed media.
- An SOP has fifteen sections. Safety and completion criteria are the two most often omitted and most often marked.

EXAM CORNER — UNIT 2

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). There is no ABQ from this unit.

Likely short answer questions (5 marks)

1. Classify Shodhana procedures as physical, chemical and biological, with examples.
2. Define Swedana, Mardana, Nirvapa and Bharjana.
3. Write a short note on Dolayantra and its use in Shodhana.
4. Describe the Samanya Shodhana of metals with the media and the order.
5. Describe the Shodhana of Gandhaka and explain its purpose.
6. Write a short note on Dhanyabhraka.
7. Describe the Shodhana of Haratala.
8. Describe the Shodhana of Vatsanabha.
9. Write a short note on Patana and its types.
10. Enumerate the sections of a standard operating procedure.
11. Write the safety precautions to be observed during Nirvapa.
12. Write a short note on biological Shodhana with examples.

Likely long answer questions (10 marks)

1. Classify the types of Shodhana. Describe the methods of Shodhana in detail with classical examples.
2. Write a complete standard operating procedure for the Samanya Shodhana of a metal.
3. Describe the Shodhana of Parada in detail, with its purpose and the media used.
4. Describe the Shodhana of any three Rasa Dravyas of your choice, with the procedure, purpose and precautions of each.
5. Discuss the importance of a standard operating procedure in Shodhana, and describe its essential sections with reference to Schedule T.

How to attempt

- For every procedure, state the classical reference. An unsourced procedure loses marks at postgraduate level.
- Always give quantity, duration, temperature and number of repetitions. Vague answers are marked as vague.
- Always include a safety paragraph.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **122, 123**

1. **122.** Ilanchezhian R, Roshy Joseph C, Acharya R. Concept of Shodhana (purification or processing) and its importance. 2011.
2. **123.** Vikram ENT, Kamaraj R, Ilavarasan R, Kusuma G, Sundharamoorthy S, Narasimhaji ChV, Marimuthu G. Understanding of Shodhana procedures.

Classical sources: Rasatarangini (reference 1), Rasa Ratna Samuccaya (references 2 and 3), Ayurveda Prakasha (reference 7), Rasendra Chudamani (reference 9). Standards: Ayurvedic Pharmacopoeia of India (reference 156) and the Ayurvedic Formulary of India.

UNIT 3
SOP OF MARANA PROCESS

SYLLABUS POINTS

3.1 Types of Marana procedures (incineration, calcination, etc.)
3.2 Methods of Marana (oxidation, thermal decomposition, etc.)
3.3 SOPs for Marana procedures

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe types of Marana procedures	1	Lecture	Knows how
Discuss methods of Marana and its SOPs	1	Lecture	Knows how
Demonstrate Marana procedure of Rasa Dravyas with different methods	10	Practical Training 4.2	Shows how
Perform Marana procedure of Rasa Dravyas with different methods	10	Experiential Learning 4.2	Shows how

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	Yes

3.1 Types of Marana procedures

3.1.1 Classification by the agent used

Type	Agent	Note
Rasa Marana	Parada, with or without Gandhaka	Traditionally held to give the best Bhasma
Gandhaka Marana	Gandhaka alone	Forms sulphides; widely used
Aushadhi Marana	Herbal juices and decoctions alone	No Parada or Gandhaka; suitable where these are to be avoided
Mixed Marana	Combination of the above	Commonest in practice

3.1.2 Classification by the heating arrangement

Type	Description
Puta Marana	The standard method; the material is sealed in a Sharava Samputa and fired in a pit of cow dung cakes
Kupipakwa	Heating in a glass bottle in a Valuka Yantra; used for Rasa Sindura and similar preparations
Aputa or Aputapaka Marana	Bhasma prepared without a formal Puta, by direct heating
Bhanda Puta	Heating in a closed vessel
Valuka Puta	Heating in a sand bath
Bhudhara Puta	The vessel is placed in a pit in the ground and fired

3.1.3 The types of Puta

The type of Puta is defined by the size of the pit and the quantity of cow dung cakes, and therefore by the temperature reached and the duration of heating.

Puta	Relative size and fuel	Approximate temperature range reported	Typical use
Maha Puta	Largest pit, greatest quantity of fuel	Highest	Ratna and hard Dravyas
Gaja Puta	A pit of about one Hasta each way, filled with cow dung cakes	Roughly 900 to 1000 degrees Celsius	The standard Puta for most metals
Varaha Puta	Smaller than Gaja Puta	Roughly 700 to 800 degrees Celsius	Where less heat is required
Kukkuta Puta	Still smaller	Roughly 600 to 700 degrees Celsius	Softer Dravyas
Kapota Puta	Small	Roughly 400 to 500 degrees Celsius	Delicate materials
Gorakha Puta	Small, using cow dung	Low	As prescribed
Lavaka Puta	Very small	Lowest	Very small quantities

> **Treat the temperature figures with care.** These ranges come from experimental studies in which thermocouples were placed in the pit. They are indicative, not fixed. The temperature actually reached depends on the quantity and dryness of the cow dung cakes, the size and depth of the pit, the weather, the wind and the arrangement of the Sharava. This variability is precisely why the number of Putas prescribed classically is not a fixed number for all conditions, and why the completion is judged by **Bhasma Pariksha** and not by counting alone. Say this in the examination.

3.1.4 Related terms

Term	Meaning
Sharava Samputa	Two shallow earthen saucers, one inverted over the other, enclosing the material
Sandhi Bandhana	Sealing the junction of the two Sharavas with cloth smeared in clay, in several layers, and drying it
Chakrika or Vataka	The material formed into small flat cakes before Puta
Swangasheeta	Self-cooling; the Puta must cool on its own, without being opened or quenched
Ardhachandra	A half-moon shaped arrangement in which the Sharava is placed
Putapaka	The general term for the heating cycle

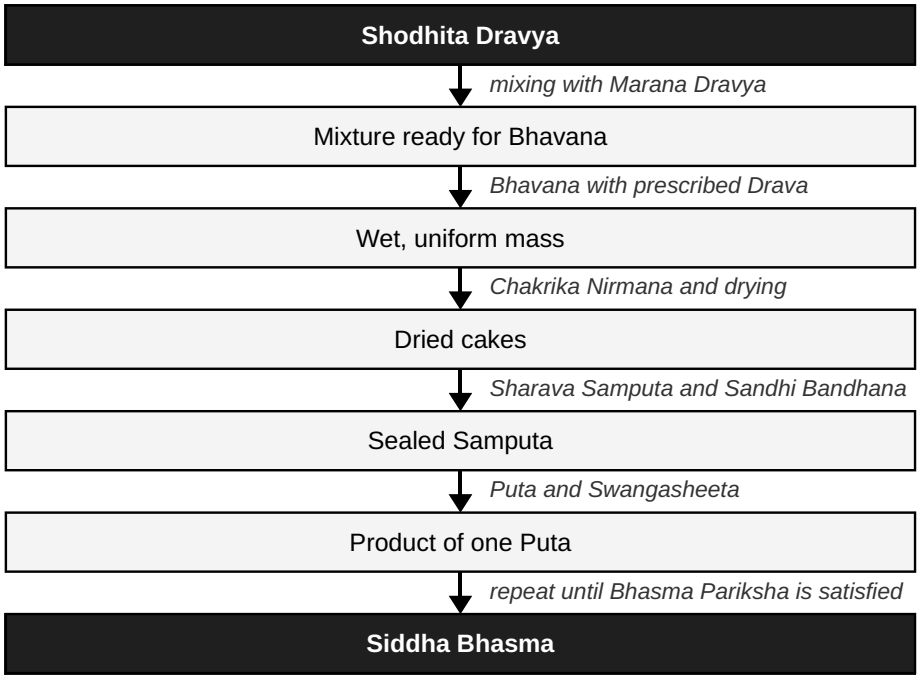
3.2 Methods of Marana

3.2.1 The chemical methods involved

The curriculum names oxidation and thermal decomposition. Both occur, and so do others.

Chemical method	What happens	Example
Oxidation	The metal combines with oxygen to form an oxide	Tamra to copper oxide; Naga to lead oxide; Yashada to zinc oxide
Sulphide formation	The metal combines with sulphur	Parada with Gandhaka to mercuric sulphide
Thermal decomposition	A compound breaks down on heating, giving off gas	Carbonates giving carbon dioxide; sulphates decomposing
Reduction	Organic matter reduces a compound	Where reducing media are used
Sintering and agglomeration	Fine particles fuse partially at high temperature	Occurs when the Puta is too intense
Burning off of organic matter	The Bhavana material chars and burns away	Every Puta
Progressive size reduction	Repeated grinding and firing gives finer particles	The cumulative effect of many Putas

3.2.2 The general Marana cycle



3.2.3 Bhasma Pariksha — the tests of a completed Bhasma

This is one of the most frequently asked topics in the whole course. Learn every test with its method and its scientific meaning.

Test	Method	What it demonstrates	Modern interpretation
Rekhapurnata	The Bhasma fills the lines of the finger when rubbed between thumb and forefinger	Extreme fineness	Very small particle size
Varitara	A pinch sprinkled on still water floats	Fineness and low density; surface property	Fine particles with surface hydrophobicity from organic residue
Unama	A grain of rice placed on the floating Bhasma continues to float	Greater fineness than Varitara alone	Stronger surface film
Nishchandratva	No lustre or shine seen, even under a lens	The metallic state is gone	Absence of free metal
Apunarbhava	The Bhasma, heated with the prescribed reducing agents such as Mitra Panchaka, does not revert to the metal	The metal cannot be recovered	The metal is present as a stable compound, not as free metal

Test	Method	What it demonstrates	Modern interpretation
Niruttha	The Bhasma heated with Rajata or the prescribed metal shows no increase in the weight of that metal	No free metal remains	Confirms complete conversion
Nirdhuma	No fumes are seen on heating	No volatile or organic matter remains	Organic residue is within limits
Nishwadatva	Tasteless	Absence of soluble salts and reactive species	No soluble ionic residue
Slakshnatva and Sukshmatva	Smooth and fine to the touch	Fineness	Small, uniform particle size
Varna	The colour is the one prescribed for that Bhasma	The correct compound has formed	Correct phase and oxidation state
Gata Rasatva	Loss of the original taste of the metal	Complete transformation	Complete conversion
Avami	Does not cause nausea or vomiting	Biological acceptability	Absence of irritant residue
Dantagre Kacha Kacha Shabda	No gritty sound between the teeth	Absence of coarse particles	Uniform fine particle size

> **Apunarbhava and Niruttha are the two decisive tests.** Both are, in modern terms, tests for the absence of free metal — that is, tests that the conversion is chemically complete. If asked which single test proves a Bhasma is finished, name these two and explain why. They test the chemistry, whereas the others chiefly test the physical state.

3.3 Standard Operating Procedure for Marana

3.3.1 A worked SOP — Marana of a metal by Puta

Use the same fifteen-section format given in Unit 2, section 2.3.1. The content specific to Marana is set out below.

Objective. To convert the Shodhita metal into a Bhasma satisfying all the prescribed Bhasma Pariksha.

Reference. The classical text with chapter and verse; the Ayurvedic Formulary of India where applicable.

Materials. Shodhita metal of stated weight. Marana Dravya — Parada, Gandhaka or as prescribed — in the stated proportion. Bhavana Dravya of stated identity and quantity, freshly prepared. Cow dung cakes of stated quantity, well dried. Sharava of stated size. Cloth and clay for Sandhi Bandhana.

Equipment. Khalva Yantra or mechanical end-runner mill; weighing balance; drying tray; Puta pit of stated dimensions; thermocouple or pyrometer if available; sieve of stated mesh; storage containers.

Safety precautions. Work in the open or under a fume hood where the Dravya releases fumes. Wear a mask, gloves and eye protection. Never open a Puta before it has cooled completely on its own. Keep water away from the hot pit. Where mercury or arsenic Dravyas are handled, follow the departmental protocol strictly and use dedicated equipment. Dispose of residues as prescribed, never into the drain.

Procedure

1. Weigh the Shodhita metal accurately. Record the weight.
2. Weigh the Marana Dravya in the prescribed proportion.
3. Triturate the two together in the Khalva Yantra until uniform.
4. Add the Bhavana Dravya in the prescribed quantity and triturate until the prescribed end point is reached.
5. Continue trituration until the mass reaches the consistency at which Chakrika can be formed.
6. Prepare Chakrika of uniform size and thickness. Record the number and the average weight.
7. Dry the Chakrika completely in shade or as prescribed.
8. Place the dried Chakrika in the lower Sharava. Cover with the upper Sharava.
9. Carry out Sandhi Bandhana with the prescribed number of layers of clay-smeared cloth. Dry completely.
10. Arrange the Puta pit as prescribed for the type of Puta required. Record the pit dimensions and the number of cow dung cakes.
11. Place the Samputa, cover with the remaining cow dung cakes, and ignite from the prescribed points.
12. Record the time of ignition. If instrumentation is available, record the temperature profile.
13. Allow the Puta to undergo Swangasheeta. Record the time at which it is opened.
14. Open the Samputa. Record the appearance, colour and condition of the material.
15. Weigh the product. Record the weight and calculate the loss.
16. Triturate the product finely.
17. Carry out the Bhasma Pariksha applicable at this stage and record the result.
18. If the tests are not satisfied, repeat from step 4 for the next Puta.
19. When all the prescribed tests are satisfied, record the total number of Putas.

20. Carry out Amritikarana if prescribed.
21. Sieve, weigh, and store in a clean, dry, labelled container.

In-process checks. After each Puta record the colour, the weight, the appearance, and the result of every Bhasma Pariksha carried out.

Completion criteria. All the prescribed Bhasma Pariksha are satisfied, in particular Rekhapurnata, Varitara, Nishchandratva, Apunarbhava and Niruttha. Where facilities exist, XRD confirms the expected phase and no free metal, and particle size analysis confirms the expected range.

Yield and reconciliation. Record the weight before and after every Puta, the cumulative loss and the final percentage yield.

> **The record after every Puta is the heart of this SOP.** A batch record that states only "twenty-one Putas given" is worthless. A record that gives the colour, weight and test result after each of the twenty-one is a research document, and it is what makes your work publishable.

KEY POINTS TO REMEMBER

- Marana types by agent: Rasa Marana, Gandhaka Marana, Aushadhi Marana, and mixed.
- Puta types by size and fuel: Maha, Gaja, Varaha, Kukkuta, Kapota, Gorakha, Lavaka.
- Gaja Puta is the standard for most metals, at roughly 900 to 1000 degrees Celsius; Kukkuta roughly 600 to 700; Kapota roughly 400 to 500.
- Temperature figures are indicative. Completion is judged by Bhasma Pariksha, not by counting Putas.
- Sharava Samputa is the two-saucer arrangement; Sandhi Bandhana is the clay-cloth sealing; Swangasheeta is self-cooling.
- Chemical methods: oxidation, sulphide formation, thermal decomposition, reduction, sintering, burning off of organic matter.
- Bhasma Pariksha: Rekhapurnata, Varitara, Unama, Nishchandratva, Apunarbhava, Niruttha, Nirdhuma, Nishwadatva, Slakshnatva, Sukshmatva, Varna, Gata Rasatva, Avami.
- **Apunarbhava and Niruttha are the two tests that prove chemical completion — the absence of free metal.**

EXAM CORNER — UNIT 3

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). There is no ABQ from this unit.

Likely short answer questions (5 marks)

1. Classify Marana procedures by the agent used.
2. Enumerate the types of Puta with their approximate temperature ranges.
3. Write a short note on Gaja Puta.
4. Define Sharava Samputa, Sandhi Bandhana and Swangasheeta.

5. Write a short note on Chakrika and its purpose.
6. Enumerate the chemical methods involved in Marana.
7. Write a short note on Rekhapurnata and Varitara.
8. Explain Apunarbhava and Niruttha, and state their scientific significance.
9. Enumerate the Bhasma Pariksha.
10. Write a short note on Kupipakwa Rasayana as a method of Marana.
11. Write the safety precautions to be observed during Puta.
12. Why is the number of Putas not fixed? Explain.

Likely long answer questions (10 marks)

1. Describe the types and methods of Marana in detail, with the chemical changes involved in each.
2. Write a complete standard operating procedure for the Marana of a metal by Puta.
3. Describe the types of Puta with their construction, temperature and application. Add a note on the factors affecting the temperature attained.
4. Describe the Bhasma Pariksha in detail, with the method and the scientific interpretation of each test.
5. Discuss the standardisation of the Marana process, and explain how classical tests and modern analytical methods can be used together.

REFERENCES USED IN THIS UNIT

NCISM reference number for this unit: **124**

1. **124.** Harisharananand. Bhasma Vigyana. Ayurveda Vigyan Granthamala. Amritsar: Panjab Ayurvedic Press; 1954.

Classical sources: Rasatarangini (reference 1), Rasa Ratna Samuccaya (references 2 and 3), Ayurveda Prakasha (reference 7), Rasendra Chudamani (reference 9). Standards: Ayurvedic Pharmacopoeia of India (reference 156) and the Ayurvedic Formulary of India.

UNIT 4
CHEMICAL CHANGES INVOLVED IN SHODHANA AND MARANA PROCESSES

SYLLABUS POINTS

4.1 Chemical changes involved in Shodhana procedures
4.2 Chemical changes involved in Marana procedures

Session learning objective (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Discuss chemical changes involved in Shodhana and Marana procedures	1	Lecture	Knows how

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

> This unit is short in the syllabus but heavy in the examination. It is one of only two units in this module that can be set as a twenty-mark Analytical Based Question. Everything you learned in Module 3 is applied here.

4.1 Chemical changes involved in Shodhana

4.1.1 The categories of change

Category	What happens	Where it is seen
Removal of soluble impurities	Salts and soluble matter dissolve into the medium and are discarded with it	Washing, Swedana, Prakshalana
Acid-base reaction	Acidic media dissolve basic surface oxides and carbonates; alkaline media act on acidic impurities	Nirvapa in Kanji; Swedana in Gomutra
Chelation	Organic acids, tannins and amino acids in the media bind metal ions at two or more points	All herbal and fermented media
Reduction or oxidation	Phenolics and reducing sugars change the oxidation state at the surface	Bhavana with herbal juices

Category	What happens	Where it is seen
Thermal shock and micro-cracking	Rapid heating and quenching cracks the metal and exposes fresh surface	Nirvapa
Physical delamination	Layered minerals split along their planes	Abhraka
Removal of volatile impurities	Volatile matter is driven off	Heating, Patana
Formation of a new compound	The Dravya reacts to give a different, safer compound	Parada with Gandhaka
Surface impregnation	Organic molecules from the medium attach to the exposed surface	Every medium used

4.1.2 Worked examples

Dravya	Classical procedure	Chemical account
Parada	Graded Samskaras with Kanji, Kshara, Amla, Lashuna, Saindhava and other media	The metallic impurities described as Naga and Vanga form amalgams with mercury; the prescribed treatments separate them. Acidic and alkaline media remove surface oxides and adherent matter. Trituration with organic material disperses the mercury finely and increases contact area
Gandhaka	Melted in Ghrita and poured through cloth into Godugdha	Sulphur melts at about 115 degrees Celsius. Earthy and mineral impurities do not melt, and are retained on the cloth. Rapid quenching in milk gives a fine, brittle solid rather than large crystals. Lipid from the Ghrita is retained on the surface
Hingula	Levigated with Nimbu Swarasa or Ardraka Swarasa	Hingula is mercuric sulphide. Citric acid dissolves adherent carbonate and oxide impurities and chelates free metal ions. Trituration reduces particle size
Abhraka	Repeated heating and quenching, then rubbing with paddy	Mica is a layered sheet silicate. Rapid heating drives off interlayer water and cracks the sheets apart. Mechanical rubbing then separates the delaminated flakes from the matrix

Dravya	Classical procedure	Chemical account
Tamra	Nirvapa in the prescribed media	The surface oxide layer, chiefly copper oxides, dissolves in the acidic media. Thermal shock cracks the metal. Fresh copper surface is exposed and partly bound by organic constituents of the media
Loha	Nirvapa in the five media, then Triphala Kwatha	Iron oxide scale is removed by the acidic media. Tannins and gallic acid of Triphala are strong iron binders and form dark iron-polyphenol complexes; the darkening observed classically is exactly this
Haratala and Manahshila	Swedana in Dolayantra with Kushmanda Swarasa or the prescribed media	These are arsenic sulphides. Prolonged contact with the hot aqueous medium removes soluble arsenic species and adherent impurities. The bulk sulphide, which is poorly soluble, is retained
Naga and Vanga	Melted and poured repeatedly into the prescribed medium	Dross and oxide skim off on melting. Pouring the molten metal into a liquid gives fine, brittle granules instead of a solid mass
Shankha and Shukti	Swedana with Kanji or Nimbu Swarasa	These are calcium carbonate. The acid attacks the surface, removes adherent organic matter, and makes the material friable

> **The Triphala and iron example is worth memorising.** Tannins and gallic acid form intensely coloured complexes with iron. This is the same chemistry as iron-gall ink. When the classical text says that Loha treated with Triphala turns black, it is describing a real and predictable chemical reaction, and you can name it. Examples of this quality are what raise an answer.

4.2 Chemical changes involved in Marana

4.2.1 The categories of change

Category	What happens
Oxidation	The metal combines with atmospheric oxygen to form an oxide
Sulphide formation	The metal combines with sulphur from Gandhaka
Thermal decomposition	Carbonates, sulphates and hydroxides break down, releasing gas
Reduction	Charring organic matter can reduce a compound to a lower oxide
Combustion of organic matter	The Bhavana material chars and burns away with each Puta
Particle size reduction	Repeated grinding and firing gives progressively finer particles
Sintering	If the Puta is too intense, fine particles fuse into lumps — an unwanted change
Phase transformation	One crystalline form converts to another
Surface association	Residual organic material remains associated with the particle surface

4.2.2 The reported chemistry of important Bhasmas

The table below gives the phases most commonly reported in published characterisation studies using XRD and related methods. The exact phases obtained depend on the method, the media and the temperature, so results differ between studies.

Bhasma	Starting material	Marana Dravya generally used	Phases commonly reported
Tamra Bhasma	Copper	Gandhaka, with or without Parada	Copper sulphide and copper oxides
Lauha Bhasma	Iron	Herbal media, Gandhaka	Iron oxides — chiefly ferric oxide, sometimes mixed oxide
Yashada Bhasma	Zinc	Herbal media	Zinc oxide
Naga Bhasma	Lead	Herbal media, Gandhaka	Lead oxide, lead sulphide
Vanga Bhasma	Tin	Herbal media	Tin oxide
Abhraka Bhasma	Mica	Herbal media, with repeated Puta	Complex silicate phases, with iron and magnesium
Mandura Bhasma	Iron rust	Herbal media	Iron oxides
Swarna Bhasma	Gold	Parada and Gandhaka	Gold, in very fine particulate form
Rajata Bhasma	Silver	Herbal media, Gandhaka	Silver and silver sulphide
Kajjali	Parada with Gandhaka	Trituration alone, no fire	Black mercuric sulphide
Rasa Sindura	Kajjali	Kupipakwa	Red mercuric sulphide

> **The Kajjali to Rasa Sindura change is the clearest single example of a phase transformation in Rasashastra, and it is worth learning in detail.**
>
> Mercuric sulphide exists in two crystalline forms. The **black** form is what results from trituration of Parada with Gandhaka at room temperature — this is Kajjali. On controlled heating in the Kachakupi during Kupipakwa, it sublimes and redeposits as the **red** form, which is the same phase as natural cinnabar, Hingula — this is Rasa Sindura.
>
> Same chemical formula. Different crystal structure. Different colour. This is textbook polymorphism, and the classical process brings it about by exactly the means a chemist would choose — sublimation and controlled recondensation. Note also that Swarna Bhasma is one case where the metal itself, not a compound, is the reported product; it is chemically very unreactive, and the change is chiefly one of physical state and particle size.

4.2.3 The particle size change

This is one of the most consistently reported findings in Bhasma research.

Stage	Typical particle size
Raw metal or mineral	Bulk, in millimetres or larger
After Shodhana	Coarse powder, in hundreds of micrometres
After the early Putas	Tens of micrometres
After the prescribed number of Putas	Micrometre to sub-micrometre, with a nanometre-scale fraction commonly reported

The reduction is progressive, and it comes from the combination of repeated wet grinding during Bhavana and repeated firing. Each Puta breaks down aggregates further, and the organic material from the Bhavana media, coating the particles, limits their re-growth and their clumping. This is the same capping mechanism discussed in Module 3, Unit 3.

4.2.4 The weight change and what it tells you

A common examination question, and a common point of confusion.

Process	Effect on weight	Reason
Metal converting to oxide	**Increase**	Oxygen is added to the metal
Metal converting to sulphide	**Increase**	Sulphur is added
Organic Bhavana matter burning off	**Decrease**	Carbon dioxide and water are lost
Volatile constituent lost	**Decrease**	Material leaves as vapour
Material lost in handling	**Decrease**	Physical loss at each transfer

The net weight change is the sum of these, and it varies with the Dravya and the method. **Record the weight after every Puta.** A weight that stops changing is one indication that the chemical conversion is approaching completion, and it is a useful in-process control that costs nothing.

4.2.5 What the analytical methods show

Method	What it demonstrates about the transformation
XRD	Which crystalline phases are present before and after; loss of the metal peaks; appearance of oxide or sulphide peaks; crystallite size from peak broadening
SEM	Particle shape, size and surface texture
EDAX	Which elements are present, and in what proportion
TEM	Particle size at the nanometre scale
FTIR	Metal-oxygen and metal-sulphur bonds; the organic functional groups remaining
TGA and DTA	Weight loss on heating; the temperature at which organic matter burns off; thermal stability
Particle size analyser	The size distribution of the finished Bhasma
ICP-MS or AAS	Elemental content, including any residual heavy metal contaminant
XPS	The oxidation state of the metal at the particle surface

> **What to write when asked to prove the chemical change.** No single method proves it. XRD shows the phase change. TGA shows the organic content. SEM and TEM show the size. EDAX gives the elements. FTIR shows the bonding. State that a set of methods is required, say what each one contributes, and say clearly what each one cannot show. That answer is worth far more than a list of instruments.

KEY POINTS TO REMEMBER

- Shodhana changes: removal of soluble impurities, acid-base reaction, chelation, redox, thermal shock, delamination, removal of volatiles, new compound formation, surface impregnation.
- Marana changes: oxidation, sulphide formation, thermal decomposition, reduction, combustion of organic matter, size reduction, sintering, phase transformation, surface association.
- Triphala darkens iron because tannins and gallic acid form coloured iron-polyphenol complexes — the chemistry of iron-gall ink.
- Gandhaka Shodhana works because sulphur melts at about 115 degrees Celsius while its earthy impurities do not.
- Abhraka Shodhana works by driving off interlayer water and delaminating the mica sheets.
- **Kajjali is black mercuric sulphide; Rasa Sindura is the red form. Same formula, different crystal structure — polymorphism brought about by sublimation and recondensation.**
- Oxidation and sulphide formation increase weight; loss of organic matter and volatiles decreases it.
- Particle size falls progressively with each Puta, into the micrometre and sub-micrometre range.

- No single analytical method proves the transformation. A set is required.

EXAM CORNER — UNIT 4

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Enumerate the chemical changes occurring during Shodhana.
2. Enumerate the chemical changes occurring during Marana.
3. Explain the chemical basis of the Shodhana of Gandhaka.
4. Explain the chemical basis of the Shodhana of Abhraka.
5. Explain why Loha turns black when treated with Triphala.
6. Explain the chemical change from Kajjali to Rasa Sindura.
7. Write a short note on the particle size change during Marana.
8. Explain the weight changes observed during Marana.
9. Write a short note on sintering and why it is undesirable.
10. What does XRD show about a Bhasma, and what does it not show?
11. Write a short note on the role of Bhavana Dravya in the chemical change of Marana.
12. Explain the chemical basis of the Shodhana of Haratala.

Likely analytical based questions (20 marks)

1. Analyse in detail the chemical changes taking place at every stage from the raw metal to the finished Bhasma. Justify each with the underlying chemistry, and state which analytical method would demonstrate each change.
2. A Bhasma passes Rekhapurnata and Varitara but fails Apunarbhava. Analyse what this combination of results means chemically, identify the probable causes, and describe the corrective action and the additional analysis you would carry out.
3. Analyse the classical Shodhana media used for metals in the light of their chemical constituents. Explain what each medium contributes, why the prescribed order is chemically sensible, and design an experiment to determine whether the order matters.

REFERENCES USED IN THIS UNIT

NCISM reference number for this unit: **125**

1. **125.** Maurya SK, Seth A, Laloo D, Singh NK, Gautam DN, Singh AK. Shodhana: an Ayurvedic process for detoxification and modification of therapeutic activities of poisonous medicinal plants.

Additional sources: Bhasma Vigyana (reference 124); Rasatarangini (reference 1); Rasa Ratna Samuccaya (references 2 and 3); Ayurvedic Pharmacopoeia of India (reference 156); Protocol for Testing of Ayurvedic, Siddha and Unani Medicines, Department of AYUSH (reference 163); and the inorganic and crystallography texts of Module 3 (references 105 to 108).

UNIT 5
SHODHANA AND MARANA — CASE STUDIES, TROUBLESHOOTING AND EMERGING TRENDS

SYLLABUS POINTS

5.1 Case studies of Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing
5.2 Troubleshooting and optimisation of Shodhana and Marana procedures
5.3 Future directions and emerging trends in Shodhana and Marana procedures

Session learning objectives (Table 3 of the curriculum)

Objective	Hours	Mode	Level
Describe troubleshooting and optimisation of Shodhana and Marana procedures	1	Lecture	Knows how
Discuss future directions and emerging trends in Shodhana and Marana procedures	1	Lecture	Knows how
Analyse and write Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing through case studies	6	Experiential Learning 4.3	Knows how

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

5.1 Case studies

5.1.1 What a case study is, in this context

A case study here means a documented account of an actual Shodhana or Marana carried out in a Rasashala or a manufacturing unit, analysed for what happened, what varied, and what was learned.

5.1.2 How to write a case study

Section	Content
Title	The Dravya, the procedure and the text followed
Objective	What was to be achieved
Materials	Source, quantity, and Grahya Pariksha of the raw material
Method	The procedure as actually carried out, not as it should have been
Observations	Recorded at every stage, with dates, weights and appearance
Analytical findings	Bhasma Pariksha results and instrumental data
Deviations	Every departure from the SOP, with the reason
Discussion	Why the result was what it was
Conclusion	What was achieved and what would be done differently

> **The most valuable case study is one that did not go smoothly.** A batch that failed a test, or lost more weight than expected, or took more Putas than the text prescribes, teaches far more than a batch that worked. Record the failures fully. Most published Rasashastra research reports only successes, and that is precisely why the literature is weaker than it should be.

5.1.3 Suggested case studies for your record book

Case study	What it demonstrates
Shodhana of Parada, full Samskara sequence	The most elaborate Shodhana; weight loss at each stage
Shodhana of Gandhaka	A simple procedure with a clear chemical basis and measurable yield
Shodhana and Marana of Tamra	A complete metal-to-Bhasma sequence with clear colour change
Marana of Abhraka with a large number of Putas	The relationship between Puta number and particle size
Preparation of Kajjali and its conversion to Rasa Sindura	Phase transformation, demonstrable by colour and by XRD
Preparation of Lauha Bhasma	Oxidation state and the effect of Triphala
Comparative study of the same Bhasma by two different classical methods	The effect of method on the finished product
A batch that failed a Bhasma Pariksha	Troubleshooting in practice

5.2 Troubleshooting and optimisation

5.2.1 Problems in Shodhana

Problem	Probable cause	Corrective action
Metal does not become brittle	Insufficient Nirvapa; metal not heated to full red heat; pieces too thick	Increase the number of Nirvapas; check the furnace temperature; use thinner sheets
Excessive weight loss	Metal splashing out during quenching; too much oxidation; loss in transfer	Use a deeper vessel; reduce exposure to air between furnace and quench; weigh at every stage
Medium exhausted or contaminated	Same medium reused too many times	Prepare fresh medium as prescribed; record the number of uses
Product still shows impurities	Insufficient repetitions; wrong medium; poor washing	Repeat as prescribed; verify the medium against the text; wash thoroughly
Fumes and irritation during Nirvapa	Normal with oil and organic media, but excessive if ventilation is poor	Work in the open or under extraction; use a mask; never lean over the vessel
Variable results between batches	Media not standardised; quantities not measured; operator variation	Fix the SOP; standardise media preparation; train and record

5.2.2 Problems in Marana

Problem	Probable cause	Corrective action
Bhasma fails Varitara	Particle size still too coarse; insufficient Putas; sintering has occurred	Continue with further Bhavana and Puta; if sintered, reduce the Puta intensity
Bhasma fails Nishchandratva — lustre still visible	Free metal remains; conversion incomplete	Give further Putas; check the quantity of Marana Dravya
Bhasma fails Apunarbhava or Niruttha	Free metal present; conversion chemically incomplete	Give further Putas; review the Marana Dravya proportion and the Puta temperature
Colour not as prescribed	Wrong temperature; wrong phase formed; contamination	Verify the Puta type; check the pit dimensions and fuel quantity; check the raw material
Material sintered into hard lumps	Puta too intense; too many cow dung cakes; pit too small	Use a lower Puta; increase the pit size; reduce the fuel
Chakrika crumble before firing	Insufficient Bhavana; mass too dry when formed	Continue Bhavana to the correct consistency; form the Chakrika at the right stage

Problem	Probable cause	Corrective action
Sharava cracks during Puta	Chakrika not fully dried; Sandhi Bandhana incomplete; heating too rapid	Dry completely; apply the prescribed number of clay-cloth layers and dry each; allow gradual ignition
Excessive weight loss per Puta	Material escaping through a crack; over-intense Puta	Check the seal; reduce the Puta intensity
Number of Putas far exceeds the classical figure	Cow dung cakes of poor quality or damp; pit too shallow; weather	Use well-dried fuel; standardise the pit; record the conditions
Batches differ between operators	Judgement-based steps not defined	Convert every judgement-based step into a measurable one in the SOP

5.2.3 Optimisation — turning judgement into measurement

The classical procedure relies on trained judgement. Optimisation means converting each judgement into something that can be measured and recorded, without abandoning the classical requirement.

Classical judgement	Measurable equivalent to record
“Heat until red hot”	Temperature measured with a pyrometer
“Bhavana until properly done”	Time of trituration, and quantity of medium used
“Chakrika of suitable size”	Diameter, thickness and weight recorded
“Gaja Puta”	Pit dimensions, number and weight of cow dung cakes, temperature profile if available
“Until Bhasma Pariksha is satisfied”	Each test recorded after each Puta, with the result
“Fine powder”	Sieve number passed, or particle size analysis
“Correct colour”	Recorded against a colour standard, or photographed under fixed lighting

> **This table is the single most useful thing in Unit 5.** It is the answer to almost any question on optimisation or standardisation, and it is also the framework for your own dissertation work. The classical instruction is not replaced. It is made recordable, so that the same batch can be made again and so that another worker can verify it.

5.3 Future directions and emerging trends

Trend	What it involves	What it offers, and what to watch
Instrumental characterisation as routine	XRD, SEM-EDAX, TEM, FTIR, TGA-DTA, PSA, ICP-MS applied to every batch	Makes the product verifiable and publishable. Cost and access remain the obstacle
Nanoscience approach	Studying Bhasma as a nanomaterial, with the tools of nanoscience	A strong and productive research direction, developed in Module 8

Trend	What it involves	What it offers, and what to watch
Electric muffle furnace with a programmed profile	Replacing the traditional Puta with a controlled electrical heating cycle	Gives reproducible temperature and time. But the classical Puta involves a particular heating and cooling profile and a particular atmosphere, and whether the product is equivalent must be demonstrated, not assumed
Mechanisation of Bhavana	End-runner mills and ball mills in place of the Khalva Yantra	Consistency and reduced labour. The effect on the finished product must be studied and reported
Standardised SOPs across manufacturers	Common, published SOPs for each Bhasma	Reduces batch variation. Requires agreement between texts, which differ
Safety and toxicity studies	Systematic preclinical and clinical safety data	Essential for acceptance and for regulatory approval
Pharmacokinetic and bioavailability studies	Studying absorption, distribution and excretion of Bhasma	The largest gap in the present evidence base
Pharmacovigilance	Systematic collection of adverse event data	Required by regulation and by good practice
Green chemistry correlation	Relating classical Bhavana to modern green synthesis	Strengthens the scientific standing of the classical method
Heavy metal limits and export requirements	Meeting the limits set by importing countries	A real commercial and regulatory constraint on metal-containing formulations, which must be addressed openly and with data
Digital documentation and traceability	Electronic batch records with audit trails	Applies Module 2 directly to the Rasashala
Reference standards and reference Bhasma	Establishing authenticated reference samples	Would allow any laboratory to check its product against a standard

How to answer the "future directions" question well. Do not simply list instruments. Group your answer under three headings: **standardisation** of the process, **characterisation** of the product, and **evidence** of safety and efficacy. Say plainly that the evidence heading is where the largest gap lies. That is an honest and well-informed answer, and it is the one a postgraduate examiner wants.

KEY POINTS TO REMEMBER

- A case study records what actually happened, including deviations and failures.
- Failure of Varitara indicates coarse particles or sintering. Failure of Nishchandratva, Apunarbhava or Niruttha indicates free metal and incomplete conversion.
- Sintering is caused by too intense a Puta and produces hard lumps.
- Sharava cracking is caused by damp Chakrika, incomplete Sandhi Bandhana or too rapid heating.

- Optimisation means converting each classical judgement into a measurable, recordable quantity.
- Future directions fall under three headings: standardisation, characterisation, and evidence of safety and efficacy.
- Replacing the Puta with an electric furnace must be validated by comparison, not assumed to be equivalent.
- Pharmacokinetic data on Bhasma is the largest gap in the present evidence base.

EXAM CORNER — UNIT 5

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Describe how a case study of a Marana procedure should be written.
2. Enumerate the common problems encountered during Shodhana and their remedies.
3. Enumerate the common problems encountered during Marana and their remedies.
4. What are the probable causes if a Bhasma fails Varitara?
5. What does failure of Apunarbhava indicate, and what is the corrective action?
6. Write a short note on sintering and how it is prevented.
7. Why does a Sharava crack during Puta, and how is it prevented?
8. Write a short note on the optimisation of the Marana process.
9. How would you convert judgement-based classical steps into measurable ones?
10. Write a short note on emerging trends in Shodhana and Marana.
11. Discuss the use of an electric muffle furnace in place of the traditional Puta.
12. What are the present gaps in the evidence base for Bhasma preparations?

REFERENCES USED IN THIS UNIT

Sources for this unit: Rasatarangini (reference 1); Rasa Ratna Samuccaya (references 2 and 3); Bhasma Vigyana (reference 124); Protocol for Testing of Ayurvedic, Siddha and Unani Medicines, Department of AYUSH (reference 163); Ayurvedic Pharmacopoeia of India, PCIM&H, Ministry of AYUSH (reference 156); CCRAS Guideline of Drug Development (reference 49); and Schedule T of the Drugs and Cosmetics Rules, 1945.

MODULE 4
PRACTICAL TRAINING ACTIVITIES

Module 4 carries **20 hours of practical training**, in two extended activities, as listed in Table 4 of the curriculum.

No	Practical name	Hours
4.1	Shodhana procedure of Rasa Dravyas using various methods	10
4.2	Marana procedure of Rasa Dravyas with different methods	10
	Total	**20**

Practical Training 4.1 — Shodhana of Rasa Dravyas by various methods

Aim. To perform the Shodhana of different Rasa Dravyas by the prescribed methods, and to record the process and the changes observed.

Requirements. Raw Dravyas selected in consultation with the teacher, covering different methods; Dolayantra; Khalva Yantra; Musha; Koshthi or furnace; Sandamsha; vessels for the media; Tila Taila, Takra, Gomutra, Kanji, Kulattha Kwatha and the specific media required; weighing balance; drying trays; storage containers; safety equipment.

Selection of Dravyas. Choose at least five, so that different methods are covered:

Dravya	Method demonstrated
A metal such as Tamra or Loha	Nirvapa
Gandhaka	Melting, straining, quenching
Abhraka	Nirvapa and Dhanyabhraka preparation
A Visha Dravya such as Vatsanabha	Swedana in Dolayantra
Shankha or Shukti	Swedana with acidic medium

Procedure

1. For each Dravya, first record the Grahya Pariksha — the classical characters of a genuine sample.
2. Record the source, the quantity and the initial weight.
3. Locate the procedure in the prescribed classical text. Record the text, chapter and verse.
4. Write the SOP before beginning the work, not after.
5. Prepare all the media required, fresh, and record their quantity and method of preparation.
6. Carry out the procedure exactly as written.
7. Record the observations at every stage — appearance, colour, odour, the state of the medium, any fumes, and the weight.
8. On completion, wash, dry, weigh and record.
9. Calculate the percentage loss or gain.

10. Record the characters of the Shodhita Dravya and compare with the classical description of a properly purified sample.
11. Store in a labelled container with the date and batch number.

Observation tables to draw. One per Dravya, with columns for stage, medium, duration, observation and weight. One summary table comparing all five Dravyas.

Result. Record, for each Dravya, whether Shodhana was completed as prescribed, the yield, and the observed change.

Precautions. Read Unit 2 section 2.3.2 before beginning. Work in the open or with extraction when quenching. Never lean over the vessel. Wear a face shield, gloves and apron. **Handle Parada, Haratala, Manahshila and other toxic Dravyas only under direct supervision, with dedicated equipment, and dispose of all residues as directed. Never wash such residues into the sink.** Keep a record of every quantity taken and every quantity discarded.

Practical Training 4.2 — Marana of Rasa Dravyas by different methods

Aim. To perform the Marana of Rasa Dravyas by different methods, and to record the process, the changes and the Bhasma Pariksha at each stage.

Requirements. Shodhita Dravyas from Practical 4.1; Marana Dravya as prescribed; Bhavana Dravya; Khalva Yantra or end-runner mill; Sharava; cloth and clay for Sandhi Bandhana; Puta pit; cow dung cakes; Kachakupi and Valuka Yantra if Kupipakwa is being demonstrated; weighing balance; sieves; safety equipment.

Selection of methods. Cover at least three:

Method	Suggested Dravya
Puta Marana with Gandhaka	Tamra or Loha
Puta Marana with herbal media only	Abhraka or Mandura
Kajjali preparation, and Kupipakwa if permitted	Parada and Gandhaka

Procedure

1. Verify that the Shodhana of each Dravya is complete, and record the verification.
2. Write the SOP for each Marana before starting.
3. Weigh the Shodhita Dravya and the Marana Dravya in the prescribed proportion.
4. Triturate to a uniform mixture.
5. Add the Bhavana Dravya and triturate as prescribed. Record the time and quantity.
6. Prepare Chakrika of uniform size. Record the number, diameter and weight.
7. Dry completely and record the drying time.
8. Arrange the Sharava Samputa and carry out Sandhi Bandhana with the prescribed number of layers. Dry.
9. Prepare the Puta pit. **Record the pit dimensions and the number and weight of cow dung cakes.**
10. Fire the Puta. Record the time of ignition and, if instrumentation permits, the temperature profile.
11. Allow Swangasheeta. Record the time of opening.

12. Open, observe, weigh and record.
13. Triturate the product and carry out the applicable Bhasma Pariksha. Record every result.
14. Repeat from step 5 for each subsequent Puta, recording all of the above every time.
15. Continue until all the prescribed Bhasma Pariksha are satisfied.
16. Record the total number of Putas.
17. Carry out Amritikarana if prescribed.
18. Sieve, weigh, record the final yield, and store in a labelled container.

Observation table to draw

Puta number	Bhavana Dravya and quantity	Weight before	Weight after	Colour	Rekhapurnata	Varitara	Nishchandratva	Apunarbhava	Niruttha	Re

Result. Record the total number of Putas required, the final yield, and the complete Bhasma Pariksha profile of the finished product.

Precautions. Never open a Puta before it has cooled completely on its own. Keep water away from the hot pit. Wear a mask when opening the Samputa and when triturating the fired product, as fine particles become airborne. Where Parada or arsenic Dravyas are used, work only under supervision with dedicated equipment, and account for every gram taken and every gram recovered. Dispose of residues as directed by the department.

MODULE 4
EXPERIENTIAL LEARNING ACTIVITIES

Module 4 carries **30 hours** under experiential learning, including modular assessment. Three activities are prescribed in Table 5 of the curriculum.

No	Experiential learning activity	Hours
4.1	Perform Shodhana of Rasa Dravyas with various methods	10
4.2	Perform Marana procedure of Rasa Dravyas with different methods	10
4.3	Analyse and write Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing through case studies	6
	Total	**26**

Experiential Learning 4.1 — Perform Shodhana of Rasa Dravyas

What you have to do. Move from performing Shodhana under demonstration to performing it independently, and to accounting for what you observe.

Method. Take responsibility for a complete Shodhana of at least two Dravyas, from writing the SOP to storing the finished material. Work independently, with the teacher supervising rather than directing.

Questions your report must answer

- What was the percentage loss at each stage, and what accounts for it?
- Did the appearance of the medium change from the first Nirvapa to the last? What does that tell you?
- Where did your procedure differ from the text, and why?
- If you repeated this Shodhana, what would you change?
- How would you convince an examiner, from your records alone, that the Shodhana was properly done?

How to record it. Your written SOP, the complete batch record, the observation tables, the yield calculation, photographs at each stage, and the written answers.

Experiential Learning 4.2 — Perform Marana of Rasa Dravyas

What you have to do. Carry a Marana through to completion, independently, with a full record after every Puta.

Method. Take one Dravya through its complete Marana. Do not stop at a fixed number of Putas. Continue until the Bhasma Pariksha are satisfied, and record how many Putas that actually took.

Questions your report must answer

- How many Putas were needed, and how does that compare with the classical figure?

- Which test was the last to be satisfied, and what does that tell you about the process?
- How did the colour and weight change from Puta to Puta? Plot both.
- Did the weight stabilise before the tests were satisfied, or after?
- What conditions in your Puta could have caused it to differ from the classical prescription?

> **Plot the weight and the colour against the Puta number.** Two simple graphs turn a routine practical record into a piece of data. Very few students do this, and it is immediately noticed by an examiner.

How to record it. The SOP, the per-Puta record table, the two graphs, the complete Bhasma Pariksha profile, photographs, and the written answers.

Experiential Learning 4.3 — Case study analysis and writing

What you have to do. Analyse published and departmental case studies of Shodhana and Marana, then write your own to the same standard.

Method

1. Obtain at least four documented accounts — from published papers, from departmental records, or from a manufacturing unit.
2. For each, extract the Dravya, the text followed, the method, the number of Putas, the yield, the tests applied and the analytical data reported.
3. Prepare a comparison table.
4. Identify what each account records well and what it omits.
5. Note in particular whether deviations and failures were reported at all.
6. Now write up one of your own batches to a standard higher than any of them.

Comparison table to prepare

Case	Dravya	Text followed	Method	Putas	Yield	Tests applied	Analytical data	Deviations reported

How to record it. The comparison table, a written critique of the four accounts, and your own case study written in full.

MODULE 4
VIVA POINTERS AND QUESTION BANK

Viva pointers

On concepts

- Define Shodhana. Give all four of its senses.
- Differentiate Samanya and Vishesha Shodhana.
- Why is Marana called “killing”?
- Which Dravyas undergo Shodhana but not Marana?
- What are the Doshas of Parada?

On procedures, at the bench

- Show me the Dolayantra. What is it used for?
- Show me the Khalva Yantra. How do you judge that Bhavana is complete?
- What are the five media of Samanya Shodhana, and in what order?
- How is Gandhaka Shodhana done, and why is it strained through cloth?
- What is Dhanyabhraka?
- Demonstrate Sandhi Bandhana. How many layers, and why?
- What is Swangasheeta, and why must it not be hurried?

On Puta and Bhasma Pariksha

- Name the types of Puta in order of temperature.
- What are the dimensions of a Gaja Puta pit?
- Show me Rekhapurnata. Show me Varitara.
- What does Apunarbhava test for? What does Niruttha test for?
- Which two tests prove the chemistry is complete, and why?
- Why is the number of Putas not fixed?

On chemistry

- What chemical change occurs during Marana of Tamra?
- Why does Loha turn black with Triphala?
- What is the difference between Kajjali and Rasa Sindura?
- Why does the weight sometimes increase during Marana?
- What would XRD of this Bhasma show?

On safety

- What precautions do you take when handling Parada?
- What do you do with the residue after Shodhana of Haratala?
- Why must a Puta never be opened before it cools?

Module 4 question bank

Analytical Based Questions (20 marks) — from Units 1 and 4

1. A pharmacy proposes to shorten the Shodhana of a metal by omitting Nirvapa and grinding mechanically instead. Analyse the proposal on classical and chemical grounds, predict the effect on the finished Bhasma, and justify your conclusion.
2. Analyse the principles underlying Shodhana and Marana, correlate each with the concept of Samskara, and discuss how these principles guide the design of an SOP.
3. Two Rasashalas prepare the same Bhasma from the same raw material and obtain products that differ. Analyse every stage at which a difference could arise, and design a control system to prevent it.
4. Analyse in detail the chemical changes from the raw metal to the finished Bhasma, justify each with the underlying chemistry, and state which analytical method demonstrates each change.
5. A Bhasma passes Rekhapurnata and Varitara but fails Apunarbhava. Analyse what this means chemically, identify the probable causes, and describe the corrective action and further analysis.
6. Analyse the classical Shodhana media for metals in the light of their chemical constituents, explain why the prescribed order is chemically sensible, and design an experiment to test whether the order matters.

Long Answer Questions (10 marks) — from Units 2 and 3

1. Classify the types of Shodhana. Describe the methods of Shodhana in detail with classical examples.
2. Write a complete standard operating procedure for the Samanya Shodhana of a metal.
3. Describe the Shodhana of Parada in detail, with its purpose and the media used.
4. Describe the Shodhana of any three Rasa Dravyas, with the procedure, purpose and precautions of each.
5. Discuss the importance of an SOP in Shodhana, with reference to Schedule T.
6. Describe the types and methods of Marana in detail, with the chemical changes involved.
7. Write a complete standard operating procedure for the Marana of a metal by Puta.
8. Describe the types of Puta with their construction, temperature and application, and the factors affecting the temperature attained.
9. Describe the Bhasma Pariksha in detail, with the method and scientific interpretation of each.
10. Discuss the standardisation of the Marana process, using classical tests and modern analytical methods together.

Short Answer Questions (5 marks) — from all five units

From Unit 1

1. Define Shodhana and explain its four senses.
2. Differentiate Samanya and Vishesha Shodhana.
3. Define Marana. Why is it so called?
4. Enumerate the importance of Shodhana.
5. Enumerate the importance of Marana.
6. Write a short note on Samskara.
7. Write a short note on the Doshas of Parada.
8. Which Dravyas undergo Shodhana but not Marana, and why?

From Unit 2

1. Classify Shodhana as physical, chemical and biological, with examples.
2. Define Swedana, Mardana, Nirvapa and Bharjana.
3. Describe the Samanya Shodhana of metals with media and order.
4. Describe the Shodhana of Gandhaka and its purpose.
5. Write a short note on Dhanyabhraka.
6. Enumerate the sections of a standard operating procedure.
7. Write the safety precautions during Nirvapa.

From Unit 3

1. Enumerate the types of Puta with approximate temperature ranges.
2. Define Sharava Samputa, Sandhi Bandhana and Swangasheeta.
3. Enumerate the chemical methods involved in Marana.
4. Explain Apunarbhava and Niruttha and their significance.
5. Enumerate the Bhasma Pariksha.
6. Why is the number of Putas not fixed?

From Unit 4

1. Enumerate the chemical changes during Shodhana.
2. Enumerate the chemical changes during Marana.
3. Explain the chemical basis of Gandhaka Shodhana.
4. Explain why Loha turns black with Triphala.
5. Explain the chemical change from Kajjali to Rasa Sindura.
6. Explain the weight changes observed during Marana.
7. What does XRD show about a Bhasma, and what does it not show?

From Unit 5

1. Enumerate the common problems in Marana and their remedies.
2. What are the causes if a Bhasma fails Varitara?

3. Write a short note on sintering and its prevention.
4. Write a short note on the optimisation of the Marana process.
5. Write a short note on emerging trends in Shodhana and Marana.
6. Discuss the use of an electric muffle furnace in place of the traditional Puta.

> **Note on the marks range.** Table 6F places Module 4 in the marks range of **5 to 20**. Units 1 and 4 are ABQ-eligible; Units 2 and 3 carry the long answers; Unit 5 carries short answers only. Prepare Unit 4 with particular care — it is short in the syllabus but carries a twenty-mark question, and it is where the chemistry of Module 3 is put to work.

MODULE 5
PRINCIPLES, SPECIFICATION AND SOPs OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS

Credits	Lectures	Practical Training	Experiential Learning	Total hours	Modular assessment
3	15	30	45	90	75 marks

About this module

This module deals with the principles, specifications and standard operating procedures of both classical and advanced instruments used in Rasashastra and Bhaishajya Kalpana. It begins with Puta and Musha, the two classical instruments on which the whole of Marana depends, and proceeds to chromatography, spectroscopy, microscopy, calibration and the advanced characterisation instruments used in the standardisation of Ayurvedic medicines.

Module Learning Objectives

At the end of this module, the student should be able to:

1. Describe the principles, specifications and SOPs for handling classical instruments such as Puta and Musha, and their modern aspects.
2. Discuss the principles, types and applications of chromatography, spectroscopy and microscopy in the standardisation of Ayurvedic formulations.
3. Demonstrate the use of advanced characterisation instruments such as PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA and VSM analysis.
4. Develop SOPs for calibration, troubleshooting and maintenance of various instruments.
5. Integrate the knowledge of instrumental analysis for the standardisation and quality control of Ayurvedic medicines.

Units in this module

Unit	Name	ABQ	SAQ	LAQ
M5U1	Puta and its modern aspects	No	Yes	Yes
M5U2	Musha and its modern aspects	No	Yes	No
M5U3	Chromatography	Yes	Yes	Yes
M5U4	Spectroscopy	Yes	Yes	Yes
M5U5	Microscopy	Yes	Yes	No
M5U6	Calibration techniques	Yes	Yes	No
M5U7	Overview of characterisation instruments	Yes	Yes	Yes

Module 5 is placed in the marks range of **5 to 20**. It is the largest module in the course, and **five of its seven units are ABQ-eligible**. It also carries the most marks in the modular assessment, at 75.

How to approach this module

This module asks for two different kinds of knowledge, and students often prepare only one of them.

The first is **how the instrument works** — the principle. The second is **what it can and cannot tell you** — its application and its limits. The second is where the marks are, and it is where most answers are weak.

For every instrument in this module, be able to state four things in one line each: the principle, what it measures, what it is used for in your subject, and what it cannot show. If you can do that for all of them, you can answer any question in this module.

> **A note on the curriculum text.** The syllabus point 2.1 of Unit 2 is printed as “Introduction, principles, and specifications of Puta.” From the unit title, from the remaining points of that unit, and from the session learning objectives, this clearly refers to **Musha**. It has been read as Musha throughout this book.

UNIT 1
PUTA AND ITS MODERN ASPECTS

SYLLABUS POINTS

1.1 Introduction, principles and specifications of Puta
1.2 SOPs for handling Puta
1.3 Modern aspects of Puta (automation, computerisation, etc.)
1.4 Troubleshooting and maintenance of Puta

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	Yes

1.1 Introduction, principles and specifications

1.1.1 Definition

Puta is the classical system of controlled heating in Rasashastra, in which the material, sealed in a Sharava Samputa, is subjected to a defined quantity of heat for a defined period, by burning a measured quantity of fuel in a pit of defined dimensions.

The word means to enclose or to cover. The enclosure is essential — it is what distinguishes a Puta from ordinary heating.

1.1.2 The principle

Three variables determine the result of any heating operation: **temperature, duration and atmosphere.** The classical Puta controls all three, without a thermometer.

Variable	How the classical Puta controls it
Temperature	By the pit dimensions and the quantity of fuel, both specified for each type of Puta
Duration	By the burning time of that quantity of fuel, and by the requirement of Swangasheeta
Atmosphere	By the sealed Sharava Samputa, which restricts the access of air to the material

This is why Puta is properly described as a **calibrated heating cycle** and not merely as firing.

1.1.3 Types of Puta with specifications

Different texts give different figures. The values below are the commonly taught ones; always cite the text you follow.

Puta	Pit dimensions	Fuel quantity	Approximate temperature	Application
Maha Puta	About 2 Hasta each way	Largest	Highest	Ratna and very hard Dravyas
Gaja Puta	About 1 Hasta length, breadth and depth	About 1000 dried cow dung cakes	Roughly 900 to 1000 degrees Celsius	The standard Puta for most metals
Varaha Puta	Somewhat smaller than Gaja	Fewer cakes	Roughly 700 to 800 degrees Celsius	Where less heat is needed
Kukkuta Puta	About 1 Vitasti each way	Correspondingly fewer	Roughly 600 to 700 degrees Celsius	Softer Dravyas, Ratna Bhasma in some methods
Kapota Puta	Small	Small quantity	Roughly 400 to 500 degrees Celsius	Delicate materials
Gorakha Puta	Small, with cow dung	Small	Low	As prescribed
Lavaka Puta	Smallest	Least	Lowest	Very small quantities
Bhanda Puta	Heating in a closed vessel	—	Moderate	Where a vessel is used instead of a pit
Bhudhara Puta	Vessel buried in a pit in the ground	—	Moderate	As prescribed
Valuka Puta	Sand bath	—	Controlled and gradual	Kupipakwa preparations
Ardhachandra Puta	Half-moon arrangement	Small	Low	Small quantities

> **The measurement units.** One **Hasta** is the length from the elbow to the tip of the middle finger, taken as about 18 to 24 inches. One **Vitasti** is a span, about half a Hasta. These are body-based measures, and this is precisely why the classical figures need to be converted into fixed measurements in a modern SOP.

1.1.4 Factors that affect the temperature actually reached

This is the most examinable part of the unit, and it is the basis of every troubleshooting question.

Factor	Effect
Quantity of fuel	More fuel gives a higher peak and a longer burn
Dryness of the cow dung cakes	Damp fuel gives a much lower temperature and incomplete burning
Quality and thickness of the cakes	Denser cakes burn longer and hotter
Pit dimensions and depth	A deeper pit retains heat; a shallow one loses it
Arrangement of the Samputa within the pit	The centre is hotter than the periphery
Number of ignition points	Ignition from several points gives more even burning
Wind and weather	Wind raises the temperature but shortens the burn; rain and humidity lower it
Season	Winter and monsoon give lower temperatures than summer
Size and thickness of the Sharava	Thicker walls slow the heat transfer to the material
Quantity of material inside	A larger charge heats more slowly

The conclusion to draw, and to write. Because so many variables act at once, the number of Putas prescribed in a text is a guide and not a constant. Completion is judged by **Bhasma Pariksha**. Any modernisation of Puta must aim at controlling these variables, not at abolishing the classical requirement.

1.2 SOP for handling Puta

Use the fifteen-section SOP format given in Module 4, Unit 2. The Puta-specific content is below.

Objective. To deliver a defined and reproducible heating cycle to the material sealed in the Sharava Samputa.

Materials and equipment. Sharava of specified diameter and thickness; cloth and clay for Sandhi Bandhana; dried cow dung cakes of specified number, weight and moisture content; the Puta pit of specified dimensions; ignition material; thermocouple or pyrometer with data logger if available; tongs; protective equipment.

Safety precautions. Site the pit in the open, away from buildings and combustible material. Keep sand and a fire extinguisher within reach. **Never use water on a hot Puta pit.** Never open a Puta before Swangasheeta is complete. Wear a mask when opening the Samputa. Where toxic Dravyas are being processed, stand upwind and follow the departmental protocol.

Procedure

1. Verify that the Chakrika are completely dry and that Sandhi Bandhana is complete and dry.
2. Weigh the sealed Samputa. Record the weight.
3. Prepare the pit to the specified dimensions. Record the actual dimensions measured.

4. Weigh the cow dung cakes. Record the number and total weight. Record their condition.
5. Lay half the cakes in the pit.
6. Place the Samputa centrally. If instrumentation is available, position the thermocouple.
7. Cover with the remaining cakes.
8. Ignite from the prescribed points. Record the time of ignition and the weather conditions.
9. Record the temperature profile if instrumentation is available.
10. Allow the fire to burn out completely.
11. Allow Swangasheeta. Record the time at which the pit reaches ambient temperature.
12. Recover the Samputa. Record its appearance and whether it is intact.
13. Open, observe, weigh and record.

In-process record. Date, weather, pit dimensions, fuel number and weight, ignition time, burn-out time, opening time, temperature profile, appearance of the Samputa, weight before and after.

Completion criteria. The Samputa is intact, the material is uniformly fired, and the appearance and weight are as expected for that stage.

1.3 Modern aspects of Puta

1.3.1 The electric muffle furnace

The **electric muffle furnace** is the principal modern substitute. The material is heated in an insulated chamber by electrical resistance elements, with the temperature controlled by a thermostat or a programmable controller.

Feature	What it provides
Temperature control	A set temperature held within a few degrees
Programmable profile	Rate of rise, holding temperature, holding time and rate of cooling all set in advance
PID controller	Continuous correction to hold the set temperature
Thermocouple feedback	Actual temperature measured and displayed continuously
Data logging	The whole profile recorded automatically
Repeatability	The same cycle delivered every time, in any weather
Cleanliness	No smoke, ash or fuel handling

1.3.2 Comparison of traditional Puta and electric muffle furnace

Point	Traditional Puta	Electric muffle furnace
Temperature control	Indirect, by fuel and pit size	Direct and precise
Reproducibility	Variable with weather and fuel	High
Heating profile	Slow rise, long soak, very slow self-cooling	As programmed; can be made to match, but usually is not
Atmosphere	Reducing to partly reducing, from burning organic fuel, within a sealed Samputa	Ordinarily air, unless a controlled atmosphere furnace is used
Fuel-derived effects	Ash, smoke and gases contact the Samputa	Absent
Documentation	Manual	Automatic
Cost and space	Low cost, needs open ground	Higher cost, needs power, fits indoors
Classical acceptability	Prescribed by the texts	Requires justification and validation

> **How to answer the question “can the muffle furnace replace Puta?”**
>
> State the advantages honestly — control, reproducibility, documentation, cleanliness. Then state the differences that matter: the traditional Puta has a characteristic slow rise and very slow self-cooling, and it takes place in the atmosphere created by burning organic fuel. A muffle furnace set to a peak temperature alone does not reproduce either of these.
>
> The correct position is that a muffle furnace **can** be used **if** the whole cycle is programmed to reproduce the classical profile, and **if** the resulting Bhasma is shown by Bhasma Pariksha and by instrumental analysis to be equivalent. Equivalence must be demonstrated, not assumed. That is a defensible answer and it is also a genuine research question you could take up.

1.3.3 Other modern developments

Development	Description
Thermocouples and data loggers in the traditional pit	Keeps the classical method but records the actual profile
Programmable temperature controller	Sets rate of rise, soak and cooling
Gas-fired furnace	Cleaner than cow dung, more controllable, still a combustion atmosphere
Solar furnace	Experimental; free fuel, but weather dependent
Controlled atmosphere furnace	Allows an inert or reducing atmosphere to be maintained deliberately
Automated Sharava handling	Reduces breakage and operator exposure
Software-based batch records	Links the temperature profile directly to the batch record

1.4 Troubleshooting and maintenance of Puta

1.4.1 Troubleshooting the traditional Puta

Problem	Probable cause	Action
Fuel does not burn fully	Damp cakes; poor ventilation; pit too deep and packed too tightly	Use well-dried fuel; allow air channels; check the pit depth
Temperature too low, material underfired	Insufficient or damp fuel; shallow pit; wet weather	Increase fuel; deepen the pit; postpone in bad weather; record conditions
Material sintered, temperature too high	Excess fuel; pit too small for the charge; strong wind	Reduce fuel; enlarge the pit; shield from wind
Sharava cracked	Damp Chakrika; incomplete Sandhi Bandhana; rapid heating	Dry fully; apply the prescribed layers; ignite gradually
Material lost or contaminated with ash	Seal failed	Improve Sandhi Bandhana; check the Sharava for flaws before use
Uneven firing across the charge	Samputa placed off-centre; uneven fuel	Place centrally; distribute fuel evenly; ignite from several points
Results vary between batches	Fuel and conditions not recorded or standardised	Standardise fuel by weight and moisture; record weather every time

1.4.2 Troubleshooting and maintenance of the electric muffle furnace

Problem	Probable cause	Action
Furnace does not reach the set temperature	Heating element failed or aged; door seal worn; controller fault	Check and replace the element; replace the seal; service the controller
Displayed temperature differs from the actual	Thermocouple drift or damage	Recalibrate against a reference; replace the thermocouple
Temperature fluctuates	Faulty controller or loose connection	Service the PID controller; check wiring
Uneven heating within the chamber	Overloading; poor placement; element failure on one side	Reduce the load; centre the sample; check all elements
Chamber lining cracked	Thermal shock; mechanical damage	Repair or reline; avoid rapid cooling
Cycle not recorded	Data logger not enabled	Enable logging; verify before every run

Routine maintenance. Clean the chamber after each use. Inspect the elements and the lining periodically. Check and record the door seal. **Calibrate the thermocouple at the interval fixed by the department and record it** — this is a Schedule T requirement and is dealt with in Unit 6. Maintain a logbook for every run, with date, operator, programme used and material processed.

KEY POINTS TO REMEMBER

- Puta controls three variables: temperature, duration and atmosphere. The sealed Samputa controls the third.
- Gaja Puta uses a pit of about one Hasta each way with about 1000 cow dung cakes, reaching roughly 900 to 1000 degrees Celsius.
- One Hasta is elbow to fingertip, about 18 to 24 inches; one Vitasti is a span, about half a Hasta.
- Temperature actually reached depends on fuel quantity, dryness, pit size, arrangement, wind and season.
- The muffle furnace gives control and reproducibility but does not by itself reproduce the classical heating profile or atmosphere. Equivalence must be demonstrated.
- Never open a Puta before Swangasheeta is complete. Never use water on a hot pit.
- Thermocouple calibration is a documented requirement, not an optional refinement.

EXAM CORNER — UNIT 1

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). There is no ABQ from this unit.

Likely short answer questions (5 marks)

1. Define Puta and explain its principle.
2. Enumerate the types of Puta with their specifications.
3. Describe Gaja Puta with its dimensions, fuel and temperature.

4. Enumerate the factors affecting the temperature attained in a Puta.
5. Write a short note on Valuka Puta.
6. Write the safety precautions in handling a Puta.
7. Write a short note on the electric muffle furnace.
8. Write a short note on the PID controller and its use.
9. Enumerate the troubleshooting steps if a Sharava cracks during Puta.
10. Write a short note on the maintenance of a muffle furnace.
11. What is Swangasheeta and why must it not be hurried?
12. Write a short note on the modern aspects of Puta.

Likely long answer questions (10 marks)

1. Describe the types of Puta with their principles, specifications and applications. Add a note on the factors affecting the temperature attained.
2. Write a complete standard operating procedure for handling a Puta.
3. Discuss the modern aspects of Puta. Compare the traditional Puta with the electric muffle furnace and state the conditions under which the latter may be used.
4. Discuss the troubleshooting and maintenance of Puta, both traditional and modern.
5. Explain how the classical Puta may be standardised without departing from the classical requirement.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **1, 2, 10**

1. **1.** Sharma S, Shastri HD. Rasatarangini, 11th edition. Delhi: Motilal Banarasidas; 1979.
2. **2.** Vagbhatacharya S. Rasa Ratna Samuccaya, Hindi translation edited by Kulkarni D. New Delhi: Meharchand Lachhmandas Publication; reprint 2006.
3. **10.** Yashodhara. Rasaprakasha Sudhakara, Hindi translation by Mishra Siddhinandan. Varanasi: Chaukhambha Orientalia; third edition.

UNIT 2
MUSHA AND ITS MODERN ASPECTS

SYLLABUS POINTS

2.1 Introduction, principles and specifications of Musha
2.2 SOPs for handling Musha
2.3 Modern aspects of Musha (automation, computerisation, etc.)
2.4 Troubleshooting and maintenance of Musha

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

2.1 Introduction, principles and specifications

2.1.1 Definition

Musha is the crucible of Rasashastra — a vessel made chiefly of specially prepared clay, in which Dravyas are heated to high temperature for Satvapatana, Druti, Jarana, melting and certain Marana procedures.

Its modern counterpart is the **refractory crucible**.

2.1.2 The principle

A Musha must do four things at once:

1. **Withstand high temperature** without melting or softening.
2. **Withstand thermal shock** — sudden heating and cooling — without cracking.
3. **Remain chemically inert** towards the charge, so that it does not contaminate the product.
4. **Retain the charge** without leaking, and allow it to be recovered.

The classical composition is designed for exactly these four requirements.

2.1.3 Materials used in preparing a Musha

Material	Function in modern terms
Mrittika — clay, especially from a termite mound or river bank	The refractory body; plastic when wet, hard when fired
Valuka — sand	Grog; reduces shrinkage and cracking
Bhusa or Tusha — chaff and husk	Burns out on firing, leaving pores that absorb thermal shock
Kesha or Loma — hair	Fibre reinforcement; burns out leaving fine channels
Karpasa — cotton	Fibre reinforcement
Lauha Kitta — iron slag	Raises the refractoriness
Guda — jaggery	Binder in the green state
Kadali Kanda and similar mucilaginous materials	Binder and plasticiser
Gomaya — cow dung	Organic filler; burns out leaving porosity

> **This composition is a genuinely sophisticated refractory formulation.** Modern refractory practice uses exactly this combination of principles — a clay body, a non-plastic grog to control shrinkage, an organic burnout material to create controlled porosity, and fibre for green strength. The classical Musha is not a crude pot. Say this, with the reasons, and the answer is immediately a postgraduate answer.

2.1.4 Types of Musha

Different texts describe different Mushas, named after their shape or their use.

Musha	Description and use
Vrintaka Musha	Shaped like a brinjal; general purpose, widely used
Gostani Musha	Shaped like a cow’s teat; used for Satvapatana
Vajra Musha	Very strong, made with additional reinforcement; used for hard Dravyas and severe heat
Maha Musha	Large size; used for larger charges
Yoga Musha	Used for specific Yogas as prescribed
Malla Musha	Made of two earthen cups joined mouth to mouth; used for closed heating
Gara Musha	Used for Gara and specific preparations
Varta or Vartula Musha	Round shaped
Pakwa Musha	Pre-fired before use
Musha for Satvapatana	Deeper form, allowing the Satva to collect at the bottom

2.1.5 Musha Lakshana — the qualities of a good Musha

Quality	Reason
Uniform thickness	Uneven walls crack from differential expansion
Free from cracks and air pockets	A flaw becomes a failure at high temperature
Smooth inner surface	The product does not stick and can be recovered
Adequately dried before firing	Trapped moisture turns to steam and bursts the wall
Pre-fired where prescribed	Removes water and organic matter in advance
Appropriate size for the charge	An overfilled Musha spills; an oversized one wastes heat
Chemically inert to the charge	Prevents contamination of the product
Able to withstand the required temperature	Must exceed the working temperature with a margin

2.2 SOP for handling Musha

Objective. To prepare, condition, use and store a Musha so that it performs reliably and does not contaminate the charge.

Procedure for preparation

1. Select the clay as prescribed. Remove stones and vegetable matter.
2. Dry, powder and sieve the clay to the specified mesh.
3. Weigh each ingredient in the prescribed proportion. Record.
4. Mix thoroughly in the dry state.
5. Add water or the prescribed binder liquid gradually and knead to a uniform, plastic mass.
6. Allow the mass to mature for the prescribed period.
7. Shape the Musha to the specified form, size and wall thickness.
8. Smooth the inner surface.
9. Dry in shade slowly and completely. Record the drying time.
10. Pre-fire if prescribed, raising the temperature gradually.
11. Inspect for cracks. Reject any that are flawed.
12. Label with the date and batch, and store in a dry place.

Procedure for use

1. Inspect the Musha for cracks before every use. Reject if in doubt.
2. Pre-heat gradually. Never place a cold Musha directly into a hot furnace.
3. Charge to the specified level, leaving the prescribed headspace.
4. Heat gradually to the working temperature.
5. Do not quench or cool rapidly.
6. Allow to cool on its own before recovering the product.
7. Record the number of times each Musha has been used.
8. Retire the Musha after the number of uses fixed by the department, or earlier if it shows deterioration.

Safety precautions. Use long tongs of appropriate design. Wear heat-resistant gloves, apron and face shield. Never carry a hot Musha over an unprotected surface. Have sand available. Never handle a Musha containing molten material without a second person present.

2.3 Modern aspects of Musha

2.3.1 Modern refractory crucibles

Material	Maximum working temperature, approximate	Advantages	Limitations
Fireclay	Up to about 1300 degrees Celsius	Cheap, widely available	Lower refractoriness; reacts with some melts
Silica	About 1600 degrees Celsius	Cheap, good thermal shock resistance	Attacked by alkalis
Alumina	About 1700 to 1800 degrees Celsius	Very refractory, chemically stable	Costly; poorer thermal shock resistance
Graphite and clay-graphite	Very high	Excellent for metal melting; good conductivity	Oxidises in air; not for oxidising conditions
Silicon carbide	Very high	Strong, good thermal conductivity and shock resistance	Costly
Zirconia	Very high	Extremely refractory and inert	Very costly
Platinum	Very high	Chemically inert; used in analysis	Extremely costly; not for base metals
Porcelain	About 1200 degrees Celsius	Clean, used in laboratories	Limited temperature

2.3.2 Selecting a crucible

Select on four criteria, in this order:

1. **Maximum temperature required**, with a margin above the working temperature.
2. **Chemical compatibility** with the charge — an alkaline melt will attack silica, an oxidising atmosphere will consume graphite.
3. **Thermal shock resistance**, if rapid heating or cooling is involved.
4. **Cost and availability**, once the first three are satisfied.

2.3.3 Automation and instrumentation

Development	Description
Standardised commercial crucibles	Fixed dimensions and known specification, replacing hand-made variation
Crucible furnaces with programmed control	The heating profile is set and recorded
Mechanical tongs and pouring systems	Reduce operator exposure and breakage
Controlled atmosphere crucible furnace	Inert or reducing atmosphere maintained deliberately
Non-contact temperature measurement	Infrared pyrometer reads the charge temperature without contact
Crucible usage logs	Each crucible tracked by number of uses and by material processed

> **A caution worth stating in any answer.** Modern crucibles are chosen to be as inert as possible. The classical Musha is not inert — its composition includes Lauha Kitta and organic material, and it takes part in the process to some degree. Whether replacing a Musha with an inert alumina crucible changes the finished product is a real question, and it should be answered by comparison, not by assumption. This is the same argument as for the muffle furnace in Unit 1.

2.4 Troubleshooting and maintenance of Musha

Problem	Probable cause	Action
Musha cracks on first heating	Incomplete drying; too thick or uneven wall; heating too rapid	Dry completely; control the wall thickness; raise the temperature gradually
Musha cracks after several uses	Fatigue from repeated thermal cycling	Fix a maximum number of uses and retire on schedule
Charge leaks through the wall	Porosity too high; a flaw not detected	Improve the mix; inspect before every use
Product sticks to the Musha	Inner surface rough; wrong material for the charge	Smooth the inner surface; select a compatible material
Product contaminated	The Musha reacted with the charge	Select a chemically compatible refractory; pre-fire the Musha
Musha softens or slumps	Working temperature exceeded its refractoriness	Select a higher grade; verify the actual furnace temperature
Yield lower than expected	Material retained in a porous or rough wall	Smooth the surface; account for the retained material in the reconciliation

Routine maintenance. Store in a dry place, off the floor. Inspect before every use. Clean out residues after each use. Number every Musha and maintain a usage record. Retire on schedule rather than on failure.

KEY POINTS TO REMEMBER

- Musha is the crucible; its modern counterpart is the refractory crucible.
- Four requirements: high temperature resistance, thermal shock resistance, chemical inertness, retention of the charge.
- Classical composition uses clay as the body, sand as grog, chaff and cow dung as burnout for porosity, hair and cotton as fibre reinforcement, iron slag to raise refractoriness, and jaggery as binder. This matches modern refractory practice.
- Types include Vrintaka, Gostani, Vajra, Maha, Yoga, Malla, Gara, Vartula and Pakwa Musha.
- Gostani Musha is used for Satvapatana; Vajra Musha for severe heat.
- Never place a cold Musha into a hot furnace; never quench a hot one.
- Crucible selection order: temperature, chemical compatibility, thermal shock resistance, cost.
- Graphite crucibles oxidise in air and must not be used under oxidising conditions.

EXAM CORNER — UNIT 2

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Define Musha and state its principle.
2. Enumerate the materials used in the preparation of a Musha, with the function of each.
3. Enumerate the types of Musha with their uses.
4. Write a short note on Gostani Musha and Vajra Musha.
5. Enumerate the Musha Lakshana of a good Musha.
6. Describe the preparation of a Musha.
7. Write the safety precautions in handling a Musha.
8. Enumerate the modern refractory materials with their working temperatures.
9. On what criteria is a crucible selected?
10. Why do graphite crucibles fail in an oxidising atmosphere?
11. Enumerate the common problems with a Musha and their remedies.
12. Write a short note on the maintenance of a Musha.

How to attempt

- Always give the modern function of each classical ingredient. That single table is what distinguishes a postgraduate answer here.
- Close with the caution that a Musha is not inert, and that substituting an inert crucible must be validated.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **1, 2, 3**

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2. **2 and 3.** Rasa Vagbhata. Rasa Ratna Samuccaya, edited by Kulkarni DA. New Delhi: Meharchand Lachhmandas Publications; reprint 2006.

UNIT 3
CHROMATOGRAPHY

SYLLABUS POINTS

3.1 Scope and applications of chromatography in herbal drug standardisation — identification of active compounds, adulterant detection, phytochemical analysis and quality assessment
3.2 Principle and types of chromatography, including TLC, HPTLC, HPLC, LC-MS and GC-MS
3.3 Instrumentation of chromatography
3.4 Data analysis in chromatography, including interpretation of chromatograms and spectra
3.5 Chromatography's limitations, challenges and prospects in pharmaceutical and phytochemical research

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	Yes

3.1 Scope and applications in herbal drug standardisation

3.1.1 Why chromatography is central

Preliminary phytochemical screening, learned in Module 3, tells you that a class of constituent is present. It cannot tell you **which** compound, or **how much**.

Chromatography separates a mixture into its components before detecting them. That single step turns a qualitative test into an identification and an assay. This is why every pharmacopoeial monograph for a herbal drug now includes a chromatographic test.

3.1.2 Applications

Application	How chromatography serves it
Identification of active compounds	Compares the sample against a reference standard by Rf or retention time
Fingerprinting	The whole chromatogram serves as a profile of the drug, without identifying every peak
Adulterant and substitute detection	An extra peak, or a missing one, reveals the substitution
Quantitative assay	Peak area is proportional to concentration
Marker-based standardisation	The content of the chosen marker is fixed within limits
Stability studies	The decline of the marker is followed over time
Detection of adulteration with synthetic drugs	A serious problem in some herbal products; chromatography detects it
Pesticide residue analysis	GC and LC methods
Purity of an isolated compound	A single sharp peak indicates purity
Process monitoring	Following the change in constituents during Shodhana, Bhavana and Paka

> **Fingerprinting deserves emphasis for your subject.** A classical formulation contains many drugs and hundreds of compounds. Identifying every one is neither possible nor necessary. What is needed is a **reproducible profile** — the same pattern of spots or peaks, batch after batch. HPTLC fingerprinting does exactly this and is the single most practical standardisation tool available to an Ayurvedic manufacturer. The PCIM&H has published a TLC Atlas for this purpose, and it is reference 134 of your list.

3.2 Principle and types of chromatography

3.2.1 The general principle

Chromatography separates the components of a mixture by their **differential distribution between two phases** — a **stationary phase** that does not move, and a **mobile phase** that moves through or over it.

A component that interacts strongly with the stationary phase moves slowly. One that prefers the mobile phase moves quickly. Given enough distance, they separate.

Term	Meaning
Stationary phase	The fixed phase — silica gel, alumina, a bonded column packing, or paper
Mobile phase	The moving phase — a solvent or a carrier gas
Eluent	The mobile phase as it passes through
Eluate	What comes out at the end
Analyte	The substance being separated
Retention	How strongly a component is held back
Resolution	How well two adjacent components are separated

3.2.2 Classification by mechanism

Mechanism	Basis of separation	Example
Adsorption	Adhesion to a solid surface	TLC, column chromatography on silica
Partition	Differential solubility between two liquids	Paper chromatography, reverse-phase HPLC
Ion exchange	Exchange of ions with a charged resin	Ion exchange chromatography
Size exclusion or gel filtration	Molecular size	Gel permeation chromatography
Affinity	Specific biological binding	Purification of enzymes and antibodies

3.2.3 The types named in your syllabus

A. Thin Layer Chromatography (TLC)

- **Principle.** Adsorption, on a thin layer of silica gel or alumina coated on a glass, aluminium or plastic plate.
- **Procedure.** The sample is spotted near one edge; the plate is placed in a chamber with the mobile phase; the solvent rises by capillary action; the plate is dried and the spots visualised.
- **Visualisation.** Under ultraviolet light at 254 and 366 nanometres, or by spraying a reagent such as anisaldehyde-sulphuric acid, or by iodine vapour.
- **Result expressed as R_f** — the retardation factor.

R_f = distance travelled by the spot ÷ distance travelled by the solvent front

R_f always lies between 0 and 1, and is characteristic of a compound under stated conditions.

- **Advantages.** Simple, rapid, cheap, needs little equipment, several samples run side by side.
- **Limitations.** Semi-quantitative at best; resolution limited; R_f varies with temperature, humidity, chamber saturation and plate quality.

B. High Performance Thin Layer Chromatography (HPTLC)

An instrumental refinement of TLC. The plate has a finer, more uniform particle size; the sample is applied by an automatic applicator as a band; development is in a controlled chamber; and the plate is read by a densitometric scanner.

- **Advantages over TLC.** Quantitative; reproducible; produces a densitogram as well as an image; many samples per plate; low solvent consumption.
- **The principal tool for fingerprinting Ayurvedic formulations.**

C. High Performance Liquid Chromatography (HPLC)

- **Principle.** The mobile phase, a liquid, is pumped at high pressure through a column packed with very fine particles.
- **Modes. Normal phase** has a polar stationary phase and a non-polar mobile phase. **Reverse phase**, which is the commoner, has a non-polar stationary phase, typically C18, and a polar mobile phase.
- **Elution. Isocratic** keeps the mobile phase composition constant; **gradient** changes it during the run.
- **Result expressed as retention time**, the time from injection to the peak maximum.
- **Advantages.** High resolution, quantitative, automated, suitable for non-volatile and thermolabile compounds.
- **Limitations.** Costly; needs skilled operation; large solvent consumption; column care is critical.

D. Gas Chromatography (GC)

- **Principle.** The mobile phase is an inert carrier gas — nitrogen, helium or hydrogen. The sample is vaporised in a heated injector and carried through a column held in an oven.
- **Suitable for.** Volatile and thermally stable compounds. **Volatile oils are the obvious application in your subject.**
- **Not suitable for.** Non-volatile or heat-labile compounds, unless they are derivatised first.

E. LC-MS and GC-MS

These couple a chromatograph to a **mass spectrometer**, so that each separated component is also identified by its mass.

System	Chromatograph	Suitable for	Gives
GC-MS	Gas chromatograph	Volatile compounds — volatile oils, pesticide residues	Retention time plus mass spectrum
LC-MS	Liquid chromatograph	Non-volatile compounds — glycosides, alkaloids, flavonoids	Retention time plus mass spectrum

Why coupling matters. Chromatography alone tells you that a component is present and how much. Mass spectrometry tells you what it is. Together they identify and quantify in a single run, and they can do so at very low concentrations. This is what makes them the reference methods for residue and contaminant analysis.

3.2.4 Other types worth knowing

Type	Note
Paper chromatography	Partition on cellulose; historical and teaching value
Column chromatography	Preparative; used to isolate a compound in quantity
Flash chromatography	Column chromatography under pressure; faster
Ion exchange	For charged species
Size exclusion	For separating by molecular size

3.3 Instrumentation

3.3.1 TLC and HPTLC

Component	Function
Plate	Glass, aluminium or plastic coated with the stationary phase
Applicator	Capillary in TLC; automatic band applicator in HPTLC
Development chamber	Twin trough or flat bottom; saturated with the mobile phase
Mobile phase	The chosen solvent system
Visualisation unit	UV cabinet at 254 and 366 nanometres; spray reagents
Densitometric scanner	In HPTLC; measures absorbance along each track
Documentation system	Photographs the plate under each light source
Software	Processes the densitogram and calculates results

3.3.2 HPLC

Component	Function
Solvent reservoir	Holds the mobile phase; solvents must be degassed and filtered
Degasser	Removes dissolved gas that would form bubbles
Pump	Delivers the mobile phase at constant high pressure
Injector	Introduces a fixed volume of sample; usually an autosampler
Column	The separation takes place here; commonly C18
Column oven	Holds the column at a constant temperature
Detector	UV-visible, photodiode array, fluorescence, refractive index, or mass spectrometer
Data system	Records and integrates the chromatogram
Waste collection	Collects the eluate

The photodiode array detector deserves a mention. It records the whole ultraviolet spectrum at every point of the run, so each peak carries a spectrum as well as a retention time. This helps confirm identity and check peak purity.

3.3.3 GC

Component	Function
Carrier gas supply	Nitrogen, helium or hydrogen, with regulators
Injector	Heated; vaporises the sample; split or splitless
Column	Packed or, more usually, capillary
Column oven	Programmable temperature; the programme is a key variable
Detector	Flame ionisation, thermal conductivity, electron capture, or mass spectrometer
Data system	Records and integrates

3.4 Data analysis and interpretation

3.4.1 The chromatogram

A chromatogram plots detector response against distance, in TLC, or against time, in HPLC and GC.

Feature	What it tells you
Position — R_f or retention time	Identity, when compared with a standard under identical conditions
Peak height and area	Quantity; area is the more reliable
Peak shape	A sharp symmetrical peak indicates good chromatography; tailing indicates a problem
Number of peaks	How many components were separated
Baseline	Should be flat and stable; drift indicates a problem
Resolution between peaks	How completely two components are separated

3.4.2 The key calculations

Quantity	How it is obtained
R_f	Distance of spot ÷ distance of solvent front
Retention time	Read directly from the chromatogram
Relative retention	Retention of the analyte ÷ retention of a reference peak
Peak area	Integrated by the software
Content of the analyte	By comparison with a calibration curve of the reference standard, or by a single-point comparison
Resolution	Calculated from the separation of two peaks and their widths
Theoretical plates	A measure of column efficiency
Tailing factor	A measure of peak symmetry

3.4.3 How to read a fingerprint

1. Confirm that the reference standard gives the expected R_f or retention time in this run.
2. Compare the sample track against the standard track.
3. Note every spot or peak — its position, its intensity and its colour under each light source.
4. Compare against the reference fingerprint for that drug.
5. Note any **extra** feature — a possible adulterant.
6. Note any **missing** feature — a possible substitution, or degradation.
7. State the conditions in full. **A chromatographic result quoted without its conditions is meaningless.**

> **The conditions you must always record.** For TLC and HPTLC: the stationary phase and plate, the mobile phase with exact proportions, the chamber and its saturation, the application volume, the development distance, the visualisation method, and the temperature and humidity. For HPLC and GC: the column with its dimensions and particle size, the mobile phase or carrier gas, the flow rate, the gradient or oven programme, the injection volume, the detector and its wavelength, and the column temperature. Omitting these is the commonest fault in student dissertations.

3.5 Limitations, challenges and prospects

3.5.1 Limitations

Limitation	Explanation
Detects only what the detector responds to	A compound with no chromophore is invisible to a UV detector
Identification depends on a reference standard	Without an authentic standard, a peak cannot be named
Reference standards for Ayurvedic markers are scarce and costly	A real and practical obstacle
Co-elution	Two compounds may emerge together and be mistaken for one
R_f is condition-dependent	Values are not transferable between laboratories without care
Sample preparation affects everything	A poorly prepared extract gives a poor chromatogram, however good the instrument
Cost and skill	HPLC, LC-MS and GC-MS need investment and trained staff
Does not detect inorganic constituents	Of direct relevance to Bhasma and to metal-containing formulations

> **The last limitation is the one specific to your speciality, and it is often forgotten.** Chromatography separates molecules in solution. A Bhasma is an insoluble inorganic solid. Chromatography has almost nothing to say about it. For a Bhasma you need XRD, SEM-EDAX, ICP and the other methods of Units 4 and 7. For the herbal part of a formulation, chromatography is the right tool. **Choose the method to match the material.** Writing this sentence in an ABQ answer shows real understanding.

3.5.2 Challenges in herbal analysis specifically

- Natural variation between batches of the same drug
- Very large numbers of compounds, most of them unidentified
- Constituents present at very low concentration
- Absence of a single “active” compound in most Ayurvedic drugs
- Processing may have changed the constituents completely
- Lack of validated methods for many classical formulations
- Lack of agreed reference fingerprints

3.5.3 Prospects

Development	What it offers
UPLC — ultra performance liquid chromatography	Higher resolution, much shorter runs, less solvent
High resolution mass spectrometry	Identifies unknown compounds from accurate mass
Chemometrics	Statistical treatment of whole fingerprints to classify samples
Metabolomics	Profiling the entire small-molecule content of a drug
Two-dimensional chromatography	Far greater separating power for complex mixtures
Green chromatography	Reduced solvent use and safer solvents
Hyphenated HPTLC	HPTLC coupled to mass spectrometry
Shared reference fingerprint databases	Would allow any laboratory to compare against a common standard

KEY POINTS TO REMEMBER

- Chromatography separates by differential distribution between a stationary and a mobile phase.
- Mechanisms: adsorption, partition, ion exchange, size exclusion, affinity.
- R_f = distance travelled by the spot ÷ distance travelled by the solvent front. It lies between 0 and 1.
- HPTLC is quantitative, reproducible and the principal tool for fingerprinting Ayurvedic formulations.
- Reverse phase HPLC has a non-polar stationary phase, usually C18, and a polar mobile phase.
- GC needs volatile, thermally stable compounds; volatile oils are the obvious application.
- GC-MS and LC-MS add identification by mass to separation by chromatography.
- A photodiode array detector gives a full UV spectrum for every peak.
- Always record the full chromatographic conditions. A result without conditions is meaningless.
- **Chromatography cannot analyse a Bhasma. Match the method to the material.**

EXAM CORNER — UNIT 3

This unit can be asked as ABQ (20 marks), SAQ (5 marks) and LAQ (10 marks).

Likely short answer questions (5 marks)

1. Define chromatography and state its principle.
2. Classify chromatography on the basis of mechanism, with examples.
3. Define R_f. State its significance and the factors affecting it.
4. Write a short note on TLC.
5. Differentiate between TLC and HPTLC.

6. Write a short note on HPLC instrumentation.
7. Differentiate between normal phase and reverse phase HPLC.
8. Write a short note on GC and its applications in Ayurveda.
9. Write a short note on GC-MS and LC-MS.
10. Write a short note on fingerprinting of herbal formulations.
11. Enumerate the chromatographic conditions that must be recorded.
12. Enumerate the limitations of chromatography in the analysis of Ayurvedic formulations.

Likely long answer questions (10 marks)

1. Define chromatography. Describe its types with the principle and application of each.
2. Describe HPLC in detail — principle, instrumentation, modes and applications.
3. Describe HPTLC in detail and discuss its role in the standardisation of Ayurvedic formulations.
4. Discuss the scope and applications of chromatography in herbal drug standardisation.
5. Discuss the interpretation of a chromatogram, and the limitations and future prospects of chromatography in phytochemical research.

Likely analytical based questions (20 marks)

1. You are given a market sample of a classical Ayurvedic formulation and asked to establish whether it is genuine. Analyse the problem, select the analytical methods you would use with reasons, describe the procedure, and explain how you would interpret each possible result.
2. A herbal formulation shows an extra peak in its HPLC chromatogram that is absent from the reference. Analyse the possible explanations, describe how you would establish which is correct, and state the action to be taken in each case.
3. Design a complete chromatographic standardisation protocol for a classical Kwatha Churna, from sample preparation to acceptance criteria, and justify every choice you make.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **129, 130, 131, 132, 133, 134**

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2. **130.** Chromatography and Its Applications. IntechOpen; 2012.
3. **131.** Vitha MF. Chromatography: Principles and Instrumentation. Wiley; 2016.
4. **132.** Chemometrics and Data Analysis in Chromatography. IntechOpen; 2019.
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6. **134.** Thin Layer Chromatography (TLC) Atlas. Pharmacopoeia Commission for Indian Medicine and Homoeopathy, Ministry of AYUSH.

UNIT 4
SPECTROSCOPY

SYLLABUS POINTS

4.1 Introduction, principles and types (IR, UV, NMR, AAS, ICP, etc.) of spectroscopy
4.2 Specifications of spectroscopy instruments
4.3 SOPs for handling spectroscopy instruments
4.4 Troubleshooting and maintenance of spectroscopy instruments

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	Yes

4.1 Introduction, principles and types

4.1.1 The basic principle

Spectroscopy is the study of the interaction between matter and electromagnetic radiation.

When radiation meets a molecule or an atom, energy may be absorbed, emitted or scattered. The energy involved corresponds to a particular change within the substance. By measuring which wavelengths are affected, and by how much, we learn what the substance is and how much of it is present.

Region of the spectrum	Energy	Change produced	Technique
Ultraviolet and visible	High	Electronic transitions between orbitals	UV-Visible spectroscopy
Infrared	Moderate	Vibration of bonds — stretching and bending	IR and FTIR
Microwave	Low	Molecular rotation	Microwave spectroscopy
Radiofrequency	Very low	Nuclear spin flip in a magnetic field	NMR
X-ray	Very high	Inner shell electron transitions	XRF, XPS

4.1.2 UV-Visible spectroscopy

- **Principle.** Absorption of ultraviolet or visible light causes electrons to move from a lower to a higher energy orbital. Compounds with conjugated double bonds and aromatic rings absorb strongly.
- **Wavelength range.** Ultraviolet about 200 to 400 nanometres; visible about 400 to 800 nanometres.
- **Key terms.** A **chromophore** is the group responsible for absorption. An **auxochrome** is a group that shifts the absorption when attached to a chromophore. λ_{max} is the wavelength of maximum absorption.

The Beer-Lambert law is the basis of every quantitative measurement.

$A = \varepsilon c l$

where A is the absorbance, ε the molar absorptivity, c the concentration and l the path length, usually one centimetre. Absorbance is directly proportional to concentration, so a calibration curve of absorbance against concentration allows an unknown to be measured.

- **Applications in your subject.** Assay of a marker compound; determination of total phenolic and total flavonoid content; dissolution studies; following the progress of an extraction.
- **Limitations.** The compound must absorb; the sample must be in clear solution; overlapping absorption from other constituents interferes.

4.1.3 Infrared and FTIR spectroscopy

- **Principle.** Absorption of infrared radiation causes bonds to vibrate — to stretch and to bend. A bond absorbs only if the vibration changes the dipole moment of the molecule. Each type of bond absorbs at a characteristic frequency.
- **Units.** Wavenumber, in reciprocal centimetres. The usual range is 4000 to 400.
- **FTIR** — Fourier Transform Infrared — collects all frequencies at once using an interferometer and converts the result mathematically. It is faster, more sensitive and now standard.

The two regions of an IR spectrum

Region	Range	What it shows
Functional group region	4000 to 1500 reciprocal centimetres	Identifies the functional groups present
Fingerprint region	1500 to 400 reciprocal centimetres	Complex and unique to each compound; used for identification by comparison

Characteristic absorptions worth remembering

Bond or group	Approximate wavenumber
O—H, broad	3200 to 3600
N—H	3300 to 3500
C—H	2850 to 3000
C≡N	about 2250
C=O	1650 to 1750
C=C aromatic	1450 to 1600
Metal—oxygen	Generally below 1000

> **Why FTIR matters directly to Rasashastra.** Two things:
> **1.** Metal-oxygen and metal-sulphur bonds absorb at low wavenumber, generally below 1000. Their appearance in the spectrum of a Bhasma is evidence that the metal has been converted into an oxide or sulphide.
> **2.** Bands in the functional group region show the **organic material** remaining associated with the Bhasma from the Bhavana media. FTIR is one of the few accessible methods that demonstrates this directly. Recall the discussion in Module 3, Unit 2.

4.1.4 Nuclear Magnetic Resonance spectroscopy

- **Principle.** Certain nuclei, notably hydrogen-1 and carbon-13, behave as tiny magnets. Placed in a strong magnetic field and irradiated with radiofrequency radiation, they absorb energy and flip their spin. The exact frequency depends on the electronic environment of the nucleus.
- **Chemical shift**, expressed in parts per million, is measured relative to tetramethylsilane, which is set at zero.
- **What an NMR spectrum gives.** The number of signals shows how many different environments exist; the chemical shift shows what kind of environment; the integration shows how many nuclei of each type; the splitting pattern shows how many neighbours each has.
- **Application.** Structural elucidation of an isolated phytoconstituent. It is the definitive method for determining an unknown organic structure.
- **Limitation.** Requires a pure compound in reasonable quantity. It is of little use on a crude extract, and of no use on a Bhasma.

4.1.5 Atomic Absorption Spectroscopy (AAS)

- **Principle.** The sample is atomised in a flame or a graphite furnace. Light from a hollow cathode lamp, made of the element being measured, passes through the atom cloud. The free atoms absorb light at their own characteristic wavelength. The absorbance is proportional to the concentration of that element.
- **Key point.** One lamp per element. AAS is essentially a **single-element** technique.
- **Types.** Flame AAS for higher concentrations; graphite furnace AAS for very low concentrations.

Why AAS is directly relevant to you. It is the standard method for determining **heavy metals** — lead, cadmium, mercury and arsenic — in Ayurvedic products. The Ayurvedic Pharmacopoeia of India prescribes limits for these, and any manufacturer exporting a product must demonstrate compliance. Mercury requires the cold vapour technique, and arsenic the hydride generation technique, because of their volatility.

4.1.6 Inductively Coupled Plasma (ICP)

- **Principle.** The sample is introduced into an argon plasma at a very high temperature. At that temperature atoms are excited and ionised.

Technique	What is measured	Character
ICP-OES or ICP-AES	Light emitted by the excited atoms	Multi-element, rapid, wide concentration range
ICP-MS	The ions themselves, sorted by mass	Multi-element, extremely sensitive, gives isotope information

- **Advantages over AAS.** Many elements measured at once; far greater sensitivity in the case of ICP-MS; a much wider working range.
- **Application in your subject.** Complete elemental profiling of a Bhasma; heavy metal determination at trace level; verifying that the elements present are those expected from the raw material and the media used.
- **Limitation.** ICP tells you **which elements** are present and how much. It does **not** tell you in what chemical form they are present. For that you need XRD or XPS.

> **This distinction is examinable and important.** An elemental technique such as AAS or ICP will report the total mercury in a sample. It cannot distinguish free mercury from mercuric sulphide. Since the whole argument of Rasashastra rests on the **form** in which the metal is present, elemental analysis alone can never settle the question. It must be combined with a phase-identifying method. Write this in any answer on heavy metals in Ayurvedic medicine.

4.1.7 Other spectroscopic methods in brief

Method	Principle	Use
Mass spectrometry	Ionisation and separation by mass to charge ratio	Molecular weight and structure; coupled to GC or LC
Fluorescence spectroscopy	Emission after absorption	Very sensitive assay of fluorescent compounds
Raman spectroscopy	Inelastic scattering of light	Complementary to IR; works through glass and water
Flame photometry	Emission from atoms in a flame	Sodium, potassium, calcium, lithium
XPS	Ejection of core electrons by X-rays	Oxidation state at the particle surface

4.2 Specifications of spectroscopy instruments

When purchasing, or when writing about an instrument, these are the specifications that matter.

Instrument	Key specifications
UV-Visible spectrophotometer	Wavelength range; spectral bandwidth; wavelength accuracy and repeatability; photometric range and accuracy; stray light; single or double beam; detector type; cuvette holder options
FTIR	Spectral range; resolution in reciprocal centimetres; signal to noise ratio; scan speed; detector type; sampling accessories such as ATR, KBr pellet and diffuse reflectance
NMR	Magnetic field strength in tesla, or proton frequency in megahertz; probe type; nuclei observable; shim system; temperature control
AAS	Wavelength range; flame and furnace options; number of lamp positions; background correction method; detection limits; hydride and cold vapour accessories
ICP-OES or ICP-MS	Wavelength or mass range; resolution; detection limits; plasma power; sample introduction system; interference removal, such as a collision cell in ICP-MS

4.3 SOP for handling spectroscopy instruments

A general SOP outline, to be adapted to each instrument. Use the fifteen-section format from Module 4.

1. **Before use.** Check that the calibration is current and the label is valid. Check the log book. Switch on and allow the prescribed warm-up time. Verify performance with a check standard.
2. **Sample preparation.** Prepare as per the validated method. Filter where required. Record the weights, volumes and dilutions.
3. **Blank and standard.** Run the blank. Run the reference standard and confirm it is within limits before running any sample.
4. **Measurement.** Run the samples. Run a check standard at intervals and at the end.
5. **Recording.** Record every reading, the conditions, the operator and the date. Save the raw data file.
6. **After use.** Clean cuvettes, cells or the sample compartment. Shut down as prescribed. Complete the log book.
7. **Waste.** Dispose of solvents and residues as prescribed. Never into the sink.

Instrument-specific points

Instrument	Specific requirements
UV-Visible	Handle cuvettes by the frosted faces only; match cuvettes; no bubbles; blank with the same solvent
FTIR	Keep the desiccant dry; prepare KBr pellets with dry KBr; clean the ATR crystal after every sample; collect a background scan before each set
AAS	Check the gas supply and exhaust; align the lamp; ignite and extinguish the flame in the prescribed order; **never leave a lit flame unattended**
ICP	Check argon supply and pressure; check the exhaust; allow the plasma to stabilise; verify the torch and nebuliser are clean
NMR	**The magnet is always on.** Keep all ferromagnetic objects, and anyone with a pacemaker or implant, outside the marked exclusion zone

4.4 Troubleshooting and maintenance

Problem	Probable cause	Action
UV-Visible: noisy or drifting baseline	Lamp ageing; instrument not warmed up; dirty cuvette	Allow full warm-up; clean or replace the cuvette; check lamp hours and replace
UV-Visible: absorbance not reproducible	Bubbles; cuvette not matched or wrongly seated; sample not homogeneous	Degas; use matched cuvettes; mix the sample

Problem	Probable cause	Action
UV-Visible: absorbance out of range	Concentration too high	Dilute to bring the absorbance into the reliable range
FTIR: weak or noisy spectrum	Sample too thin or too thick; poor ATR contact; detector or desiccant problem	Adjust the sample; apply proper pressure on the ATR; replace the desiccant
FTIR: broad water bands appearing	Moist KBr or damp sample	Dry the KBr; dry the sample; check the purge
AAS: low or unstable signal	Lamp misaligned or ageing; burner slot blocked; nebuliser blocked; flame conditions wrong	Realign; clean the burner and nebuliser; adjust the fuel to oxidant ratio
AAS: high blank reading	Contaminated reagents, water or glassware	Use high-purity reagents; clean glassware with acid
ICP: plasma will not ignite or is unstable	Argon supply low; torch dirty or damaged; poor gas flow	Check the supply; clean or replace the torch; verify the flows
ICP: poor sensitivity	Nebuliser or cones blocked; sample uptake poor	Clean the nebuliser and cones; check the peristaltic pump tubing
NMR: poor resolution	Poor shimming; sample not spinning; insoluble matter present	Reshim; check spinning; filter the sample
All instruments: result differs from the expected	Calibration overdue; wrong method loaded; standard degraded	Verify calibration status; check the method; prepare a fresh standard

Routine maintenance for all. Keep a log book for every use. Follow the calibration schedule and record it. Keep the environment within specification for temperature, humidity and dust. Maintain an uninterrupted power supply. Keep a service contract and record every service. Keep spare lamps, cells and consumables in stock.

KEY POINTS TO REMEMBER

- UV-Visible involves electronic transitions; IR involves bond vibrations; NMR involves nuclear spin; AAS and ICP involve free atoms.
- **Beer-Lambert law: A = εcl.** Absorbance is proportional to concentration.
- IR functional group region is 4000 to 1500; fingerprint region is 1500 to 400 reciprocal centimetres.
- Metal-oxygen bonds absorb below 1000 reciprocal centimetres — the basis for detecting oxide formation in a Bhasma by FTIR.
- NMR chemical shift is measured in parts per million against tetramethylsilane.
- AAS uses one hollow cathode lamp per element and is a single-element technique. Mercury needs cold vapour; arsenic needs hydride generation.
- ICP-OES measures emitted light; ICP-MS measures ions by mass and is far more sensitive.
- **Elemental methods give the element, not its chemical form. They must be combined with a phase-identifying method such as XRD.**

- The NMR magnet is always on. The exclusion zone must be respected at all times.

EXAM CORNER — UNIT 4

This unit can be asked as ABQ (20 marks), SAQ (5 marks) and LAQ (10 marks).

Likely short answer questions (5 marks)

1. Define spectroscopy. Relate each region of the electromagnetic spectrum to the change it produces.
2. State the Beer-Lambert law and explain its application.
3. Define chromophore, auxochrome and λ_{max}.
4. Write a short note on FTIR and its application to Bhasma analysis.
5. Explain the two regions of an infrared spectrum.
6. Write a short note on NMR and its applications.
7. Write a short note on AAS and its principle.
8. How are mercury and arsenic determined by AAS?
9. Differentiate between ICP-OES and ICP-MS.
10. Why can elemental analysis alone not settle the question of the form of a metal in a Bhasma?
11. Write the safety precautions for handling an NMR instrument.
12. Enumerate the troubleshooting steps for an unstable AAS signal.

Likely long answer questions (10 marks)

1. Define spectroscopy. Describe its types with the principle and application of each.
2. Describe UV-Visible spectroscopy in detail, including the Beer-Lambert law, instrumentation and applications.
3. Describe infrared spectroscopy, its instrumentation and its application in the standardisation of Ayurvedic formulations.
4. Describe the atomic spectroscopic methods, AAS and ICP, and discuss their role in heavy metal analysis of Ayurvedic medicines.
5. Write a standard operating procedure for a spectroscopic instrument, and discuss its troubleshooting and maintenance.

Likely analytical based questions (20 marks)

1. A batch of Bhasma is reported to contain a high level of a heavy metal by an elemental method. Analyse what this result does and does not prove, describe the further investigation you would carry out, and state what conclusion each possible finding would support.
2. You are required to establish the complete analytical profile of a newly prepared Bhasma. Analyse which spectroscopic and other methods you would apply, in what order, what each would contribute, and how you would integrate the findings into a single conclusion.

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NCISM reference numbers for this unit: **11, 12, 13, 14, 15, 16, 135, 136, 137**

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UNIT 5
MICROSCOPY

SYLLABUS POINTS

5.1 Introduction, principles and types of microscopy (optical, petrological, electron, etc.)
5.2 Specifications of microscopy instruments
5.3 SOPs for handling microscopy instruments
5.4 Use of micrometer in microscopy

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

5.1 Introduction, principles and types

5.1.1 The two fundamental concepts

Concept	Definition	Note
Magnification	The ratio of the apparent size of the image to the actual size of the object	Total magnification = objective × eyepiece
Resolution or resolving power	The smallest distance between two points at which they are still seen as separate	This, not magnification, is the real limit

Magnifying beyond the resolving power gives a larger but no clearer image. This is called **empty magnification**, and it is a favourite examination point.

Resolution depends on the wavelength of the illuminating radiation and the numerical aperture of the objective. Shorter wavelength gives better resolution. This is exactly why the electron microscope, using electrons of very short wavelength, resolves far more than any light microscope.

Term	Meaning
Numerical aperture	A measure of the light-gathering ability of the objective; higher gives better resolution
Working distance	Distance from the objective to the specimen when in focus
Depth of field	The thickness of specimen in acceptable focus at one time
Immersion oil	Placed between specimen and objective to raise the numerical aperture at high power

5.1.2 Types of microscopy

A. Optical or light microscopy

Type	Principle	Use
Simple microscope	A single lens	Hand lens; preliminary examination
Compound microscope	Objective and eyepiece in series	The standard laboratory instrument; powder microscopy of herbal drugs
Stereo or dissecting microscope	Two separate optical paths giving a three-dimensional view at low power	Examining whole drugs, crystals, mineral specimens
Dark field	Illuminates obliquely; the specimen appears bright on a dark ground	Unstained and transparent specimens
Phase contrast	Converts phase differences into brightness differences	Living, unstained cells
Fluorescence	The specimen fluoresces under excitation light	Located fluorescent constituents
Polarising or petrological microscope	Uses polarised light and a rotating stage	**Minerals and crystalline material**
Confocal	Optical sectioning by laser scanning	Three-dimensional imaging
Trinocular microscope	A compound microscope with a third port for a camera	Documentation and measurement

B. Electron microscopy

Type	Principle	Gives
SEM — Scanning Electron Microscope	A focused electron beam scans the surface; secondary and backscattered electrons are collected	Surface topography, particle shape and size, in three-dimensional appearance
TEM — Transmission Electron Microscope	Electrons pass through an ultra-thin specimen	Internal structure at the nanometre scale; the highest resolution

Both are dealt with further in Unit 7.

5.1.3 The petrological microscope — why it matters to you

The **petrological or polarising microscope** is specifically named in your syllabus, and it is the instrument of mineral identification.

Component	Function
Polariser	Below the stage; produces plane polarised light
Analyser	Above the objective; a second polariser, inserted or removed
Rotating stage	Graduated in degrees; the specimen is rotated to observe optical changes
Bertrand lens	Used to view the interference figure
Accessory slot	For gypsum, mica and quartz wedge plates

Two modes of observation

Mode	Analyser	What is observed
Plane polarised light	Out	Colour, pleochroism, relief, cleavage, habit, shape
Crossed polars	In	Extinction, birefringence and interference colours, twinning, isotropic or anisotropic character

Application in Rasashastra. Mineral Dravyas can be identified by their optical properties in a thin section or a grain mount. An **isotropic** mineral stays dark under crossed polars in all positions of the stage; an **anisotropic** one brightens and darkens as the stage is rotated. Together with the crystal system, refractive index and interference colours, this allows a mineral to be identified without destroying it. It is the natural companion to the mineral databases you learned in Module 2.

5.2 Specifications of microscopy instruments

Instrument	Key specifications
Compound microscope	Objectives available, with magnification and numerical aperture; eyepiece magnification and field number; head type — monocular, binocular, trinocular; illumination type and intensity control; condenser type and numerical aperture; stage type and travel; focus, coarse and fine, with the fine division value
Stereo microscope	Zoom range; working distance; illumination, incident and transmitted
Petrological microscope	Strain-free objectives; rotating graduated stage; polariser and analyser; Bertrand lens; accessory plates
Trinocular with camera	Camera resolution in megapixels; sensor size; C-mount adapter magnification; software for capture and measurement
SEM	Resolution; accelerating voltage range; magnification range; detectors fitted; EDAX attachment; chamber size; vacuum system
TEM	Accelerating voltage; point and line resolution; magnification range; specimen holder types

5.3 SOP for handling microscopy instruments

Before use

1. Check the log book and the calibration status.
2. Remove the dust cover. Check that the lenses are clean.
3. Set the illumination to minimum before switching on.
4. Place the specimen and secure it on the stage.

During use

1. **Always begin with the lowest power objective.** Focus with the coarse adjustment, then the fine.
2. Move to higher power only after focusing at low power. Use only the fine adjustment at high power.
3. Adjust the condenser and the iris diaphragm for the objective in use. Set up Köhler illumination where the instrument permits.
4. **Never use the coarse adjustment with a high power objective in place.** The objective will strike the slide.
5. For the oil immersion objective, apply a single drop of immersion oil, and use only the fine adjustment.
6. Record what is observed. Photograph where a trinocular head is available.

After use

1. Lower the stage and return to the lowest power objective.
2. Clean the oil immersion objective immediately with lens tissue and the prescribed solvent.
3. Reduce the illumination to minimum, then switch off.
4. Cover the instrument and complete the log book.

Care and precautions. Carry the microscope with one hand on the arm and one under the base. Clean lenses only with lens tissue, never with cloth or paper. Keep the instrument away from moisture, dust and direct sunlight. Store with a desiccant in a humid climate to prevent fungal growth on the lenses. Never dismantle an objective.

5.4 Use of the micrometer in microscopy

Micrometry is the measurement of an object under the microscope. It requires two components.

Component	Description
Stage micrometer	A slide bearing a scale of known length, usually 1 millimetre divided into 100 divisions, so that **one division equals 0.01 millimetre, that is 10 micrometres**
Ocular or eyepiece micrometer	A glass disc with an arbitrary scale, placed inside the eyepiece. Its divisions have no fixed value until calibrated

5.4.1 Calibration procedure

1. Insert the ocular micrometer into the eyepiece.
2. Place the stage micrometer on the stage.
3. Select the objective to be calibrated.
4. Focus on the stage micrometer scale.
5. Rotate the eyepiece so that the two scales lie parallel.
6. Align the zero of the ocular scale with a line of the stage scale.
7. Find a point further along where a line of the ocular scale coincides exactly with a line of the stage scale.
8. Count the number of divisions on each scale between the two coincidence points.
9. Calculate the calibration factor.

Calibration factor, in micrometres per ocular division =

(number of stage micrometer divisions × 10) ÷ (number of ocular micrometer divisions)

A worked example. Suppose 8 stage divisions coincide with 25 ocular divisions.

Calibration factor = (8 × 10) ÷ 25 = 80 ÷ 25 = **3.2 micrometres per ocular division**

So an object measuring 15 ocular divisions is 15 × 3.2 = **48 micrometres**.

> **Two points that are always asked.**
> **1.** The calibration factor is different for every objective. Calibrate each one separately and record all of them.
> **2.** Remove the stage micrometer before measuring the specimen. Then measure the object in ocular divisions and multiply by the factor for that objective.

5.4.2 Applications in your subject

- Measuring particle size of a Churna or a Bhasma under the light microscope
- Measuring the dimensions of starch grains, fibres, stone cells and trichomes in powder microscopy of herbal drugs
- Measuring crystal size in a mineral preparation
- Demonstrating the progressive reduction in particle size with successive Putas

KEY POINTS TO REMEMBER

- Resolution, not magnification, is the real limit. Magnifying beyond it gives empty magnification.
- Resolution improves with shorter wavelength and higher numerical aperture. This is why electron microscopes resolve so much more.
- The petrological microscope has a polariser, an analyser, a rotating graduated stage, a Bertrand lens and accessory plates.
- Under crossed polars, an isotropic mineral remains dark in every position; an anisotropic one brightens and darkens on rotation.
- Always begin at the lowest power. Never use the coarse adjustment with a high power objective.
- Stage micrometer: 1 millimetre in 100 divisions, so **one division is 10 micrometres**.
- **Calibration factor = (stage divisions × 10) ÷ ocular divisions**, in micrometres per ocular division.
- The calibration factor differs for every objective and must be determined separately for each.

EXAM CORNER — UNIT 5

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Differentiate between magnification and resolution. What is empty magnification?

2. Enumerate the types of microscopy with the principle of each.
3. Write a short note on the compound microscope.
4. Write a short note on the petrological microscope and its components.
5. Differentiate between observation in plane polarised light and under crossed polars.
6. Differentiate between isotropic and anisotropic minerals under the microscope.
7. Write a short note on the trinocular microscope.
8. Differentiate between SEM and TEM.
9. Write the SOP for handling a compound microscope.
10. Describe the calibration of an ocular micrometer with the formula and a worked example.
11. Write a short note on the care and maintenance of a microscope.
12. What are the applications of micrometry in Rasashastra and Bhaishajya Kalpana?

Likely analytical based questions (20 marks)

1. You are given an unknown mineral sample said to be a Rasa Dravya. Analyse how you would establish its identity using microscopy together with the other methods available to you, describe each step, and state what conclusion each observation would support.
2. Design a study to demonstrate the progressive reduction in particle size of a Bhasma with successive Putas. Analyse which microscopic and other methods you would use, how you would calibrate and control them, and how you would present the data.

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UNIT 6
CALIBRATION TECHNIQUES

SYLLABUS POINTS

6.1 Introduction, principles and types of calibration (instrument calibration, method calibration, etc.)
6.2 SOPs for calibration
6.3 Troubleshooting and maintenance of calibration instruments

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

6.1 Introduction, principles and types

6.1.1 Definitions

Term	Definition
Calibration	The comparison of the reading of an instrument against a reference standard of known accuracy, and the determination of the deviation
Verification	Confirming that an instrument still performs within its specification
Qualification	Documented evidence that equipment is installed, operates and performs as intended — DQ, IQ, OQ, PQ
Validation	Documented evidence that a process or method consistently produces the intended result
Traceability	An unbroken chain of comparisons linking a measurement to a national or international standard
Reference standard	A material or instrument of established accuracy used for comparison
Tolerance or acceptance criterion	The maximum deviation that is acceptable
Uncertainty	The range within which the true value is expected to lie
Drift	Gradual change in an instrument’s reading over time
Calibration interval	The period after which recalibration is required

> **Calibration is not the same as adjustment.** Calibration **measures** the error. Adjustment **corrects** it. An instrument may be calibrated and found within tolerance, needing no adjustment at all. Getting this distinction right is worth a mark or two every time.

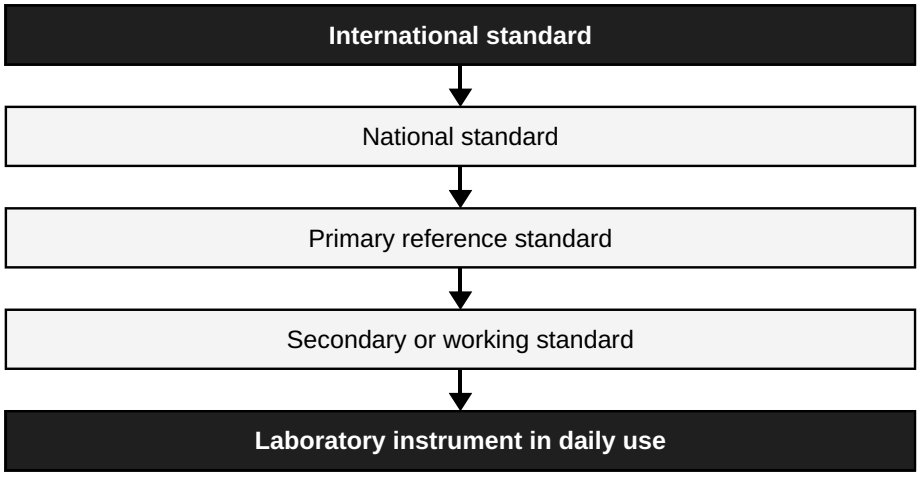
6.1.2 Why calibration matters

1. **Accuracy of results.** An uncalibrated balance makes every weight in the batch wrong.
2. **Regulatory requirement.** GMP, including Schedule T, requires that equipment be calibrated and that records be kept.
3. **Traceability.** Results can be defended only if the instruments are traceable to a recognised standard.
4. **Comparability.** Two laboratories can compare results only if both are calibrated.
5. **Detection of drift and failure.** Regular calibration finds a fault before it spoils a batch.
6. **Patient safety.** Every error in measurement ends up in the medicine.

6.1.3 Types of calibration

Type	Description
Instrument calibration	Checking the instrument against a reference standard — a balance against certified weights, a thermometer against a reference thermometer
Method calibration	Establishing the relationship between the instrument response and the quantity being measured — for example a calibration curve of absorbance against concentration
Internal calibration	Carried out in-house against secondary standards
External calibration	Carried out by an accredited external agency, giving a certificate
Single point calibration	Checked at one value
Multi-point calibration	Checked across the working range
Static calibration	Under steady conditions
Dynamic calibration	Under changing conditions

6.1.4 Hierarchy of standards



Each level is calibrated against the level above it. This unbroken chain is what **traceability** means.

6.1.5 Calibration of the instruments in a Rasashastra department

Instrument	Calibrated against	Typical parameters checked
Analytical balance	Certified standard weights of the appropriate class	Accuracy at several points across the range; repeatability; linearity; eccentricity
pH meter	Standard buffer solutions, commonly pH 4.0, 7.0 and 9.2	Slope and offset; two or three point calibration before every use
UV-Visible spectrophotometer	Holmium oxide or didymium filter for wavelength; potassium dichromate solution for absorbance	Wavelength accuracy; photometric accuracy; stray light; resolution
Thermometer, furnace, oven, incubator	Reference thermometer or calibrated thermocouple	Accuracy and uniformity across the chamber
Muffle furnace	Calibrated thermocouple	Set point accuracy; temperature distribution
Hot air oven	Reference thermometer or data logger	Temperature mapping at several positions
Sieves	Certified test sieves or reference glass beads	Aperture size; wire condition
Volumetric glassware	Gravimetric method, by weighing water	Volume delivered or contained
Microscope	Stage micrometer	Ocular micrometer calibration factor for each objective
Melting point apparatus	Certified melting point standards	Accuracy at several temperatures
Disintegration and dissolution apparatus	Calibrated tachometer, thermometer, and reference tablets	Speed, temperature, basket or paddle height

Instrument	Calibrated against	Typical parameters checked
HPLC and GC	System suitability standards	Retention time repeatability; peak area repeatability; resolution; tailing; theoretical plates

6.2 SOP for calibration

Objective. To ensure that every measuring instrument in the department gives readings within the defined tolerance, traceable to a recognised standard.

Responsibility. The instrument in-charge performs the calibration; the head of department approves; records are retained as prescribed.

Procedure

1. Maintain a **master list of every instrument** requiring calibration, with its identification number, location, and calibration interval.
2. Maintain a **calibration schedule** and review it monthly.
3. Before calibration, check that the reference standard itself is within its own validity period and has a valid certificate.
4. Ensure the environment is within specification for temperature, humidity and vibration.
5. Allow the instrument the prescribed stabilisation time.
6. Take readings at the prescribed number of points across the working range.
7. Repeat each point the prescribed number of times.
8. Record every reading. **Record the readings as found, before any adjustment.**
9. Calculate the deviation at each point.
10. Compare each deviation against the acceptance criterion.
11. If within tolerance, record “passed” and affix the calibration label.
12. If outside tolerance, **do not simply adjust and move on.** Record the failure, investigate the impact on all results produced since the last successful calibration, adjust or repair, recalibrate, and record the whole sequence.
13. Affix a calibration label showing the instrument number, the date of calibration, the due date, and the signature of the person responsible.
14. File the calibration certificate or record.
15. Update the master list and the schedule.

Records to maintain. Master instrument list; calibration schedule; individual calibration records with as-found and as-left readings; certificates from external agencies; out-of-calibration investigation reports; and the log book of every instrument.

> **The “as found” reading is the most important single entry in a calibration record.** It tells you how far the instrument had drifted before it was corrected. Without it, you cannot judge whether the results produced since the last calibration are reliable. An inspector will look for this entry, and its absence is a serious observation.

6.3 Troubleshooting and maintenance

Problem	Probable cause	Action
Balance readings unstable	Vibration; draught; unlevel bench; static charge; moisture	Use an anti-vibration table; close the draught shield; level the balance; use an antistatic device; control humidity
Balance fails calibration at high load	Overloaded in the past; mechanism damaged	Have it serviced; check the load history
pH meter slope out of range	Electrode ageing, dried out, or contaminated	Clean and recondition the electrode; replace if the slope remains out of range
pH readings drift	Electrode junction blocked; buffers degraded or contaminated	Clean the junction; use fresh buffers; store the electrode in the correct solution, never in distilled water
Furnace temperature not uniform	Element failure on one side; overloading; door seal worn	Check every element; reduce the load; replace the seal; carry out a temperature mapping
Thermocouple reading drifts	Ageing, contamination or mechanical damage	Recalibrate; replace the thermocouple
Spectrophotometer wavelength inaccurate	Lamp ageing; optics disturbed or dusty	Service; check with a holmium filter; replace the lamp
Instrument passes today, fails tomorrow	Environment out of control; interval too long	Control the environment; shorten the calibration interval
Calibration overdue and instrument used	Schedule not followed	Quarantine the instrument; investigate all results produced; record the deviation

Preventive maintenance. Follow the manufacturer’s schedule. Keep the environment within specification. Keep consumables and spares in stock. Train every user and record the training. Maintain a service contract for major instruments. Above all, **keep a log book for every instrument** — date, user, purpose, condition observed, and signature.

KEY POINTS TO REMEMBER

- Calibration measures the error; adjustment corrects it. They are not the same thing.
- Traceability is an unbroken chain of comparisons to a national or international standard.
- Types: instrument and method calibration; internal and external; single point and multi-point; static and dynamic.
- A balance is calibrated with certified weights; a pH meter with standard buffers, commonly 4.0, 7.0 and 9.2.
- A UV spectrophotometer is checked for wavelength with a holmium oxide filter and for absorbance with potassium dichromate.

- **Always record the as-found reading before any adjustment.**
- If an instrument fails calibration, investigate the impact on every result since the last successful calibration.
- Calibration and its records are a Schedule T and GMP requirement, not an optional refinement.

EXAM CORNER — UNIT 6

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Define calibration, verification, qualification and validation.
2. Differentiate between calibration and adjustment.
3. Define traceability and explain the hierarchy of standards.
4. Enumerate the types of calibration.
5. Why is calibration important? Give five reasons.
6. Describe the calibration of an analytical balance.
7. Describe the calibration of a pH meter.
8. Describe the calibration of a UV-Visible spectrophotometer.
9. Describe the calibration of a muffle furnace.
10. Write the SOP for calibration.
11. What is an as-found reading and why is it important?
12. What action is taken when an instrument fails calibration?

Likely analytical based questions (20 marks)

1. During a routine check, the analytical balance in your department is found to be reading outside tolerance. Analyse the consequences for the work carried out since the last calibration, describe the investigation you would conduct, and set out the corrective and preventive actions.
2. Design a complete calibration programme for a Rasashastra department, covering every instrument in use. Analyse the intervals, the standards, the records and the responsibilities, and justify your choices with reference to Schedule T.

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NCISM reference numbers for this unit: **143, 144, 145**

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UNIT 7
OVERVIEW OF CHARACTERISATION INSTRUMENTS

SYLLABUS POINTS

7.1 Overview of advanced characterisation instruments and their significance in the standardisation of Ayurvedic medicines
7.2 Principles and applications of working mechanism and applications of instrumental methods: PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA, VSM analysis

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	Yes

7.1 Overview and significance

7.1.1 Why these instruments matter to Rasashastra

The classical Bhasma Pariksha you learned in Module 4 are excellent tests. They are also **qualitative and operator-dependent**. Rekhapurnata tells you the particles are fine, but not how fine. Apunarbhava tells you no free metal remains, but does not say what compound has formed.

The instruments in this unit answer those questions numerically. They do not replace Bhasma Pariksha. They **complete** it.

Classical test	What it establishes	Instrument that quantifies it
Rekhapurnata, Varitara, Slakshnatva	Fineness	PSA, SEM, TEM
Nishchandratva, Apunarbhava, Niruttha	No free metal remains	XRD, XPS
Varna	The correct compound formed	XRD, FTIR
Nirdhuma	No organic or volatile residue	TGA
Gata Rasatva, Nishwadatva	Complete transformation	XRD, ICP
—	Elemental composition	XRF, EDAX, ICP
—	Magnetic character	VSM

7.2 The instruments

7.2.1 PSA — Particle Size Analyser

- **Principle.** In the laser diffraction method, particles suspended in a liquid or air scatter a laser beam. Large particles scatter at small angles, small particles at large angles. The pattern of scattering is converted into a size distribution. In dynamic light scattering, used for the nanometre range, the fluctuation of scattered light caused by Brownian motion gives the size.
- **Gives.** The particle size distribution — commonly reported as D10, D50 and D90, where D50 is the median size.
- **Application.** Measuring the fineness of a Bhasma or a Churna; demonstrating the reduction in size with successive Putas; comparing batches.
- **Limitation.** Measures aggregates as single particles unless the sample is properly dispersed. Sample preparation is critical.

7.2.2 XRD — X-Ray Diffraction

- **Principle.** X-rays incident on a crystalline solid are diffracted by the planes of atoms. Reinforcement occurs at angles satisfying **Bragg's law, $n\lambda = 2d \sin \theta$**. Each crystalline phase gives a unique set of d-spacings.
- **Gives.** Identification of the crystalline phases present; an estimate of crystallite size from peak broadening, by the Scherrer relationship; the degree of crystallinity.
- **Application. The single most important instrument for Bhasma characterisation.** It shows which compound has formed, whether free metal remains, and how the phase changes with each Puta.
- **Limitation.** Detects only crystalline material. Amorphous content is not seen. Phases present below a few per cent may be missed. It gives no information about organic matter.

7.2.3 XRF — X-Ray Fluorescence

- **Principle.** X-rays eject inner shell electrons. As outer electrons fall into the vacancies, X-rays of characteristic energy are emitted. The energy identifies the element; the intensity gives the quantity.
- **Gives.** Elemental composition, qualitative and quantitative.
- **Application.** Rapid elemental analysis of a Bhasma or a mineral; screening for heavy metal contaminants; verifying that the elements present are those expected.
- **Advantage.** Non-destructive, needs little sample preparation, covers many elements at once.
- **Limitation.** Poor for the lightest elements. Gives the element, **not the compound**.

7.2.4 SEM-EDAX

- **SEM principle.** A focused electron beam scans the specimen surface. Secondary electrons give topographic contrast; backscattered electrons give compositional contrast. The image has a three-dimensional appearance and a large depth of field.
- **EDAX principle.** X-rays emitted from the point where the beam strikes are analysed by energy. This identifies the elements present at that exact point.
- **Together they give.** Particle shape, size and surface texture, together with the elemental composition of the very region being viewed.

- **Application.** Examining the morphology of Bhasma particles; showing whether particles are separate or agglomerated; mapping the distribution of elements across the sample.
- **Limitation.** Surface information only. Non-conducting samples need a conductive coating. EDAX is semi-quantitative and poor for light elements.

> **SEM-EDAX is the most informative single technique for a Bhasma, because it gives you shape, size and composition of the same particle at the same time.** If your department has access to only one advanced instrument, this is the one to use. Pair it with XRD, which supplies the phase information that EDAX cannot.

7.2.5 TEM — Transmission Electron Microscopy

- **Principle.** Electrons pass through an ultra-thin specimen and are imaged. Because the wavelength of electrons is extremely short, resolution reaches the atomic scale.
- **Gives.** Internal structure, particle size and shape at the nanometre scale; electron diffraction patterns showing crystal structure.
- **Application.** Confirming the presence of nanoparticles in a Bhasma; measuring their size directly. Central to the discussion in Module 8.
- **Limitation.** Demanding sample preparation; a very small area is examined, so the field seen may not represent the whole; costly.

7.2.6 TGA — Thermogravimetric Analysis

- **Principle.** The weight of the sample is recorded continuously as it is heated at a controlled rate.
- **Gives.** A curve of weight against temperature, showing at what temperature weight is lost and how much.
- **Application.** Determining moisture content; determining the amount of organic material associated with a Bhasma and the temperature at which it burns off; assessing thermal stability; detecting decomposition of carbonates and hydrates.
- **In your subject specifically.** TGA is the standard way of quantifying the organic residue on a Bhasma. It gives a number where FTIR gives only a presence.

7.2.7 DTA — Differential Thermal Analysis

- **Principle.** The temperature difference between the sample and an inert reference is recorded as both are heated at the same rate. A difference appears whenever the sample absorbs or releases heat.
- **Gives.** Endothermic peaks for melting, dehydration and decomposition; exothermic peaks for crystallisation, oxidation and combustion.
- **Application.** Detecting phase transitions and thermal events; identifying the temperature at which a change occurs.
- **Note.** TGA and DTA are usually run together on the same sample. TGA shows *how much* weight is lost; DTA shows *what kind of event* caused it. Together they are far more informative than either alone. **DSC**, differential scanning calorimetry, is a related technique that measures the heat flow itself.

7.2.8 VSM — Vibrating Sample Magnetometer

- **Principle.** The sample is vibrated in a magnetic field. The changing magnetic flux induces a voltage in pickup coils, from which the magnetic moment of the sample is determined.
- **Gives.** The magnetisation curve, from which the saturation magnetisation, the coercivity and the remanence are obtained; and the classification of the material as diamagnetic, paramagnetic, ferromagnetic or superparamagnetic.
- **Application in your subject. Iron-containing preparations** — Lauha Bhasma, Mandura Bhasma and Abhraka Bhasma. The magnetic behaviour depends on which iron oxide phase has formed and on the particle size. Very small particles of certain iron oxides show superparamagnetism, and VSM detects this.
- **Why it matters.** It provides an independent confirmation of the phase and the particle size, complementing XRD, and it is one of the few techniques specifically suited to the iron preparations that are so important in Ayurvedic practice.

7.2.9 Summary table

Instrument	Principle in one line	What it gives	Chief limitation
PSA	Laser scattering by particles	Particle size distribution	Measures aggregates unless well dispersed
XRD	Diffraction by crystal planes	Crystalline phases, crystallite size	Blind to amorphous and organic material
XRF	Characteristic X-ray emission	Elemental composition	Gives element, not compound
SEM	Scanning electron beam, surface electrons	Surface morphology and size	Surface only; needs coating
EDAX	X-rays from the beam spot	Elemental composition at a point	Semi-quantitative; poor for light elements
TEM	Electrons transmitted through a thin sample	Nanoscale internal structure and size	Very small area; demanding preparation
TGA	Weight during controlled heating	Weight loss against temperature	Does not identify what is lost
DTA	Temperature difference against a reference	Thermal events, endothermic and exothermic	Does not quantify weight change
VSM	Magnetic moment in a varying field	Magnetic properties	Useful mainly for magnetic materials

The answer that ties this unit together. No instrument characterises a Bhasma by itself. A complete characterisation needs: **XRD** for the phase, **SEM-EDAX** for morphology and elements, **TEM or PSA** for size, **FTIR** for bonding and organic residue, **TGA-DTA** for thermal behaviour and organic content, **ICP or AAS** for trace elements and heavy metals, and **VSM** where the material is magnetic. State this set, say what each contributes, and then say that the results must be read alongside the classical Bhasma Pariksha, not instead of them.

KEY POINTS TO REMEMBER

- These instruments quantify what the classical Bhasma Pariksha establish qualitatively. They complete the classical tests, not replace them.
- PSA gives the size distribution, reported as D10, D50 and D90.
- XRD identifies crystalline phases by Bragg's law and estimates crystallite size from peak broadening.
- XRF and EDAX give elements, not compounds.
- SEM-EDAX gives shape, size and composition of the same particle at once — the most informative single technique for a Bhasma.
- TEM reaches the nanometre scale and confirms nanoparticles.
- TGA quantifies weight loss with temperature; DTA identifies the thermal event causing it. They are run together.
- VSM measures magnetic properties and is specifically useful for Lauha, Mandura and Abhraka Bhasma.
- A complete Bhasma characterisation requires a set of instruments, not one.

EXAM CORNER — UNIT 7

This unit can be asked as ABQ (20 marks), SAQ (5 marks) and LAQ (10 marks).

Likely short answer questions (5 marks)

1. Write a short note on the particle size analyser.
2. Write a short note on XRD and its application in Bhasma standardisation.
3. Differentiate between XRD and XRF.
4. Write a short note on SEM-EDAX.
5. Differentiate between SEM and TEM.
6. Write a short note on TGA.
7. Differentiate between TGA and DTA.
8. Write a short note on VSM and its application in Ayurvedic preparations.
9. Which instruments would you use to demonstrate that a Bhasma contains no free metal?
10. Which instruments would you use to determine the particle size of a Bhasma?
11. Correlate the classical Bhasma Pariksha with modern instrumental methods.

12. Why can no single instrument characterise a Bhasma completely?

Likely long answer questions (10 marks)

1. Give an overview of the advanced characterisation instruments and discuss their significance in the standardisation of Ayurvedic medicines.
2. Describe the principle, working and applications of XRD, SEM-EDAX and TEM.
3. Describe the thermal methods of analysis, TGA and DTA, with their principles and applications.
4. Discuss how classical Bhasma Pariksha and modern instrumental methods can be integrated into a single standardisation protocol.
5. Describe the instruments you would use to characterise a Lauha Bhasma completely, with the contribution of each.

Likely analytical based questions (20 marks)

1. You are required to establish, for publication, that a Bhasma prepared in your department is properly made. Analyse the complete set of classical and instrumental evidence you would generate, justify the inclusion of each method, and explain how you would present and interpret the combined data.
2. Two samples of the same Bhasma give identical XRD patterns but differ markedly in their SEM images and particle size distribution. Analyse what this means, what could have caused it, and what the therapeutic implications might be.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **14, 15, 17, 18, 20, 26**

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MODULE 5
PRACTICAL TRAINING ACTIVITIES

Module 5 carries **30 hours of practical training**, over seven activities.

No	Practical name	Hours
5.1	Demonstration of selected Puta to give the knowledge of standard operating procedure	4
5.2	Demonstration of the handling of Musha following the SOP	4
5.3	Demonstration of chromatography techniques to separate, identify components and interpret results	6
5.4	Demonstration of spectroscopic instruments through audiovisual aids	3
5.5	Demonstration of handling of petrological and trinocular microscopes	6
5.6	Demonstration of calibration of various instruments present in the department	3
5.7	Demonstration of characterisation instruments with audiovisual aids	4
	Total	**30**

Practical Training 5.1 — Demonstration of a selected Puta

Aim. To perform a selected Puta following a written SOP, and to record the complete cycle.

Requirements. Sharava, material for Sandhi Bandhana, dried cow dung cakes, Puta pit, thermocouple with data logger if available, weighing balance, measuring tape, safety equipment.

Procedure

1. Select the type of Puta and locate its specification in the classical text. Record the reference.
2. Write the SOP before beginning.
3. Measure and record the actual pit dimensions.
4. Weigh and count the cow dung cakes. Record their condition and moisture.
5. Prepare and seal the Sharava Samputa. Weigh and record.
6. Position the thermocouple if available.
7. Arrange, ignite, and record the time and the weather.
8. Record the temperature profile at intervals.

9. Allow Swangasheeta. Record the total cooling time.
10. Recover and examine the Samputa. Weigh and record.
11. Plot the temperature against time.
12. Compare the observed profile with the classical expectation.

Observation to record. Pit dimensions, fuel weight and number, weather, ignition and burnout times, the temperature-time table, and the plotted curve.

Precautions. As per Unit 1, section 1.2. **Never open the pit before Swangasheeta is complete. Never use water on the pit.**

Practical Training 5.2 — Handling of Musha following the SOP

Aim. To prepare, condition and use a Musha according to a written SOP.

Procedure

1. Write the SOP first.
2. Select and prepare the clay. Record the source.
3. Weigh each ingredient in the prescribed proportion. Record.
4. Mix, knead, and allow the mass to mature.
5. Shape the Musha to the specified form and wall thickness. Record the dimensions.
6. Smooth the inner surface. Dry slowly in shade. Record the drying time.
7. Pre-fire, raising the temperature gradually.
8. Inspect for cracks. Record the pass or reject decision with the reason.
9. Number the Musha and begin its usage record.
10. Use it for a prescribed operation, following the handling SOP.
11. Record the performance and any deterioration observed.

Observation to record. The composition with quantities, the dimensions, the drying and firing schedule, the inspection result, and the performance in use.

Precautions. Long tongs, heat-resistant gloves, apron and face shield. Never place a cold Musha into a hot furnace. Never quench.

Practical Training 5.3 — Chromatography

Aim. To separate, identify and interpret the components of a herbal drug or formulation by chromatography.

Requirements. TLC plates or HPTLC plates; development chamber; mobile phase solvents; applicator; UV cabinet at 254 and 366 nanometres; spray reagents; reference standards; the extracts prepared in Module 3; HPTLC or HPLC system if available.

Procedure

1. Prepare the sample extract and the reference standard solution.
2. Select the mobile phase, from the pharmacopoeia or by trial.
3. Saturate the chamber for the prescribed period.
4. Apply the sample and the standard on the same plate.
5. Develop to the marked distance. Record the time.
6. Dry the plate. Observe under daylight, then at 254 nanometres, then at 366 nanometres. Record each.
7. Spray the derivatising reagent and heat as prescribed. Record again.
8. Measure and calculate the R_f of every spot.
9. Compare the sample track against the standard track.
10. If HPTLC is available, scan densitometrically and record the densitogram.
11. If HPLC is available, run the same sample and record the chromatogram and retention times.
12. Interpret: identify the marker; note any extra or missing feature.

Observation table to draw

Track	Spot number	Distance travelled	R_f	Colour in daylight	At 254 nm	At 366 nm	After spray	Inference

Result. Record the identity established and the fingerprint obtained. **Record every chromatographic condition in full.**

Precautions. Handle solvents in a fume hood. Many spray reagents contain concentrated sulphuric acid — wear gloves and eye protection. **Never look directly at the ultraviolet source.** Dispose of solvents as prescribed.

Practical Training 5.4 — Spectroscopic instruments through audiovisual aids

Aim. To understand the construction, working and application of spectroscopic instruments.

Procedure

1. View the demonstration or video for each instrument — UV-Visible, FTIR, NMR, AAS and ICP.
2. For each, draw a labelled block diagram from the eyepiece or source to the detector.
3. Record the function of every component.
4. Record the specifications discussed.
5. Study a sample spectrum from each and interpret it.
6. Record which questions about an Ayurvedic sample each instrument could answer, and which it could not.

Observation to record. Labelled diagrams, component function tables, and the interpretation of one specimen spectrum from each instrument.

Practical Training 5.5 — Petrological and trinocular microscopes

Aim. To handle the petrological and trinocular microscopes, and to examine mineral and powder specimens.

Procedure

1. Identify and record every part of each microscope.
2. Follow the SOP in Unit 5, section 5.3, for setting up.
3. Calibrate the ocular micrometer for each objective. Record the calibration factor for each.
4. Examine a herbal powder under the trinocular microscope. Identify the diagnostic characters. Measure them.
5. Photograph the field and record the magnification.
6. Mount a mineral specimen on the petrological microscope.
7. Observe in plane polarised light. Record colour, pleochroism, relief and cleavage.
8. Insert the analyser. Rotate the stage through 360 degrees. Record whether the specimen is isotropic or anisotropic, and note the extinction and interference colours.
9. Attempt an identification, and verify it against a mineral database.
10. Examine a Bhasma sample and measure its particle size in ocular divisions. Convert to micrometres.

Observation tables to draw. One for the calibration factor of each objective. One for the powder microscopy characters with measurements. One for the mineral optical properties.

Precautions. Follow the handling SOP exactly. Never use the coarse adjustment at high power. Clean the oil immersion objective immediately after use.

Practical Training 5.6 — Calibration of instruments in the department

Aim. To perform the calibration of the instruments available, and to prepare the records.

Procedure

1. Prepare the master list of instruments in the department.
2. For each instrument selected, obtain the reference standard and verify its validity.
3. Follow the calibration SOP in Unit 6, section 6.2.
4. **Record the as-found readings before any adjustment.**
5. Calculate the deviation at each point and compare against the acceptance criterion.
6. Record pass or fail with the reason.
7. Prepare and affix a calibration label.
8. Complete the calibration record and file it.
9. For at least one instrument, prepare a full calibration certificate.

Observation table to draw

Instrument	ID number	Reference standard used	Point checked	As-found reading	Deviation	Acceptance criterion	Pass or fail

Practical Training 5.7 — Characterisation instruments through audiovisual aids

Aim. To understand the working and application of the advanced characterisation instruments.

Procedure

1. View the demonstration or video for PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA and VSM.
2. For each, draw a labelled schematic and record the working principle in your own words.
3. Study one specimen output from each — a particle size distribution, an XRD pattern, an EDAX spectrum, an SEM image, a TGA curve, a DTA curve and a magnetisation curve.
4. Interpret each output.
5. Prepare a table mapping each instrument to the classical Bhasma Pariksha it corresponds to.
6. Design, on paper, a complete characterisation protocol for one Bhasma.

Observation to record. The schematics, the interpretation of each specimen output, the mapping table, and the designed protocol.

MODULE 5
EXPERIENTIAL LEARNING ACTIVITIES

Module 5 carries **45 hours** under experiential learning, including modular assessment. Seven activities are prescribed.

No	Experiential learning activity	Hours
5.1	Developing standard operating procedures for a given Puta	8
5.2	Developing a standard operating procedure for handling Musha	8
5.3	Performance of chromatography to separate and identify components, and interpretation of results	8
5.4	Analysis of published articles on spectroscopic instruments	4
5.5	Hands-on practice on the trinocular microscope	4
5.6	Analysis of articles related to challenges in the calibration of equipment	3
5.7	Analysis of published articles on advanced instruments used for characterisation	4
	Total	**39**

Experiential Learning 5.1 — Developing an SOP for a given Puta

What you have to do. Write a complete, usable SOP for a specified Puta, and then test it by having another person follow it.

Method. Write all fifteen sections. Convert every classical judgement into a measurable quantity, using the table in Module 4, Unit 5. Then hand the SOP to a colleague who has not seen your work, and ask them to perform the Puta from your document alone.

The test of a good SOP. If your colleague has to ask you a single question, the SOP is incomplete. Record every question asked, and revise the document until none remain.

How to record it. The SOP, the record of questions asked during the trial, the revised SOP, and the batch record from the trial run.

Experiential Learning 5.2 — Developing an SOP for handling Musha

What you have to do. Write a complete SOP covering the preparation, conditioning, use, inspection, retirement and storage of a Musha.

Method. As above. Include the composition with exact proportions, the drying and pre-firing schedule, the inspection criteria for acceptance and rejection, the maximum permitted number of uses, and the record formats.

Questions your SOP must answer without ambiguity

- Exactly how much of each ingredient, by weight?
- Exactly how thick should the wall be?
- How long to dry, and how is dryness confirmed?
- What flaw makes a Musha unfit for use?
- After how many uses is it retired?

Experiential Learning 5.3 — Performance and interpretation of chromatography

What you have to do. Move from following a chromatographic method to developing one and interpreting the result independently.

Method. Take one classical formulation. Develop a TLC or HPTLC method for it from first principles — select the extraction, trial several mobile phases, optimise the separation, select the visualisation. Then establish its fingerprint, and compare three market samples of the same formulation against it.

Questions to answer. Which mobile phase gave the best resolution, and why? Do the three market samples give the same fingerprint? If not, what could explain the difference? What would you recommend as the acceptance criterion for this formulation?

How to record it. The method development record including the failed trials, the final validated conditions, the plate images, the densitograms, the comparison of market samples, and your recommendation.

Experiential Learning 5.4 — Analysis of published articles on spectroscopic instruments

What you have to do. Read and critically analyse published studies that used spectroscopy on Ayurvedic materials.

Method. Obtain at least four papers. For each, record the material studied, the instruments used, the conditions reported, the results, and the conclusion drawn.

Then answer. Were the conditions reported completely enough for you to repeat the work? Did the conclusion follow from the data, or did the authors claim more than they showed? Was an elemental method used to support a claim about chemical form? What would you have done differently?

Experiential Learning 5.5 — Hands-on practice on the trinocular microscope

What you have to do. Become independently competent on the trinocular microscope, including photography and measurement.

Method. Calibrate every objective. Prepare and examine at least ten specimens — herbal powders, a formulation, a mineral grain mount, and a Bhasma. Photograph each with the scale recorded. Measure the diagnostic characters. Build a photomicrograph album with full details of magnification and measurement for each.

Deliverable. The calibration record for every objective, and the annotated photomicrograph album.

Experiential Learning 5.6 — Challenges in calibration of equipment

What you have to do. Analyse published and practical accounts of calibration problems, and relate them to your own department.

Method. Read the available literature on calibration challenges. Then audit your own department — list every instrument, check whether each has a current calibration, examine the records, and identify the gaps.

Deliverable. A written audit report with the master list, the compliance status of each instrument, the gaps found, and your recommendations with a proposed schedule.

Experiential Learning 5.7 — Analysis of published articles on characterisation instruments

What you have to do. Critically analyse published characterisation studies of Bhasma and other Ayurvedic preparations.

Method. Obtain at least four studies that used XRD, SEM-EDAX, TEM, TGA or VSM on Ayurvedic materials. Extract the methods and the findings into a comparison table.

Then answer. Which combinations of instruments were used, and were they sufficient? What did each study claim, and did the data support it? Were the classical Bhasma Pariksha reported alongside the instrumental data? What is the commonest weakness across the four studies?

Deliverable. The comparison table, the critique, and a proposed model characterisation protocol that avoids the weaknesses you identified.

MODULE 5
VIVA POINTERS AND QUESTION BANK

Viva pointers

On Puta and Musha

- Name the types of Puta in order of temperature.
- What are the dimensions of a Gaja Puta pit, and how many cow dung cakes?
- What factors change the temperature actually reached?
- Can a muffle furnace replace a Puta? Justify your answer.
- What is a Musha made of, and what does each ingredient do?
- Which Musha is used for Satvapatana?
- Why must a cold Musha never enter a hot furnace?

On chromatography

- Define R_f. What are its limits?
- Differentiate TLC from HPTLC.
- What is reverse phase HPLC?
- Which technique would you use for volatile oil? Why?
- Can chromatography analyse a Bhasma? Why not?

On spectroscopy

- State the Beer-Lambert law.
- What are the two regions of an IR spectrum?
- Where do metal-oxygen bonds absorb?
- How is mercury determined by AAS?
- Differentiate ICP-OES from ICP-MS.
- Does ICP tell you the chemical form of a metal?

On microscopy and calibration

- Differentiate magnification from resolution.
- Show me the polariser and the analyser.
- How do you distinguish an isotropic from an anisotropic mineral?
- What is the value of one stage micrometer division?
- Calibrate this ocular micrometer and show me the calculation.
- What is the difference between calibration and adjustment?
- What is an as-found reading?

On characterisation instruments

- Which instrument tells you the phase present in a Bhasma?
- Which gives shape and composition together?
- What does TGA measure? What does DTA measure?
- Why is VSM used for Lauha Bhasma?
- Why can no single instrument characterise a Bhasma?

Module 5 question bank

Analytical Based Questions (20 marks) — from Units 3, 4, 5, 6 and 7

1. You are given a market sample of a classical Ayurvedic formulation and asked to establish whether it is genuine. Analyse the problem, select your methods with reasons, and interpret every possible result.
2. A herbal formulation shows an extra peak in HPLC that is absent from the reference. Analyse the possible explanations and the action for each.
3. Design a complete chromatographic standardisation protocol for a classical Kwatha Churna, and justify every choice.
4. A Bhasma is reported to contain a high level of a heavy metal by an elemental method. Analyse what this proves and does not prove, and describe the further investigation required.
5. Establish the complete analytical profile of a newly prepared Bhasma. Analyse which methods, in what order, and how the findings are integrated.
6. An unknown mineral sample is presented as a Rasa Dravya. Analyse how you would establish its identity using microscopy and the other methods available.
7. Design a study to demonstrate the reduction in particle size of a Bhasma with successive Putas.
8. The analytical balance is found outside tolerance. Analyse the consequences and set out the corrective and preventive action.
9. Design a complete calibration programme for a Rasashastra department, with reference to Schedule T.
10. Establish for publication that a Bhasma is properly made. Analyse the complete classical and instrumental evidence required.
11. Two samples of the same Bhasma give identical XRD patterns but differ in SEM images and particle size. Analyse what this means and what could have caused it.

Long Answer Questions (10 marks) — from Units 1, 3, 4 and 7

1. Describe the types of Puta with their principles, specifications and applications, and the factors affecting the temperature attained.
2. Write a complete standard operating procedure for handling a Puta.
3. Discuss the modern aspects of Puta and compare the traditional Puta with the electric muffle furnace.
4. Define chromatography and describe its types with the principle and application of each.
5. Describe HPLC in detail — principle, instrumentation, modes and applications.
6. Describe HPTLC and discuss its role in the standardisation of Ayurvedic formulations.
7. Define spectroscopy and describe its types with principle and application.
8. Describe UV-Visible spectroscopy including the Beer-Lambert law, instrumentation and applications.

9. Describe the atomic spectroscopic methods and their role in heavy metal analysis of Ayurvedic medicines.
10. Give an overview of advanced characterisation instruments and their significance in standardisation.
11. Describe the principle, working and applications of XRD, SEM-EDAX and TEM.
12. Discuss how classical Bhasma Pariksha and modern instrumental methods can be integrated into one protocol.

Short Answer Questions (5 marks) — from all seven units

From Unit 1

1. Define Puta and explain its principle.
2. Enumerate the types of Puta with their specifications.
3. Enumerate the factors affecting the temperature attained in a Puta.
4. Write a short note on the electric muffle furnace.
5. What is Swangasheeta and why must it not be hurried?

From Unit 2

1. Enumerate the materials used in preparing a Musha with the function of each.
2. Enumerate the types of Musha with their uses.
3. Enumerate the Musha Lakshana of a good Musha.
4. Enumerate the modern refractory materials with their working temperatures.
5. On what criteria is a crucible selected?

From Unit 3

1. Define R_f and state the factors affecting it.
2. Differentiate between TLC and HPTLC.
3. Differentiate between normal phase and reverse phase HPLC.
4. Write a short note on GC-MS and LC-MS.
5. Write a short note on fingerprinting of herbal formulations.
6. Enumerate the limitations of chromatography in analysing Ayurvedic formulations.

From Unit 4

1. State the Beer-Lambert law and explain its application.
2. Explain the two regions of an infrared spectrum.
3. Write a short note on FTIR and its application to Bhasma analysis.
4. Write a short note on AAS and its principle.
5. Differentiate between ICP-OES and ICP-MS.
6. Why can elemental analysis alone not establish the form of a metal in a Bhasma?

From Unit 5

1. Differentiate between magnification and resolution. What is empty magnification?
2. Write a short note on the petrological microscope and its components.

3. Differentiate between isotropic and anisotropic minerals under crossed polars.
4. Describe the calibration of an ocular micrometer with the formula and an example.
5. Write a short note on the care and maintenance of a microscope.

From Unit 6

1. Define calibration, verification, qualification and validation.
2. Differentiate between calibration and adjustment.
3. Define traceability and explain the hierarchy of standards.
4. Describe the calibration of an analytical balance and of a pH meter.
5. What is an as-found reading and why is it important?

From Unit 7

1. Write a short note on XRD and its application in Bhasma standardisation.
2. Write a short note on SEM-EDAX.
3. Differentiate between TGA and DTA.
4. Write a short note on VSM and its application in Ayurvedic preparations.
5. Correlate the classical Bhasma Pariksha with modern instrumental methods.

> **Note on the marks range.** Table 6F places Module 5 in the marks range of **5 to 20**, and it carries the largest modular assessment in the course at 75 marks. Five of its seven units are ABQ-eligible — Units 3, 4, 5, 6 and 7. Units 2, 5 and 6 carry no long answer. Prepare Units 3, 4 and 7 most thoroughly, since these three carry all three question types.

MODULE 6
ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE

Credits	Lectures	Practical Training	Experiential Learning	Total hours	Modular assessment
1	5	10	15	30	25 marks

About this module

This module provides a comprehensive understanding of animal house management and animal handling techniques. It covers the principles of animal care and welfare, animal house design and layout, record-keeping, animal behaviour, species-specific handling techniques, health and disease prevention, and animal welfare and ethics.

Module Learning Objectives

At the end of this module, the student should be able to:

1. Describe the principles of animal care and welfare, and apply them to animal house management.
2. Demonstrate handling techniques for various animal species, including small and large animals.
3. Identify and prevent common diseases in laboratory animals, and implement disease control measures.
4. Apply advanced animal handling techniques and promote animal welfare and ethics in research.
5. Develop skills in animal behaviour observation, enrichment techniques, and humane endpoints and euthanasia methods.

Units in this module

Unit	Name	ABQ	SAQ	LAQ
M6U1	Introduction to animal house management	No	Yes	No
M6U2	Study of animal behaviour	No	Yes	No
M6U3	Species-specific handling techniques	No	Yes	No
M6U4	Animal health and disease prevention	No	Yes	Yes
M6U5	Animal welfare and ethics	No	Yes	No

Module 6 is placed in the marks range of **5 to 15**. **No unit of this module is set as an Analytical Based Question.** Only Unit 4 carries a ten-mark long answer. Everything else is short answer. This is the smallest module in the course and the easiest to score in, provided you learn the definitions, the lists and the legal names accurately.

Why this module is in your syllabus

Rasashastra administers metals and minerals to human beings. Before any new preparation, or any modified preparation, can be given to a patient, its safety must be established. Rule 158-B, which you studied in Module 1, requires proof of safety and effectiveness for several categories of Ayurvedic medicine. For many of those categories, that proof begins with animal studies.

You will therefore encounter the animal house in three situations:

1. **Acute and sub-acute toxicity studies** on a Bhasma or a new formulation.
2. **Pharmacological screening** to establish an activity before a clinical study.
3. **Bioavailability and pharmacokinetic studies**, which are the largest gap in the present evidence base for Bhasma.

You cannot do any of this without knowing the law, the ethics and the practice of animal experimentation.

> **An important legal point that most textbooks still get wrong.**
>
> The regulatory body was formerly called the **Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA)**. *By order dated 6 January 2023 the Ministry of Fisheries, Animal Husbandry and Dairying changed its name to the* ***Committee for Control and Supervision of Experiments on Animals (CCSEA)****. The functions and powers of the committee remain the same.*
>
> Many books, guidelines and even registration numbers still carry the old abbreviation. Use **CCSEA** in your answers, and mention that it was formerly CPCSEA. That single sentence tells an examiner that your knowledge is current.

UNIT 1
INTRODUCTION TO ANIMAL HOUSE MANAGEMENT

SYLLABUS POINTS

1.1 Overview of animal house management
1.2 Animal house design and layout
1.3 Record-keeping and documentation

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

1.1 Overview of animal house management

1.1.1 Definition

An **animal house** or **laboratory animal facility** is a specialised establishment maintained exclusively for the housing, breeding and use of animals for scientific purposes, under controlled environmental conditions and under statutory supervision.

1.1.2 The legal framework in India

Element	Detail
Parent Act	*The Prevention of Cruelty to Animals Act, 1960*
Regulatory body	*The Committee for Control and Supervision of Experiments on Animals (CCSEA), a statutory committee of the Department of Animal Husbandry and Dairying, Ministry of Fisheries, Animal Husbandry and Dairying*
Former name	Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), renamed in January 2023
Principal rules	*The Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998, amended in 2001 and 2006*
Duty of the committee	*To take all measures necessary to ensure that animals are not subjected to unnecessary pain or suffering before, during or after the performance of experiments on them*
Institutional body	The Institutional Animal Ethics Committee (IAEC)

1.1.3 Registration and the IAEC

Every establishment that breeds, trades in, or experiments upon laboratory animals must be **registered with CCSEA**. Registration is granted after inspection of the facility, and it carries a registration number that must be quoted in every publication and every protocol.

The Institutional Animal Ethics Committee (IAEC) is constituted within the institution. It reviews and approves every proposal to use animals, before any work begins. Its composition is prescribed and includes:

Member	Role
Biological scientist	Chairperson, usually
Scientist from a different biological discipline	Member
Veterinarian	Responsible for animal health and welfare
Scientist in charge of the animal facility	Member Secretary, usually
Nominee of CCSEA — main nominee	Statutory external member
Link nominee and scientist from outside the institute	External members
Socially aware member	Represents the public interest

No animal experiment may begin without prior written IAEC approval. For certain categories, including large animals, approval of CCSEA itself is also required.

1.1.4 Objectives of animal house management

1. To provide housing that meets the biological needs of each species.
2. To maintain a controlled and monitored environment.
3. To prevent, detect and control disease.
4. To ensure the health, comfort and welfare of every animal.
5. To produce reliable and reproducible experimental data.
6. To protect the staff from injury and from zoonotic infection.
7. To comply fully with the law and to maintain the records that prove it.

> **The scientific argument for good animal husbandry, which is worth writing.** A stressed, diseased or poorly housed animal gives variable and unreliable results. Welfare and science are not in conflict here — they point the same way. Good husbandry is a requirement of valid data, not merely a requirement of kindness.

1.1.5 Personnel

Personnel	Responsibility
Veterinarian	Health, welfare, treatment, advice on procedures
Scientist in charge of the facility	Overall management and compliance
Animal house technician	Daily care, feeding, cleaning, observation
Animal attendant	Cleaning, waste handling, assistance
Investigator	The experiment itself, and the welfare of animals on the study

All personnel must be trained, and the training must be recorded.

1.2 Animal house design and layout

1.2.1 Location and general requirements

- Away from noise, dust, smoke and traffic
- Away from the main hospital and from human traffic
- Well drained, with an assured water and power supply
- Adequate space for expansion
- Secure against entry by unauthorised persons and by wild animals, insects and rodents

1.2.2 The barrier concept

A modern animal house is arranged so that clean and dirty materials never cross. Movement is **unidirectional** — from clean to dirty, never the reverse. Staff enter through a change room, and materials enter through a defined route.

1.2.3 Areas of an animal house

Area	Purpose
Quarantine room	Newly received animals are isolated and observed before entering the colony
Stabilisation or acclimatisation room	Animals adapt to the new environment before use
Animal housing rooms	Separate rooms for each species; separate for stock and for experimental animals
Breeding room	Where breeding colonies are maintained
Isolation or sick room	For animals showing signs of disease
Experimental or procedure room	Where procedures are carried out, away from the housing rooms
Surgery and post-operative room	Where surgical procedures and recovery take place

Area	Purpose
Feed and bedding store	Dry, pest-proof, off the floor
Washing and sterilisation area	Cage washing, autoclave
Store for equipment and consumables	—
Staff facilities	Change room, wash room, rest area
Records and office area	—
Waste disposal and carcass disposal area	Including a freezer for carcasses and an incinerator or an approved disposal arrangement

1.2.4 Environmental parameters

These figures are prescribed in the CCSEA guidelines and are frequently asked.

Parameter	Generally prescribed range
Temperature, rodents	About 22 degrees Celsius, plus or minus 3
Temperature, rabbit	About 20 degrees Celsius, plus or minus 3
Relative humidity	About 30 to 70 per cent
Light and dark cycle	12 hours light and 12 hours dark
Light intensity at cage level	Sufficient for observation without causing distress; excessive intensity is avoided, particularly for albino animals
Air changes	About 10 to 15 per hour, with fresh air, without recirculation
Noise	Kept low; sudden loud noise is a significant stressor
Ventilation	Filtered supply and exhaust

Why these matter. Temperature and humidity outside the range cause stress and alter physiology. An irregular light cycle disturbs circadian rhythm and hormone levels. Inadequate air change allows ammonia to accumulate from urine, which damages the respiratory tract. Every one of these changes the experimental result.

1.2.5 Caging and housing

Requirement	Detail
Floor space per animal	Prescribed by species and body weight; overcrowding must be avoided
Cage material	Non-toxic, non-absorbent, resistant to corrosion, easy to clean and autoclave
Bedding	Absorbent, non-toxic, dust-free; paddy husk and corn cob are commonly used; changed at a defined frequency
Feed	Balanced pelleted diet appropriate to the species; stored properly and within its date
Water	Clean drinking water, always available, in clean bottles or an automatic system
Identification	Cage card and individual animal identification
Enrichment	Nesting material, shelter, and companions for social species

1.3 Record-keeping and documentation

Records are a legal requirement, not merely good practice. An inspection will examine them first.

Record	Content
Animal acquisition record	Source, date, species, strain, sex, number, age, weight, health certificate
Animal stock register	Animals in the facility, with identification
Breeding records	Mating, litters, weaning, pedigree
Health monitoring records	Daily observation, clinical signs, veterinary examination, treatment given
Environmental monitoring records	Daily temperature and humidity; light cycle; air change verification
Feed and water records	Batch, date of receipt, expiry, consumption
Cleaning and sanitation records	Schedule and completion, for every room
Experimental records	Protocol number, IAEC approval reference, animals used, procedures, observations
IAEC records	Proposals, minutes, approvals, annual reports
CCSEA records	Registration, inspection reports, correspondence, returns filed
Disposal records	Method and date of disposal of every carcass
Training records	For every member of staff

Record	Content
Equipment records	Maintenance and calibration

KEY POINTS TO REMEMBER

- The regulatory body is **CCSEA**, formerly CPCSEA, renamed in January 2023. It is a statutory committee under the Prevention of Cruelty to Animals Act, 1960.
- The rules are the Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998, amended in 2001 and 2006.
- Every establishment must be registered with CCSEA. Every experiment needs prior IAEC approval.
- The IAEC includes a veterinarian, external scientists, a CCSEA nominee and a socially aware member.
- Design follows the barrier concept, with unidirectional flow from clean to dirty.
- Environmental parameters: about 22 ± 3 degrees Celsius for rodents, 30 to 70 per cent humidity, 12 hours light and 12 hours dark, 10 to 15 air changes per hour.
- Good husbandry is a requirement of valid data, not only of kindness.

EXAM CORNER — UNIT 1

This unit is asked as SAQ only (5 marks).

Likely short answer questions (5 marks)

1. Define an animal house. Enumerate the objectives of animal house management.
2. Write a short note on CCSEA and its functions.
3. Write a short note on the Institutional Animal Ethics Committee and its composition.
4. Enumerate the legal provisions governing animal experimentation in India.
5. Describe the layout of an animal house with its various areas.
6. Write a short note on the barrier concept in animal house design.
7. Enumerate the environmental parameters to be maintained in an animal house.
8. Why is quarantine necessary? Describe the quarantine procedure.
9. Write a short note on caging and bedding requirements.
10. Enumerate the records to be maintained in an animal house.
11. Write a short note on the personnel of an animal house and their responsibilities.
12. Explain why poor animal husbandry affects experimental results.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **28, 29**

1. **28.** Instructions of CPCSEA for the Animal House Facilities, 2019.

2. **29.** CPCSEA. Guidelines for Reuse and Rehabilitation of Large Animals post experimentation; 2020.

Additional verified sources: CCSEA, Ministry of Fisheries, Animal Husbandry and Dairying, Government of India — Acts, Rules and Guidelines (NCISM reference 152); the Prevention of Cruelty to Animals Act, 1960; and the Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998, as amended.

UNIT 2
STUDY OF ANIMAL BEHAVIOUR

SYLLABUS POINTS

2.1 Introduction to animal behaviour
2.2 Types of animal behaviour
2.3 Animal communication and their body language

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

2.1 Introduction to animal behaviour

2.1.1 Definition

Animal behaviour is the response of an animal to internal and external stimuli. **Ethology** is the scientific study of animal behaviour, particularly in natural conditions.

2.1.2 Why a researcher must understand behaviour

1. **To recognise normal behaviour**, and therefore to recognise abnormality.
2. **To detect pain, distress and illness early**, since animals conceal weakness.
3. **To handle animals safely**, for the animal and for yourself.
4. **To reduce stress**, which otherwise distorts every physiological measurement.
5. **To design humane housing and enrichment.**
6. **To interpret behavioural experiments correctly.**
7. **To recognise a humane endpoint** when it is reached.

2.1.3 The four questions of Tinbergen

Any behaviour can be studied from four angles. This framework is worth knowing by name.

Question	What it asks
Causation	What immediate stimulus and internal mechanism produced the behaviour?
Development	How does the behaviour change as the animal grows?
Function	What is its survival or reproductive value?
Evolution	How did the behaviour arise in the species?

2.2 Types of animal behaviour

2.2.1 Innate and learned behaviour

Type	Meaning	Examples
Innate or inherited	Present from birth; not learned; the same in every member of the species	Reflex, taxis, kinesis, instinct, fixed action pattern
Learned or acquired	Modified by experience	Habituation, classical conditioning, operant conditioning, imprinting, insight learning

Forms of innate behaviour

Form	Description
Reflex	An immediate, involuntary response to a stimulus
Taxis	Directed movement towards or away from a stimulus
Kinesis	Undirected change in the rate of movement in response to a stimulus
Instinct	A complex, unlearned, stereotyped sequence
Fixed action pattern	Once triggered, the sequence runs to completion

Forms of learned behaviour

Form	Description
Habituation	A response weakens on repeated harmless stimulation. **The basis of gentling and handling training**
Classical conditioning	A neutral stimulus becomes associated with a significant one
Operant conditioning	Behaviour is shaped by reward or punishment
Imprinting	Rapid learning at a critical early period
Insight learning	Solving a new problem without trial and error

2.2.2 Functional categories of behaviour

Category	Description
Ingestive	Feeding and drinking, including food preference and hoarding
Eliminative	Urination and defaecation, often at particular sites
Grooming and comfort	Self-grooming and mutual grooming; **a sudden loss of grooming is an early sign of illness**

Category	Description
Social	Dominance, hierarchy, huddling, group living
Sexual and reproductive	Courtship, mating
Maternal	Nest building, nursing, retrieval of pups
Agonistic	Aggression, threat, submission, flight
Exploratory	Investigation of the environment; strongly reduced by fear
Territorial	Defence of an area, often marked by scent
Circadian and rhythmic	Activity cycles; **most laboratory rodents are nocturnal**
Play	Particularly in young animals
Stereotypic	Repetitive, invariant, apparently purposeless behaviour such as bar-biting or circling. **An indicator of poor welfare**

> **Two points that carry marks every time.**
> **1.** Rats and mice are **nocturnal**. They are most active in the dark phase. Testing them during the light phase, when they would normally be asleep, is a common design fault and affects the result.
> **2. Stereotypic behaviour is not normal.** It indicates that the housing or the environment is inadequate, and it is a recognised welfare indicator.

2.3 Animal communication and body language

2.3.1 Channels of communication

Channel	Description	Example
Visual	Posture, movement, colour, facial expression	Piloerection, arched back, tail position
Auditory	Sounds, including those above the human range	Rodents use ultrasonic calls that we cannot hear
Olfactory	Scent marks, urine, pheromones	Territory marking; recognition of individuals
Tactile	Touch, grooming, huddling	Social bonding

> **Ultrasonic vocalisation is important and easily forgotten.** Rats and mice communicate largely in the ultrasonic range, above human hearing. A room that seems quiet to you may be full of alarm calls. Distress calls are emitted at frequencies you cannot hear. This is one reason why observing behaviour matters more than listening.

2.3.2 Body language and signs of pain or distress

General signs of pain, distress or illness in a laboratory animal

- Hunched posture
- Piloerection — the coat standing up, giving a rough or staring appearance
- Reduced or absent grooming, leading to an unkempt coat
- Reduced food and water intake; loss of body weight
- Reduced activity and reduced exploration
- Isolation from cage mates
- Abnormal gait or reluctance to move
- Half-closed eyes; porphyrin staining around the eyes and nose in rats
- Vocalisation on handling or on palpation
- Guarding or licking a particular area
- Abnormal respiration
- Aggression in a normally docile animal

Species-specific body language

Species	Signs of fear or distress	Signs of contentment
Mouse	Freezing, escape attempts, tail rattling, urination, defaecation	Normal exploration, grooming, nest building
Rat	Piloerection, porphyrin staining, teeth chattering, freezing, biting	Relaxed exploration, grooming, social contact
Rabbit	Thumping the hind foot, crouching flat, wide eyes, rapid breathing, struggling	Relaxed lying with legs extended, grooming, chewing
Guinea pig	Freezing, scattering, high-pitched squealing	Contented chattering, exploring, eating

2.3.3 Environmental enrichment

Enrichment is the provision of stimuli and structures that allow an animal to express its natural behaviour.

Type	Examples
Physical	Nesting material, shelter or hide, tubes, gnawing blocks, platforms
Social	Housing social species in compatible groups
Occupational	Objects to manipulate; foraging opportunities
Sensory	Varied but not excessive stimuli
Nutritional	Varied presentation of food

Enrichment reduces stereotypic behaviour, reduces stress and improves the quality of the data obtained. It is expected under current welfare standards.

KEY POINTS TO REMEMBER

- Ethology is the scientific study of animal behaviour.
- Tinbergen's four questions: causation, development, function, evolution.
- Innate behaviour includes reflex, taxis, kinesis, instinct and fixed action pattern.
- Learned behaviour includes habituation, classical and operant conditioning, imprinting and insight.
- Habituation is the basis of gentling and handling training.
- Laboratory rodents are nocturnal, and rats and mice communicate by ultrasonic vocalisation.
- Loss of grooming and piloerection are early signs of illness or distress.
- Stereotypic behaviour indicates poor welfare and is not normal.
- Enrichment reduces stress and improves the quality of experimental data.

EXAM CORNER — UNIT 2

This unit is asked as SAQ only (5 marks).

Likely short answer questions (5 marks)

1. Define animal behaviour and ethology. Why must a researcher understand behaviour?
2. Write a short note on Tinbergen's four questions.
3. Differentiate between innate and learned behaviour with examples.
4. Enumerate the forms of innate behaviour.
5. Enumerate the forms of learned behaviour.
6. Write a short note on habituation and its practical use in the animal house.
7. Enumerate the functional categories of animal behaviour.
8. Write a short note on stereotypic behaviour and its significance.

9. Enumerate the channels of animal communication with examples.
10. Write a short note on ultrasonic vocalisation in rodents.
11. Enumerate the signs of pain and distress in a laboratory animal.
12. Write a short note on environmental enrichment and its types.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **146, 147, 148**

1. **146.** Breed MD, Moore J. Of Cockroaches and Wolves: Framing Animal Behavior.
2. **147.** Exploring Animal Behavior in Laboratory and Field: An Hypothesis-testing Approach.
3. **148.** Huntingford F. The Study of Animal Behaviour. Springer Netherlands; 2012.

UNIT 3
SPECIES-SPECIFIC HANDLING TECHNIQUES

SYLLABUS POINTS

3.1 Rodents (mice, rats, etc.)
3.2 Non-human primates
3.3 Aquatic animals
3.4 Other specialised species (e.g., birds, reptiles)

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

3.1 General principles of handling

Before any species-specific technique, these principles apply to all.

1. **Handle only if trained and authorised.** Handling is a skill; it is learned under supervision.
2. **Approach quietly and calmly.** Sudden movement and noise trigger a flight response.
3. **Allow the animal to become aware of you** before touching it.
4. **Handle confidently but gently.** Hesitant handling causes struggling; rough handling causes injury.
5. **Restrain for the shortest time necessary.**
6. **Support the body weight.** Never suspend an animal by a single part.
7. **Never cause pain in order to restrain.**
8. **Wear appropriate protective equipment** — gloves, gown, mask, eye protection as required.
9. **Wash hands before and after** handling, and between animals where required.
10. **Return the animal gently** to its cage; never drop it.
11. **Observe the animal afterwards** for any sign of injury or distress.
12. **Record every handling procedure.**

Handling reduces the reliability of your data if it causes stress. Regular gentle handling, begun before the experiment, habituates the animal and reduces this effect.

3.2 Rodents

Rodents are the animals you are most likely to work with in Ayurvedic pharmacology and toxicology.

3.2.1 The mouse

Step	Technique
Lifting	Grasp the **base of the tail** — never the tip — and lift briefly onto a surface the animal can grip
Restraint	With the animal gripping a grid, take a firm fold of loose skin over the neck and shoulders between thumb and forefinger, and secure the tail against the palm with the little finger
Modern refinement	**Tunnel handling and cupping** are now preferred where possible, because they cause markedly less anxiety than tail handling
Never	Lift by the tip of the tail; suspend for long; squeeze the chest

3.2.2 The rat

Step	Technique
Approach	Rats respond well to gentle regular handling and become docile
Lifting	Place the hand over the back, encircling the thorax behind the forelimbs, with the thumb under the jaw if needed; support the hindquarters with the other hand
Restraint	Firm but not compressing; the chest must be free to move
Never	**Lift a rat by the tail alone.** The skin of the tail can strip. Never lift by the tip
Devices	A restrainer may be used for injections and blood sampling, for the minimum period

3.2.3 Guinea pig and rabbit

Species	Technique	Warning
Guinea pig	Support the body with one hand under the chest and the other under the hindquarters. Guinea pigs are easily frightened and scatter	Never lift by a limb; always support the hindquarters
Rabbit	Grasp the loose skin over the shoulders with one hand while **supporting the hindquarters firmly with the other**. Hold against the body	**Never lift a rabbit by the ears.** A struggling rabbit can fracture its own spine by kicking; the hindquarters must always be supported

3.3 Non-human primates

Non-human primates are used only in exceptional circumstances, and their use is very tightly restricted.

Point	Position
Regulatory position	Their use requires specific approval and is permitted only where no alternative species will answer the question
Justification	The scientific justification must be exceptional, and the IAEC and CCSEA scrutiny is correspondingly strict
Zoonotic risk	**Very high**, in both directions. Herpes B virus, tuberculosis, hepatitis and enteric infections are serious hazards
Handling	Only by specially trained personnel, with full protective equipment, and usually under chemical restraint rather than physical restraint
Housing	Requires large, complex, socially appropriate enclosures with substantial enrichment
Practical position for you	**You are extremely unlikely to use non-human primates in Ayurvedic research, and should not propose them.** Know the principles and the restrictions for the examination

3.4 Aquatic animals

Zebrafish is now a widely used laboratory species, and fish are covered by specific CCSEA guidance.

Aspect	Requirement
Housing	Recirculating systems with controlled water quality
Water quality	Temperature, pH, dissolved oxygen, ammonia, nitrite and nitrate monitored and recorded
Handling	**Minimise removal from water.** Transfer using a soft net or, better, in a container of water
Duration out of water	As short as possible; the gills collapse in air and the skin and mucus layer are easily damaged
Anaesthesia	Administered in the water, by an approved agent, under veterinary guidance
Advantages of zebrafish	Small, rapid development, transparent embryos, low cost, large numbers, and a lower position on the ethical scale than mammals

3.5 Other specialised species

Species	Handling points
Birds	Restrain the wings against the body to prevent injury. **Do not compress the chest** — birds have no diaphragm and breathe by movement of the sternum, so chest compression causes suffocation. Support the legs
Reptiles	Support the whole body length. Temperature-dependent physiology means the environment must be correct before and after handling. Some species can shed the tail if grasped by it
Amphibians	Handle with wet gloves or wet hands. The skin is permeable and easily damaged; dry hands remove the protective mucus
Invertebrates	Not covered by the same statutory provisions in most jurisdictions, but humane treatment is still expected. Increasingly used as replacement models

> **The replacement argument, which belongs in every answer on this unit.** Zebrafish, invertebrates, cell culture and computational models are all **Replacement** alternatives under the 3Rs, which you will study in Unit 5. For much early screening work in Ayurvedic pharmacology, one of these can answer the question without using a mammal at all. Proposing the lowest appropriate model is both good ethics and good science, and it will be looked for by your IAEC.

KEY POINTS TO REMEMBER

- Handle only if trained and authorised; restrain for the shortest possible time; always support the body weight.
- Mice may be lifted by the **base** of the tail, never the tip. Tunnel handling and cupping cause less anxiety and are preferred.
- **Never lift a rat by the tail alone** — the tail skin can strip.
- **Never lift a rabbit by the ears**, and always support the hindquarters, or the animal may fracture its own spine.
- Never compress the chest of a bird — birds have no diaphragm.
- Amphibians must be handled with wet hands or wet gloves.
- Non-human primates are very tightly restricted, carry a high zoonotic risk in both directions, and should not be proposed for Ayurvedic research.
- Fish should be handled in water wherever possible, and time out of water kept to a minimum.

EXAM CORNER — UNIT 3

This unit is asked as SAQ only (5 marks).

Likely short answer questions (5 marks)

1. Enumerate the general principles of handling laboratory animals.
2. Describe the handling and restraint of a mouse.
3. Describe the handling and restraint of a rat. What must never be done?
4. Describe the handling of a rabbit and the precautions to be observed.
5. Describe the handling of a guinea pig.
6. Write a short note on the restrictions on the use of non-human primates.
7. Write a short note on the handling of aquatic animals.
8. Why is zebrafish a preferred laboratory species?
9. Write the precautions in handling birds and the reason for each.
10. Write a short note on the handling of amphibians and reptiles.
11. Why does handling stress affect experimental results, and how is it reduced?
12. Write a short note on tunnel handling and cupping in mice.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **30, 31**

1. **30.** Guidelines of CPCSEA for Reuse and Rehabilitation of Large Animals.
2. **31.** FASEB. Invertebrates in Research — fact sheet.

Additional verified sources: CCSEA Guidelines for Experimentation on Fishes, 2021, and CCSEA Guidelines for Poultry and Birds Facility, 2020 (NCISM reference 152).

UNIT 4
ANIMAL HEALTH AND DISEASE PREVENTION

SYLLABUS POINTS

4.1 Common diseases in laboratory animals
4.2 Vaccination and immunisation programs
4.3 Parasite control and management

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	Yes

> This is the **only unit in Module 6 carrying a ten-mark long answer question**. Prepare it more thoroughly than the others.

4.1 Common diseases in laboratory animals

4.1.1 Why disease control matters

1. **Animal welfare** — a sick animal suffers.
2. **Data validity** — subclinical infection alters immune function, organ weights, enzyme levels and body weight, and can invalidate an entire study without the investigator ever knowing.
3. **Loss of the colony** — an outbreak can destroy years of breeding.
4. **Human health** — several laboratory animal diseases are **zoonotic**.
5. **Legal compliance** — health monitoring is a requirement.

4.1.2 Microbiological categories of laboratory animals

Category	Definition
Conventional	Bred and maintained in an open system; microbial status undefined
Specified Pathogen Free (SPF)	Free of specified pathogens; maintained under barrier conditions and regularly screened
Gnotobiotic	All microorganisms present are known and defined
Germ free or axenic	No demonstrable microorganisms; maintained in isolators

4.1.3 Common diseases by cause

Cause	Examples	Common signs
Bacterial	Respiratory infections; enteric infections; abscesses; salmonellosis; pasteurellosis in rabbits, causing snuffles	Sneezing, nasal discharge, chattering, diarrhoea, weight loss, abscesses
Viral	Various rodent viral infections, many of them subclinical but capable of altering research results	Often no visible sign; detected only by serological screening
Fungal	Ringworm, or dermatophytosis	Circular areas of hair loss, scaling; **zoonotic**
Parasitic	Ectoparasites — mites, lice, fleas; endoparasites — pinworms, tapeworms, protozoa such as coccidia and Giardia	Itching, hair loss, scratching, poor coat, weight loss, diarrhoea
Nutritional	Deficiency of vitamin C in guinea pigs, which cannot synthesise it	Weakness, joint swelling, poor healing
Environmental and management	Ammonia-induced respiratory damage; heat and cold stress; trauma from fighting; malocclusion of incisors in rodents and rabbits	Respiratory signs, wounds, difficulty eating, weight loss

> **Two points worth remembering precisely.**
> **1. The guinea pig cannot synthesise vitamin C** and must receive it in the diet. This is a favourite examination question.
> **2.** Many rodent viral infections are **subclinical**. The animals look perfectly healthy while the infection alters the very parameters you are measuring. This is precisely why routine serological health monitoring exists, and why SPF animals are used.

4.1.4 Zoonoses — diseases transmissible to humans

Disease	Source
Ringworm	Rodents, rabbits
Salmonellosis	Rodents, reptiles
Leptospirosis	Rodents
Rat bite fever	Rats
Lymphocytic choriomeningitis	Rodents
Tuberculosis and herpes B	Non-human primates
Allergy to animal proteins	All species; **the commonest occupational hazard in an animal house**

Protection of staff. Protective clothing, gloves and masks; hand washing; prompt attention to any bite or scratch; adequate ventilation; regular health checks; and training in recognising the risks.

4.1.5 Health monitoring

Level	Activity
Daily	Visual observation of every animal by trained staff; any abnormality recorded and reported
Regular clinical examination	By the veterinarian
Periodic laboratory screening	Serology, bacteriology, parasitology on sentinel animals
Sentinel animals	Healthy animals deliberately exposed to the colony environment and screened at intervals to detect infection early
Post-mortem examination	Of any animal that dies unexpectedly

4.2 Vaccination and immunisation

4.2.1 The position in laboratory rodents

This point is often misunderstood, and stating it correctly is worth marks.

Laboratory rodents are generally not vaccinated. Disease is controlled instead by **prevention** — barrier housing, SPF status, quarantine, strict hygiene and health monitoring. There are two reasons:

1. Vaccination would produce antibodies that **interfere with serological screening**, making it impossible to tell an infected colony from a vaccinated one.
2. The immune response induced could **alter the experimental result**, which is unacceptable in a research animal.

4.2.2 Where vaccination is used

Situation	Note
Larger species such as dogs, cats, sheep and goats	Vaccinated according to the standard veterinary schedule for that species
Rabbits, in some facilities	As advised by the veterinarian and permitted by the protocol
Animals to be rehabilitated after experimentation	Vaccinated before release, as required under the reuse and rehabilitation guidelines
Personnel	Tetanus, and other vaccination as advised. **Protecting the staff is part of the programme**

4.2.3 Principles of an immunisation programme

1. Vaccinate on the advice of the veterinarian, and only where it is permitted by the study protocol.
2. Follow the manufacturer’s schedule, route and dose.
3. Maintain the cold chain for every vaccine.
4. Record the vaccine name, batch number, expiry, date, route and the animal identification.
5. Observe every animal for adverse reaction after vaccination.
6. **Never vaccinate an animal that is already on a study, without approval**, because the immune response will confound the results.

4.3 Parasite control and management

4.3.1 Types of parasites

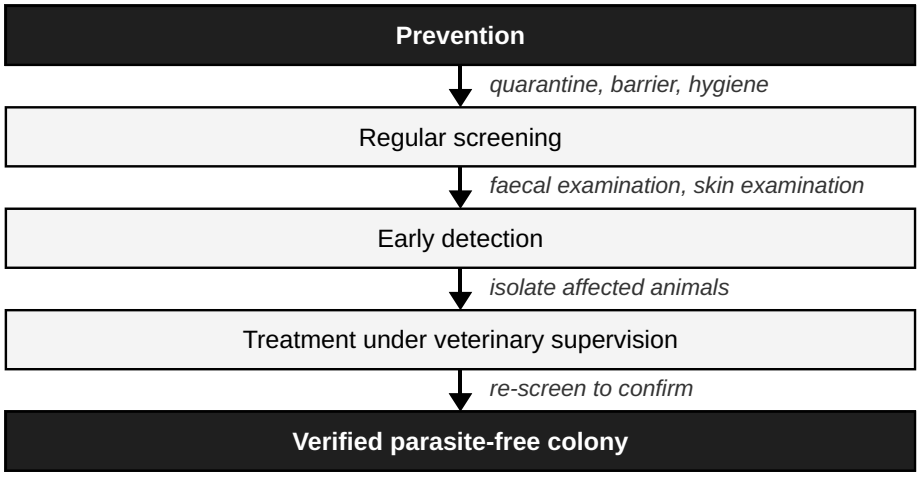
Type	Examples	Effect
Ectoparasites	Mites, lice, fleas, ticks	Itching, scratching, hair loss, dermatitis, anaemia in heavy infestation, restlessness
Endoparasites — helminths	Pinworms, tapeworms	Weight loss, poor condition, intestinal irritation; often subclinical
Endoparasites — protozoa	Coccidia, Giardia, Entamoeba	Diarrhoea, weight loss, poor growth

4.3.2 Detection

Parasite	Method of detection
Ectoparasites	Direct examination of the coat; skin scraping examined microscopically; adhesive tape test
Helminths	**Faecal examination** — direct smear and flotation; perianal tape test for pinworm eggs; examination of intestinal contents at post-mortem
Protozoa	Direct wet smear of fresh faeces; staining; flotation

Faecal flotation is the standard screening method. A faecal sample is mixed with a solution of higher specific gravity than the eggs, so that the eggs float to the surface and are collected on a coverslip for microscopic examination. This is the basis of Practical 6.4.

4.3.3 Control and management



Preventive measures

- Quarantine and screen every new animal before it enters the colony
- Obtain animals only from approved, screened sources
- Maintain barrier housing where possible
- Sterilise bedding, feed and water
- Clean and disinfect cages on a defined schedule
- Control wild rodents and insects in and around the facility
- Do not allow personnel to move from a contaminated room to a clean one
- Screen the colony routinely, using sentinel animals

Treatment. Only under veterinary supervision, with an approved agent, at the correct dose and duration, and with the treatment recorded. **Any treatment given to an animal on a study must be reported to the investigator and the IAEC**, because it may affect the result.

After treatment. Re-screen to confirm elimination. Treat the environment as well as the animals, or reinfestation will follow.

KEY POINTS TO REMEMBER

- Subclinical infection alters research data without any visible sign. This is the main scientific reason for health monitoring.
- Microbiological categories: conventional, SPF, gnotobiotic, germ free.
- Guinea pigs cannot synthesise vitamin C and must receive it in the diet.
- Pasteurellosis causes snuffles in rabbits.
- Zoonoses include ringworm, salmonellosis, leptospirosis and rat bite fever. Allergy to animal proteins is the commonest occupational hazard.
- Sentinel animals are used for early detection of infection in a colony.
- **Laboratory rodents are generally not vaccinated**, because vaccination interferes with serological screening and alters experimental results.
- Parasite screening is by faecal flotation, skin scraping and the perianal tape test.
- Treat the environment as well as the animals, or reinfestation follows.

EXAM CORNER — UNIT 4

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). There is no ABQ.

Likely short answer questions (5 marks)

1. Why is disease control important in a laboratory animal facility?
2. Enumerate the microbiological categories of laboratory animals.
3. Enumerate the common bacterial and viral diseases of laboratory animals.
4. Write a short note on zoonoses in the animal house and their prevention.
5. Write a short note on health monitoring and sentinel animals.
6. Why are laboratory rodents generally not vaccinated?
7. Write the principles of an immunisation programme.
8. Enumerate the ectoparasites and endoparasites of laboratory animals.
9. Describe the methods of detecting parasites in laboratory animals.
10. Write a short note on faecal flotation.
11. Enumerate the preventive measures for parasite control.
12. Why must guinea pigs receive vitamin C in the diet?

Likely long answer questions (10 marks)

1. Describe the common diseases of laboratory animals, their signs, and the measures for their prevention and control.
2. Discuss health monitoring in a laboratory animal facility, including the categories of animals, the methods used, and the role of sentinel animals.
3. Describe parasite control and management in a laboratory animal facility, including detection, treatment and prevention.

4. Discuss vaccination and immunisation in laboratory animals. Explain why rodents are generally not vaccinated and what is done instead.
5. Discuss the effect of animal health on the validity of experimental data, and describe the measures taken to safeguard it.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **149, 150, 151**

1. **149.** Management of Animal Care and Use Programs in Research, Education, and Testing. CRC Press; 2017.
2. **150.** Use of Laboratory Animals in Biomedical and Behavioral Research. National Academy Press; 1988.
3. **151.** Essentials of Laboratory Animal Science: Principles and Practices. Springer Nature Singapore; 2021.

UNIT 5
ANIMAL WELFARE AND ETHICS

SYLLABUS POINTS

5.1 Animal welfare principles and ethics
5.2 Humane endpoints and euthanasia methods
5.3 Case studies and group discussions on animal welfare and ethics

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

5.1 Animal welfare principles and ethics

5.1.1 The Five Freedoms

The **Five Freedoms** are the internationally accepted statement of what an animal in human care is entitled to.

Freedom	Provided by
Freedom from hunger and thirst	Ready access to fresh water and an appropriate diet
Freedom from discomfort	A suitable environment, including shelter and a comfortable resting area
Freedom from pain, injury and disease	Prevention, rapid diagnosis and treatment
Freedom to express normal behaviour	Sufficient space, proper facilities, and company of the animal’s own kind
Freedom from fear and distress	Conditions and treatment that avoid mental suffering

5.1.2 The 3Rs

The **3Rs** were introduced by **Russell and Burch in 1959** and are the foundation of all modern animal research ethics. *CCSEA guidelines give particular significance to the ethical obligation of replacing, reducing and refining the use of animals in research.*

R	Meaning	Practical application in your field
Replacement	Using a non-animal method, or a species of lower sentience, wherever possible	Cell culture, in vitro dissolution studies, computational models, zebrafish and invertebrates for early screening of a Bhasma
Reduction	Using the smallest number of animals that will give a valid result	Sound statistical design; proper power calculation; sharing tissue between studies; using the best available data before starting
Refinement	Minimising pain, suffering and distress, and improving welfare	Better handling, analgesia, anaesthesia, enrichment, non-invasive techniques, and humane endpoints

> **Reduction is the R most often misunderstood.** Using too few animals is not Reduction — it is bad science, because the study cannot reach a valid conclusion and the animals are wasted entirely. Reduction means using the **correct** number, determined statistically before the study begins. Say this in the examination.

5.1.3 Ethical principles

Principle	Meaning
Necessity	The work must have a genuine scientific or medical purpose
Justification	The expected benefit must outweigh the harm to the animals
Cost-benefit assessment	Formally carried out by the IAEC before approval
Competence	Only trained and authorised persons may perform procedures
Responsibility	The investigator is personally responsible for the welfare of every animal on the study
Transparency	Full and honest reporting, including of adverse outcomes
Compliance	Full adherence to the law and to institutional approval

5.1.4 The Ayurvedic perspective

Ayurveda and Indian tradition place a strong emphasis on **Ahimsa**, non-injury to living beings, and this is worth stating in an ethics answer.

At the same time, the classical texts do describe the use of animal products — Godugdha, Ghrita, Gomutra, Madhu, Mamsa, Kasturi and others. The tradition is not one of absolute avoidance, but of use with restraint, purpose and gratitude.

For a modern researcher the practical position is clear: **use the 3Rs rigorously; use an animal only when no alternative will answer the question; and when animals are used, treat them with the care that both the law and Ahimsa require.** This is a genuine and defensible position, and it can be written in an examination without difficulty.

5.2 Humane endpoints and euthanasia

5.2.1 Humane endpoint

A **humane endpoint** is the earliest point at which an animal's pain or distress can be ended, prevented or relieved, while still meeting the scientific objectives of the study.

The principle. Death should never be the endpoint of a study if an earlier, humane endpoint will answer the question. Waiting for an animal to die when the outcome is already certain adds suffering without adding information.

Criteria commonly used as humane endpoints

Criterion	Typical threshold
Body weight loss	A defined percentage loss from the starting weight
Inability to reach food or water	Immediate endpoint
Failure to respond to stimulation; prolonged inactivity	Immediate endpoint
Severe or persistent clinical signs	As defined in the protocol
Laboured or difficult breathing	Immediate endpoint
Persistent self-mutilation	Immediate endpoint
Signs of unrelieved pain despite analgesia	Immediate endpoint
Tumour size or ulceration, in relevant studies	As defined in the protocol

Every humane endpoint must be defined in the protocol, in writing, before the study begins, and approved by the IAEC. The person authorised to act on it, and the action to be taken, must also be stated.

5.2.2 Euthanasia

Euthanasia means a humane death — one brought about with the minimum of pain, distress and anxiety to the animal.

When it is carried out

- When the humane endpoint of a study is reached
- At the scheduled end of a study, where tissue is required
- When an animal is suffering from an untreatable disease or injury
- When an animal is surplus to requirement and cannot be rehomed or rehabilitated

Requirements of an acceptable method

1. Rapid loss of consciousness, followed by cardiac or respiratory arrest.
2. Minimum pain, distress and anxiety to the animal.
3. Reliability and irreversibility.
4. Safety for the personnel carrying it out.
5. Compatibility with the scientific purpose, since some agents alter the tissue.
6. Appropriateness to the species, the age and the size of the animal.

Classification of methods

Category	Description
Acceptable	Methods that consistently produce a humane death when used correctly. These include approved inhalant agents and approved injectable agents administered by a competent person
Conditionally acceptable	Physical methods that are humane only when performed by a properly trained and skilled operator, and only in species and sizes for which they are appropriate. They are used where an agent would interfere with the scientific objective
Unacceptable	Methods that cause pain or distress, or that are unreliable. These must never be used

The governing requirements, whichever method is used

- The method must be **specified in the approved protocol** and permitted by the IAEC.
- It must be carried out **only by a trained and authorised person**.
- It should be performed **away from other animals**, since they detect distress.
- **Death must be confirmed** before the body is disposed of. Confirmation is by a defined secondary check, not by assumption.
- The carcass must be **disposed of by the approved route**, and the disposal recorded.

- Every euthanasia must be **recorded** with the date, the animal identification, the method, the reason and the operator.

> **How to write about euthanasia in an examination.** Give the definition, the situations in which it is carried out, the six requirements of an acceptable method, the three categories, and the governing requirements above. Emphasise that the method must be in the approved protocol, performed by a trained person, and confirmed. Detailed agents and doses are veterinary matters, decided by the veterinarian and the IAEC for each protocol — and saying so is itself the correct answer.

5.2.3 Disposal and rehabilitation

Situation	Action
Carcass disposal	By incineration or another approved method; recorded
Large animals after experimentation	*CCSEA has issued specific guidelines for the reuse and rehabilitation of large animals after experimentation*; where an animal has not suffered lasting harm, rehabilitation rather than euthanasia is required
Rehoming	Where permitted and appropriate

5.3 Case studies and group discussion

The curriculum prescribes case studies and group discussion. The following are suitable and cover the main ethical themes.

Case study theme	Questions to discuss
A study proposal uses 60 animals where a power calculation shows 24 would suffice	Which R is breached? What should the IAEC do?
An investigator wishes to use death as the endpoint of a toxicity study	Is this justified? What humane endpoint could replace it?
Animals on a study develop a parasitic infestation requiring treatment	Should they be treated? What happens to the data? Who must be informed?
A researcher proposes a mammalian model for an early screening question	Could a Replacement model answer it? How would you justify either choice?
An animal reaches a humane endpoint at night, when no trained person is present	What should the protocol have provided for?
Findings are unfavourable and the investigator wishes not to publish them	What is the ethical obligation? What is lost if unfavourable results are suppressed?

How to conduct the discussion. State the facts. Identify the ethical principle at stake. Identify who is responsible. State what should have been done, and what must now be done. Record the group’s conclusion.

KEY POINTS TO REMEMBER

- The Five Freedoms: from hunger and thirst; from discomfort; from pain, injury and disease; to express normal behaviour; from fear and distress.
- **The 3Rs were introduced by Russell and Burch in 1959** — Replacement, Reduction, Refinement.
- Reduction means the correct number determined statistically, not the fewest possible.
- A humane endpoint is the earliest point at which suffering can be ended while still meeting the scientific objective.
- Death should never be the endpoint if an earlier humane endpoint will answer the question.
- Every humane endpoint must be defined in the protocol before the study begins.
- Euthanasia methods are acceptable, conditionally acceptable, or unacceptable. Only an approved method, by a trained person, with death confirmed.
- CCSEA requires reuse and rehabilitation of large animals after experimentation where appropriate.
- Ahimsa is the Ayurvedic ethical anchor, and it aligns closely with the 3Rs.

EXAM CORNER — UNIT 5

This unit is asked as SAQ only (5 marks).

Likely short answer questions (5 marks)

1. Enumerate and explain the Five Freedoms.
2. Write a short note on the 3Rs and their application in Ayurvedic research.
3. Explain Replacement with examples relevant to your subject.
4. Explain Reduction. Why is using too few animals not Reduction?
5. Explain Refinement with examples.
6. Enumerate the ethical principles governing animal experimentation.
7. Define a humane endpoint. Enumerate the criteria commonly used.
8. Define euthanasia and enumerate the requirements of an acceptable method.
9. Enumerate the categories of euthanasia methods.
10. What are the governing requirements for carrying out euthanasia?
11. Write a short note on the reuse and rehabilitation of large animals.
12. Discuss the concept of Ahimsa in relation to animal experimentation in Ayurvedic research.

REFERENCES USED IN THIS UNIT

NCISM reference number for this unit: **152**

1. **152.** CCSEA. Acts, Rules and Guidelines. Committee for Control and Supervision of Experiments on Animals, Ministry of Fisheries, Animal Husbandry and Dairying, Government of India.

Additional verified sources: Russell WMS and Burch RL, The Principles of Humane Experimental Technique, 1959; the Prevention of Cruelty to Animals Act, 1960; the Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998, as amended; and CCSEA Guidelines for Reuse and Rehabilitation of Large Animals, 2020 (NCISM references 29 and 30).

MODULE 6
PRACTICAL TRAINING ACTIVITIES

Module 6 carries **10 hours of practical training**, over five activities.

No	Practical name	Hours
6.1	Demonstration of animal house model	2
6.2	Demonstration of animal body language	2
6.3	Demonstration of species-specific handling techniques for laboratory animals	2
6.4	Demonstration of microscopic examination of faecal samples for parasites	2
6.5	Evaluation of animal welfare and ethics in a research setting	2
	Total	**10**

Practical Training 6.1 — Demonstration of an animal house model

Aim. To study the design, layout and functioning of an animal house.

Procedure

1. Study the model or the plan of an animal house.
2. Identify and label every area — quarantine, stabilisation, housing, breeding, isolation, procedure, surgery, stores, washing, staff and waste disposal.
3. Trace the movement of personnel, of animals, of clean material and of waste. Mark the direction of each on the plan.
4. Identify where the barrier lies and where the change rooms are placed.
5. Record the environmental parameters prescribed for each species.
6. Examine the caging, bedding, feeding and watering arrangements.
7. Examine the record formats maintained.
8. Draw a labelled plan of an ideal animal house.

Observation to record. The labelled plan with the flow of material marked; the table of environmental parameters; the list of records maintained.

Practical Training 6.2 — Demonstration of animal body language

Aim. To observe and interpret the body language of laboratory animals.

Procedure

1. Observe animals quietly, without disturbing them, for a defined period. Record the time and the light phase.
2. Record the normal behaviours seen — feeding, drinking, grooming, exploration, social contact, resting.
3. Note the posture, coat condition, eyes, ears and tail position of each animal.
4. Observe the response when a person approaches, and record the change in behaviour.
5. Identify and record any sign of fear, distress or illness.
6. Identify any stereotypic behaviour and record it.
7. Compare the behaviour of animals in enriched and unenriched cages, if both are available.

Observation table to draw

Animal ID	Species	Posture	Coat	Activity	Social behaviour	Signs of distress	Interpretation

Precautions. Observe quietly and at a distance first. Do not disturb the animals unnecessarily. Record the time of day, because most laboratory rodents are nocturnal.

Practical Training 6.3 — Species-specific handling techniques

Aim. To learn and demonstrate the correct handling and restraint of laboratory animals.

Procedure

1. Observe the demonstration by the trained handler for each species.
2. Record the correct technique step by step for the mouse, the rat, the guinea pig and the rabbit.
3. Record what must never be done for each species, and the reason.
4. Practise under direct supervision, one species at a time.
5. Observe the animal after handling and record any sign of distress.
6. Record the time for which each animal was restrained.

Observation table to draw

Species	Correct lifting technique	Correct restraint	What must never be done	Reason	Signs observed after handling

Precautions. Handle only under direct supervision and only after the demonstration. Wear the prescribed protective equipment. Wash hands before and after, and between species. Report any bite or scratch immediately. Restrain for the shortest possible time. If an animal struggles severely, return it to the cage and try again later rather than forcing it.

Practical Training 6.4 — Microscopic examination of faecal samples for parasites

Aim. To detect parasite ova and cysts in faecal samples by microscopic examination.

Requirements. Fresh faecal samples, flotation solution, test tubes, stirring rod, strainer, slides, coverslips, microscope, gloves, disinfectant.

Procedure

1. Wear gloves. Collect a fresh faecal sample and record the animal identification.
2. **Direct smear.** Mix a small quantity with a drop of saline on a slide, apply a coverslip and examine under low and then high power.
3. **Flotation.** Mix the sample thoroughly with the flotation solution. Strain to remove coarse debris.
4. Transfer to a test tube and fill to form a slight positive meniscus.
5. Place a coverslip on top and allow it to stand for the prescribed time, so that the eggs rise and adhere to the coverslip.
6. Lift the coverslip straight up and place it on a slide.
7. Examine systematically under the microscope, in a regular pattern so that no area is missed.
8. Identify and record any ova, cysts or larvae seen. Measure them using the calibrated ocular micrometer from Module 5.
9. Draw what is seen.

Observation table to draw

Sample ID	Species	Method used	Structures seen	Measurement	Identification	Remarks

Precautions. Faecal samples may contain zoonotic organisms. Wear gloves throughout. Do not eat, drink or touch the face. Disinfect the bench and all glassware after use. Dispose of the sample by the approved route. Wash hands thoroughly.

Practical Training 6.5 — Evaluation of animal welfare and ethics in a research setting

Aim. To evaluate the welfare standards and ethical compliance of an animal research facility.

Procedure

1. Prepare an assessment checklist under the headings of the **Five Freedoms** and the **3Rs**.
2. Add a section on legal compliance — CCSEA registration, IAEC constitution and approvals, records.
3. Visit or study the facility. Record observations against each item.
4. Assess housing, space, enrichment, feeding, watering, health monitoring and staff training.
5. Examine a sample protocol for its humane endpoints and its statistical justification of animal numbers.
6. Identify where the 3Rs have been applied, and where they could be applied further.
7. Record every observation as compliant or as an area for improvement.
8. Prepare a written report with recommendations.

Observation table to draw

Item assessed	Standard expected	Observation	Compliant	Recommendation

Precautions. Obtain permission before any visit. Do not photograph without written permission. Maintain confidentiality. Report facts, and keep your recommendations separate from your observations.

MODULE 6
EXPERIENTIAL LEARNING ACTIVITIES

Module 6 carries **15 hours** under experiential learning, including modular assessment. Three activities are prescribed.

No	Experiential learning activity	Hours
6.1	Visit to an animal house	4
6.2	Analysis of online resources to develop critical thinking and research skills, and analysis of studies on experimental animal behaviour	5
6.3	Preparation and presentation on parasite control in laboratory animals	4
	Total	**13**

Experiential Learning 6.1 — Visit to an animal house

What you have to do. Visit a registered animal house and produce a full report on its design, functioning and compliance.

Before the visit. Obtain permission. Study the CCSEA guidelines. Prepare your checklist and your questions in advance.

During the visit. Observe the layout and the flow. Record the environmental parameters actually maintained, from the monitoring records. Examine the caging, the feed and water arrangements, and the enrichment provided. Examine every category of record. Speak to the veterinarian and to the technician about the difficulties they face.

Questions your report must answer

- How is the barrier maintained, and where does it break down in practice?
- Which environmental parameters are recorded, how often, and by whom?
- How is disease detected, and how quickly?
- What enrichment is provided, and for which species?
- How are humane endpoints defined and acted upon?
- Which of the 3Rs are being applied, and which are not?
- What would you recommend, and what would it cost?

Deliverable. A written report with the labelled layout plan, the observation tables, and your recommendations.

Experiential Learning 6.2 — Analysis of studies on experimental animal behaviour

What you have to do. Read and critically analyse published studies on animal behaviour, and develop the judgement to assess their quality.

Method. Obtain at least five published studies that used behavioural observation or behavioural tests. For each, extract the species and strain, the housing conditions, the light phase during testing, the sample size and how it was justified, the behavioural measures used, the handling method, and the ethical approval reported.

Then answer these questions

- Was the study conducted in the light phase or the dark phase? For a nocturnal species, does this matter for the result reported?
- Was the number of animals justified statistically?
- Was habituation to handling carried out before testing?
- Was ethical approval reported with the committee and the approval number?
- Were the housing and enrichment described well enough for the work to be repeated?
- Which of the 3Rs did the authors apply, and did they say so?
- What is the commonest weakness across the five studies?

Deliverable. A comparison table of the five studies, your critique of each, and a short written statement of the standards you will apply to your own work.

Experiential Learning 6.3 — Parasite control in laboratory animals

What you have to do. Prepare and deliver a presentation on parasite control, based on both the literature and your own practical findings.

Method. Combine the results of your faecal examinations from Practical 6.4 with a review of the literature. Cover the parasites of each laboratory species, the methods of detection, the control measures, the treatment principles, and the effect of parasitism on experimental data.

Points your presentation must cover

- Which parasites affect which species, and what signs each produces
- How each is detected, with your own micrographs where you have them
- Why treating the animals alone is not enough
- How parasitism can invalidate a study without being noticed
- What a complete parasite control programme for a facility would look like
- Who must be informed when animals on a study are treated

Deliverable. The presentation, the handout, and the record of questions asked and your answers.

MODULE 6
VIVA POINTERS AND QUESTION BANK

Viva pointers

On law and ethics

- What is the full form of CCSEA? What was it called before, and when did it change?
- Under which Act is it constituted?
- What are the rules governing animal experimentation in India?
- What is the IAEC, and who sits on it?
- Can you begin an experiment before IAEC approval?
- What are the Five Freedoms?
- What are the 3Rs, and who introduced them, and in what year?

On the animal house

- Show me the quarantine area. Why is it needed?
- What temperature and humidity are maintained here?
- How many air changes per hour?
- Why does the light cycle matter?
- Which records are maintained, and who signs them?

On behaviour and handling

- Are rats nocturnal or diurnal?
- What does piloerection indicate?
- What is stereotypic behaviour, and what does it mean?
- Show me how you would pick up a mouse. A rat. A rabbit.
- What must you never do when lifting a rat? A rabbit? A bird?
- What is tunnel handling?

On health and parasites

- Why are laboratory rodents not vaccinated?
- What is a sentinel animal?
- Why must guinea pigs get vitamin C?
- Name three zoonoses of the animal house.
- Show me how you perform a faecal flotation.
- What did you find in this preparation?

On welfare

- What is a humane endpoint?
- Should death be an endpoint? Why not?
- What are the requirements of an acceptable method of euthanasia?
- Who decides the method, and where is it recorded?
- Is using fewer animals always better?

Module 6 question bank

Long Answer Questions (10 marks) — from Unit 4 only

1. Describe the common diseases of laboratory animals, their signs, and the measures for their prevention and control.
2. Discuss health monitoring in a laboratory animal facility, including the categories of animals, the methods used, and the role of sentinel animals.
3. Describe parasite control and management in a laboratory animal facility, including detection, treatment and prevention.
4. Discuss vaccination and immunisation in laboratory animals, and explain why rodents are generally not vaccinated.
5. Discuss the effect of animal health on the validity of experimental data, and the measures taken to safeguard it.

Short Answer Questions (5 marks) — from all five units

From Unit 1

1. Define an animal house and enumerate the objectives of its management.
2. Write a short note on CCSEA and its functions.
3. Write a short note on the IAEC and its composition.
4. Describe the layout of an animal house with its areas.
5. Enumerate the environmental parameters to be maintained in an animal house.
6. Enumerate the records to be maintained in an animal house.
7. Write a short note on the barrier concept.

From Unit 2

1. Define animal behaviour and ethology.
2. Write a short note on Tinbergen’s four questions.
3. Differentiate between innate and learned behaviour with examples.
4. Write a short note on habituation and its practical use.
5. Write a short note on stereotypic behaviour and its significance.
6. Enumerate the channels of animal communication.
7. Enumerate the signs of pain and distress in a laboratory animal.
8. Write a short note on environmental enrichment.

From Unit 3

1. Enumerate the general principles of handling laboratory animals.
2. Describe the handling and restraint of a mouse and a rat.
3. Describe the handling of a rabbit and the precautions observed.
4. Write a short note on the restrictions on the use of non-human primates.
5. Write a short note on the handling of aquatic animals.
6. Write the precautions in handling birds and the reason for each.

From Unit 4

1. Enumerate the microbiological categories of laboratory animals.
2. Write a short note on zoonoses in the animal house and their prevention.
3. Write a short note on sentinel animals.
4. Why are laboratory rodents generally not vaccinated?
5. Describe the methods of detecting parasites in laboratory animals.
6. Write a short note on faecal flotation.
7. Why must guinea pigs receive vitamin C in the diet?

From Unit 5

1. Enumerate and explain the Five Freedoms.
2. Write a short note on the 3Rs and their application in Ayurvedic research.
3. Explain Reduction. Why is using too few animals not Reduction?
4. Define a humane endpoint and enumerate the criteria used.
5. Define euthanasia and enumerate the requirements of an acceptable method.
6. Write a short note on the reuse and rehabilitation of large animals.
7. Discuss Ahimsa in relation to animal experimentation in Ayurvedic research.

> **Note on the marks range.** Table 6F places Module 6 in the marks range of **5 to 15**. No unit is ABQ-eligible, and only Unit 4 carries a long answer. Everything else is short answer. Learn the legal names, the numbered lists — Five Freedoms, 3Rs, environmental parameters, microbiological categories — and the species-specific handling rules exactly, and this module will give you marks easily.

MODULE 7
CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS

Credits	Lectures	Practical Training	Experiential Learning	Total hours	Modular assessment
2	10	20	30	60	50 marks

About this module

This module covers the classical and advanced quality control techniques used for Ayurvedic formulations. It deals with quality control and quality assurance, the classical quality control parameters described in the texts, and the specific quality control parameters applied to solid, semi-solid, liquid and gaseous dosage forms.

Module Learning Objectives

At the end of this module, the student should be able to:

1. Describe the importance of quality control and quality assurance in Ayurveda product manufacturing.
2. Describe classical quality control parameters for Ayurvedic formulations, including Mana, Avaleha Siddhi Lakshana and Bhasma Pariksha.
3. Apply quality control parameters for various Ayurvedic dosage forms, including solid, semi-solid and liquid.
4. Identify regulatory requirements for quality control, and ensure documentation and record-keeping compliance.
5. Integrate classical and advanced quality control techniques to ensure the quality and safety of Ayurvedic formulations.

Units in this module

Unit	Name	ABQ	SAQ	LAQ
M7U1	Quality Control and Quality Assurance	Yes	**No**	Yes
M7U2	Classical aspects of the quality control parameter for Ayurvedic formulations	No	Yes	No
M7U3	Quality control parameter for solid dosage forms	Yes	Yes	No
M7U4	Quality control parameter for semi-solid dosage forms	Yes	Yes	No
M7U5	Quality control parameter for liquid and gaseous dosage forms	Yes	Yes	No

Module 7 is placed in the marks range of **5 to 20**.

> **Note the unusual pattern in Unit 1.** It is the only unit in the entire course that carries **ABQ and LAQ but no SAQ**. So it will never appear as a five-mark question. When it comes, it comes as a ten-mark essay or a twenty-mark analytical question. Prepare it as a whole subject, not as a set of short notes.

How this module connects to the rest of the course

This module is where the earlier modules are put to work.

Module	What it contributed
Module 1	GMP, Schedule T, QC and QA as concepts, packaging and stability
Module 3	Phytochemistry, extraction, phytochemical screening
Module 4	Bhasma Pariksha and the chemistry of Shodhana and Marana
Module 5	Chromatography, spectroscopy, microscopy and the characterisation instruments
Module 7	**Applies all of them, dosage form by dosage form**

UNIT 1
QUALITY CONTROL AND QUALITY ASSURANCE

SYLLABUS POINTS

1.1 Importance of quality control in Ayurveda
1.2 Quality assurance of Ayurveda products
1.3 Quality control parameters (identity, purity, etc.)
1.4 Regulatory requirements for quality control
1.5 Documentation and record-keeping

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	No	Yes

1.1 Importance of quality control in Ayurveda

1.1.1 Why quality control is harder in Ayurveda than in modern pharmacy

A modern tablet contains one active ingredient of known purity. An Ayurvedic formulation may contain fifteen plants, a mineral and a metal, each of which varies. This makes quality control genuinely more difficult, and a postgraduate answer should say so plainly.

Difficulty	Explanation
Natural variability	The same plant differs with season, region, soil, altitude, age and part used
Multi-ingredient formulations	The final product is a mixture of hundreds of compounds
No single active principle	The action is attributed to the whole, not to one molecule
Processing changes the material	Shodhana, Bhavana, Paka and Puta all alter the constituents
Substitution and adulteration	Common in the raw drug trade, particularly for scarce drugs
Inorganic ingredients	Bhasma and mineral preparations cannot be analysed by the usual chromatographic methods
Absence of reference standards	Authentic markers for many Ayurvedic drugs are scarce or unavailable
Classical variability	Different texts prescribe different methods for the same formulation

1.1.2 Why it must be done nonetheless

1. **Patient safety.** The patient cannot judge the medicine; the manufacturer must guarantee it.
2. **Therapeutic reliability.** A medicine that varies from batch to batch gives results that vary from patient to patient.
3. **Legal obligation.** The Drugs and Cosmetics Act applies to ASU drugs in full.
4. **Detection of adulteration and substitution.**
5. **Control of contaminants** — heavy metals, pesticide residues, aflatoxins and microbial load.
6. **Export requirement.** Importing countries demand documented quality.
7. **Scientific credibility.** Research on an uncharacterised material cannot be replicated, and therefore cannot be believed.
8. **Preservation of the tradition.** A tradition whose products cannot be relied upon will not survive.

1.2 Quality assurance of Ayurveda products

The distinction between QC and QA was given in Module 1, Unit 4. In brief: **QC tests the product; QA builds the system that makes the product right.** QC is a part of QA.

1.2.1 The elements of quality assurance in an Ayurvedic unit

Element	What it means in practice
Approved suppliers and source control	Raw drugs bought only from verified sources, with the botanical identity confirmed
Written specifications	For every raw material, in-process stage and finished product
Master formula record	The authorised composition and method for each product
Batch manufacturing record	A complete record of what was actually done for every batch
Standard operating procedures	For every operation, as developed in Modules 4 and 5
In-process controls	Checks during manufacture, not only at the end
Trained personnel	With training recorded
Qualified equipment, calibrated on schedule	As covered in Module 5, Unit 6
Controlled premises	Schedule T requirements
Change control	No change to a process without written authorisation
Deviation and out-of-specification handling	Every departure investigated and recorded
Internal audit or self-inspection	Regular, documented
Product recall system	Tested, not merely written
Complaint handling and pharmacovigilance	Recorded and analysed
Retention samples	Kept for the shelf life plus the prescribed period

Element	What it means in practice
Stability programme	Ongoing, with data supporting the shelf life claimed

> **The single most important idea in this unit.** Quality cannot be tested into a product. It must be **built into** it. Testing the finished Bhasma tells you whether this batch is acceptable; it does not make the next batch acceptable. Only the system does that. Write this sentence in any answer on QA.

1.3 Quality control parameters

1.3.1 The four questions every QC test answers

Question	Parameter	Meaning
Is it the right material?	**Identity**	Confirms the drug is what it claims to be
Is anything unwanted present?	**Purity**	Absence of foreign matter, contaminants and adulterants
Is enough of it present?	**Strength or assay**	Quantity of the active or marker constituent
Will it perform as intended?	**Quality and performance**	Disintegration, dissolution, uniformity, stability

1.3.2 Identity tests

Level	Test
Organoleptic	Colour, odour, taste, touch, appearance — the classical Varna, Gandha, Rasa, Sparsha
Macroscopic	Shape, size, surface, fracture, of the whole drug
Microscopic	Diagnostic characters of the powder — starch grains, fibres, stone cells, trichomes, crystals
Chemical	Preliminary phytochemical screening; specific identity tests
Chromatographic	TLC or HPTLC fingerprint compared against a reference
Instrumental	For minerals and Bhasma: XRD phase identification, EDAX elemental profile
Molecular	DNA barcoding, where morphology cannot decide

1.3.3 Purity and contaminant tests

A. Physicochemical parameters — these are the standard set, and they are asked constantly.

Parameter	What it indicates
Foreign matter	Extraneous material — soil, stones, other plant parts, insect damage
Loss on drying, or moisture content	Excess moisture promotes microbial growth and degradation
Total ash	Total inorganic residue after incineration; indicates the mineral content plus any earthy adulteration
Acid insoluble ash	The portion of ash not dissolved by dilute hydrochloric acid; **chiefly silica, so it indicates contamination with sand and earth**
Water soluble ash	The water-soluble portion of the ash
Alcohol soluble extractive	Amount extracted by alcohol; indicates the alcohol-soluble constituents present
Water soluble extractive	Amount extracted by water; a low value may indicate exhausted or substituted material
pH	Relevant for liquids and semi-solids
Specific gravity and refractive index	For liquids and oils
Volatile oil content	For aromatic drugs, measured in a Clevenger apparatus
Swelling index and foaming index	For mucilaginous and saponin-containing drugs

> **Acid insoluble ash is the parameter to understand properly.** Total ash tells you how much inorganic material is present, but some of that is genuine — the plant's own minerals. Acid insoluble ash is largely silica, which the plant does not contain in quantity. **A high acid insoluble ash therefore points to contamination with sand or earth**, and is the standard indicator of poor collection and cleaning. This distinction is a very likely question.

B. Contaminants

Contaminant	Why it matters	Method
Heavy metals — lead, arsenic, mercury, cadmium	Toxic; also an export barrier	AAS, ICP-OES or ICP-MS
Pesticide residues	Toxic; residues from cultivation	GC, GC-MS, LC-MS
Aflatoxins — B1, B2, G1, G2	Produced by fungi on stored material; carcinogenic	HPTLC, HPLC with fluorescence detection
Microbial load	Infection risk; product spoilage	Total plate count, yeast and mould count
Specific pathogens — E. coli, Salmonella, S. aureus, P. aeruginosa	Must be absent	Selective culture
Adulterants and substitutes	Reduce efficacy; may be harmful	Microscopy, chromatography, DNA barcoding

> **An important point on heavy metal limits, stated carefully.**
>
> Limits for lead, arsenic, mercury and cadmium are prescribed for Ayurvedic, Siddha and Unani products, and the Ministry of AYUSH requires heavy metal testing, particularly for export. Your department will have the current figures; **always verify them against the current edition of the Ayurvedic Pharmacopoeia of India and the current AYUSH notifications**, because they are revised.
>
> There is a further point that a postgraduate student must understand and be able to explain. These limits are framed for products in which the metal is a **contaminant**. In a Bhasma or a Rasaushadhi the metal is a **declared, intended ingredient**. The two situations are not the same, and applying a contaminant limit to a declared ingredient is a category error. The regulatory treatment of metal-containing ASU preparations therefore rests on the licensing route under Rule 158-B, on proof of safety, and on correct preparation and dosage — not on the contaminant limits alone. Being able to state this distinction clearly is exactly what is expected of you at this level.

1.3.4 Assay and performance

Test	Applies to
Assay of a marker compound	Herbal formulations, by HPTLC or HPLC
Assay of the metal content	Bhasma and metallic preparations, by titrimetry or ICP
Uniformity of weight	Tablets, capsules, unit-dose packs
Disintegration time	Tablets, Vati, Gutika
Dissolution	Where prescribed
Hardness and friability	Tablets
Sterility	Ophthalmic and parenteral products
Particle size	Churna, Bhasma, Anjana

1.4 Regulatory requirements for quality control

Requirement	Provision
Governing law	Drugs and Cosmetics Act, 1940 and Rules, 1945, Chapter IVA
GMP	**Schedule T** — Part I gives GMP; Part II gives machinery and premises
Licensing	**Rule 158-B**, with proof of safety and effectiveness by category
Standards	**Ayurvedic Pharmacopoeia of India** and the **Ayurvedic Formulary of India**
Authoritative texts	The **First Schedule** books
Poisonous substances	**Schedule E(1)**
Licensing and enforcement authority	State Licensing Authority and State Drug Controller
Standards body	**PCIM&H**
Advisory bodies	ASUDTAB and ASUDCC
Testing laboratories	Government-approved drug testing laboratories for ASU drugs
Voluntary marks	AYUSH Premium Mark and AYUSH Standard Mark, operated by the Quality Council of India
Export	WHO Certificate of Pharmaceutical Products; the requirements of the importing country

Schedule T requirements specific to quality control. A licensed unit must have its own quality control section with adequate space and equipment, or an arrangement with an approved laboratory; qualified technical staff; written specifications and testing procedures; and complete records of every test performed.

1.5 Documentation and record-keeping

1.5.1 Why documentation is the heart of quality

If it is not written down, it did not happen. This is the rule an inspector applies, and it is the rule you should apply to your own research.

1.5.2 The documents required

Document	Content
Master Formula Record (MFR)	The authorised formula, method, specifications and yields for a product. The parent document from which every batch record is made
Batch Manufacturing Record (BMR)	A complete record of one batch as actually made, with every quantity, observation, and signature
Batch Packaging Record	Packaging and labelling of that batch
Specifications	For raw materials, packaging materials, in-process stages and finished product
Standard Testing Procedures (STP)	How each test is performed
Standard Operating Procedures (SOP)	For every operation and every instrument
Analytical records and raw data	The actual readings, chromatograms and spectra, not only the results
Certificate of Analysis	Issued for each batch released
Calibration and maintenance records	As in Module 5, Unit 6
Training records	For every person
Cleaning and sanitation records	—
Deviation and OOS records	Every departure, with investigation and conclusion
Change control records	—
Stability study records	—
Distribution records	Needed for recall
Complaint and recall records	—
Retention sample register	—

1.5.3 Good documentation practice

1. Record at the time of the activity, not afterwards.
2. Write legibly, in indelible ink.
3. Never overwrite or erase. Strike through with a single line, write the correct entry, and **sign and date the correction with the reason**.
4. Never leave a blank. Enter "not applicable" where a field does not apply.
5. Sign and date every entry.
6. Never sign for work you did not perform or witness.
7. Follow the **ALCOA** principles from Module 2, Unit 5 — Attributable, Legible, Contemporaneous, Original, Accurate.
8. Retain records for the period prescribed.

KEY POINTS TO REMEMBER

- Quality control in Ayurveda is harder because of natural variability, multi-ingredient formulations, processing changes, and the absence of reference standards.
- **Quality cannot be tested into a product; it must be built into it.**
- The four QC questions: identity, purity, strength, and quality or performance.
- **Acid insoluble ash is chiefly silica and indicates contamination with sand and earth.**
- Contaminants tested: heavy metals, pesticide residues, aflatoxins, microbial load and specific pathogens.
- Heavy metal limits are framed for metals as **contaminants**. In a Bhasma the metal is a declared ingredient, and the regulatory route is Rule 158-B with proof of safety.
- Schedule T requires a quality control section, qualified staff, written procedures and complete records.
- Master Formula Record is the parent; the Batch Manufacturing Record is the record of one batch as actually made.
- **If it is not written down, it did not happen.**

EXAM CORNER — UNIT 1

This unit is asked as ABQ (20 marks) and LAQ (10 marks). It is never asked as a five-mark question.

Likely long answer questions (10 marks)

1. Discuss the importance of quality control in Ayurveda, and the difficulties peculiar to Ayurvedic formulations.
2. Describe quality assurance of Ayurvedic products, with its essential elements and its relation to Schedule T.
3. Describe the quality control parameters applied to Ayurvedic formulations under the headings of identity, purity, strength and performance.

4. Discuss the regulatory requirements for quality control of Ayurvedic products in India.
5. Describe the documentation and record-keeping required in an Ayurvedic manufacturing unit, and explain the principles of good documentation practice.
6. Discuss the physicochemical parameters used in the quality control of Ayurvedic drugs, and explain what each indicates.

Likely analytical based questions (20 marks)

1. A batch of a classical Ayurvedic formulation fails its finished product specification. Analyse every stage at which the failure could have originated, describe the investigation you would conduct, and set out the corrective and preventive actions with reference to the quality system.
2. You are asked to establish a complete quality control system for a new Ayurvedic manufacturing unit. Analyse the requirements under Schedule T, design the specifications, procedures and records needed, and justify your design.
3. A metal-containing Ayurvedic preparation is rejected by an importing country on the ground of heavy metal content. Analyse the scientific and regulatory position, explain what evidence you would generate, and set out how the matter should be addressed.

How to attempt

- Because there is no SAQ from this unit, every question will be long. Plan the answer before writing, and use headings throughout.
- Always distinguish QC from QA early in the answer. It frames everything that follows.
- Cite Schedule T, Rule 158-B and the Ayurvedic Pharmacopoeia of India by name.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **153, 154, 155, 156**

1. **153.** CCRAS. Guideline of Drug Development; 2024.
2. **154.** CCRAS. Guideline of Clinical Evaluation; 2024.
3. **155.** Mukherjee PK. Quality Control and Evaluation of Herbal Drugs.
4. **156.** Ayurvedic Pharmacopoeia of India, Part I, Volumes I to X. PCIM&H, Ministry of AYUSH.

Additional sources: Drugs and Cosmetics Act 1940 and Rules 1945, Schedule T and Rule 158-B; Protocol for Testing of Ayurvedic, Siddha and Unani Medicines, Department of AYUSH (reference 163); WHO Quality Control Methods for Medicinal Plant Materials (reference 129).

UNIT 2
CLASSICAL ASPECTS OF THE QUALITY CONTROL PARAMETER FOR AYURVEDIC FORMULATIONS

SYLLABUS POINTS

2.1 Mana and their application in quality control
2.2 Avaleha Siddhi Lakshana
2.3 Sneha Paka Siddhi Lakshana
2.4 Sandhana Siddhi Lakshana
2.5 Guggulu Paka Lakshana
2.6 Bhasma and Pishti Pariksha

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

> This unit is the classical heart of the module. Every Siddhi Lakshana is an **in-process quality control test**, devised at a time when no instrument existed. Learn each one, and learn what it measures in modern terms. That pairing is what earns marks.

2.1 Mana and its application in quality control

2.1.1 What Mana is

Mana means measurement. The classical texts prescribe exact quantities for every ingredient, and a system of units to express them. Without a fixed system of measurement there can be no reproducible formulation, so Mana is the foundation of all quality control.

2.1.2 The two systems

System	Relation
Magadha Mana	The system in general use, and the one followed in most texts and in the Ayurvedic Formulary of India
Kalinga Mana	The values are **half** those of Magadha Mana

Always state which system you are following. A formula quoted without the system is ambiguous by a factor of two.

2.1.3 The Magadha Mana table

Unit	Equivalent	Approximate metric value
1 Gunja or Raktika	—	125 mg
8 Gunja	1 Masha	1 g
12 Masha	1 Karsha or Tola	12 g
2 Karsha	1 Shukti	24 g
2 Shukti	1 Pala	48 g
2 Pala	1 Prasruti	96 g
2 Prasruti	1 Kudava	192 g
2 Kudava	1 Prastha	384 g
4 Prastha	1 Adhaka	1.536 kg
4 Adhaka	1 Drona	6.144 kg
100 Pala	1 Tula	4.8 kg
20 Tula	1 Bhara	96 kg

> **Two cautions worth stating in an answer.**
> **1.** Different texts give slightly different equivalents, and different regions used different local values. **Always cite the text you follow.**
> **2.** The Gunja seed, from which the smallest unit is taken, varies naturally in weight. The value of 125 mg is a standardised convention, not a measurement of any particular seed. Modern practice uses the metric equivalent and weighs on a calibrated balance.

2.1.4 Measures of volume, time and fire

Category	Examples
Drava Mana — liquid measure	Anjali, Kudava, Prastha, Adhaka, Drona, expressed by volume
Kala Mana — time	Muhurta, Yama, Ahoratra, Paksha, Masa, Ritu, Ayana, Varsha
Agni Mana — fire intensity	Mridu, Madhyama, Tikshna or Khara Agni; and the Puta types of Module 5

Application to quality control. Every one of these is a process parameter. Quantity, duration and temperature are exactly the three variables a modern SOP must fix. The classical system fixed all three, in its own units.

2.2 Avaleha Siddhi Lakshana

Avaleha is a semi-solid confection prepared by cooking drugs with sugar or jaggery. Its completion is judged by the following signs.

Siddhi Lakshana	Method and observation	Modern interpretation
Tantuvat	A drop taken between the thumb and forefinger and drawn apart forms **a single unbroken thread**	The sugar solution has reached the correct concentration and viscosity
Apsu majjanam	A drop placed in cold water **sinks without dissolving or dispersing**	Sufficient concentration; the water content has been reduced adequately
Darvi lepa or Darvi pralepa	The mass **coats the ladle** and does not run off	Correct consistency and viscosity
Pidite mudrikavat	When pressed between the fingers it **retains the impression**, like a ring	Correct plasticity and solid content
Gandha, Varna and Rasa utpatti	The characteristic **odour, colour and taste** of the finished Avaleha appear	Chemical changes of cooking are complete
Phena shanti	Frothing subsides	Free water has evaporated
Ghana bhava	The mass becomes uniformly thick and does not stick to the vessel	Correct end point

Modern equivalent tests. Total solids, moisture content, reducing and non-reducing sugars, pH, viscosity, and specific gravity. These are dealt with in Unit 4.

2.3 Sneha Paka Siddhi Lakshana

Sneha Kalpana is the preparation of medicated Ghrita and Taila by cooking the Sneha with Kalka and Drava Dravya.

2.3.1 The stages of Paka

Stage	Characteristics	Use
Ama Paka	Cooking incomplete; moisture remains	Not used
Mridu Paka	Kalka is soft and waxy; can be rolled but is sticky	**Nasya**
Madhyama Paka	Kalka forms a **Varti**, a roll or wick, without sticking to the fingers	**Pana, Basti, Nasya** — the most widely useful stage
Khara Paka	Kalka becomes hard, dry and crumbly	**Abhyanga and external use**
Dagdha Paka	Burnt; colour and odour spoilt	**Discarded**

2.3.2 The Siddhi Lakshana

Lakshana	Observation	Modern interpretation
Varti bhavati	The Kalka rolls into a wick without sticking to the fingers	The Kalka has lost its free moisture
Shabda heenata	The crackling sound of boiling **ceases**	All the water has evaporated; only oil remains
Phena	**In Ghrita Paka, froth appears at completion. In Taila Paka, the froth subsides at completion.**	The behaviour of the froth reflects the difference between the two Snehas
Agni Pariksha	A small quantity of Kalka thrown into fire **burns without crackling**	No water remains
Varna, Gandha and Rasa	Characteristic colour, odour and taste appear	Extraction of the active constituents is complete
Sneha darshana	The Sneha becomes clear and separates from the Kalka	The process is complete

> **The froth test is the classic examination point of this unit, and students reverse it constantly.**
>
> **Ghrita Paka — froth APPEARS at completion.**
> **Taila Paka — froth SUBSIDES at completion.**
>
> Remember it by the first letters: **G**hrita — froth **G**athers. **T**aila — froth **T**erminates.

2.4 Sandhana Siddhi Lakshana

Sandhana Kalpana covers Asava and Arishta, prepared by fermentation, in which alcohol is generated within the preparation itself.

Siddhi Lakshana	Observation	Modern interpretation
Cessation of bubbling	Effervescence stops	Fermentation is complete; carbon dioxide is no longer produced
Deepa or Agni Pariksha	A lighted lamp or stick held above the surface **does not extinguish**	Carbon dioxide is no longer being evolved
Madya Gandha	The characteristic alcoholic odour is present	Alcohol has been generated
Rasa	The taste becomes Katu, Kashaya and slightly Amla	Sugar has been converted; acids have formed
Varna	The characteristic colour appears	Extraction and fermentation are complete
Sedimentation	The sediment settles and the liquid clears	Yeast and solids have settled
Vessel test	The preparation floats a specified object, in some texts	Specific gravity has reached the expected value

Modern equivalent tests. Alcohol content, specific gravity, pH, total solids, reducing and non-reducing sugars, methanol test, and microbial limits. These are dealt with in Unit 5.

2.5 Guggulu Paka Lakshana

Guggulu is an oleo-gum-resin, and Guggulu Kalpana requires it to be purified and then processed with other drugs into a workable mass.

Stage	Lakshana
Shodhana	Guggulu is subjected to Swedana in Dolayantra with the prescribed medium — commonly Triphala Kwatha, Godugdha, Gomutra or Nirgundi Kwatha, as specified for the particular Yoga
Correct Paka	The mass becomes **soft, uniform and glossy**; it can be rolled between the fingers **without sticking**; it does not crumble
Under-processed	Sticky, uneven, adheres to the fingers and to the vessel; excess moisture remains
Over-processed	Hard, dry, brittle; will not form a proper Vati; the volatile constituents are lost
Addition of other drugs	The prescribed Churna is added and triturated in until the mass is homogeneous
Final test	The Vati formed holds its shape, is uniform in weight, and disintegrates within the prescribed time

Modern equivalent tests. Loss on drying, uniformity of weight, hardness, disintegration time, and assay of the marker.

2.6 Bhasma and Pishti Pariksha

2.6.1 Bhasma Pariksha

The complete Bhasma Pariksha, with the method and the modern interpretation of each test, is given in **Module 4, Unit 3, section 3.2.3**. Revise it from there rather than learning it twice. In summary, the tests are:

Rekhapurnata, Varitara, Unama, Nishchandratva, Apunarbhava, Niruttha, Nirdhuma, Nishwadatva, Slakshnatva and Sukshmatva, Varna, Gata Rasatva, Avami, and the absence of a gritty sound between the teeth.

Apunarbhava and Niruttha remain the two decisive tests, because they alone demonstrate that no free metal remains.

2.6.2 Pishti Pariksha

Pishti is prepared by trituration of a Dravya with a prescribed liquid, generally without fire. Common examples are Mukta Pishti, Pravala Pishti, Akika Pishti and Godanti Pishti.

Point	Pishti	Bhasma
Preparation	Trituration with a liquid, usually **without Puta**	Repeated Bhavana and **Puta**
Nature of change	Chiefly physical — extreme particle size reduction	Physical and **chemical** — a new compound is formed
Appearance	Retains the colour of the parent Dravya, generally white or pale	Colour characteristic of the compound formed
Potency and action	Considered Sheeta in Virya; used where cooling action is required	Varies with the Dravya

Pishti Pariksha

Test	Observation
Slakshnatva	Extremely smooth and fine to the touch
Rekhapurnata	Fills the lines of the fingers
Varna	The prescribed colour, generally white and lustreless
Nishchandratva	No lustre remains
Absence of grittiness	No gritty sound between the teeth
Varitara	Applied in some texts

Modern tests for both. Particle size analysis, XRD for phase, SEM for morphology, EDAX or ICP for elements, TGA for organic residue. These are from Module 5, Unit 7.

KEY POINTS TO REMEMBER

- Every Siddhi Lakshana is an in-process quality control test devised before instruments existed.
- Kalinga Mana values are **half** those of Magadha Mana. Always state which system you follow.
- 1 Gunja = 125 mg; 8 Gunja = 1 Masha = 1 g; 12 Masha = 1 Karsha or Tola = 12 g; 4 Pala × 25 = 1 Tula = 4.8 kg.
- Avaleha Siddhi: Tantuvat, sinking in water, coating the ladle, retaining an impression, characteristic Varna Gandha Rasa.
- Sneha Paka stages: Ama, Mridu for Nasya, Madhyama for Pana Basti Nasya, Khara for Abhyanga, Dagdha discarded.
- **Ghrita Paka — froth appears. Taila Paka — froth subsides.**
- Sandhana Siddhi: bubbling ceases, a flame held above is not extinguished, alcoholic odour, sediment settles.
- Guggulu is correctly processed when it is soft, glossy and rolls without sticking.
- Pishti is made by trituration without Puta; the change is chiefly physical. Bhasma involves Puta and a chemical change.

EXAM CORNER — UNIT 2

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Define Mana. Differentiate between Magadha and Kalinga Mana.
2. Write the Magadha Mana table from Gunja to Tula.
3. Write a short note on the application of Mana in quality control.
4. Enumerate the Avaleha Siddhi Lakshana with the modern interpretation of each.
5. Write a short note on Tantuvat and its significance.
6. Enumerate the stages of Sneha Paka with their uses.
7. Enumerate the Sneha Paka Siddhi Lakshana.
8. Differentiate the froth behaviour in Ghrita Paka and Taila Paka.
9. Enumerate the Sandhana Siddhi Lakshana with the modern interpretation of each.
10. Write a short note on Guggulu Paka Lakshana.
11. Differentiate between Bhasma and Pishti.
12. Enumerate the Pishti Pariksha.

How to attempt

- For every Siddhi Lakshana question, give the classical observation **and** the modern interpretation, in a two-column table. This is what the unit is testing.
- Cite the classical text for the Lakshana you quote.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **14, 15, 157, 158, 159, 160, 161, 162**

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5. **159.** Sharangadhara. Sharangadhara Samhita with the Dipika of Adhamalla and the Gudartha Dipika of Kashirama.
6. **160.** Mishra S. Bhaishajya Ratnavali. Varanasi: Chaukhamba Surbharati Prakashan; 2015.
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UNIT 3
QUALITY CONTROL PARAMETER FOR SOLID DOSAGE FORMS

SYLLABUS POINTS

3.1 Churna, Kwatha Churna, Lepa
3.2 Kshara, Lavana, Satva
3.3 Tablets, pills and Varti
3.4 Khalveeya, Parpati and Kupipakwa Rasayana
3.5 Lauha and Mandura

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

3.0 The general scheme for any dosage form

Before the individual forms, learn this scheme. **Every quality control answer in Units 3, 4 and 5 can be built on it**, and the examiner is looking for the structure as much as for the content.

Group of tests	Content
1. Organoleptic	Varna, Gandha, Rasa, Sparsha — colour, odour, taste, touch, appearance
2. Physical	Particle size or sieve number, weight, uniformity, hardness, consistency, specific gravity, viscosity, as appropriate to the form
3. Physicochemical	Loss on drying, total ash, acid insoluble ash, water soluble ash, alcohol and water soluble extractive, pH
4. Chemical and chromatographic	Phytochemical screening, TLC or HPTLC fingerprint, assay of the marker
5. Contaminants	Heavy metals, pesticide residues, aflatoxins, microbial limits, specific pathogens
6. Performance	Disintegration, dissolution, spreadability, sterility, as appropriate
7. Classical	The Siddhi Lakshana prescribed for that Kalpana
8. Packaging and stability	Container suitability, storage condition, shelf life

3.1 Churna, Kwatha Churna and Lepa

3.1.1 Churna

Churna is a fine powder of one or more dried drugs.

Parameter	Requirement or method
Organoleptic	Characteristic colour, odour and taste of the constituent drugs
Fineness	**Churna for internal use should pass through a number 80 sieve.** The API prescribes the sieve number for each product
Foreign matter	Within the prescribed limit
Loss on drying	Within the prescribed limit; excess moisture causes caking and microbial growth
Total ash	Within the prescribed limit
Acid insoluble ash	Low; a high value indicates sand and earth
Alcohol soluble extractive	As prescribed
Water soluble extractive	As prescribed
pH	Where applicable
TLC or HPTLC	Fingerprint compared against the reference
Assay	Of the marker, where prescribed
Microbial limits	Total plate count, yeast and mould, and absence of specified pathogens
Heavy metals, pesticides, aflatoxins	Within the prescribed limits
Packing	Airtight container; Churna is hygroscopic

3.1.2 Kwatha Churna

Kwatha Churna is the coarse powder from which a decoction is prepared. **It is deliberately coarse, not fine.**

Parameter	Requirement
Fineness	**Yavakuta** — coarse, comparable to a crushed barley grain; passes a coarse sieve and is retained on a finer one, as prescribed
Why coarse	A coarse powder allows water to penetrate and to be strained off. A fine powder would form a paste, block the strainer and be impossible to filter
Other parameters	As for Churna — foreign matter, loss on drying, ash values, extractives, TLC, contaminants
Additional	Yield and total solids of the Kwatha prepared from it

3.1.3 Lepa Churna

Lepa Churna is for external application, and must be finer than Churna for internal use.

Parameter	Requirement
Fineness	**Very fine — Shlakshna.** It should pass through a number 100 sieve or finer, as prescribed. A gritty Lepa injures the skin
Absence of grittiness	Tested by rubbing between the fingers
Consistency on mixing	Should form a smooth, uniform paste with the prescribed liquid
Adhesiveness and spreadability	Should adhere to the skin without cracking off too soon
pH	Should be compatible with skin
Other parameters	Foreign matter, loss on drying, ash values, microbial limits

> **The comparison worth memorising.**
> **Kwatha Churna — coarse (Yavakuta)**, because it must be strained.
> **Churna for internal use — fine, number 80 sieve.**
> **Lepa Churna — very fine, number 100 sieve or finer**, because it goes on the skin.
> Three forms, three different fineness requirements, each for a reason. This is a very likely short answer question.

3.2 Kshara, Lavana and Satva

3.2.1 Kshara

Kshara is an alkaline preparation obtained from the ash of a drug, by dissolving, filtering and evaporating.

Parameter	Requirement or method
Appearance	White or greyish-white, as prescribed
pH	**Strongly alkaline**, as prescribed for that Kshara
Total alkalinity	Determined by titration against a standard acid; the principal assay
Water soluble ash	High, since the active portion is water soluble
Acid insoluble ash	Low
Loss on drying	Low; Kshara is hygroscopic
Solubility	Readily soluble in water
Chloride, sulphate and carbonate content	As prescribed
Taste	Katu and Kshara
Packing	**Airtight container. Kshara absorbs moisture and carbon dioxide from the air and loses its alkalinity**

3.2.2 Lavana

Lavana covers the salts used in Ayurveda — Saindhava, Samudra, Vida, Sauvarchala and Romaka.

Parameter	Requirement or method
Appearance and colour	Characteristic of the particular Lavana
Taste	Lavana Rasa
Assay of sodium chloride	By titration; the principal assay
Solubility in water	Readily soluble
Loss on drying	Low
Acid insoluble matter	Low; indicates sand
Sulphate, calcium and magnesium content	As prescribed
pH of solution	As prescribed
Heavy metals	Within limits

3.2.3 Satva

Satva is the extracted essence — either the starch-like extract of a plant such as Guduchi Satva, or the metallic essence obtained by Satvapatana as studied in Module 3.

Type	Parameters
Plant Satva (Guduchi Satva, Amrita Satva)	Appearance — white and light; solubility; loss on drying; ash values; starch and carbohydrate content; TLC; microbial limits
Metallic Satva (Satvapatana product)	Appearance and lustre; **percentage yield**; assay of the metal content; specific gravity; melting behaviour; XRD phase; EDAX elemental profile; freedom from unreacted ore and flux

3.3 Tablets, pills and Varti

3.3.1 Vati, Gutika and tablets

Parameter	Method and requirement
Description	Shape, size, colour, surface, absence of defects
Uniformity of weight	Twenty units weighed individually; deviation within the prescribed limit
Hardness	Monsanto or equivalent tester; sufficient to withstand handling
Friability	Roche friabilator; generally not more than 1 per cent
Disintegration time	Disintegration test apparatus, in the prescribed medium; within the time prescribed in the API for that product
Dissolution	Where prescribed
Assay	Of the marker or the metal content
Loss on drying	Within limits
Ash values	Within limits
TLC or HPTLC	Fingerprint
Microbial limits and contaminants	Within limits
For Guggulu Vati	Additionally: correct Paka Lakshana of the Guggulu, and absence of stickiness

3.3.2 Varti

Varti is a wick or suppository-like form, used for rectal, vaginal, urethral, nasal or ophthalmic application.

Parameter	Requirement
Shape and dimensions	Uniform length and diameter, as prescribed
Uniformity of weight	Within the prescribed deviation
Surface	Smooth, free from cracks and air pockets
Hardness and mechanical strength	Should withstand handling and insertion without breaking
Melting or softening behaviour	Should soften at body temperature where a systemic or local release is intended
Disintegration or dissolution	As prescribed
Assay	Of the active
Sterility	**Required for ophthalmic Varti such as Anjana Varti**
Absence of irritation	Particularly for ophthalmic and urethral use

3.4 Khalveeya, Parpati and Kupipakwa Rasayana

These are the Chaturvidha Rasayana forms studied in Module 3, Unit 2. Each has its own quality profile.

3.4.1 Khalveeya Rasayana

Prepared in the Khalva Yantra by trituration, without fire. Kajjali is the base of most.

Parameter	Requirement
Colour	**Kajjali should be uniformly jet black, lustreless**
Nishchandratva	No globule of free mercury visible, even under a lens
Rekhapurnata and Slakshnatva	Extremely fine; fills the lines of the fingers
Absence of grittiness	No gritty sound between the teeth
Absence of free mercury	The decisive test; verified chemically
Assay	Mercury and sulphur content
Loss on drying	Low
XRD	Confirms the phase present
Uniformity	The mixture must be homogeneous throughout

> **The presence of a single visible globule of free mercury means the Kajjali is not ready.** Trituration must continue. This is the most important single quality check in Khalveeya Rasayana, and it is checked by eye and by lens before anything else.

3.4.2 Parpati Rasayana

Prepared by melting Kajjali and pouring it out to set as a thin flake.

Parameter	Requirement
Appearance	Thin, flake-like, of uniform thickness
Colour	As prescribed for that Parpati
Fracture	Breaks with a characteristic clean fracture
Lustre	Absent on the surface
Uniformity of thickness	Consistent across the flake
Powdering behaviour	Powders readily into a fine, uniform powder
Loss on drying	Low
Assay	Of the constituents
Absence of free mercury	Verified
XRD	Phase confirmation

3.4.3 Kupipakwa Rasayana

Prepared by controlled heating in a Kachakupi within a Valuka Yantra.

Parameter	Requirement
Site of collection	Whether the product was collected from the neck or the bottom of the Kupi, as prescribed for that product
Colour	Characteristic and prescribed. **Rasa Sindura is a bright red**
Appearance	Crystalline or as prescribed
Nishchandratva	No lustre; no free mercury
Rekhapurnata and Varitara	Applied where prescribed
Loss on drying	Low
Assay	Mercury and sulphur content
XRD	Confirms the expected phase and the absence of unreacted material
Percentage yield	Recorded, since it reflects the control of the process
Absence of free sulphur	Verified by the prescribed test

3.5 Lauha and Mandura

Lauha preparations are made from iron; **Mandura** from purified iron rust.

Parameter	Requirement or method
Bhasma Pariksha	The full set, particularly Rekhapurnata, Varitara, Nishchandratva, **Apunarbhava and Niruttha**
Colour	The prescribed colour for that Bhasma; reddish to reddish-black for Lauha Bhasma
Assay of total iron	By titrimetry or by instrumental method; the principal quantitative test
Particle size	By particle size analyser, SEM or TEM
XRD	Identifies the iron oxide phase formed and confirms absence of free iron
Magnetic properties	**VSM.** Directly relevant here, because iron oxide phases differ in magnetic behaviour and very fine particles may show superparamagnetism
Loss on drying and loss on ignition	Within limits
Acid insoluble ash	Low
Heavy metals other than iron	Within limits
Absence of free iron	Confirmed by magnet test and by Apunarbhava
Number of Putas recorded	Part of the batch record

Why VSM belongs specifically in this section. Metallic iron is strongly ferromagnetic. The iron oxides formed during Marana have different and characteristic magnetic behaviour, and their behaviour also changes with particle size. So a magnetic measurement independently confirms both the phase and the fineness — and it also detects free unreacted iron. For Lauha and Mandura Bhasma, VSM is not an exotic addition; it is one of the most directly informative tests available.

KEY POINTS TO REMEMBER

- Build every QC answer on the eight-group scheme: organoleptic, physical, physicochemical, chemical and chromatographic, contaminants, performance, classical, packaging and stability.
- **Kwatha Churna is coarse (Yavakuta); Churna for internal use passes a number 80 sieve; Lepa Churna is finer still, number 100 or finer.**
- Kshara is assayed by total alkalinity and must be kept in an airtight container.
- Lavana is assayed for sodium chloride content by titration.
- Vati tests: uniformity of weight, hardness, friability (not more than 1 per cent), disintegration time, assay.
- **Kajjali must be uniformly jet black and lustreless, with no visible globule of free mercury.**
- Parpati is judged by its flake form, uniform thickness and characteristic fracture.
- Kupipakwa is judged by the site of collection, the prescribed colour, and XRD phase confirmation. Rasa Sindura is bright red.

- Lauha and Mandura: total iron assay, XRD phase, and VSM for magnetic properties and free iron.

EXAM CORNER — UNIT 3

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ.

Likely short answer questions (5 marks)

1. Enumerate the quality control parameters of Churna.
2. Differentiate the fineness required for Churna, Kwatha Churna and Lepa Churna, with reasons.
3. Enumerate the quality control parameters of Kshara.
4. Why must Kshara be stored in an airtight container?
5. Enumerate the quality control parameters of Lavana.
6. Write a short note on the quality control of Satva.
7. Enumerate the quality control parameters of Vati and Gutika.
8. Write a short note on the quality control of Varti.
9. Enumerate the quality control parameters of Kajjali.
10. Enumerate the quality control parameters of Parpati.
11. Enumerate the quality control parameters of Kupipakwa Rasayana.
12. Enumerate the quality control parameters of Lauha Bhasma, and explain the role of VSM.

Likely analytical based questions (20 marks)

1. You are asked to establish a complete quality control protocol for a Kupipakwa Rasayana, from raw material to finished product. Analyse each stage, select the classical and instrumental tests for each, justify every selection, and state the acceptance criteria.
2. Two batches of Lauha Bhasma pass all the classical Bhasma Pariksha but differ in their assay and in their XRD patterns. Analyse what this means, what could have caused it, and how the quality system should be changed to prevent it.
3. A market sample of a Churna fails on acid insoluble ash but passes every other test. Analyse what this indicates, trace where in the supply chain the problem arose, and set out the corrective and preventive actions.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **161, 162, 163** and the associated pharmacopoeial volumes.

1. **161 and 162.** Ayurvedic Pharmacopoeia of India, Part II, Volumes I to X and I to IV. PCIM&H, Ministry of AYUSH.
2. **163.** Lohar D. Protocol for Testing of Ayurvedic, Siddha and Unani Medicines. Department of AYUSH, Government of India; 2008.

Additional sources: Ayurvedic Pharmacopoeia of India, Part I (reference 156); Ayurvedic Formulary of India; Rasatarangini (reference 1); Rasa Ratna Samuccaya (references 2 and 3); Bhasma Vigyana (reference 124); Mukherjee PK, Quality Control and Evaluation of Herbal Drugs (reference 155); WHO Quality Control Methods for Medicinal Plant Materials (reference 129).

UNIT 4
QUALITY CONTROL PARAMETER FOR SEMI-SOLID DOSAGE FORMS

SYLLABUS POINTS

4.1 Rasakriya, Avaleha and Paka
4.2 Anjana

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

4.1 Rasakriya, Avaleha and Paka

4.1.1 Rasakriya

Rasakriya is a semi-solid preparation obtained by evaporating a Kwatha or Swarasa to a thick consistency.

Parameter	Requirement or method
Appearance and consistency	Semi-solid, uniform, free from lumps
Colour, odour and taste	Characteristic
Total solids	The principal test; indicates the degree of concentration reached
Loss on drying	Low; excess moisture causes fermentation and mould
pH	As prescribed
Ash values	Within limits
Water and alcohol soluble extractive	As prescribed
TLC or HPTLC	Fingerprint
Assay	Of the marker, where prescribed
Microbial limits	Particularly yeast and mould, since sugar and moisture support growth
Heavy metals and pesticide residues	Within limits
Classical	The prescribed Siddhi Lakshana

4.1.2 Avaleha and Paka

Avaleha, also called Leha or Lehya, is a confection prepared by cooking drugs with sugar or jaggery, with Ghrita, Taila, Madhu and Prakshepaka Dravya added as prescribed.

Parameter	Requirement or method
Classical Siddhi Lakshana	**Tantuvat**, sinking in water, coating the ladle, retaining an impression, characteristic Varna, Gandha and Rasa. See Unit 2, section 2.2
Appearance and consistency	Uniform, semi-solid, free from crystallisation and separation
Total solids	The principal physicochemical test
Loss on drying or moisture	Critical. Excess moisture permits fermentation and mould growth, and is the commonest cause of spoilage
Reducing sugars	Determined by the prescribed method
Non-reducing sugars	Determined by the prescribed method
pH	As prescribed
Specific gravity	Where applicable
Ash values	Within limits
Alcohol and water soluble extractive	As prescribed
TLC or HPTLC	Fingerprint
Assay	Of the marker
Microbial limits	Total plate count, **yeast and mould count**, and absence of specified pathogens
Heavy metals, pesticides, aflatoxins	Within limits
Packing and storage	Wide-mouthed airtight container; stored in a cool, dry place away from light
Stability	Checked for crystallisation, separation, mould and change of taste over the shelf life

> **Why reducing and non-reducing sugars are tested, which students often cannot explain.**
>
> An Avaleha contains a large quantity of sugar or jaggery. During cooking, sucrose — a non-reducing sugar — is partly inverted to glucose and fructose, which are reducing sugars. **The ratio of reducing to non-reducing sugar therefore reflects how far the cooking went and at what temperature.** It is an indirect but sensitive measure of the Paka, and it is also relevant to the stability of the product and to its suitability for a diabetic patient. Being able to explain this is worth full marks on the question.

4.1.3 Common defects in Avaleha and their causes

Defect	Probable cause
Crystallisation of sugar	Over-cooking; insufficient invert sugar; storage at low temperature
Fermentation, souring, gas formation	Under-cooking; excess moisture; contamination; poor packing
Mould growth on the surface	Excess moisture; humid storage; container opened repeatedly
Separation of Ghrita or Taila	Insufficient mixing; incorrect Paka
Hard, dry mass	Over-cooking; Khara Paka
Loss of aroma	Prakshepaka Dravya added too early, or added while the mass was too hot
Darkening of colour	Over-cooking; caramelisation

> **A point of classical practice with a clear chemical basis.** Prakshepaka Dravya — the aromatic powders such as Twak, Ela and Patra — and Madhu are added **after the Avaleha has cooled to the prescribed temperature**, never to the hot mass. The volatile constituents would be driven off, and honey is classically held to be unsuitable when heated. The instruction and the chemistry agree.

4.2 Anjana

Anjana is a preparation applied to the eye — as a paste, a powder or a semi-solid. Because it goes into the eye, the quality requirements are the strictest of any Ayurvedic dosage form.

Parameter	Requirement or method
Particle size	**Extremely fine.** No gritty particle may be present. Verified microscopically and by the absence of grittiness on rubbing. A single coarse particle can abrade the cornea
Absence of foreign particulate matter	Examined under magnification
Sterility or microbial limits	As prescribed for an ophthalmic product. Absence of pathogens is essential
pH	Close to that of tears, to avoid irritation
Tonicity	Where the preparation is liquid or semi-liquid, it should be isotonic where possible
Absence of irritation	Tested as prescribed
Appearance, colour and consistency	Uniform and characteristic
Loss on drying	As prescribed
Assay	Of the active

Parameter	Requirement or method
Heavy metals	Within limits, with particular care
Packing	Clean, well-closed, preferably single-use or small containers to limit contamination after opening
Storage	Cool, dry, protected from light

Anjana deserves special emphasis and is a likely question. Three requirements set it apart from every other semi-solid form: **extreme fineness**, because the cornea is easily abraded; **freedom from micro-organisms**, because an eye infection can destroy sight; and **pH close to that of tears**, because the eye is intolerant of extremes. State these three first, then the general parameters.

KEY POINTS TO REMEMBER

- Rasakriya is assayed chiefly by total solids and loss on drying.
- Avaleha is judged classically by Tantuvat, sinking in water, coating the ladle and retaining an impression.
- **The ratio of reducing to non-reducing sugars reflects how far the Paka went**, because cooking inverts sucrose to glucose and fructose.
- Excess moisture is the commonest cause of spoilage in Avaleha, leading to fermentation and mould.
- Prakshepaka Dravya and Madhu are added after cooling, to preserve the volatile constituents.
- **Anjana requires extreme fineness, freedom from micro-organisms, and a pH close to that of tears.**
- A single coarse particle in an Anjana can abrade the cornea.

EXAM CORNER — UNIT 4

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ.

Likely short answer questions (5 marks)

1. Enumerate the quality control parameters of Rasakriya.
2. Enumerate the quality control parameters of Avaleha.
3. Why are reducing and non-reducing sugars determined in an Avaleha?
4. Enumerate the classical Siddhi Lakshana of Avaleha with their modern interpretation.
5. Enumerate the common defects of Avaleha and their causes.
6. Why are Prakshepaka Dravya and Madhu added after cooling?
7. Enumerate the quality control parameters of Anjana.
8. Why must Anjana be extremely fine?
9. Why is pH important in an ophthalmic preparation?

10. Write a short note on the packing and storage of Avaleha.
11. What causes crystallisation in an Avaleha, and how is it prevented?
12. Write a short note on the microbial quality requirements of semi-solid Ayurvedic preparations.

Likely analytical based questions (20 marks)

1. A batch of Avaleha develops mould within two months of manufacture, well within its stated shelf life. Analyse every possible cause, describe the investigation you would carry out, and set out the corrective and preventive actions across the whole quality system.
2. Design a complete quality control protocol for an Anjana intended for commercial manufacture. Analyse the special requirements of an ophthalmic preparation, select the tests with reasons, and state the acceptance criteria and the packaging you would specify.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **161, 162, 163** and the associated pharmacopoeial volumes.

1. **161 and 162.** Ayurvedic Pharmacopoeia of India, Part II. PCIM&H, Ministry of AYUSH.
2. **163.** Lohar D. Protocol for Testing of Ayurvedic, Siddha and Unani Medicines. Department of AYUSH; 2008.

Additional sources: Ayurvedic Pharmacopoeia of India, Part I (reference 156); Sharangadhara Samhita (reference 159); Bhaishajya Ratnavali (reference 160); Mukherjee PK, Quality Control and Evaluation of Herbal Drugs (reference 155).

UNIT 5
QUALITY CONTROL PARAMETER FOR LIQUID AND GASEOUS DOSAGE FORMS

SYLLABUS POINTS

5.1 Kashaya
5.2 Arka
5.3 Sharkara
5.4 Asava and Arishta
5.5 Netra Bindu and Karna Bindu
5.6 Aerosols and sprays

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

5.1 Kashaya

Kashaya or **Kwatha** is a decoction prepared by boiling the coarse powder of a drug with water and reducing it to the prescribed fraction.

Parameter	Requirement or method
Appearance	Clear or slightly turbid, as prescribed; free from sediment and foreign matter
Colour, odour and taste	Characteristic
Specific gravity	The principal physical test
Total solids	Indicates the concentration achieved
pH	As prescribed
Refractive index	Where prescribed
Water and alcohol soluble extractive	As prescribed
TLC or HPTLC	Fingerprint
Assay	Of the marker
Microbial limits	Critical; an aqueous preparation supports rapid microbial growth
Heavy metals and pesticide residues	Within limits

Parameter	Requirement or method
Shelf life	**Very short.** Classically a Kwatha is prepared fresh and used the same day. A commercial preparation requires a preservative and validated stability data
Packing	Clean, well-closed container; amber where the constituents are light sensitive

5.2 Arka

Arka is a distillate obtained by distilling drugs soaked in water, and collecting the volatile fraction.

Parameter	Requirement or method
Appearance	**Clear and free from turbidity and suspended matter**
Colour and odour	Characteristic; usually colourless to pale, with the aroma of the drug
Specific gravity	Close to that of water, since Arka is chiefly an aqueous distillate
Refractive index	A sensitive and simple check of composition
pH	As prescribed
Volatile oil content	Determined where prescribed; the aromatic principle is the active fraction
Total solids	**Low**, because the non-volatile matter is left behind in the still
Alcohol content	Where alcohol has been used or added as a preservative
TLC or GC	Fingerprint; **GC is appropriate here because the constituents are volatile**
Microbial limits	Within limits
Packing	**Airtight container, filled to minimise headspace**, since volatile constituents are lost
Storage	Cool, dark place

> **The distinguishing test between Kashaya and Arka is total solids.** A Kashaya is an extract, so its total solids are relatively **high**. An Arka is a distillate, so almost all the non-volatile matter has been left behind, and its total solids are very **low**. If a sample labelled Arka shows high total solids, it has not been properly distilled, or it has been adulterated with a decoction. This is a neat, examinable point.

5.3 Sharkara

Sharkara Kalpana covers the sugar-based liquid preparations — syrups and Panaka.

Parameter	Requirement or method
Appearance	Clear, free from crystals and sediment
Colour, odour and taste	Characteristic
Total sugar content	The principal assay
Reducing and non-reducing sugars	As for Avaleha, indicating the degree of inversion
Specific gravity	High, because of the sugar content
Viscosity	Should pour readily but not be watery
pH	As prescribed
Total solids	As prescribed
Assay of the marker	Where prescribed
Microbial limits	A sugar solution of insufficient concentration supports microbial growth
Preservative content	Where a preservative is used
Stability	Checked for crystallisation, fermentation, and change of colour or taste
Packing	Clean, airtight, filled to minimise headspace

5.4 Asava and Arishta

Asava and **Arishta** are prepared by fermentation, in which alcohol is generated within the preparation itself. Asava uses cold water with the drugs in powder form; Arishta uses a decoction.

Parameter	Requirement or method
Classical Siddhi Lakshana	Cessation of bubbling; a flame held above is not extinguished; alcoholic odour; sediment settled. See Unit 2, section 2.4
Appearance	Clear after filtration; characteristic colour
Odour and taste	Alcoholic odour; Katu, Kashaya and slightly Amla taste
Alcohol content	**The single most important assay.** Self-generated, and within the range prescribed for that product
Methanol	**Must be absent.** A critical safety test
Total solids	As prescribed
Reducing sugars	Indicate how far fermentation proceeded; a high residual sugar suggests incomplete fermentation
Specific gravity	As prescribed
pH	Acidic, as prescribed

Parameter	Requirement or method
Total acidity	As prescribed
TLC or HPTLC	Fingerprint
Assay of the marker	Where prescribed
Microbial limits	Within limits
Heavy metals	Within limits
Storage	Well-closed container. **Asava and Arishta are classically held to improve with age**, unlike most Kalpanas
Regulatory point	The alcohol is self-generated, which is what distinguishes these from an alcoholic beverage; the declared content must be within the permitted range

> **Two safety-critical tests, and why.**
> **1. Alcohol content** must be within the prescribed range. Too low indicates failed fermentation, and the product will not keep. Too high raises regulatory and safety questions.
> **2. Methanol must be absent.** Methanol is toxic and causes blindness and death. Its presence indicates either contamination or deliberate adulteration with unfit spirit. This test is not a formality — it is a life-safety check, and it should be named in any answer on Asava and Arishta.

5.5 Netra Bindu and Karna Bindu

Netra Bindu are eye drops; **Karna Bindu** are ear drops. Both are sterile or near-sterile preparations for a sensitive site, and their requirements are strict.

Parameter	Netra Bindu (eye)	Karna Bindu (ear)
Sterility	**Required**	Required where the ear drum is perforated or the ear is inflamed
Clarity	Must be clear, free from any visible particle	Clear or as prescribed
Particulate matter	Within the prescribed limit; a particle can abrade the cornea	Within limits
pH	**Close to that of tears**, to avoid stinging and damage	Compatible with the ear canal
Tonicity	**Isotonic with tears where possible**; a strongly hypertonic or hypotonic drop causes pain and damage	Less critical, but still relevant
Assay	Of the active	Of the active
Viscosity	Where a viscosity enhancer is used to prolong contact	Relevant for oil-based drops

Parameter	Netra Bindu (eye)	Karna Bindu (ear)
Preservative	Required for a multiple-dose container; its effectiveness must be demonstrated	As required
Absence of irritation	Tested as prescribed	Tested as prescribed
Container	Sterile dropper bottle; tamper-evident; small volume to limit use after opening	Dropper bottle
Labelling	Must carry the date of opening instruction and the period of use after opening	As prescribed
Stability	Full stability data supporting the shelf life	As prescribed

5.6 Aerosols and sprays

A gaseous or pressurised dosage form. In Ayurveda these are modified forms — an Arka spray, a nasal spray, or a throat spray.

Parameter	Requirement or method
Appearance and clarity of the solution	Free from particles that would block the valve
Leak test	The container must not lose propellant or contents
Pressure test	Internal pressure within the specified range, tested at the prescribed temperature
Net content	Weight or volume as declared
Spray pattern	Uniform, of the specified shape and distribution
Dose delivered per actuation	Uniform across the life of the container, from first to last dose
Number of actuations per container	As declared
Particle or droplet size	Critical for an inhaled product; expressed as the aerodynamic diameter and the size distribution
Valve and actuator performance	Functions reliably; no blockage
Microbial limits	Within limits; sterility where required
Assay	Of the active
Propellant identity and content	As specified
Safety	Container integrity; behaviour at raised temperature; flammability where relevant

Parameter	Requirement or method
Labelling	Directions for use, priming instructions, storage, and warnings

Why droplet or particle size decides everything in an inhaled product. Where the droplet lands in the respiratory tract depends almost entirely on its aerodynamic size. Large droplets deposit in the mouth and throat and never reach the lung; very small ones are breathed out again without depositing. Only a defined intermediate range reaches the intended site. So particle size is not one test among many — **it determines whether the product works at all.** The curriculum's experiential activity on assessing the aerodynamic properties of a gaseous dosage form is asking exactly this.

KEY POINTS TO REMEMBER

- Kashaya: specific gravity, total solids, pH, microbial limits. Classically prepared fresh; shelf life is very short.
- **Arka has low total solids because it is a distillate; Kashaya has high total solids because it is an extract.** This distinguishes them.
- Arka is packed in an airtight container with minimal headspace, because the active fraction is volatile. GC is the appropriate chromatographic method.
- Sharkara: total sugar, reducing and non-reducing sugars, specific gravity, viscosity.
- Asava and Arishta: **alcohol content is the principal assay, and methanol must be absent.** The alcohol is self-generated.
- Asava and Arishta are classically held to improve with age.
- Netra Bindu: sterility, clarity, freedom from particulate matter, pH close to tears, isotonicity, preservative for multiple-dose containers.
- Aerosols: leak test, pressure test, spray pattern, dose per actuation, and **particle or droplet size, which decides where the dose deposits**.

EXAM CORNER — UNIT 5

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ.

Likely short answer questions (5 marks)

1. Enumerate the quality control parameters of Kashaya.
2. Enumerate the quality control parameters of Arka.
3. How would you distinguish an Arka from a Kashaya analytically?
4. Why is an Arka packed in an airtight container with minimal headspace?
5. Enumerate the quality control parameters of Sharkara.
6. Enumerate the quality control parameters of Asava and Arishta.
7. Why is alcohol content the principal assay in Asava and Arishta?

8. Why must methanol be absent, and how is it tested?
9. Enumerate the quality control parameters of Netra Bindu.
10. Why must an ophthalmic preparation be isotonic and of a suitable pH?
11. Enumerate the quality control parameters of an aerosol or spray.
12. Why does droplet size determine the performance of an inhaled product?

Likely analytical based questions (20 marks)

1. A market sample of an Arishta shows a lower alcohol content than declared, and a high residual sugar. Analyse what has happened during manufacture, what the consequences are for the product, and set out the investigation and the corrective actions.
2. You are asked to design a complete quality control protocol for a newly developed Ayurvedic nasal spray. Analyse the requirements of the dosage form, select the tests with reasons, state the acceptance criteria, and justify the container and labelling you would specify.
3. A sample labelled as Arka is suspected of being adulterated with a decoction. Analyse how you would establish this, which tests would be decisive, and what each possible result would show.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **161, 162, 163** and the associated pharmacopoeial volumes.

1. **161 and 162.** Ayurvedic Pharmacopoeia of India, Part II. PCIM&H, Ministry of AYUSH.
2. **163.** Lohar D. Protocol for Testing of Ayurvedic, Siddha and Unani Medicines. Department of AYUSH; 2008.

Additional sources: Ayurvedic Pharmacopoeia of India, Part I (reference 156); Sharangadhara Samhita (reference 159); Mukherjee PK, Quality Control and Evaluation of Herbal Drugs (reference 155); WHO Quality Control Methods for Medicinal Plant Materials (reference 129); and the chromatographic references of Module 5 (references 129 to 134).

MODULE 7
PRACTICAL TRAINING ACTIVITIES

Module 7 carries **20 hours of practical training**, over eight activities.

No	Practical name	Hours
7.1	Demonstration of documentation and record-keeping processes in the quality control section	2
7.2	Demonstration of assessment of Bhasma Siddhi Lakshana	2
7.3	Demonstration of quality control parameters for powder dosage forms — Churna, Kwatha Churna, Lepa Churna, Kshara, Lavana, Satva	3
7.4	Demonstration of quality control parameters for solid dosage forms — tablets, pills and Varti	3
7.5	Demonstration of quality control parameters for Khalveeya, Parpati, Kupipakwa Rasayana, Lauha and Mandura	4
7.6	Demonstration of quality control parameters for semi-solid dosage forms as per pharmacopoeial standards	2
7.7	Demonstration of quality control parameters for liquid dosage forms — Kashaya, Arka, Sharkara, Asava-Arishta	2
7.8	Demonstration of quality control parameters for Netra Bindu, Karna Bindu and aerosols or sprays	2
	Total	**20**

Practical Training 7.1 — Documentation and record-keeping in the quality control section

Aim. To study and prepare the documents used in a quality control section.

Procedure

1. Study the format of each document listed in Unit 1, section 1.5.2.
2. Prepare a specimen **specification** for one raw material and one finished product.

3. Prepare a specimen **Standard Testing Procedure** for one test.
4. Complete a specimen **analytical record** using real data from a test you have performed.
5. Prepare a specimen **Certificate of Analysis** for a batch.
6. Study the good documentation practices and demonstrate the correct way to make a correction in a record.
7. Examine a completed Batch Manufacturing Record and identify any gap in it.

Observation to record. All specimen documents prepared, and the list of gaps found in the record examined.

Precautions. Never sign for work you did not perform. Never leave a blank field. Never overwrite — strike through once, correct, sign and date with the reason.

Practical Training 7.2 — Assessment of Bhasma Siddhi Lakshana

Aim. To perform the classical quality tests on a prepared Bhasma.

Procedure

1. Take the Bhasma prepared in Module 4.
2. Perform **Rekhapurnata** — rub between thumb and forefinger and observe whether the lines fill.
3. Perform **Varitara** — sprinkle a pinch on still water and observe whether it floats.
4. Perform **Unama** — place a grain of rice on the floating Bhasma and observe.
5. Examine for **Nishchandratva** under a lens and under the microscope.
6. Perform **Apunarbhava** as prescribed, with the reducing agents specified, and record whether the metal reappears.
7. Perform **Niruttha** as prescribed, and record any weight change in the test metal.
8. Test **Nirdhuma**, **Nishwadatva**, **Slakshnatva** and the absence of a gritty sound.
9. Record the **Varna** against the prescribed colour.
10. Where instruments are available, correlate with XRD, SEM and particle size data.

Observation table to draw

Test	Method used	Observation	Pass or fail	Modern correlate

Precautions. Handle Bhasma with a spatula, never with wet fingers. Wear a mask; the particles are extremely fine and become airborne. **Do not taste any Bhasma containing a toxic metal.** Perform Apunarbhava and Niruttha only under supervision.

Practical Training 7.3 — Quality control of powder dosage forms

Aim. To perform the quality control tests on Churna, Kwatha Churna, Lepa Churna, Kshara, Lavana and Satva.

Procedure

1. Record the organoleptic characters of each sample.

2. **Sieve analysis.** Pass each powder through the prescribed sieve. Record the percentage passing and retained. Confirm that Churna passes number 80, and Lepa Churna the finer sieve prescribed.
3. Determine **loss on drying** by the prescribed method.
4. Determine **total ash**, **acid insoluble ash** and **water soluble ash**.
5. Determine **alcohol soluble** and **water soluble extractive**.
6. Determine **pH** of an aqueous suspension or solution.
7. For **Kshara**, determine the **total alkalinity** by titration.
8. For **Lavana**, determine the **sodium chloride content** by titration.
9. For **Satva**, record appearance, solubility and yield, and assay as prescribed.
10. Perform **TLC** and record the fingerprint.
11. Compare every result against the pharmacopoeial limit.

Observation table to draw

Sample	Sieve number passed	LOD %	Total ash %	Acid insoluble ash %	Water soluble ash %	Alcohol soluble extractive %	Water soluble extractive %	pH	Pharmacopoeial limit	Complies

Precautions. Use a calibrated balance. Ignite ash in a muffle furnace at the prescribed temperature only. Handle acids with care and add acid to water, never the reverse. Wear a mask during sieving.

Practical Training 7.4 — Quality control of tablets, pills and Varti

Aim. To perform the quality control tests on solid unit dosage forms.

Procedure

1. Record the description — shape, size, colour, surface, defects.
2. **Uniformity of weight.** Weigh twenty units individually. Calculate the mean and the deviation of each. Compare against the prescribed limit.
3. **Hardness.** Test the prescribed number of units and record the mean.
4. **Friability.** Weigh, tumble in the friabilator for the prescribed revolutions, dedust, reweigh, and calculate the percentage loss.
5. **Disintegration time.** Use the disintegration apparatus in the prescribed medium at the prescribed temperature. Record the time for each unit.
6. **Dissolution**, where prescribed.
7. Determine **loss on drying** and **ash values**.
8. **Assay** the marker or the metal content.
9. For **Varti**, record dimensions, uniformity of weight, surface, mechanical strength and softening or dissolution behaviour.
10. Compare all results against the pharmacopoeial limits.

Observation tables to draw. One for weight variation with the deviation of each unit; one for hardness, friability and disintegration; one summary of compliance.

Precautions. Calibrate the balance and the apparatus before use. Maintain the prescribed medium temperature. Do not touch the tablets with bare hands where moisture would affect the weight.

Practical Training 7.5 — Quality control of Khalveeya, Parpati, Kupipakwa, Lauha and Mandura

Aim. To perform the quality control tests on Rasaushadhi.

Procedure

1. For **Kajjali**, examine the colour under good light and under a lens. Record whether it is uniformly jet black and lustreless, and whether any globule of free mercury is visible.
2. Perform Rekhapurnata and test for grittiness.
3. For **Parpati**, record the appearance, thickness, colour, lustre and fracture. Powder a portion and record how readily it powders.
4. For **Kupipakwa Rasayana**, record the site of collection, the colour, the appearance and the yield. Test for Nishchandratva.
5. For **Lauha and Mandura Bhasma**, perform the full Bhasma Pariksha.
6. Test all the Bhasma samples with a magnet and record the response.
7. Determine **loss on drying** for each.
8. **Assay** — iron content for Lauha and Mandura; mercury and sulphur for the mercurial preparations, as prescribed and under supervision.
9. Where available, correlate with XRD, SEM-EDAX, particle size and VSM data.

Observation table to draw

Preparation	Appearance	Colour	Classical tests performed	Result	Assay	Instrumental data	Complies

Precautions. Mercurial preparations must be handled only under direct supervision, with dedicated equipment, in a ventilated area. Never heat a mercurial preparation in an open vessel. Account for every quantity taken and recovered. Dispose of all residues by the approved route, never into the sink. Wear a mask throughout.

Practical Training 7.6 — Quality control of semi-solid dosage forms

Aim. To perform the quality control tests on Rasakriya, Avaleha, Paka and Anjana.

Procedure

1. Record the organoleptic characters and the consistency.
2. Perform the classical **Avaleha Siddhi Lakshana** — Tantuvat, sinking in water, coating the ladle, retaining an impression.
3. Determine **total solids** and **loss on drying**.
4. Determine **reducing sugars** and **non-reducing sugars** by the prescribed method.
5. Determine **pH**.

6. Determine **ash values** and **extractive values**.
7. Perform **TLC** and record the fingerprint.
8. Determine **microbial limits**, particularly the yeast and mould count.
9. For **Anjana**, examine the particle size microscopically using the calibrated ocular micrometer from Module 5. Record the largest particle found. Test for grittiness. Determine pH.
10. Compare all results against the pharmacopoeial limits.

Observation table to draw

Sample	Siddhi Lakshana result	Total solids %	LOD %	Reducing sugar %	Non-reducing sugar %	pH	Microbial count	Complies

Precautions. For Anjana, examine a representative sample and report the largest particle found, not the average. Maintain aseptic technique during microbial testing.

Practical Training 7.7 — Quality control of liquid dosage forms

Aim. To perform the quality control tests on Kashaya, Arka, Sharkara, Asava and Arishta.

Procedure

1. Record the appearance, clarity, colour, odour and taste of each.
2. Determine **specific gravity** using a calibrated pycnometer or hydrometer.
3. Determine **refractive index** for Arka.
4. Determine **pH** of each.
5. Determine **total solids** for each. **Compare the value for Kashaya against that for Arka and explain the difference.**
6. Determine **viscosity** for Sharkara.
7. Determine **total sugar**, **reducing** and **non-reducing sugars** for Sharkara and for Asava-Arishta.
8. Determine **alcohol content** of Asava and Arishta by the prescribed method.
9. Perform the prescribed test for **methanol** in Asava and Arishta.
10. Determine **total acidity** for Asava and Arishta.
11. Perform **TLC** on each and record the fingerprint.
12. Determine **microbial limits**.
13. Compare all results against the pharmacopoeial limits.

Observation table to draw

Sample	Appearance	Specific gravity	Refractive index	pH	Total solids %	Alcohol %	Methanol	Complies

Precautions. Bring all samples to the prescribed temperature before measuring specific gravity or refractive index. Handle distillation apparatus for alcohol determination with care. The methanol test is a safety-critical test — perform it exactly as prescribed and record it fully.

Practical Training 7.8 — Quality control of Netra Bindu, Karna Bindu and aerosols

Aim. To perform the quality control tests on ophthalmic, otic and pressurised dosage forms.

Procedure

1. Examine the **clarity** of the eye and ear drops against a black and a white background.
2. Examine for **particulate matter** under magnification.
3. Determine **pH**.
4. Determine **tonicity** or osmolarity where facilities permit.
5. Examine the **container and closure** — integrity, dropper function, tamper evidence.
6. Examine the **label** for all required particulars, including the period of use after opening.
7. Perform the **sterility test** where facilities permit, or record the procedure.
8. For the **aerosol or spray**, perform the **leak test** by the prescribed method.
9. Determine the **net content** by weight.
10. Record the **spray pattern** on a suitable surface and measure its dimensions.
11. Determine the **dose delivered per actuation** by weighing before and after a defined number of actuations, at the beginning, middle and end of the container.
12. Record the **number of actuations** obtained per container.
13. Compare all results against the specification.

Observation table to draw

Sample	Clarity	Particulate matter	pH	Container and label check	Leak test	Net content	Dose per actuation	Complies

Precautions. Never point an aerosol at the face or the eyes. Do not puncture or incinerate a pressurised container. Maintain aseptic technique for sterility testing. Do not use an ophthalmic sample that has been opened for testing on any person.

MODULE 7
EXPERIENTIAL LEARNING ACTIVITIES

Module 7 carries **30 hours** under experiential learning, including modular assessment. Eight activities are prescribed.

No	Experiential learning activity	Hours
7.1	Practice of documentation and record-keeping processes in the quality control section	2
7.2	Assessment of Bhasma Siddhi Lakshana	2
7.3	Performance of quality control procedures for powder dosage forms	4
7.4	Performance of quality control procedures for solid dosage forms	4
7.5	Performance of quality control procedures for Rasaushadhi	5
7.6	Performance of quality control tests for semi-solid dosage forms	3
7.7	Performance of quality control tests for liquid dosage forms	3
7.8	Performance of quality control tests for Netra Bindu, Karna Bindu and aerosols or sprays	3
	Total	**26**

> **The difference between the practicals and these activities.** In the practicals you follow a method under demonstration. Here you take responsibility — you write the specification, you choose the tests, you decide whether the batch passes, and you defend that decision. The deliverable in each case is a **complete analytical report with a release decision**, not merely a set of readings.

Experiential Learning 7.1 — Documentation and record-keeping in practice

What you have to do. Build the complete document set for one product, and then have it audited.

Method. Choose one Ayurvedic formulation. Prepare its Master Formula Record, its raw material and finished product specifications, its Standard Testing Procedures, its Batch Manufacturing Record format, and its Certificate of Analysis format. Then exchange your document set with a colleague and audit each other's against the requirements of Schedule T and against good documentation practice.

Deliverable. The complete document set, your colleague's audit observations on it, and your revised set.

Experiential Learning 7.2 — Assessment of Bhasma Siddhi Lakshana

What you have to do. Assess a Bhasma independently and issue a written verdict.

Method. Take at least three Bhasma samples — ideally one prepared by you, one from the departmental store, and one from the market. Perform the complete Bhasma Pariksha on each, blind where possible. Record every result. Then correlate with any instrumental data available.

Questions your report must answer

- Which sample satisfied all the tests, and which failed?
- Which test was the most difficult to judge, and why?
- Would two observers reach the same conclusion on these samples? What does that tell you about the classical tests?
- Which instrumental test would settle the point that the classical test left uncertain?

Deliverable. The full test record for each sample, and a written release-or-reject decision for each with the reasoning.

Experiential Learning 7.3 — Quality control of powder dosage forms

What you have to do. Carry out the full analysis of powder dosage forms and decide on release.

Method. Take at least four samples covering Churna, Kwatha Churna, Lepa Churna and either Kshara or Lavana. Look up the pharmacopoeial monograph for each and write out the specification **before** you begin testing. Then perform every test in that specification.

Deliverable. The written specification, the complete analytical record, the comparison against limits, and a signed release decision for each sample.

Experiential Learning 7.4 — Quality control of solid dosage forms

What you have to do. Analyse tablets, Vati and Varti to their full specification, and investigate any failure.

Method. Test at least three samples. Where a sample fails any parameter, do not stop at recording the failure — **investigate it**. Consider the formulation, the process, the storage and the packaging. Propose a cause and a correction.

Deliverable. The analytical report, and for every failure an investigation note with the probable cause and the proposed corrective action.

Experiential Learning 7.5 — Quality control of Rasaushadhi

What you have to do. Carry out the most demanding analysis in the module — the quality control of mercurial and metallic preparations.

Method. Take Kajjali, a Parpati, a Kupipakwa Rasayana and a Lauha or Mandura Bhasma. Perform the classical tests and, where available, the instrumental tests. Correlate the two.

Questions your report must answer

- Where did the classical tests and the instrumental data agree, and where did they diverge?

- Which classical test proved the most reliable?
- What did the instrumental data show that no classical test could have shown?
- What did the classical tests capture that the instruments did not?
- What minimum test set would you specify for routine release of each of these products?

> **The last question is the real work of this activity.** A full instrumental characterisation is not affordable for every batch. Deciding which few tests must be done on every batch, and which are done periodically, is exactly the judgement a quality control head exercises. Answer it with reasons and costs, not with a wish list.

Deliverable. The complete analytical record, the correlation analysis, and your proposed routine release specification for each product.

Experiential Learning 7.6 — Quality control of semi-solid dosage forms

What you have to do. Analyse Avaleha, Rasakriya and Anjana to their full specification, with attention to stability.

Method. Test the samples as in Practical 7.6. Then place a portion of the Avaleha under accelerated conditions and examine it at intervals for crystallisation, separation, mould and change of taste.

Deliverable. The analytical report, the stability observation record, and a recommendation on packaging and storage with reasons.

Experiential Learning 7.7 — Quality control of liquid dosage forms

What you have to do. Analyse Kashaya, Arka, Sharkara and Asava-Arishta, and use the data to detect adulteration.

Method. Test the samples as in Practical 7.7. Then take a deliberately prepared adulterated sample — for example an Arka to which a decoction has been added — and establish, from your data alone, that it is adulterated.

Questions to answer. Which single test detected the adulteration most clearly? Would that test have detected a smaller degree of adulteration? What test would you add to make the detection reliable?

Deliverable. The analytical report for each genuine sample, and the investigation report on the adulterated sample.

Experiential Learning 7.8 — Quality control of Netra Bindu, Karna Bindu and aerosols

What you have to do. Analyse ophthalmic, otic and pressurised preparations, and design a specification for one of them.

Method. Test the available samples as in Practical 7.8. Then take one Ayurvedic liquid — an Arka, for example — and design a complete specification for converting it into a marketable nasal spray or eye drop.

Your specification must cover. The formulation including tonicity adjustment and preservative; the sterilisation or microbial control method; every quality control test with its acceptance criterion; the container and closure; the label including the period of use after opening; and the stability study you would conduct.

Deliverable. The analytical reports, and the complete written specification with justification for every decision.

MODULE 7
VIVA POINTERS AND QUESTION BANK

Viva pointers

On quality systems

- Differentiate quality control from quality assurance.
- Can quality be tested into a product?
- What is a Master Formula Record? How does it differ from a Batch Manufacturing Record?
- Show me how you would correct an error in a record.
- What does ALCOA stand for?
- Under which schedule is GMP for ASU drugs given?

On physicochemical parameters

- What does total ash tell you? What does acid insoluble ash tell you?
- Why is acid insoluble ash the better indicator of sand?
- What is the difference between alcohol soluble and water soluble extractive?
- What does a low water soluble extractive suggest?

On classical parameters

- What is Tantuvat? Demonstrate it.
- In Ghrita Paka, does froth appear or subside? And in Taila Paka?
- Name the stages of Sneha Paka and their uses.
- How do you know an Arishta has finished fermenting?
- What is 1 Gunja in milligrams? 1 Karsha?
- Differentiate Magadha and Kalinga Mana.

At the bench

- Show me Rekhapurnata. Show me Varitara.
- Which two Bhasma Pariksha prove the chemistry is complete?
- Show me the sieve you would use for Churna. For Lepa Churna. Why different?
- Perform the friability test and calculate the result.
- What is the disintegration time limit for this Vati?
- Why is this Kajjali not ready?
- Show me how you determine specific gravity.

On liquids and specials

- How would you tell an Arka from a Kashaya analytically?
- What is the principal assay for an Arishta? Why?
- Why must methanol be absent?
- Why must an Anjana be extremely fine?
- Why must eye drops be isotonic?
- What decides where an inhaled droplet deposits?

Module 7 question bank

Analytical Based Questions (20 marks) — from Units 1, 3, 4 and 5

1. A batch of a classical Ayurvedic formulation fails its finished product specification. Analyse every stage at which the failure could have originated, describe your investigation, and set out the corrective and preventive actions.
2. Establish a complete quality control system for a new Ayurvedic manufacturing unit. Analyse the Schedule T requirements and design the specifications, procedures and records needed.
3. A metal-containing Ayurvedic preparation is rejected by an importing country on heavy metal grounds. Analyse the scientific and regulatory position and the evidence you would generate.
4. Establish a complete quality control protocol for a Kupipakwa Rasayana from raw material to finished product, with acceptance criteria.
5. Two batches of Lauha Bhasma pass all classical tests but differ in assay and XRD. Analyse the meaning, the cause and the system change required.
6. A Churna fails on acid insoluble ash but passes every other test. Analyse what this indicates and where in the supply chain it arose.
7. A batch of Avaleha develops mould well within its shelf life. Analyse the causes and set out the corrective and preventive actions.
8. Design a complete quality control protocol for an Anjana for commercial manufacture.
9. An Arishta shows lower alcohol than declared and high residual sugar. Analyse what happened and set out the investigation.
10. Design a complete quality control protocol for a new Ayurvedic nasal spray.
11. A sample labelled Arka is suspected of adulteration with a decoction. Analyse how you would establish this.

Long Answer Questions (10 marks) — from Unit 1 only

1. Discuss the importance of quality control in Ayurveda and the difficulties peculiar to Ayurvedic formulations.
2. Describe quality assurance of Ayurvedic products, its elements, and its relation to Schedule T.
3. Describe the quality control parameters under identity, purity, strength and performance.
4. Discuss the regulatory requirements for quality control of Ayurvedic products in India.
5. Describe the documentation and record-keeping required in an Ayurvedic manufacturing unit, with the principles of good documentation practice.
6. Discuss the physicochemical parameters used in the quality control of Ayurvedic drugs and what each indicates.

Short Answer Questions (5 marks) — from Units 2, 3, 4 and 5

From Unit 2

1. Define Mana and differentiate Magadha from Kalinga Mana.
2. Write the Magadha Mana table from Gunja to Tula.
3. Enumerate the Avaleha Siddhi Lakshana with their modern interpretation.
4. Enumerate the stages of Sneha Paka with their uses.
5. Differentiate the froth behaviour in Ghrita Paka and Taila Paka.
6. Enumerate the Sandhana Siddhi Lakshana.
7. Write a short note on Guggulu Paka Lakshana.
8. Differentiate between Bhasma and Pishti.

From Unit 3

1. Enumerate the quality control parameters of Churna.
2. Differentiate the fineness required for Churna, Kwatha Churna and Lepa Churna, with reasons.
3. Enumerate the quality control parameters of Kshara.
4. Enumerate the quality control parameters of Vati and Gutika.
5. Enumerate the quality control parameters of Kajjali.
6. Enumerate the quality control parameters of Kupipakwa Rasayana.
7. Enumerate the quality control parameters of Lauha Bhasma and explain the role of VSM.

From Unit 4

1. Enumerate the quality control parameters of Avaleha.
2. Why are reducing and non-reducing sugars determined in an Avaleha?
3. Enumerate the common defects of Avaleha and their causes.
4. Why are Prakshepaka Dravya and Madhu added after cooling?
5. Enumerate the quality control parameters of Anjana.
6. Why must Anjana be extremely fine?

From Unit 5

1. Enumerate the quality control parameters of Kashaya.

2. How would you distinguish an Arka from a Kashaya analytically?
3. Why is an Arka packed in an airtight container with minimal headspace?
4. Enumerate the quality control parameters of Asava and Arishta.
5. Why must methanol be absent from an Arishta?
6. Enumerate the quality control parameters of Netra Bindu.
7. Enumerate the quality control parameters of an aerosol or spray.
8. Why does droplet size determine the performance of an inhaled product?

> **Note on the marks range.** Table 6F places Module 7 in the marks range of **5 to 20**. Four of the five units are ABQ-eligible. Unit 1 alone carries the long answer and is never asked as a short answer. Unit 2 alone carries only short answers. Learn the eight-group scheme in section 3.0 — it is the skeleton of almost every answer in this module.

MODULE 8
NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA

Credits	Lectures	Practical Training	Experiential Learning	Total hours	Modular assessment
2	10	20	30	60	50 marks

About this module

This module introduces nano-science and nanotechnology and their application to Ayurvedic formulations. It covers the principles of nanotechnology, the characterisation of micro and nano particles in Ayurvedic preparations, the types of nanocarriers, the regulatory and safety aspects of nanotechnology-based formulations, and the recent advances and future directions in the field.

Module Learning Objectives

At the end of this module, the student should be able to:

1. Define nanotechnology, its history, principles and applications.
2. Describe Ayurvedic formulations with respect to nanotechnology.
3. Discuss the potential of nanotechnology in Ayurvedic formulations.
4. Identify and describe various types of nanocarriers for Ayurvedic formulations.
5. Discuss the regulatory frameworks, safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations.
6. Describe recent advances and future directions in nanotechnology-based Ayurvedic formulations.

Units in this module

Unit	Name	ABQ	SAQ	LAQ
M8U1	Introduction to Nanotechnology	No	Yes	Yes
M8U2	Ayurvedic Formulations and Nanotechnology	Yes	Yes	No
M8U3	Nanocarriers for Ayurvedic Formulations	No	Yes	No
M8U4	Regulatory Aspects and Safety Concerns	Yes	Yes	No
M8U5	Recent Advances and Future Directions	No	Yes	No

Module 8 is placed in the marks range of **5 to 20**.

The final module, and how to think about it

This is the last module of the course, and it is the one that brings everything together.

In Module 3 you learned that Bhavana reduces particle size and that plant extracts reduce and cap metal particles. In Module 4 you learned that repeated Puta reduces particle size progressively. In Module 5 you learned the instruments that measure that size. This module asks what all of it means.

> **A word of caution before you begin, which will serve you for your whole career.**
>
> It is often said that Bhasma is “ancient nanomedicine”. This claim is attractive, it is partly supported, and it is frequently overstated.
>
> **What is documented:** published characterisation studies of several Bhasmas report particles in the micrometre and, in a proportion of the material, the nanometre range; organic material is found associated with the particle surface; and the process bears a real mechanistic resemblance to green synthesis.
>
> **What is not established:** that every Bhasma is a nanomaterial; that the nano fraction is what produces the therapeutic effect; or that a Bhasma meets the modern regulatory definition of a nanopharmaceutical.
>
> Your task in this module is to hold both halves. Present the evidence fully, and mark the boundary honestly. An examiner at postgraduate level is testing exactly this discrimination, and an over-claimed answer will score lower than a careful one.

UNIT 1
INTRODUCTION TO NANOTECHNOLOGY

SYLLABUS POINTS

1.1 Definition and history of nanotechnology
1.2 Basic principles and applications of nanotechnology
1.3 Understanding nanoscale

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	Yes

1.1 Definition and history

1.1.1 Definitions

Term	Definition
Nanometre	One thousand-millionth of a metre, that is 10^{-9} metre
Nanoscale	The size range from approximately **1 to 100 nanometres**
Nanoscience	The study of phenomena and the manipulation of materials at the nanoscale, where properties differ significantly from those at a larger scale
Nanotechnology	The design, characterisation, production and application of structures, devices and systems by controlling shape and size at the nanoscale
Nanomaterial	A material with any external dimension, or an internal or surface structure, in the nanoscale
Nanoparticle	A particle with all three external dimensions in the nanoscale
Nanomedicine	The application of nanotechnology to the diagnosis, treatment and prevention of disease
Nanopharmaceutical	A pharmaceutical product in which nanoscale properties are deliberately used to achieve a therapeutic purpose

1.1.2 A sense of scale

Object	Approximate size
Human hair, diameter	80,000 to 100,000 nm
Red blood cell	About 7,000 nm
Bacterium	1,000 to 10,000 nm
Virus	20 to 400 nm
Nanoparticle range	**1 to 100 nm**
DNA double helix, diameter	About 2 nm
Water molecule	About 0.3 nm

> **An easy way to remember the scale.** If a nanometre were the size of a marble, a metre would be the size of the Earth. A sheet of paper is about 100,000 nanometres thick.

1.1.3 History of nanotechnology

Year	Event
1959	**Richard Feynman** delivers his lecture *There's Plenty of Room at the Bottom* at Caltech, describing the possibility of manipulating matter atom by atom. Regarded as the conceptual beginning of the field
1974	**Norio Taniguchi** coins the term **nanotechnology**
1981	**Gerd Binnig and Heinrich Rohrer** invent the **Scanning Tunnelling Microscope (STM)** at IBM Zurich, allowing individual atoms to be imaged. They received the Nobel Prize in Physics in 1986
1985	**Kroto, Curl and Smalley** discover the **fullerene**, C_{60}. Nobel Prize in Chemistry, 1996
1986	**Binnig, Quate and Gerber** invent the **Atomic Force Microscope (AFM)**
1986	**Eric Drexler** publishes *Engines of Creation*, popularising the concept
1991	**Sumio Iijima** reports **carbon nanotubes**
2000 onward	Rapid growth of nanomedicine, nanoelectronics and nanomaterials; national nanotechnology programmes established worldwide
2019	**Guidelines for Evaluation of Nanopharmaceuticals in India** released

The much older history. Nanoscale materials were used long before they were understood. Gold and silver nanoparticles gave colour to ancient stained glass and to the Lycurgus Cup. Carbon nanostructures have been reported in the steel of some historical blades. And in India, the classical processes of Rasashastra were producing extremely fine particulate material centuries before any of this was named. What was absent was not the material but the **means of seeing and measuring it**.

1.2 Basic principles

1.2.1 Why properties change at the nanoscale

This is the heart of the unit, and the most likely long answer question. There are two reasons.

A. The surface area to volume ratio increases enormously.

When a solid is divided into smaller and smaller particles, the total volume stays the same but the total surface area increases dramatically. A very large proportion of the atoms then lie on the surface rather than in the interior.

Consequence	Effect
More surface atoms	Higher surface energy and greater reactivity
Greater contact area with the surrounding medium	Faster dissolution
More sites for adsorption	Organic molecules can attach in quantity
Greater catalytic activity	More active sites available

> **This is the single most important principle for your subject.** Recall the Noyes-Whitney principle from Module 1, Unit 5: dissolution rate increases with surface area. A poorly soluble drug reduced to the nanoscale dissolves far faster, and therefore is absorbed far better. **This is the chemical link between Bhavana, Puta, particle size and bioavailability**, and it connects Modules 1, 3, 4 and 8 in a single sentence.

B. Quantum effects become significant.

At the nanoscale the behaviour of electrons is confined by the size of the particle. This produces size-dependent optical, electrical and magnetic properties that the bulk material does not show.

Example	Observation
Gold	Bulk gold is yellow and inert. Gold nanoparticles are red or purple depending on size, and are catalytically active
Semiconductor quantum dots	The colour of the light emitted depends on the particle size, not on the chemical composition alone
Iron oxide	Very small particles show **superparamagnetism**, which the bulk material does not
Melting point	Falls as particle size decreases

1.2.2 The two approaches to making nanomaterials

Approach	Description	Examples
Top-down	Start with bulk material and break it down	Ball milling, grinding, lithography, laser ablation, high-pressure homogenisation
Bottom-up	Build up from atoms and molecules	Chemical precipitation, sol-gel, chemical vapour deposition, self-assembly, **green synthesis**

> **Where the classical processes fit. Bhavana, Mardana and Churnikarana are top-down** — bulk material is progressively broken down by mechanical force. **Marana with herbal media has a bottom-up character** — the metal is chemically converted and new particles form in the presence of organic molecules that cap them. Most classical processes are a combination of the two, which is exactly what a modern nanomaterials chemist would recognise. This is a strong point for any answer.

1.2.3 Classification of nanomaterials by dimension

Class	Dimensions in the nanoscale	Examples
0D	All three	Nanoparticles, quantum dots, fullerenes
1D	Two	Nanotubes, nanowires, nanorods, nanofibres
2D	One	Nanofilms, nanocoatings, graphene
3D	Bulk material made of nanoscale units	Nanocomposites, nanostructured powders

1.2.4 Applications of nanotechnology

Field	Applications
Medicine and pharmacy	Targeted drug delivery, controlled release, improved bioavailability, imaging, diagnostics, tissue engineering, antimicrobial surfaces
Electronics	Smaller and faster devices, memory, displays
Energy	Solar cells, batteries, fuel cells, catalysts
Environment	Water purification, air filtration, remediation of pollutants
Agriculture	Nano-fertilisers, controlled-release pesticides, soil sensors
Textiles	Stain resistance, water repellence, antimicrobial fabric
Cosmetics	Sunscreens, delivery of actives into the skin
Food	Packaging, preservation, sensors
Construction	Stronger and self-cleaning materials

1.3 Understanding the nanoscale

1.3.1 How nanoscale material is measured and seen

Every one of these instruments was studied in Module 5. This section applies them.

Instrument	What it gives at the nanoscale
TEM	Direct image of individual nanoparticles; size, shape, internal structure
SEM	Surface morphology and particle size; with EDAX, the elemental composition
AFM	Three-dimensional surface topography at atomic resolution; works in air and in liquid
STM	Images individual atoms on a conducting surface
DLS or particle size analyser	Hydrodynamic size and size distribution in suspension; also the polydispersity index
Zeta potential analyser	Surface charge, which predicts the stability of a suspension
XRD	Crystalline phase, and crystallite size from peak broadening by the Scherrer relationship
UV-Visible spectroscopy	Surface plasmon resonance peak for metallic nanoparticles; a quick confirmation of formation
FTIR	The functional groups of the capping and stabilising molecules

Instrument	What it gives at the nanoscale
TGA	Quantity of organic capping material
BET surface area analysis	Specific surface area
VSM	Magnetic behaviour, including superparamagnetism

Two points about measurement that are frequently asked.
1. DLS gives the hydrodynamic diameter — the particle together with the layer of liquid and molecules moving with it. This is normally **larger** than the size seen by TEM, which images the solid core. The two are not contradictory; they measure different things. Report both, and say which is which.
2. Zeta potential predicts stability. A suspension in which the particles carry a high surface charge, positive or negative, resists aggregation because the particles repel one another. A value near zero means the particles will clump together. This is why zeta potential is measured in every nanoformulation study.

1.3.2 The problem of aggregation

Nanoparticles have very high surface energy, so they tend to clump together to reduce it. Once aggregated, the advantage of the small size is lost.

Method of stabilisation	How it works
Electrostatic stabilisation	The particles carry a like surface charge and repel one another
Steric stabilisation	A layer of large molecules on the surface physically prevents close approach
Capping agents	Molecules that bind to the particle surface and limit both growth and aggregation
Surfactants	Reduce interfacial energy

The classical connection. The organic material from the Bhavana media, associated with the surface of Bhasma particles, acts in exactly this way — as a **capping and steric stabilising layer**. This is the mechanism proposed in Module 3, Unit 3, and it is why repeated Bhavana produces a finer product than dry heating alone.

KEY POINTS TO REMEMBER

- One nanometre is 10^{-9} metre. The nanoscale is 1 to 100 nanometres.
- Feynman's lecture *There's Plenty of Room at the Bottom* was in 1959; Taniguchi coined "nanotechnology" in 1974.
- STM 1981, Binnig and Rohrer; fullerene 1985, Kroto Curl and Smalley; AFM 1986; carbon nanotubes 1991, Iijima.
- **Properties change at the nanoscale for two reasons: the enormous surface area to volume ratio, and quantum effects.**

- Top-down means breaking bulk material down; bottom-up means building from atoms and molecules. Classical Bhavana is top-down; Marana with herbal media has a bottom-up character.
- Nanomaterials are classified as 0D, 1D, 2D and 3D.
- **DLS gives the hydrodynamic diameter, which is larger than the TEM size.**
- **Zeta potential predicts suspension stability**; a value near zero means the particles will aggregate.
- Organic material from Bhavana media acts as a capping and steric stabilising layer.

EXAM CORNER — UNIT 1

This unit can be asked as SAQ (5 marks) and LAQ (10 marks). There is no ABQ from this unit.

Likely short answer questions (5 marks)

1. Define nanotechnology, nanoscience, nanomaterial and nanoparticle.
2. What is the nanoscale? Give a comparison of sizes to illustrate it.
3. Write a short note on the history of nanotechnology.
4. Why do properties change at the nanoscale?
5. Explain the significance of the surface area to volume ratio.
6. Write a short note on quantum effects at the nanoscale with examples.
7. Differentiate between top-down and bottom-up approaches, with examples.
8. Classify nanomaterials on the basis of dimension.
9. Enumerate the applications of nanotechnology.
10. Enumerate the instruments used to characterise nanomaterials and what each gives.
11. Differentiate between the size given by DLS and that given by TEM.
12. Write a short note on zeta potential and its significance.

Likely long answer questions (10 marks)

1. Define nanotechnology. Describe its history, basic principles and applications.
2. Explain in detail why the properties of a material change at the nanoscale, and give examples.
3. Describe the top-down and bottom-up approaches to the preparation of nanomaterials, and correlate them with classical Ayurvedic processes.
4. Describe the instruments used for the characterisation of nanomaterials, with the information each provides.
5. Discuss the nanoscale and the principles of nanoparticle stabilisation, and relate them to Bhavana in Rasashastra.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **173, 174**

Verified sources used: Feynman RP, *There's Plenty of Room at the Bottom*, 1959; Taniguchi N, on the concept of

nano-technology, 1974; the primary literature on STM, AFM, fullerenes and carbon nanotubes; Guidelines for Evaluation of Nanopharmaceuticals in India, DBT, ICMR and CDSCO, 2019; and the instrumental analysis texts of Module 5 (references 14, 15, 17, 18 and 20).

UNIT 2
AYURVEDIC FORMULATIONS AND NANOTECHNOLOGY

SYLLABUS POINTS

2.1 Overview of Ayurvedic formulations and their limitations
2.2 Potential of nanotechnology in Ayurvedic formulations
2.3 Characterisation of micro and nano particles in Bhasma
2.4 Characterisation of micro and nano particles in Kupipakwa Rasayana and Pottali
2.5 Characterisation of micro and nano sized particles generated during Bhavana
2.6 Synthesis of silver nanoparticles by the chemical precipitation method, and the fundamental principles of nanoparticle synthesis
2.7 Market survey of nano-based herbal formulations

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

2.1 Overview of Ayurvedic formulations and their limitations

Ayurvedic formulations are effective and time-tested. They also have real limitations, and a postgraduate student must be able to state them plainly. Naming a problem is the first step to solving it.

Limitation	Explanation
Poor aqueous solubility	Many phytoconstituents, notably curcumin, dissolve poorly in water
Poor permeability	Large or polar molecules cross membranes badly
Low oral bioavailability	The combined result of the two above, together with first-pass metabolism
Large dose	A Kwatha or Churna may require a substantial quantity, reducing compliance
Unpleasant taste and odour	Reduces acceptance, particularly in children
Instability	Volatile and phenolic constituents degrade on storage
Batch variability	Natural variation in the raw material
Short shelf life	Particularly for aqueous preparations
Non-specific distribution	The drug reaches all tissues, not the target
Difficulty in standardisation	Multi-ingredient formulations with no single marker

2.2 Potential of nanotechnology in Ayurvedic formulations

Limitation	What nanotechnology offers
Poor solubility	Nanosizing increases surface area and dissolution rate; solid dispersions and nanosuspensions
Poor permeability	Nanocarriers cross membranes more readily; permeation enhancers
Low bioavailability	Lipid-based nanocarriers bypass part of first-pass metabolism through lymphatic uptake
Large dose	Higher bioavailability permits a smaller dose
Unpleasant taste	Encapsulation masks taste
Instability	Encapsulation protects labile constituents from light, oxygen and moisture
Non-specific distribution	Surface modification allows targeting to a tissue or a cell type
Rapid clearance	Controlled and sustained release prolongs the effect
Poor skin penetration	Nanoemulsions and vesicular carriers penetrate the skin better, improving Lepa and Taila preparations

> **Curcumin is the standard example, and you should know it.** Curcumin from Haridra has well-documented biological activity in the laboratory but very poor oral bioavailability, because it dissolves badly, permeates poorly, is rapidly metabolised and is unstable at intestinal pH. It behaves like a BCS Class IV compound, as discussed in Module 1, Unit 5. A very large number of nanoformulation studies have been published attempting to solve exactly this problem. If a question asks you to name an Ayurvedic drug that would benefit from nanotechnology, curcumin is the safest and best-supported answer.

2.3 Characterisation of micro and nano particles in Bhasma

2.3.1 What the published work shows

Characterisation studies of Bhasma using XRD, SEM, TEM, particle size analysis, EDAX, FTIR and TGA have consistently reported the following.

Finding	Detail
Particle size	Predominantly in the micrometre and sub-micrometre range, with a proportion of material reported in the nanometre range in many studies
Progressive reduction	Particle size falls with successive Putas
Phase	Metal oxides and sulphides identified by XRD; free metal generally absent in a properly prepared Bhasma
Morphology	Irregular, often aggregated particles seen by SEM; individual fine particles seen by TEM
Elemental composition	The expected metal, with elements derived from the media
Organic material	Demonstrated by FTIR bands and quantified by weight loss on TGA
Crystallite size	Estimated from XRD peak broadening; often smaller than the particle size seen by SEM, indicating that particles are aggregates of smaller crystallites

> **The distinction between crystallite size and particle size is worth understanding properly, and it explains an apparent contradiction in the literature.**
>
> **Crystallite size**, from XRD peak broadening, is the size of the individual crystalline domain. **Particle size**, from SEM or a particle size analyser, is the size of the physical particle, which may be an aggregate of many crystallites.
>
> So a Bhasma may show a crystallite size of a few tens of nanometres by XRD while showing particles of several micrometres by SEM. Both results are correct. The material is nanocrystalline but micro-particulate — the fine crystallites have aggregated. **This is the most common source of confusion in Bhasma nano-characterisation papers**, and being able to explain it will distinguish your answer immediately.

2.3.2 The recommended characterisation set for a Bhasma

Instrument	Contribution
XRD	Phase identity; crystallite size; absence of free metal
SEM	Particle morphology and size; degree of aggregation
EDAX	Elemental composition at the point examined
TEM	Size and structure of the finest particles
Particle size analyser	Size distribution of the bulk material
FTIR	Metal-oxygen or metal-sulphur bonds; organic functional groups present
TGA and DTA	Quantity of organic material; thermal events
ICP-OES or ICP-MS	Elemental content including trace elements
VSM	Magnetic behaviour, for iron-containing preparations
Classical Bhasma Pariksha	**Always reported alongside the instrumental data, never replaced by it**

2.4 Kupipakwa Rasayana and Pottali

Preparation	Characterisation findings and requirements
Kupipakwa Rasayana	Formed by controlled sublimation and recondensation, so the product is crystalline and often more uniform than a Puta-prepared Bhasma. XRD confirms the phase — for Rasa Sindura, the red form of mercuric sulphide. Particle size is measured after trituration of the collected product, since the material is collected as a mass and then powdered
Pottali Rasayana	Prepared by heating within a bundle, under restricted air. Characterisation examines the phase formed, the completeness of reaction and the particle size after powdering. The restricted atmosphere is the variable that distinguishes it from an open Puta

The general point. The **method of preparation determines the particle characteristics**. A product formed by sublimation and recondensation differs in crystallinity and uniformity from one formed by repeated grinding and firing. Comparing the two by instrumental characterisation is a genuine and readily publishable research question, and it is exactly what Practical 8.3 and Experiential 8.2 are asking you to begin.

2.5 Particles generated during Bhavana

This section is the most directly researchable in the whole module.

What Bhavana does, in nanotechnology terms

Action	Nanotechnology equivalent
Prolonged wet trituration	Top-down mechanical size reduction, comparable to wet milling
Liquid medium present	Acts as a grinding aid and prevents re-aggregation during grinding
Organic molecules in the medium	Adsorb on the freshly created surface and act as **capping agents**
Repetition with drying between cycles	Progressive reduction; each cycle exposes fresh surface
Reducing constituents in the medium	Can alter the oxidation state at the surface

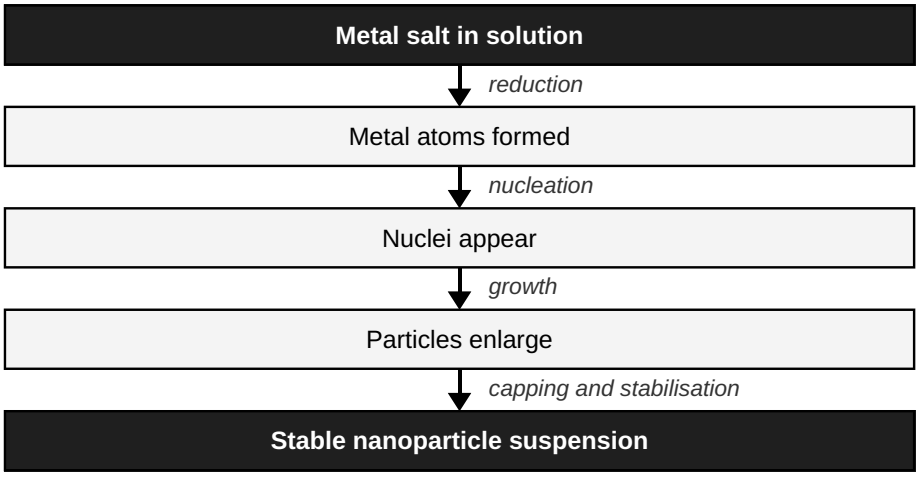
How to study it. Take samples after each Bhavana cycle. Measure particle size by particle size analyser and by SEM. Record FTIR after each cycle to follow the organic material accumulating on the surface. Plot particle size against the number of Bhavana cycles. **Very few such studies exist, and the data would be publishable.**

2.6 Synthesis of silver nanoparticles and the principles of nanoparticle synthesis

2.6.1 The three components of any chemical nanoparticle synthesis

Component	Function
Precursor	The source of the metal, usually a salt
Reducing agent	Converts the metal ion to the metal atom
Stabiliser or capping agent	Binds to the growing particle, limits its growth, and prevents aggregation

2.6.2 The stages of particle formation



The controlling principle. If **nucleation is rapid and growth is slow**, many small particles form and the product is fine and uniform. If nucleation is slow and growth is rapid, fewer and larger particles form. Every variable in a synthesis — concentration, temperature, rate of addition, pH, the stabiliser used — acts by shifting this balance.

2.6.3 Chemical precipitation synthesis of silver nanoparticles

Principle. A soluble silver salt is reduced in solution in the presence of a stabiliser, producing metallic silver as a colloidal suspension.

Outline of the method

1. Prepare a dilute aqueous solution of the silver salt precursor.
2. Prepare the reducing agent solution and the stabiliser solution, as prescribed by the method being followed.
3. Add the reducing agent to the silver salt solution slowly, with continuous vigorous stirring, under the prescribed conditions of temperature.
4. Observe the colour change. **The appearance of a yellow to brown colour indicates the formation of silver nanoparticles.**
5. Continue stirring for the prescribed period.
6. Purify by centrifugation and washing, to remove unreacted material.
7. Redisperse and store as prescribed, protected from light.

Characterisation of the product

Test	What it confirms
UV-Visible spectroscopy	A **surface plasmon resonance peak**, characteristically in the region of about 400 to 450 nanometres for silver nanoparticles. **This is the first and quickest confirmation of formation**
DLS	Hydrodynamic size and polydispersity index
Zeta potential	Surface charge and predicted stability
TEM	Direct image; size and shape
SEM-EDAX	Morphology and confirmation of the element
XRD	Crystalline phase and crystallite size
FTIR	The capping molecules present

Safety. Silver salts stain the skin and are irritant. Work with gloves and eye protection. Some reducing agents used in such syntheses are hazardous — follow the departmental protocol exactly and use only the agent and quantity prescribed by your teacher. Dispose of silver-containing waste by the approved route, never into the sink.

2.6.4 Green synthesis — and the classical parallel

Green synthesis replaces the chemical reducing agent and stabiliser with a **plant extract**, which contains constituents that perform both roles.

Component of a chemical synthesis	Provided in green synthesis by
Reducing agent	Polyphenols, flavonoids, ascorbic acid, reducing sugars in the plant extract
Capping and stabilising agent	Proteins, polysaccharides and phenolics in the same extract

The advantages are that it avoids toxic reagents, uses water as the solvent, works at or near room temperature, and produces particles already coated with biologically compatible molecules.

This is the strongest single scientific point available to you in this entire course, and it should appear in every relevant answer.

Classical **Bhavana** grinds a metal or its compound repeatedly with a plant juice before firing. Modern **green synthesis** uses a plant extract to reduce a metal salt and simultaneously cap the particles formed.

The reagents are the same. The mechanism is the same. The purpose — producing fine, stabilised, biologically compatible metal particles — is the same. The classical process was arrived at empirically, and the modern one by design, but they meet.

State it in exactly this form. Do not overstate it into "Bhavana *is* green synthesis"; say that the two rest on the same chemistry, which is both accurate and impressive.

2.7 Market survey of nano-based herbal formulations

The curriculum prescribes a market survey. This is what to look for.

Point to record	Why
Product name and manufacturer	Basic identification
The claim made — "nano", "nanoparticle", "phytosome", "liposomal"	The exact wording matters
Whether a particle size is stated	Most products claiming "nano" do not state one
The type of carrier claimed	Phytosome, liposome, nanoemulsion, nanosuspension
Composition declared	Whether the active and the carrier are both declared
Licence category	Whether licensed as an ASU drug, a nutraceutical or a cosmetic
Evidence cited	Whether any published study is referenced
Price compared with the conventional product	Whether the claim carries a premium

What the survey will teach you, and it is worth stating in your report. A large proportion of products marketed as "nano" carry no stated particle size, no characterisation data and no published evidence. The word is frequently used as a marketing term rather than a technical one. Learning to distinguish a substantiated nanoformulation from a marketing claim is one of the more valuable things this module can give you.

KEY POINTS TO REMEMBER

- Limitations of Ayurvedic formulations: poor solubility, poor permeability, low bioavailability, large dose, taste, instability, batch variability, non-specific distribution.

- Curcumin is the standard example of a poorly bioavailable Ayurvedic constituent; it behaves as a BCS Class IV compound.
- Bhasma characterisation reports particles in the micrometre and sub-micrometre range, with a nano fraction in many studies, and organic material on the surface.
- **Crystallite size from XRD is not the same as particle size from SEM.** A Bhasma may be nanocrystalline but micro-particulate, because crystallites aggregate.
- Kupipakwa products are formed by sublimation and recondensation, and differ in crystallinity from Puta-prepared Bhasma.
- Every chemical nanoparticle synthesis has three components: precursor, reducing agent, capping or stabilising agent.
- **Rapid nucleation with slow growth gives small, uniform particles.**
- Silver nanoparticle formation is confirmed first by the surface plasmon resonance peak in the UV-Visible spectrum, around 400 to 450 nanometres.
- **Green synthesis and classical Bhavana rest on the same chemistry — a plant extract that both reduces and caps.**

EXAM CORNER — UNIT 2

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Enumerate the limitations of Ayurvedic formulations.
2. Discuss the potential of nanotechnology in overcoming these limitations.
3. Why does curcumin have poor bioavailability, and how can nanotechnology help?
4. Enumerate the instruments used to characterise a Bhasma, with the contribution of each.
5. Differentiate between crystallite size and particle size.
6. Write a short note on the characterisation of Kupipakwa Rasayana.
7. Explain Bhavana in terms of nanotechnology.
8. Enumerate the three components of a chemical nanoparticle synthesis.
9. Explain the role of nucleation and growth in determining particle size.
10. Describe the synthesis of silver nanoparticles by chemical precipitation.
11. How is the formation of silver nanoparticles confirmed?
12. Write a short note on green synthesis and its parallel with Bhavana.

Likely analytical based questions (20 marks)

1. You are asked to establish whether a Bhasma prepared in your department contains nanoparticles. Analyse which methods you would use, what each would and would not prove, and state precisely what conclusion the combined data would support and what it would not.
2. Design a study to determine the effect of the number of Bhavana cycles on the particle characteristics of a Bhasma. Analyse the variables to be controlled, the measurements to be made, and how you would present and interpret the data.

3. Analyse the claim that Bhasma is “ancient nanomedicine”. Present the evidence supporting it, the evidence against it, and state a defensible position with reasons.
4. Compare, analytically, the particle characteristics expected from a Puta-prepared Bhasma and a Kupipakwa Rasayana. Explain the reasons from the method of preparation, and design a study to test your prediction.

REFERENCES USED IN THIS UNIT

NCISM reference number for this unit: **175**

Verified sources used: Ayurvedic Pharmacopoeia of India (reference 156); Protocol for Testing of Ayurvedic, Siddha and Unani Medicines, Department of AYUSH (reference 163); Rasatarangini (reference 1) and Rasa Ratna Samuccaya (references 2 and 3) for the classical processes; Bhasma Vigyana (reference 124); and the instrumental analysis references of Module 5 (14, 15, 17, 18, 20).

UNIT 3
NANOCARRIERS FOR AYURVEDIC FORMULATIONS

SYLLABUS POINTS

3.1 Types of nanocarriers (liposomes, nanoparticles, etc.)
3.2 Advantages and limitations of nanocarriers in Ayurvedic formulations
3.3 Conversion of Taila into an oil-in-water nanoemulsion and its characterisation

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

3.1 Types of nanocarriers

A **nanocarrier** is a nanoscale vehicle that carries a drug, protects it, and delivers it where it is required.

3.1.1 Lipid-based carriers

Carrier	Description	Note
Liposome	Concentric phospholipid bilayers enclosing an aqueous core	Carries both water-soluble drugs in the core and lipid-soluble drugs in the bilayer
Niosome	Similar to a liposome but made of non-ionic surfactants	Cheaper and more stable than liposomes
Phytosome	A phytoconstituent complexed with a phospholipid	**Developed specifically for herbal actives; among the most relevant to Ayurveda**
Ethosome	Vesicle containing a high proportion of alcohol	Penetrates skin deeply; useful for topical Ayurvedic preparations
Transferosome	Highly deformable vesicle with an edge activator	Squeezes through skin pores; used transdermally
Solid Lipid Nanoparticle (SLN)	Solid lipid core stabilised by surfactant	Good stability; controlled release
Nanostructured Lipid Carrier (NLC)	Mixture of solid and liquid lipids	Higher drug loading than SLN; less expulsion on storage
Nanoemulsion	Very fine oil-in-water or water-in-oil emulsion, droplets typically below 200 nanometres	**Directly applicable to Taila preparations**

3.1.2 Polymeric and other carriers

Carrier	Description
Polymeric nanoparticle	Drug entrapped in or attached to a polymer matrix, natural or synthetic
Nanocapsule and nanosphere	Drug in a central cavity, or dispersed throughout the matrix
Dendrimer	Highly branched, precisely structured polymer with cavities for the drug
Polymeric micelle	Self-assembled from block copolymers; carries poorly soluble drugs in the core
Nanosuspension	Pure drug particles reduced to the nanoscale and suspended, with a stabiliser
Nanogel	Cross-linked hydrogel network at the nanoscale
Cyclodextrin inclusion complex	The drug is enclosed within the cyclodextrin cavity; improves solubility and stability
Metallic nanoparticle	Gold, silver, iron oxide; used for delivery, imaging and antimicrobial action
Carbon nanotube	Cylindrical carbon structure; high loading capacity but toxicity concerns
Quantum dot	Semiconductor nanocrystal; used chiefly for imaging

3.1.3 Phytosome — worth knowing in detail

The phytosome deserves particular attention, because it was developed for exactly the problem Ayurveda faces.

Point	Detail
What it is	A complex formed between a phytoconstituent, usually a polyphenol, and a phospholipid such as phosphatidylcholine
The problem it solves	Many phytoconstituents are water-soluble but poorly lipid-soluble, so they cannot cross the lipid cell membrane
How it works	The phospholipid gives the complex a lipophilic exterior, allowing it to cross membranes, while the active is carried within
Result	Substantially improved absorption and bioavailability
Relevance	Directly applicable to curcumin, silymarin, boswellic acids, green tea catechins and many Ayurvedic actives

3.2 Advantages and limitations of nanocarriers

3.2.1 Advantages

1. **Improved solubility and dissolution** of poorly soluble constituents.
2. **Improved bioavailability**, and therefore a smaller effective dose.
3. **Protection of the drug** from light, oxygen, moisture, acid and enzymes.
4. **Controlled and sustained release**, reducing dose frequency.
5. **Targeted delivery** to a tissue or a cell type, reducing effects elsewhere.
6. **Masking of taste and odour.**
7. **Improved skin penetration**, particularly relevant to Lepa and Taila preparations.
8. **Reduced dose-related side effects.**
9. **Ability to carry more than one active** in a single carrier — of particular interest for multi-drug Ayurvedic formulations.

3.2.2 Limitations

1. **Cost.** Materials, equipment and characterisation are expensive.
2. **Complexity of manufacture**, and difficulty in scale-up.
3. **Physical instability** — aggregation, sedimentation, drug leakage on storage.
4. **Extensive characterisation required**, which many Ayurvedic units cannot perform.
5. **Regulatory uncertainty** for nano-modified traditional medicines.
6. **Toxicity concerns** specific to the nanoscale, dealt with in Unit 4.
7. **Loss of the classical identity of the formulation.** A nano-modified classical medicine may no longer be a classical medicine in the legal sense.
8. **Limited clinical evidence** for most nano-herbal products on the market.

> **The seventh limitation is the one that matters most for your speciality, and it is often overlooked.**
>
> Under Section 3(a) of the Drugs and Cosmetics Act, a **classical** Ayurvedic medicine is one manufactured exclusively in accordance with a formula in a First Schedule book. If you convert a classical Taila into a nanoemulsion, you have changed the method of manufacture. **It is then no longer a classical preparation**, and it must be licensed by a different route, with the proof of safety and effectiveness that Rule 158-B requires for that category.
>
> This is not a reason to avoid the work. It is a reason to plan the regulatory pathway before you begin, and to say so in any answer on nanocarriers in Ayurveda. This point connects Module 8 straight back to Module 1.

3.3 Conversion of Taila into an oil-in-water nanoemulsion

This is a syllabus point, a practical and an experiential activity, so learn it thoroughly.

3.3.1 What a nanoemulsion is

A **nanoemulsion** is an emulsion in which the dispersed droplets are typically below about 200 nanometres.

Property	Nanoemulsion	Conventional emulsion
Droplet size	Generally below 200 nm	Micrometre range
Appearance	Transparent or translucent	Milky white and opaque
Stability	Kinetically stable for long periods; resists creaming and sedimentation because Brownian motion overcomes gravity	Creams and separates in time
Surfactant required	Higher proportion	Lower
Energy of preparation	Usually higher	Lower

3.3.2 Components

Component	Function	Example for a Taila nanoemulsion
Oil phase	The medicated Taila itself	Any classical Taila
Aqueous phase	Continuous phase	Purified water
Surfactant	Reduces interfacial tension and stabilises the droplet	A pharmaceutically acceptable non-ionic surfactant, selected by HLB
Co-surfactant	Assists the surfactant; increases flexibility of the interfacial film	Short-chain alcohol or glycol, as permitted
Preservative	Prevents microbial growth in the aqueous phase	As prescribed

Recall the HLB system from Module 1, Unit 3. A high HLB surfactant, roughly 8 to 18, is required for an oil-in-water system. Selecting the surfactant by HLB is the first step in formulation.

3.3.3 Methods of preparation

Category	Method	Principle
High energy	High pressure homogenisation	Forces the mixture through a narrow gap at high pressure
	Microfluidisation	Two streams collide at high velocity
	Ultrasonication	Cavitation disrupts droplets
Low energy	Spontaneous emulsification	Emulsion forms on gentle mixing, driven by the composition
	Phase inversion temperature	The surfactant changes affinity with temperature, inverting the system
	Phase inversion composition	Inversion driven by changing the composition

High energy methods give small droplets and are widely applicable, but require costly equipment and may heat the product, which matters for a Taila containing volatile constituents. **Low energy methods** are gentler and cheaper but require a higher surfactant concentration and work only for certain systems.

3.3.4 Outline of the procedure

1. Select the Taila and record its specification.
2. Select the surfactant and co-surfactant by HLB, and confirm compatibility with the Taila.
3. Construct a pseudo-ternary phase diagram, if the facilities permit, to identify the region in which a nanoemulsion forms.
4. Prepare the oil phase — Taila with the surfactant and co-surfactant, mixed uniformly.
5. Add the aqueous phase gradually with continuous stirring.
6. Apply the chosen high-energy method for the prescribed time and intensity, controlling the temperature.
7. Allow to equilibrate.
8. Characterise the product.
9. Conduct stability studies.

3.3.5 Characterisation of the nanoemulsion

This table is a very likely short answer question.

Parameter	Method	Requirement
Appearance and clarity	Visual	Transparent to translucent
Droplet size	DLS or particle size analyser	Generally below 200 nm
Polydispersity index (PDI)	DLS	A low value indicates a narrow, uniform size distribution
Zeta potential	Zeta analyser	Higher magnitude indicates better stability against aggregation

Parameter	Method	Requirement
Morphology	TEM	Spherical, discrete droplets
pH	pH meter	Compatible with the intended route
Viscosity	Viscometer	As required for the intended use
Refractive index	Refractometer	Indicates isotropy and composition
Conductivity	Conductivity meter	**Confirms which phase is continuous. An oil-in-water system conducts; a water-in-oil system does not**
Drug content	Assay	As specified
In vitro release or diffusion	Franz diffusion cell	Release profile
Thermodynamic stability	Heating-cooling cycles, centrifugation, freeze-thaw cycles	No phase separation, creaming or cracking
Accelerated stability	Storage at specified conditions	Droplet size and drug content unchanged over the period

> **The conductivity test is the neat one to remember.** Water conducts electricity; oil does not. So an oil-in-water nanoemulsion, in which water is the continuous phase, **conducts**. A water-in-oil system does not. A single simple measurement tells you which type you have made. If you are asked how to confirm that you produced an oil-in-water and not a water-in-oil system, this is the answer.

KEY POINTS TO REMEMBER

- Lipid carriers: liposome, niosome, phytosome, ethosome, transferosome, SLN, NLC, nanoemulsion.
- **Phytosome is a phytoconstituent complexed with a phospholipid, developed specifically for herbal actives.**
- Nanoemulsion droplets are generally below 200 nanometres; the product is transparent or translucent and kinetically stable.
- High HLB surfactants, roughly 8 to 18, are used for oil-in-water systems.
- High energy methods: high pressure homogenisation, microfluidisation, ultrasonication. Low energy methods: spontaneous emulsification, phase inversion by temperature or composition.
- Characterisation: droplet size, PDI, zeta potential, TEM, pH, viscosity, refractive index, conductivity, drug content, release, stability.
- **Conductivity distinguishes oil-in-water from water-in-oil** — the oil-in-water system conducts.
- **Converting a classical Taila into a nanoemulsion changes the method of manufacture, so the product is no longer a classical medicine and needs a different licensing route.**

EXAM CORNER — UNIT 3

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Define a nanocarrier and enumerate its types.
2. Write a short note on liposomes and niosomes.
3. Write a short note on phytosomes and their relevance to Ayurveda.
4. Differentiate between SLN and NLC.
5. Write a short note on ethosomes and transferosomes.
6. Enumerate the advantages of nanocarriers in Ayurvedic formulations.
7. Enumerate the limitations of nanocarriers in Ayurvedic formulations.
8. Define a nanoemulsion and differentiate it from a conventional emulsion.
9. Enumerate the components of a nanoemulsion with the function of each.
10. Enumerate the methods of preparation of a nanoemulsion.
11. Enumerate the parameters used to characterise a nanoemulsion.
12. How would you confirm that you have prepared an oil-in-water and not a water-in-oil nanoemulsion?

How to attempt

- For any carrier, use a fixed four-line pattern: what it is made of, what it carries, what advantage it gives, and one Ayurvedic application.
- Whenever you discuss converting a classical preparation, add the regulatory sentence. It shows you are thinking as a practitioner, not only as a formulator.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **179, 180**

Verified sources used: the pharmaceutics and formulation references of Module 1 (references 4, 40, 50, 55, 56, 57, 58); Guidelines for Evaluation of Nanopharmaceuticals in India, DBT, ICMR and CDSCO, 2019; and the instrumental analysis references of Module 5.

UNIT 4
REGULATORY ASPECTS AND SAFETY CONCERNS

SYLLABUS POINTS

4.1 Regulatory frameworks for nanotechnology-based Ayurvedic formulations
4.2 Safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
Yes	Yes	No

4.1 Regulatory frameworks

4.1.1 The Indian position

*The **Guidelines for Evaluation of Nanopharmaceuticals in India** were released on 24 October 2019. They were developed by the Department of Biotechnology under the Ministry of Science and Technology, the Indian Council of Medical Research, and the Central Drugs Standard Control Organisation under the Ministry of Health and Family Welfare, as an outcome of inter-ministerial effort coordinated by DBT.*

The guidelines apply to nanopharmaceuticals both as a finished formulation and as an Active Pharmaceutical Ingredient — whether a new molecule, or an already approved molecule with altered nanoscale dimensions, properties or phenomena associated with the application of nanotechnology, intended for treatment, in vivo diagnosis, mitigation, cure or prevention of disease in human beings. Their stated purpose is to delineate the assessment of quality, safety and efficacy of novel nanoformulations, and to provide a transparent, consistent and predictable regulatory pathway.

The guidelines were developed in line with the provisions of Schedule Y of the Drugs and Cosmetics Rules, 1945 and the Second Schedule of the New Drugs and Clinical Trials Rules, 2019, with specific additional requirements for nanopharmaceuticals.

The point you must understand about the position for Ayurvedic products, and it is examinable.

These guidelines address **nanopharmaceuticals**. There is at present **no separate, dedicated regulatory guideline for nanotechnology-based Ayurvedic, Siddha and Unani formulations**.

So a nano-modified Ayurvedic product sits between two frameworks. As an ASU drug it falls under the Drugs and Cosmetics Act, Chapter IVA, Schedule T and Rule 158-B, administered through the Ministry of AYUSH and the State Licensing Authority. As a product using nanotechnology, the quality, characterisation and safety principles of the 2019 guidelines are the applicable technical standard.

The practical consequence is important: **a nano-modified classical formulation is no longer a classical formulation**, because the method of manufacture has changed. It must be licensed as a patent or proprietary medicine under Rule 158-B, with the proof of safety and effectiveness required for that category, and it should be characterised to the standard the nanopharmaceutical guidelines describe. Say this clearly, and check the current position at the time you write, because this is an area that is developing.

4.1.2 The applicable Indian framework at a glance

Aspect	Applicable provision
Nanopharmaceutical quality, safety and efficacy	Guidelines for Evaluation of Nanopharmaceuticals in India, 2019
Clinical trial and new drug requirements	New Drugs and Clinical Trials Rules, 2019; Schedule Y
ASU drug licensing	Drugs and Cosmetics Act 1940 and Rules 1945, Chapter IVA; **Rule 158-B**
ASU GMP	**Schedule T**
ASU standards	Ayurvedic Pharmacopoeia of India; PCIM&H
Regulator for nanopharmaceuticals	CDSCO
Regulator for ASU drugs	Ministry of AYUSH; State Licensing Authority
Animal studies	CCSEA, as studied in Module 6

4.1.3 International frameworks

Body	Position
USFDA	Guidance on drug products containing nanomaterials; assessed case by case under existing law rather than under a separate nano statute
EMA	Reflection papers and guidance on specific nanomedicine classes such as liposomes and iron-based colloids
ICH	Quality, safety and efficacy guidelines apply; specific nano guidance is developing
OECD	Test guidelines and the Working Party on Manufactured Nanomaterials
ISO	Terminology, measurement and characterisation standards for nanotechnology
WHO	General guidance on traditional and herbal medicine quality

4.1.4 What a regulatory submission for a nanoformulation must contain

Category	Data required
Identity and composition	Full composition including every excipient and the carrier material
Physicochemical characterisation	Particle size and distribution, polydispersity, morphology, surface charge, surface area, crystalline state, solubility, drug loading and entrapment efficiency
Surface characterisation	Surface chemistry, coating, functional groups present
In vitro release	Release profile in relevant media
Stability	Physical and chemical stability, including size and drug content over time; behaviour of the carrier
Manufacturing	Process description, in-process controls, scale-up data, batch-to-batch reproducibility
Non-clinical safety	Toxicity studies appropriate to the route and duration, including nano-specific endpoints
Biodistribution	Where the material goes and how it is cleared
Clinical data	As required by the category of the product
Justification	Why the nanoform is necessary and what advantage it provides over the conventional form

> **The stability requirement is more demanding for a nanoformulation than for a conventional one**, and this is often underestimated. It is not enough to show that the drug content has not fallen. You must also show that the **particle size has not changed**, that the particles have not aggregated, that the carrier has not degraded, and that the drug has not leaked out of the carrier. A nanoformulation can fail on stability while its assay remains perfect.

4.2 Safety concerns and toxicity

4.2.1 Why nanoscale toxicity is a separate question

This is the central idea of the unit, and it must be stated clearly.

A material that is safe in bulk form is not automatically safe at the nanoscale. The same chemical, reduced in size, may behave entirely differently in the body. The reasons are the same ones that make nanomaterials useful.

Property	Benefit	Corresponding risk
Very high surface area	Faster dissolution, better absorption	Greater surface reactivity; more interaction with cells and proteins
Very small size	Crosses membranes and barriers	May cross barriers that should exclude it, including the blood-brain barrier and the placenta
Improved absorption	Higher bioavailability	Higher systemic exposure than expected from the same dose in bulk form
Persistence of the particle	Sustained release	Accumulation in tissues; slow or absent clearance
Surface reactivity	Catalytic and biological activity	Generation of reactive oxygen species; oxidative stress

4.2.2 Routes of exposure

Route	Relevance
Oral	The main route for Ayurvedic medicines
Dermal	Lepa, Taila and topical preparations; intact skin is a good barrier, damaged skin much less so
Inhalation	Sprays, aerosols, Dhupana; **also the main occupational risk during manufacture**
Ocular	Anjana and eye drops
Parenteral	Not applicable to classical ASU drugs, but relevant in research

4.2.3 Mechanisms of nanotoxicity

Mechanism	Explanation
Oxidative stress	Generation of reactive oxygen species at the reactive particle surface; the most widely reported mechanism
Inflammation	Persistent particles provoke a sustained inflammatory response
Membrane damage	Direct physical interaction with the cell membrane
Genotoxicity	Direct interaction with DNA, or indirect damage through reactive oxygen species
Protein corona formation	Proteins adsorb onto the particle surface in biological fluid, changing how the body recognises and handles it
Mitochondrial damage	Impairment of cellular energy production
Accumulation	Deposition in liver, spleen, kidney and lung, with slow clearance
Barrier crossing	Entry into the brain or across the placenta

4.2.4 Factors determining nanotoxicity

Factor	Effect
Size	Smaller particles generally penetrate further and are more reactive
Shape	Fibrous and high-aspect-ratio particles behave differently from spherical ones
Surface charge	Influences cellular uptake and interaction with membranes
Surface coating	A biocompatible coating can substantially reduce toxicity
Chemical composition	The intrinsic toxicity of the material itself
Solubility	A soluble particle releases ions; an insoluble one persists
Dose and duration	As for any toxicant
Aggregation state	An aggregated particle behaves like a larger particle
Route of administration	Determines the tissues exposed

> **The point specific to your speciality, and it deserves a paragraph of its own in any answer.**
>
> Rasashastra administers metals. If a metal preparation contains material at the nanoscale, then the nanotoxicological questions apply to it directly — enhanced absorption, tissue accumulation, and the possibility of crossing barriers that bulk material would not cross.
>
> This cuts **both ways**, and an honest answer says so. The same properties that may explain why a Bhasma is effective at a very small dose also raise questions that the classical texts had no means of asking. The classical safeguards — correct Shodhana, correct Marana, correct Amritikarana, correct Anupana, correct Matra, and administration under a qualified physician — were arrived at empirically and are protective. But **the modern questions of biodistribution, accumulation and long-term safety have largely not been answered by systematic study**, and that is the honest position. It is also, as noted in Module 4, Unit 5, the single largest gap in the evidence base and therefore the most valuable place for a postgraduate to work.

4.2.5 Safety assessment

Level	Studies
In vitro	Cytotoxicity, oxidative stress markers, cellular uptake, genotoxicity, haemolysis
In vivo	Acute, sub-acute and chronic toxicity; biodistribution; accumulation and clearance; target organ toxicity; reproductive and developmental toxicity where relevant
Characterisation before testing	**Essential.** A toxicity result is meaningless unless the material tested was fully characterised, since the properties determine the effect
Standards followed	OECD test guidelines; Schedule Y; CCSEA requirements for animal work

4.2.6 Occupational and environmental safety

Concern	Control measure
Inhalation of airborne nanoparticles during manufacture	Enclosed handling, local exhaust ventilation, appropriate respiratory protection
Skin contact	Gloves and protective clothing
Contamination of the workplace	Wet cleaning rather than dry sweeping; defined and monitored work areas
Waste disposal	Approved route; never into ordinary drainage
Environmental release	Containment and controlled disposal
Training and health surveillance	For all personnel handling nanomaterials

KEY POINTS TO REMEMBER

- **Guidelines for Evaluation of Nanopharmaceuticals in India, 2019**, by DBT, ICMR and CDSCO, aligned with Schedule Y and the New Drugs and Clinical Trials Rules, 2019.
- There is no separate dedicated guideline for nano-based ASU formulations; the ASU framework and the nanopharmaceutical technical standards both apply.
- **A nano-modified classical formulation is no longer classical, and needs licensing under Rule 158-B as a patent or proprietary medicine.**
- Stability for a nanoformulation must show unchanged particle size and no aggregation, not merely unchanged assay.
- **A material safe in bulk is not automatically safe at the nanoscale.**
- Main mechanisms of nanotoxicity: oxidative stress, inflammation, membrane damage, genotoxicity, protein corona, accumulation, barrier crossing.
- Toxicity depends on size, shape, surface charge, coating, composition, solubility, dose, aggregation and route.
- A toxicity result is meaningless unless the material tested was fully characterised first.
- Inhalation is the principal occupational risk during manufacture.

EXAM CORNER — UNIT 4

This unit can be asked as ABQ (20 marks) and SAQ (5 marks). There is no LAQ from this unit.

Likely short answer questions (5 marks)

1. Write a short note on the Guidelines for Evaluation of Nanopharmaceuticals in India.
2. Discuss the regulatory position of a nano-modified Ayurvedic formulation in India.
3. Why is a nano-modified classical formulation no longer a classical medicine?
4. Enumerate the international regulatory bodies concerned with nanomedicine.
5. Enumerate the data required in a regulatory submission for a nanoformulation.
6. Why are stability requirements more demanding for a nanoformulation?
7. Why is a material safe in bulk not automatically safe at the nanoscale?
8. Enumerate the mechanisms of nanotoxicity.
9. Enumerate the factors determining nanotoxicity.
10. Write a short note on the protein corona.
11. Discuss the safety concerns specific to metallic Ayurvedic preparations at the nanoscale.
12. Enumerate the occupational safety measures for handling nanomaterials.

Likely analytical based questions (20 marks)

1. You propose to convert a classical Ayurvedic formulation into a nanoformulation and market it. Analyse the complete regulatory pathway you must follow, the data you must generate, and the safety studies required, and justify each step.

2. Analyse the safety questions raised by the presence of nanoscale material in classical metallic preparations. Present what is known, what is not known, and design a research programme that would answer the open questions.
3. A nanoformulation of an Ayurvedic drug shows greatly increased bioavailability in animal studies. Analyse what this finding implies for both efficacy and safety, and set out the further studies required before it could be given to human beings.

REFERENCES USED IN THIS UNIT

NCISM reference numbers for this unit: **176, 177**

Verified sources used: Guidelines for Evaluation of Nanopharmaceuticals in India, Department of Biotechnology, ICMR and CDSCO, released 24 October 2019; Press Information Bureau, Government of India, 24 October 2019; New Drugs and Clinical Trials Rules, 2019; Schedule Y and Schedule T of the Drugs and Cosmetics Rules, 1945; Rule 158-B; CCSEA guidelines (reference 152); and OECD test guidelines for the safety testing of manufactured nanomaterials.

UNIT 5
RECENT ADVANCES AND FUTURE DIRECTIONS

SYLLABUS POINTS

5.1 Recent advances in nanotechnology-based Ayurvedic formulations
5.2 Future directions and potential applications
5.3 Exploring the future of nanomedicine with Ayurvedic formulations

Exam weightage (Table 6F)

ABQ	SAQ	LAQ
No	Yes	No

5.1 Recent advances

Area	What has been achieved
Phytosome technology	Phytosome preparations of curcumin, silymarin, boswellic acids and green tea catechins, with substantially improved absorption; several are marketed
Nanoemulsions of Taila	Classical medicated oils converted into oil-in-water nanoemulsions, with improved skin penetration and stability
Green synthesis using Ayurvedic plants	Metal nanoparticles synthesised using extracts of Neem, Tulsi, Haridra, Amalaki and many others, with the plant extract acting as both reductant and capping agent
Characterisation of Bhasma	A substantial body of published work using XRD, SEM-EDAX, TEM, TGA, FTIR and particle size analysis, establishing the particle characteristics of many Bhasmas
Solid lipid nanoparticles and NLC	Applied to lipophilic phytoconstituents
Nanosuspensions	Applied to poorly soluble herbal actives
Transdermal nanocarriers	Ethosomes and transferosomes applied to Ayurvedic topical preparations
Nano-based standardisation	Particle size and phase used as additional quality parameters for classical preparations
Combination approaches	Classical preparation followed by modern characterisation and, where justified, nano-modification

5.2 Future directions

Direction	What is needed
Standardised characterisation protocol for Bhasma	An agreed minimum instrumental test set, so that results from different laboratories can be compared
Reference materials	Authenticated reference Bhasma samples against which any laboratory could check its product
Pharmacokinetic studies	Absorption, distribution, metabolism and excretion of Bhasma and of nano-Ayurvedic formulations. **The largest single gap in the field**
Systematic safety data	Acute, sub-acute and chronic toxicity, biodistribution and accumulation, generated to a common protocol
Well-designed clinical trials	Adequately powered, registered on CTRI, with characterised material
Regulatory pathway	A clear, dedicated framework for nano-modified ASU formulations
Targeted delivery	Surface-modified carriers directing an Ayurvedic active to a specific tissue
Theranostics	Carriers that both treat and allow imaging of where the drug has gone
Mechanistic studies	Establishing what the nanoscale fraction of a Bhasma actually does in the body
Green and sustainable synthesis	Scaling up plant-based synthesis for practical manufacture
Computational design	Predicting formulation behaviour before laboratory work begins
Personalised approaches	Relating formulation choice to Prakriti and to individual variation

5.3 Exploring the future of nanomedicine with Ayurvedic formulations

5.3.1 The three ways the two fields can meet

Approach	Description	Example
1. Explaining the classical	Using nanoscience to characterise and understand what classical processes already produce	Characterisation of Bhasma and of Bhavana products
2. Improving the classical	Applying nanocarrier technology to a classical formulation to overcome a known limitation	Curcumin phytosome; Taila nanoemulsion
3. Learning from the classical	Using classical process knowledge to inform modern synthesis	Green synthesis informed by the Bhavana media and methods

> **The third approach is the one least explored and most valuable, and it is worth saying so.**
>
> The first two treat Ayurveda as something to be explained or improved by modern science. The third treats it as a **source of knowledge**. The classical texts record which plant media to use with which metal, in what sequence, and for how many cycles — the accumulated result of centuries of empirical optimisation. A modern nanomaterials chemist choosing a green synthesis route has no comparable body of prior work to draw on.
>
> Testing whether the classically prescribed medium for a given metal outperforms an arbitrary plant extract in a green synthesis is a genuine, feasible, and publishable research question. It is also the kind of question only a person trained in both disciplines can even formulate — which is precisely what this course is for.

5.3.2 What is required of you

The field will be shaped by the work of people trained as you are being trained. Five things are needed.

1. **Rigour.** Characterise fully before you conclude. State conditions completely. Report negative results.
2. **Honesty about the boundary.** Distinguish what is shown from what is proposed. Never let enthusiasm outrun evidence.
3. **Respect for the classical method.** Follow it properly and cite the text. A study on improperly prepared material proves nothing about Ayurveda.
4. **Command of both languages.** Be able to explain a Bhasma to a materials scientist and a nanoparticle to a Vaidya.
5. **Patience.** The questions that matter most — pharmacokinetics, long-term safety, mechanism — take years to answer, and are worth answering.

KEY POINTS TO REMEMBER

- Recent advances: phytosomes, Taila nanoemulsions, green synthesis with Ayurvedic plants, extensive Bhasma characterisation, SLN and NLC, transdermal nanocarriers.

- Future needs: standardised characterisation protocols, reference materials, **pharmacokinetic data**, systematic safety data, well-designed clinical trials, a clear regulatory pathway.
- Pharmacokinetics is the largest single gap in the field.
- Three ways the fields meet: explaining the classical, improving the classical, and **learning from the classical**.
- The third approach is least explored and most valuable, and it requires training in both disciplines.

EXAM CORNER — UNIT 5

This unit is asked as SAQ only (5 marks). No LAQ, no ABQ.

Likely short answer questions (5 marks)

1. Enumerate the recent advances in nanotechnology-based Ayurvedic formulations.
2. Write a short note on phytosome technology and its application to Ayurvedic actives.
3. Write a short note on green synthesis using Ayurvedic plants.
4. Enumerate the future directions in nano-Ayurveda research.
5. What is the largest gap in the present evidence base, and why does it matter?
6. Write a short note on targeted drug delivery in the Ayurvedic context.
7. Write a short note on theranostics.
8. Describe the three ways in which nanoscience and Ayurveda can meet.
9. Why is a standardised characterisation protocol for Bhasma needed?
10. Write a short note on reference materials for Ayurvedic preparations.
11. What would a well-designed clinical trial of a nano-Ayurvedic formulation require?
12. Discuss the prospects of integrating nanomedicine with Ayurvedic formulations.

REFERENCES USED IN THIS UNIT

NCISM reference number for this unit: **178**

Verified sources used: Guidelines for Evaluation of Nanopharmaceuticals in India, DBT, ICMR and CDSCO, 2019; CCRAS Guideline of Drug Development (reference 49) and Guideline of Clinical Evaluation (reference 154); Ayurvedic Pharmacopoeia of India (reference 156); and the published characterisation literature on Bhasma, read alongside the instrumental methods of Module 5.

MODULE 8
PRACTICAL TRAINING ACTIVITIES

Module 8 carries **20 hours of practical training**, over five activities.

No	Practical name	Hours
8.1	Demonstration of the measurement of micro-particles using relevant equipment	4
8.2	Demonstration of particle size of Bhasma prepared with different numbers of Puta	4
8.3	Demonstration of particle size of various Kupipakwa Rasayana samples	4
8.4	Synthesis of silver nanoparticles by the chemical precipitation method	4
8.5	Conversion of Taila into an oil-in-water nanoemulsion and its characterisation	4
	Total	**20**

Practical Training 8.1 — Measurement of micro-particles

Aim. To measure the particle size of Ayurvedic preparations using the equipment available.

Requirements. Trinocular microscope with calibrated ocular micrometer, stage micrometer, particle size analyser if available, sieves of graded mesh, samples of Churna and Bhasma, slides, dispersing medium.

Procedure

1. Calibrate the ocular micrometer for every objective, as learned in Module 5, Unit 5. Record each calibration factor.
2. Prepare a dispersed sample on a slide, using a suitable medium so that the particles are separated and not clumped.
3. Examine under the microscope. Measure at least one hundred particles in a systematic manner, moving across the field in a regular pattern.
4. Record every measurement in ocular divisions and convert to micrometres.
5. Prepare a frequency distribution and plot a histogram.
6. Calculate the mean, the median and the range.
7. Perform **sieve analysis** on a larger quantity of the same sample and record the percentage retained on each sieve.
8. Where a particle size analyser is available, run the same sample and compare its result with your microscopic result.

9. Explain any difference between the two.

Observation to record. Calibration factors, the raw measurement table, the histogram, the calculated statistics, the sieve analysis, and the comparison.

Precautions. Disperse the sample properly — an aggregate measured as one particle gives a falsely large result. Measure a sufficient number of particles; fewer than a hundred is not representative. Wear a mask when handling fine powders.

Practical Training 8.2 — Particle size of Bhasma with different numbers of Puta

Aim. To demonstrate the effect of the number of Putas on the particle size of a Bhasma.

Procedure

1. Obtain or retain samples of the same Bhasma after different numbers of Putas — for example after 3, 7, 11, 15 and 21 Putas — from the Marana carried out in Module 4.
2. Record the colour and appearance of each sample.
3. Measure the particle size of each by the methods of Practical 8.1.
4. Perform the classical Bhasma Pariksha on each and record the results.
5. Where available, obtain XRD for each and record the phases and the crystallite size.
6. **Plot particle size against the number of Putas.**
7. **Plot the Bhasma Pariksha results against the number of Putas**, marking at which Puta each test was first satisfied.
8. Compare the two plots and interpret.

Observation table to draw

Puta number	Colour	Mean particle size	Range	Rekhapurnata	Varitara	Nishchandratva	Apunarbhava	Crystallite size (XRD)

Result. Record the relationship established between the number of Putas, the particle size and the satisfaction of the classical tests.

> **This practical is the most valuable in the module, and it should be done carefully.** Very few published studies plot particle size against Puta number for the same batch. If your data are properly collected and the conditions fully recorded, this is publishable work — and it directly tests the classical instruction about the number of Putas.

Practical Training 8.3 — Particle size of Kupipakwa Rasayana samples

Aim. To measure and compare the particle characteristics of Kupipakwa preparations.

Procedure

1. Obtain samples of different Kupipakwa Rasayana preparations, and if possible the same product from different sources.
2. Record the colour, appearance and site of collection for each.
3. Triturate each to a uniform powder under identical conditions, recording the trituration time.
4. Measure the particle size of each by the methods of Practical 8.1.
5. Where available, obtain XRD for each and record the phase and the crystallite size.
6. Compare the particle characteristics of the Kupipakwa products with those of a Puta-prepared Bhasma from Practical 8.2.
7. Explain the differences from the method of preparation.

Observation to record. The comparison table across all samples, and a written explanation relating the particle characteristics to the method by which each was prepared.

Precautions. Trituration time strongly affects the measured particle size of a Kupipakwa product, because it is collected as a mass. **Triturate every sample for exactly the same time**, or the comparison is invalid. Handle mercurial preparations only under supervision, with a mask, and account for all material.

Practical Training 8.4 — Synthesis of silver nanoparticles

Aim. To synthesise silver nanoparticles by chemical precipitation and to confirm their formation.

Requirements. Silver salt precursor, reducing agent and stabiliser as prescribed by the departmental method, volumetric glassware, magnetic stirrer with hot plate, centrifuge, UV-Visible spectrophotometer, amber storage bottles, personal protective equipment.

Procedure

1. Read the departmental method fully and write out the procedure before beginning.
2. Prepare all solutions freshly, using purified water, and record every concentration and volume.
3. Set up the reaction with continuous vigorous stirring at the prescribed temperature.
4. Add the reducing agent at the prescribed rate.
5. **Observe and record the colour change and the time at which it appears.**
6. Continue stirring for the prescribed period.
7. Record a UV-Visible spectrum of the product and **identify the surface plasmon resonance peak**.
8. Purify by centrifugation and washing as prescribed.
9. Redisperse and store in an amber bottle, protected from light.
10. Characterise further with the instruments available — DLS, zeta potential, TEM, XRD, FTIR.

11. Repeat the synthesis with one variable changed — for example the concentration of the reducing agent — and compare the resulting spectra and particle sizes.

Observation to record. The full method as performed, the time and nature of the colour change, the UV-Visible spectrum with the peak wavelength marked, and all characterisation data. For the repeat synthesis, the variable changed and its effect.

Precautions. Wear gloves, a laboratory coat and eye protection throughout. **Silver salts stain skin and clothing permanently and are irritant.** Some reducing agents used in such syntheses are hazardous — use only the agent, concentration and quantity prescribed by your teacher, and follow the departmental protocol exactly. Work in a fume hood where directed. Never pipette by mouth. **Dispose of all silver-containing waste by the approved route, never into the sink.**

Practical Training 8.5 — Conversion of Taila into a nanoemulsion

Aim. To convert a classical Taila into an oil-in-water nanoemulsion and to characterise it.

Requirements. The Taila, a pharmaceutically acceptable non-ionic surfactant, a co-surfactant, purified water, magnetic stirrer, high-shear homogeniser or probe ultrasonicator, DLS or particle size analyser, pH meter, viscometer, refractometer, conductivity meter, centrifuge.

Procedure

1. Record the specification of the Taila used.
2. Select the surfactant by HLB, for an oil-in-water system. Record the calculation.
3. Determine the surfactant to co-surfactant ratio by trial, and construct a pseudo-ternary phase diagram if facilities permit.
4. Prepare the oil phase by mixing the Taila with the surfactant and co-surfactant.
5. Add the aqueous phase gradually with continuous stirring.
6. Apply the high-energy method for the prescribed time and amplitude, controlling the temperature with an ice bath where the equipment generates heat.
7. Allow the product to equilibrate.
8. **Characterise:** appearance and clarity; droplet size and polydispersity index by DLS; zeta potential; pH; viscosity; refractive index; **conductivity, to confirm the oil-in-water type**; and drug content.
9. Perform **thermodynamic stability testing** — heating and cooling cycles, centrifugation, and freeze-thaw cycles. Record any phase separation.
10. Compare the nanoemulsion with the original Taila for appearance, spreadability and, where facilities permit, in vitro release.

Observation table to draw

Parameter	Original Taila	Nanoemulsion	Method used	Remark

Result. Record whether a stable oil-in-water nanoemulsion was obtained, its droplet size and polydispersity index, and the outcome of the stability tests.

Precautions. The probe ultrasonicator generates considerable heat — use an ice bath and work in intervals, or the volatile constituents of the Taila will be lost. Wear ear protection when using an ultrasonicator. Do not exceed the prescribed surfactant concentration; excess surfactant is irritant. Record the temperature reached at every stage.

MODULE 8
EXPERIENTIAL LEARNING ACTIVITIES

Module 8 carries **30 hours** under experiential learning, including modular assessment. Six activities are prescribed.

No	Experiential learning activity	Hours
8.1	Measurement of particle size of Bhasma prepared with different numbers of Puta	4
8.2	Measurement of particle size of various Kupipakwa Rasayana samples	4
8.3	Review of articles regarding the green synthesis of metallic nanoparticles	2
8.4	Experiencing the impact of Bhavana	4
8.5	Market survey of nano-based Ayurvedic formulations	6
8.6	Integration of nanomedicine with Ayurvedic formulations and its prospects	6
	Total	**26**

Experiential Learning 8.1 — Particle size against number of Puta

What you have to do. Turn Practical 8.2 into a complete piece of research work with a written conclusion.

Method. Take the data from Practical 8.2 and extend it. Ensure every condition was controlled and recorded. Plot the data properly. Then interpret.

Questions your report must answer

- At which Puta did the particle size stop falling appreciably?
- At which Puta was each classical test first satisfied?
- Did the classical tests become satisfied before, at, or after the point where particle size stabilised?
- How does the number of Putas you required compare with the number prescribed in the text?
- What conditions in your Puta could account for any difference?
- Does your data support the classical instruction, and in what respect?

Deliverable. A full report written in the format of a research paper — introduction, materials and methods with every condition stated, results with tables and graphs, discussion, and conclusion.

Experiential Learning 8.2 — Particle size of Kupipakwa samples

What you have to do. Extend Practical 8.3 into a comparative study across preparation methods.

Questions your report must answer

- Do Kupipakwa products differ systematically from Puta-prepared Bhasma in particle size or crystallinity?
- Do samples of the same product from different sources agree?
- What does any difference tell you about the control of the process in each source?
- Which method gives the more uniform product, and why would you expect that from the mechanism?

Deliverable. A comparative report with the data and a reasoned conclusion.

Experiential Learning 8.3 — Review of green synthesis articles

What you have to do. Read and critically analyse published work on the green synthesis of metallic nanoparticles.

Method. Obtain at least six papers, preferably including some that used plants of Ayurvedic importance. For each, record the plant and part used, the extract preparation, the metal salt, the conditions, the characterisation performed, and the particle size and shape obtained.

Then answer these questions

- Which plant constituents were proposed as the reducing agent and as the capping agent?
- Was the extract characterised, or merely described as “plant extract”?
- Was a voucher specimen deposited? Recall Module 6, Unit 4.
- How complete was the characterisation? Was zeta potential reported? Was FTIR used to show the capping molecules?
- **Which of these papers, if any, drew a connection with classical Bhavana? Was that connection stated accurately or overstated?**
- What would you do differently?

Deliverable. The comparison table, your critique, and a written statement of the standards you would apply to your own green synthesis work.

Experiential Learning 8.4 — Experiencing the impact of Bhavana

What you have to do. Establish, by your own measurement, what Bhavana actually does.

Method. Take one Dravya. Divide it into portions. Subject the portions to different numbers of Bhavana cycles — for example 0, 1, 3, 5 and 7 — with the same medium, the same quantity, and the same trituration time per cycle. Dry each identically.

Measure for every portion

- Particle size, by microscopy and by particle size analyser
- Appearance and colour
- Loss on drying
- FTIR, to follow the organic material accumulating on the surface
- Where available, XRD and SEM

Then answer

- How does particle size change with the number of Bhavana cycles?
- Does the change level off, and at what point?
- Does the FTIR show increasing organic material with each cycle?
- What does this tell you about the classical instruction to give a specified number of Bhavanas?
- Does the medium matter? Design the experiment that would answer that question.

> **This is the single most directly researchable activity in the course.** It requires no expensive equipment beyond what your department already uses, it tests a specific classical instruction, and almost nobody has published it properly. If you do this work carefully and record every condition, you will have something worth publishing before your dissertation is finished.

Deliverable. A full research-format report with the data, the graphs, and a conclusion on what Bhavana does and how many cycles are needed.

Experiential Learning 8.5 — Market survey of nano-based Ayurvedic formulations

What you have to do. Survey the market, and learn to distinguish a substantiated nanoformulation from a marketing claim.

Method. Survey at least twenty products claiming a nano, phytosome, liposomal or similar technology. Visit retail outlets and examine online listings. Record the details listed in Unit 2, section 2.7 for each.

Then analyse

- How many stated a particle size?
- How many named the carrier technology precisely?
- How many cited any published evidence?
- Under which licence category was each marketed — ASU drug, nutraceutical, or cosmetic?
- What price premium did the claim carry?
- **How many of the claims could you actually verify?**

Deliverable. The survey table for all twenty products, the analysis, and a written conclusion on the state of the market with recommendations on what a consumer, a prescriber and a regulator should each look for.

Experiential Learning 8.6 — Integration of nanomedicine with Ayurvedic formulations

What you have to do. The final activity of the course. Design a complete research proposal that integrates nanoscience with an Ayurvedic formulation.

Your proposal must contain

Section	Content
Title	Specific and precise
Background	What is known, with references; the classical position and the modern position
Rationale	Why this work is needed
The gap	What specifically is not known
Aim and objectives	Stated so that each can be tested
Materials	The formulation, its classical reference, its source and authentication
Methods	Preparation, characterisation, and every condition specified
Characterisation plan	Which instruments, in what order, and what each will show
Safety assessment	What studies, under what approval
Regulatory pathway	Under which rule the product would be licensed
Statistical plan	Sample size and analysis
Ethical clearance	IAEC or IEC as applicable
Timeline and budget	Realistic
Expected outcome	And what a negative result would mean
Limitations	Stated honestly

Note the two sections most students leave out, and which carry the most credit.

The **regulatory pathway** — because a formulation that cannot be licensed cannot reach a patient, however elegant the science.

And **what a negative result would mean** — because a study designed so that only a positive result is interpretable is not a study at all. Being willing to state in advance what would count as your idea failing is the mark of a scientist.

This activity is designed to be the seed of your dissertation. Treat it as such.

Deliverable. The complete written proposal, presented to the department and defended.

MODULE 8
VIVA POINTERS AND QUESTION BANK

Viva pointers

On principles

- What is a nanometre? What is the nanoscale?
- Who gave the lecture *There's Plenty of Room at the Bottom*, and when?
- Who coined the term nanotechnology?
- Why do properties change at the nanoscale? Give both reasons.
- Differentiate top-down from bottom-up. Which is Bhavana?
- Why is gold red at the nanoscale?

On measurement

- Which instrument gives the size of a single nanoparticle?
- Why is the DLS size larger than the TEM size?
- What is zeta potential, and what does it predict?
- What is the difference between crystallite size and particle size?
- Calibrate this ocular micrometer and measure this particle.

On Ayurveda and nano

- Is Bhasma a nanomedicine? Defend your answer.
- What does the published characterisation of Bhasma actually show?
- Explain Bhavana in nanotechnology terms.
- What is green synthesis? What is its parallel in Rasashastra?
- Is that parallel exact? Where does it break down?
- Which Ayurvedic constituent is the standard example of poor bioavailability?

On carriers and formulation

- What is a phytosome? Why does it work?
- Differentiate SLN from NLC.
- What droplet size defines a nanoemulsion?
- How would you confirm you made an oil-in-water and not a water-in-oil system?
- What is the polydispersity index?

On safety and regulation

- Which guidelines govern nanopharmaceuticals in India, and who issued them?
- Is a nano-modified classical Taila still a classical medicine? Why not?
- Why is a material safe in bulk not automatically safe at the nanoscale?
- Name three mechanisms of nanotoxicity.
- What is the protein corona?
- What is the largest gap in the evidence base for Bhasma?

Module 8 question bank

Analytical Based Questions (20 marks) — from Units 2 and 4

1. Establish whether a Bhasma prepared in your department contains nanoparticles. Analyse which methods you would use, what each would and would not prove, and state precisely what conclusion the combined data would support.
2. Design a study to determine the effect of the number of Bhavana cycles on the particle characteristics of a Bhasma, with the variables to be controlled and the interpretation.
3. Analyse the claim that Bhasma is “ancient nanomedicine”. Present the evidence for and against, and state a defensible position.
4. Compare analytically the particle characteristics expected from a Puta-prepared Bhasma and a Kupipakwa Rasayana, and design a study to test your prediction.
5. You propose to convert a classical formulation into a nanoformulation and market it. Analyse the complete regulatory pathway, the data required, and the safety studies.
6. Analyse the safety questions raised by nanoscale material in classical metallic preparations, and design a research programme to answer the open questions.
7. A nanoformulation of an Ayurvedic drug shows greatly increased bioavailability in animals. Analyse the implications for efficacy and safety and the further studies required.

Long Answer Questions (10 marks) — from Unit 1 only

1. Define nanotechnology. Describe its history, basic principles and applications.
2. Explain in detail why the properties of a material change at the nanoscale, with examples.
3. Describe the top-down and bottom-up approaches, and correlate them with classical Ayurvedic processes.
4. Describe the instruments used for the characterisation of nanomaterials and the information each provides.
5. Discuss the nanoscale and the principles of nanoparticle stabilisation, and relate them to Bhavana.

Short Answer Questions (5 marks) — from all five units

From Unit 1

1. Define nanotechnology, nanoscience, nanomaterial and nanoparticle.
2. Write a short note on the history of nanotechnology.
3. Explain the significance of the surface area to volume ratio.
4. Differentiate top-down from bottom-up, with examples.
5. Classify nanomaterials on the basis of dimension.
6. Differentiate the size given by DLS from that given by TEM.
7. Write a short note on zeta potential and its significance.

From Unit 2

1. Enumerate the limitations of Ayurvedic formulations.
2. Why does curcumin have poor bioavailability, and how can nanotechnology help?
3. Differentiate between crystallite size and particle size.
4. Explain Bhavana in terms of nanotechnology.
5. Enumerate the three components of a chemical nanoparticle synthesis.
6. Describe the synthesis of silver nanoparticles and how formation is confirmed.
7. Write a short note on green synthesis and its parallel with Bhavana.

From Unit 3

1. Define a nanocarrier and enumerate its types.
2. Write a short note on phytosomes and their relevance to Ayurveda.
3. Differentiate between SLN and NLC.
4. Enumerate the advantages and limitations of nanocarriers in Ayurvedic formulations.
5. Define a nanoemulsion and differentiate it from a conventional emulsion.
6. Enumerate the parameters used to characterise a nanoemulsion.
7. How would you confirm an oil-in-water rather than a water-in-oil system?

From Unit 4

1. Write a short note on the Guidelines for Evaluation of Nanopharmaceuticals in India.
2. Why is a nano-modified classical formulation no longer a classical medicine?

3. Why are stability requirements more demanding for a nanoformulation?
4. Why is a material safe in bulk not automatically safe at the nanoscale?
5. Enumerate the mechanisms of nanotoxicity.
6. Enumerate the factors determining nanotoxicity.
7. Write a short note on the protein corona.

From Unit 5

1. Enumerate the recent advances in nanotechnology-based Ayurvedic formulations.
2. Enumerate the future directions in nano-Ayurveda research.
3. What is the largest gap in the present evidence base, and why does it matter?
4. Describe the three ways in which nanoscience and Ayurveda can meet.

> **Note on the marks range.** Table 6F places Module 8 in the marks range of **5 to 20**. Units 2 and 4 are ABQ-eligible; Unit 1 alone carries the long answer; Units 3 and 5 carry short answers only. Unit 2 is the most important in the module, because it is where your own subject and this one actually meet.

A CLOSING WORD

You have now come to the end of **AYPG-AB-RB — Applied Basics of Rasashastra and Bhaishajya Kalpana**: eight modules, forty-two units, four hundred and eighty notional hours.

Look back at what the course has done. It began with pharmaceutics and ended with nanoscience, and in between it asked you to learn databases, chemistry, metallurgy, instrumentation, animal ethics and quality control. At first sight these look like eight separate subjects bolted together.

They are not. Read the course backwards and you will see it. Module 8 asks what happens at the nanoscale. Module 5 gives you the instruments to see it. Module 4 gives you the process that produces it. Module 3 gives you the chemistry that explains it. Module 7 gives you the standards to control it. Module 1 gives you the pharmaceutics and the law that govern it. Module 2 gives you the sources to verify it, and Module 6 gives you the ethics to test it responsibly.

The subject of this course is a single question, asked in eight ways: **what exactly are we making, and how do we know?**

Our Acharyas answered it with the tools they had, and answered it well. Rekhapurnata is a particle size test. Varitara is a surface property test. Apunarbhava is a test for free metal. Tantuvat is a test of concentration. The Musha is a refractory formulation. Mitra Panchaka is a smelting charge. They were not approximating a science they did not have; they were doing science with what they had.

Your generation has been given instruments they did not have, and a language in which to explain what they did. That is a privilege and it carries an obligation. Use it to **verify**, not merely to decorate. When the evidence supports the classical position, say so with confidence and with data. When it does not yet, say that too, and then go and find out.

Prepare well. Answer honestly. And when you enter the Rasashala, remember that a great deal of careful work over a great many centuries has been handed to you to carry a little further.

PRINCIPAL SOURCES AND REFERENCES

Every reference cited in this book is taken from the official NCISM reference list supplied for the course **AYPG-AB-RB — Applied Basics of Rasashastra and Bhaishajya Kalpana**. The reference numbers used at the end of each unit are the same numbers used in the curriculum.

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GLOSSARY OF AYURVEDIC TERMS

This glossary covers the Sanskrit and Ayurvedic terms used in this book. Terms are given in plain Roman letters, without diacritical marks, in the same form used throughout the text. Where a term carries more than one sense, the sense used in this book is given first.

A

Term	Meaning
Agni	Fire; also the digestive and metabolic capacity
Agni Mana	Measurement of the intensity of fire
Ahimsa	Non-injury to living beings
Amla	Sour; an acidic substance
Amlavetasa	A sour drug used as an acidic medium in Shodhana
Amritikarana	A Samskara performed after Marana to remove residual harmful quality and stabilise the Bhasma
Anjana	A preparation applied to the eye
Anupana	The vehicle or after-drink taken with a medicine
Apunarbhava	Test in which a Bhasma, heated with reducing agents, does not revert to the metal
Arishta	A fermented preparation made from a decoction
Arka	A distillate
Asava	A fermented preparation made from cold water and powdered drugs
Avaleha	A semi-solid confection; also called Leha or Lehya
Avami	Not causing nausea or vomiting
Avapa	Adding a substance into a molten mass

B

Term	Meaning
Basti	Medicated enema
Bhallataka	Semecarpus anacardium; an Upavisha requiring Shodhana
Bharjana	Frying
Bhasma	The fine, lustreless powder obtained by Marana of a metal or mineral
Bhasma Pariksha	The set of classical tests confirming a properly prepared Bhasma
Bhavana	Wet trituration with a prescribed liquid, repeated and dried between cycles
Bhudhara Puta	A Puta in which the vessel is placed in a pit in the ground
Bhaishajya Kalpana	The Ayurvedic science of dosage form design

C

Term	Meaning
Chakrika	Small flat cakes formed from the mixture before Puta; also Vataka
Chaturvidha Rasayana	The four principal Rasa preparations — Kharaliya, Parpati, Kupipakwa and Pottali
Churna	A fine powder of one or more dried drugs
Churnikarana	Reduction of a substance to powder

D

Term	Meaning
Dagdha Paka	Burnt stage of Sneha Paka; discarded
Dhalana	Pouring a molten or heated substance into a liquid; see Nirvapa
Dhanyabhraka	Shodhita Abhraka rubbed with paddy in a cloth bundle
Dhatu	Metal; also the seven bodily tissues
Dhavana	Repeated washing, generally with a stream of water
Dhupana	Fumigation
Dolayantra	An apparatus in which a drug tied in a cloth is suspended in a boiling liquid
Dosha	An impurity or fault; also the three bodily humours
Drava Mana	Measurement of liquids
Dravaka Gana	Substances that promote melting and flow; flux
Dravya	A substance or drug
Druti	Liquefaction of a metal or mineral

G

Term	Meaning
Gaja Puta	The standard Puta, using a pit of about one Hasta each way
Gandha	Odour
Gandhaka	Sulphur
Gata Rasatva	Loss of the original taste of the metal in a finished Bhasma
Gairika	Red ochre; haematite
Ghrita	Clarified butter; a medicated ghee preparation
Godugdha	Cow’s milk
Gomutra	Cow’s urine
Gostani Musha	A crucible shaped like a cow’s teat, used for Satvapatana
Grahya Lakshana	The characters of a genuine and acceptable drug
Guda	Jaggery
Guggulu	An oleo-gum-resin; also the dosage form prepared from it

Term	Meaning
Gunja	Abrus precatorius; also the smallest unit of weight, about 125 mg
Gutika	A pill

H

Term	Meaning
Haratala	Orpiment, arsenic trisulphide
Hasta	A unit of length, from elbow to fingertip, about 18 to 24 inches
Hima	A cold infusion
Hingula	Cinnabar, mercuric sulphide

J to K

Term	Meaning
Jarana	Assimilation of a metal or mineral into Parada
Kajjali	The black powder formed by trituration of Parada with Gandhaka, without fire
Kala Mana	Measurement of time
Kalka	A paste of a drug
Kalpana	A dosage form or preparation
Kanji	A fermented sour liquid used as an acidic medium
Kanta Loha	Magnetite; magnetic iron ore
Kapota Puta	A small Puta of low temperature
Karsha	A unit of weight, about 12 g; also called Tola
Kasisa	Melanterite, ferrous sulphate
Khalva Yantra	The mortar and pestle used for trituration
Khara Paka	The hard stage of Sneha Paka, suitable for external use
Kharaliya Rasayana	A Rasa preparation made by trituration alone, without fire
Kshara	An alkaline preparation obtained from the ash of a drug
Koshthi Yantra	The furnace used for Satvapatana
Kukkuta Puta	A Puta smaller than Varaha Puta

Term	Meaning
Kupipakwa Rasayana	A preparation made by controlled heating in a glass bottle in a sand bath
Kulattha	Horse gram; its decoction is used in Shodhana
Kwatha	A decoction; also called Kashaya

L to M

Term	Meaning
Lavana	Salt
Lepa	An external application; a paste applied to the skin
Loha	Iron; also a general term for metals
Madhu	Honey
Madhyama Paka	The middle stage of Sneha Paka, suitable for oral, Basti and Nasya use
Mana	Measurement
Manahshila	Realgar, an arsenic sulphide
Mandura	Purified iron rust
Mardana	Trituration
Marana	The process converting a metal or mineral into a Bhasma
Matra	Dose
Makshika	Chalcopyrite, copper iron sulphide
Mitra Panchaka	The five substances added to assist Satvapatana
Mridu Paka	The soft stage of Sneha Paka, suitable for Nasya
Musha	The crucible of Rasashastra

N to P

Term	Meaning
Naga	Lead
Namarupa Vijnana	The science of the names and forms of drugs; botanical identification
Nasya	Administration of medicine through the nose
Nirdhuma	Absence of fumes on heating a Bhasma

Term	Meaning
Nirvapa	Heating a metal to red heat and quenching it in a prescribed liquid
Nischandratva	Absence of lustre in a finished Bhasma
Nishwadatva	Tastelessness of a finished Bhasma
Niruttha	Test in which a Bhasma heated with a prescribed metal causes no increase in its weight
Paka	Cooking or processing by heat
Panaka	A sweetened liquid preparation
Parada	Mercury
Parpati	A thin flake-like preparation made by melting Kajjali and pouring it out
Phanta	A hot infusion
Pishti	A preparation made by trituration with a liquid, generally without fire
Pottali	A preparation made by heating ingredients tied in a bundle
Prakshalana	Washing
Prakshepaka Dravya	Powders added at the end of a preparation for taste and effect
Prastha	A unit of weight, about 384 g
Puta	The classical system of controlled heating

R

Term	Meaning
Rasa	Taste; also mercury; also the essence of a substance
Rasakriya	A semi-solid preparation made by evaporating a decoction
Rasaka	Calamine; a zinc ore
Rasashastra	The Ayurvedic science of metals, minerals and their processing
Rasa Sindura	A Kupipakwa preparation; the red form of mercuric sulphide
Rasaushadhi	A medicine prepared from Rasa Dravyas
Rekhapurnata	Test in which a Bhasma fills the lines of the fingers when rubbed
Rajata	Silver

S

Term	Meaning
Sandhana Kalpana	Preparations made by fermentation — Asava and Arishta
Sandhi Bandhana	Sealing the junction of the two Sharavas with clay-smeared cloth
Samanya Shodhana	General purification common to a group of Dravyas
Samskara	The imparting of new qualities to a substance by processing
Satva	The extracted essence of a plant or the metal extracted from an ore
Satvapatana	The classical process of extracting the metallic essence from an ore
Saviryata Avadhi	The period for which a preparation retains its potency; shelf life
Sharava Samputa	Two shallow earthen saucers, one inverted over the other
Sharkara	Sugar; a sugar-based liquid preparation
Shodhana	Purification, detoxification, modification and potentiation of a Dravya
Siddhi Lakshana	The signs indicating that a preparation is correctly completed
Slakshnatva	Smoothness and fineness of a Bhasma
Sneha Kalpana	Preparation of medicated Ghrita and Taila
Sparsha	Touch
Sphatika	Alum
Swangasheeta	Self-cooling; allowing a Puta to cool without being opened
Swarasa	Fresh expressed juice of a drug
Swedana	Heating a drug suspended in a boiling liquid medium
Swarna	Gold

T to Y

Term	Meaning
Taila	Oil; a medicated oil preparation
Takra	Buttermilk
Tamra	Copper

Term	Meaning
Tankana	Borax
Tantuvat	Thread formation; a Siddhi Lakshana of Avaleha
Tila Taila	Sesame oil
Tola	A unit of weight, about 12 g; see Karsha
Trikatu	The combination of Shunthi, Maricha and Pippali
Tuttha	Chalcanthite, copper sulphate
Upadhatu	Secondary metals and their products
Uparasa	A group of minerals in Rasashastra
Upavisha	Semi-poisonous drugs requiring Shodhana
Vaikranta	A Maharasa whose mineral identity is disputed
Vanga	Tin
Varaha Puta	A Puta smaller than Gaja Puta
Varitara	Test in which a Bhasma floats on still water
Varna	Colour
Varti	A wick or suppository-like dosage form
Vatsanabha	Aconitum species; a Visha requiring Shodhana
Vati	A tablet or pill
Visha	Poisonous drugs requiring Shodhana
Vitasti	A unit of length, a span, about half a Hasta
Yantra	An apparatus or instrument
Yashada	Zinc
Yoga	A formulation
Yogavahi	A substance that carries and enhances the action of other drugs

ABOUT THE AUTHOR

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Alongside his academic work he has been in clinical practice for more than a decade at Suvarna Ayurveda Clinic, Vadodara.

His teaching and research interests lie in the pharmaceutical processing of metals and minerals, the standardisation of classical Ayurvedic preparations, and the application of modern analytical methods to the verification of classical pharmaceutical claims. This book grew out of his teaching of the NCISM postgraduate curriculum, and is his first full-length work.

Correspondence and corrections. This is a first edition. The author would welcome corrections, comments and suggestions from teachers and scholars, so that later editions may be improved. Errors of fact, outdated regulatory references and omissions are of particular interest, and every such report will be acknowledged.

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